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Matej Brešar

# Introduction to Noncommutative Algebra

*Second Edition*

Springer

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# Introduction to Noncommutative Algebra

Second Edition

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*In memory of my father*

# Preface to the Second Edition

During the process of writing of the first edition—now more than a decade ago—I developed a deep appreciation for classical theory. Over time, I began incorporating it, both consciously and unconsciously, into my research. This gradually altered my view of the significance of certain topics, and I came to regret having omitted them from the book. This was the primary motivation behind preparing a second edition.

Among other enhancements, this edition includes a broader treatment of division algebras, Wedderburn's principal theorem and related topics, and a more comprehensive coverage of the Wedderburn-Artin theory. The basic Wedderburn's structure theory is now presented in three different ways: in Chap. 2, in its most classical form without making use of modules; in Chap. 3, via modules and chain conditions; and in Chap. 4, through the Jacobson density theorem. This provides various options for using the book as a textbook for a course.

I have also revised several proofs, added new exercises, rearranged parts of the material (notably, Chaps. 4 and 5 have been swapped), and made numerous small improvements throughout. The second edition is now somewhat longer. I hope and trust that it will also prove more useful.

Ljubljana and Maribor, Slovenia — Matej Brešar
May 2025

# Preface to the First Edition

The purpose of this book is to give a gentle introduction to noncommutative rings and algebras that requires fewer prerequisites than most other books on the subject. It is based on a series of lectures given to master's students at the University of Ljubljana and is intended to serve as a first reading on noncommutative algebra for beginning graduate and advanced undergraduate students. The first chapters, in which some results of historic importance are derived by rudimentary tools, can have a wider audience. For example, they can be used in a course for future high-school teachers, or for an undergraduate seminar. Mathematicians working in areas that only have some interactions with noncommutative algebra may also benefit from this book in which important, classical themes are presented in a simple manner, avoiding excessive generality.

The necessary background needed to follow this text is a standard knowledge of linear algebra and a basic knowledge about groups, rings, and fields. To make precise what we mean by this, the book begins with a survey of prerequisites. This is followed by Chap. 1 which considers finite dimensional division algebras. Its most prominent results are Frobenius' theorem on real division algebras and Wedderburn's theorem on finite division rings. Chapter 2 is devoted to the structure of finite dimensional algebras, featuring the classical Wedderburn's theory. After these warm-up chapters, which mostly deal with results known for a hundred years, the next two, Chaps. 3 and 4, introduce and study more abstract notions: modules, vector spaces over division rings, and tensor products. The double centralizer theorem and the Skolem-Noether theorem are included therein. Chapter 5 considers the structure theory of rings. The main themes are primitive rings, the Jacobson density theorem, and the Jacobson radical. If Chaps. 1–5 survey the "greatest hits" of noncommutative algebra, the last two, Chaps. 6 and 7, are slightly more specialized, reflecting my personal taste and interests. They treat polynomial identities and related notions, such as free algebras and rings of quotients.

The order of the topics in the book is quite close to the chronological order of their development. This actually occurred unintentionally, probably as a result of my attempt to follow the principle that an advanced concept should be introduced only when truly needed, and not at a higher level of generality than necessary.

Not because I do not appreciate abstract concepts; after all, they are what makes mathematics beautiful. But it is because they can be better understood and valued when given some evidence that they are indispensable. Another principle I have followed, especially in the early chapters, is to choose proofs that appear to be the simplest. This may not always be the same as the shortest, but rather proofs that seem very natural and easy to memorize. It has been my desire to write the book in such a way that readers would not get an impression that one needs supernatural abilities to create mathematics, but that even they themselves may be able, with some luck and courage, to discover a little piece of it.

One challenge in the exposition of the book was to find new proofs of classical theorems that would fulfill the simplicity criteria described above. This has been in fact my little passion over the last years, which has eventually yielded some results. The proofs of the following theorems are different from those in the standard sources: Frobenius' theorem (Sect. 1.1), the Skolem-Noether theorem (Sect. 1.6), Wedderburn's structure theorems (Sect. 2.9), Martindale's theorem on prime GPI-rings (Sect. 7.7), Posner's theorem on prime PI-rings (Sect. 7.9), and the Formanek-Razmyslov theorem on central polynomials (Sect. 7.10). How much originality is there in these proofs? Frankly, I do not know. I have not found such proofs in the literature, and have published them in mathematical journals through a series of papers. On the other hand, so many mathematicians have known these theorems, especially the first three listed, for so many years that one hardly imagines that something entirely new about them can still be found. It is not so rare that some mathematical ideas are discovered, forgotten, and rediscovered. Anyway, I hope that these proofs provide interesting alternatives to those from other sources. Two remarks must be added. First, the proof of Frobenius' theorem was obtained in a joint work with Peter Šemrl and Špela Špenko. Second, shortly after the publication of my paper on alternative proofs of Wedderburn's structure theorems Edmund Puczylowski informed me that he had used them in his class, but with a modification—which I immediately liked. I am now happy to use this modified version in the book.

The chapters end with exercises of varying difficulty levels. They are sorted by topics, not by difficulty. Some of the harder ones are accompanied with hints, but others are not, in order to give the student the opportunity to fully enjoy the pleasure of discovering the solution. A fair amount of exercises are original, but there are also some that appear, either as exercises or as theorems with proofs, in almost every book on the subject. I tried to avoid long, theoretical exercises, and gave preference to those that I had found entertaining and appealing.

The list of references at the end of the book is very short. It includes only books which are explicitly referred to on relatively rare occasions when details are not provided (say, when closing the discussion on a topic with additional information, or when omitting tedious details in some example). Most of these books are textbooks on noncommutative rings and algebras, but written at a more advanced level than this one. An exception is perhaps J. Beachy's *Introductory Lectures on Rings and Modules* [1] whose emphasis, however, differs from ours. At the time of writing I was frequently consulting many of the listed books, but mainly T. Y. Lam's *A First*

*Course in Noncommutative Rings* [6] (together with the accompanying problem book [7]) and L. H. Rowen's *Graduate Algebra: Noncommutative View* [12]. I warmly recommend these two texts as further reading to students who want to dig deeper into the field of noncommutative algebra.

English is not my mother tongue and I apologize to the reader if this is sometimes too obvious. "Translating" the English stuck in my mind into the English as it should be has been a big and often frustrating challenge for me during the writing process. The bright side of this is that it gives me the opportunity to thank my sons Jure and Martin for their continuous help in this matter.

The most difficult step in writing a book is forcing yourself to write the first sentence. I am thankful to my friend and colleague Peter Šemrl for giving me the necessary encouragement to make it.

Rough drafts of most chapters were first read by Špela Špenko. Her criticism has been very helpful (as always). Aljaž Zalar has solved most of the exercises and made me think of modifying some. Igor Klep provided many helpful suggestions concerning Chaps. 1 and 6. The book has been read cover to cover by Nik Stopar and Janez Šter. Their comments have contributed to the improvement of the text and saved me from some errors. My sincere thanks go to all of them, as well as to the Springer staff for their professional assistance.

Ljubljana and Maribor, Slovenia Matej Brešar
March 2014

# Declarations

**Competing Interests** The author has no competing interests to declare that are relevant to the content of this manuscript.

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# Prerequisites

This preliminary chapter aims to give an accurate survey of the background material needed to follow this text. It is supposed to be used as a handbook, only when there is a need. We will therefore write it in a concise and condensed manner, paying less attention to motivation and insight than in the rest of the book. Some simpler proofs that are as much a part of understanding as definitions and theorems will be outlined, but deeper results will be just stated as facts. The reader should be warned that the topics will be presented in a slightly unbalanced manner. Some will be examined carefully, while other equally or even more important will be only touched upon or even omitted. Of course, the emphasis will be on topics that are needed later.

Through this chapter we also indicate what knowledge the reader is assumed to have. We gain different knowledge at different places, so there might be at least something new for many of the readers. Maybe we will also mention a couple of things for which it would be pretentious to say that everyone should have learned them in one of the undergraduate courses. Some notational and terminological conventions will also be included. Therefore, even well-prepared readers may occasionally consult this chapter, say in case they find something ambiguous in the text. Most of the conventions, however, are standard. Let us just point out two issues before we start.

Formally, the term "noncommutative ring" means a not *necessarily* commutative ring. But we shall use this term in a loose manner. By saying that this book is about noncommutative rings we mean that it considers rings that can be commutative or not, while by saying that some particular ring is noncommutative we shall mean that this ring is *not* commutative. At any rate, even when we use this term in its precise sense, the stress is on rings that are not commutative. The goals and the means of the theory of commutative rings are somewhat different.

We will not assume that every ring must have a unity 1, which may be a subject of dispute. Not that this is uncommon, but the fact is that many mathematicians believe that, at least in a textbook, the existence of 1 should be added to the axioms of a ring. This has advantages, but also disadvantages. There are some interesting and important classes of non-unital rings, e.g., nilpotent rings, nil rings, radical rings, and simple rings consisting of finite rank operators. In the non-unital setting we

can treat ideals as (sub)rings and we can avoid the unitization process for rings that seem more natural without unity. On the other hand, sometimes the non-unital situation is not essentially more interesting, but causes technical difficulties which are distracting for a newcomer. At such instances we will simply add the assumption that a ring has a unity. Of course, some price has to be paid for omitting a possible axiom; admittedly, here and there our exposition would be slightly simpler if the presence of unity was automatically assumed. The readers who believe that all rings should have unities can simply skip parts of the text where the lack of unity creates some problems. There are not that many anyway.

## Sets and Relations

First, a word about notation. By the set of natural numbers, $\mathbb{N}$, we mean the set of all positive integers:

$$\mathbb{N} = \{1, 2, \dots\}.$$

We assume there is no need to explain the meaning of the symbols $\emptyset, \cap, \cup, \subseteq, \supseteq$. By $\subsetneq$ and $\supsetneq$ we denote the strict inclusions, by $S \setminus T$ the set $\{s \in S \mid s \notin T\}$, and by $|S|$ the cardinality of the set $S$.

If $I$ is a (possibly infinite) index set and $\{S_i \mid i \in I\}$ is a family of sets, then $\Pi_{i\in I} S_i$ denotes the **Cartesian product** of the sets $S_i$. The elements of $\Pi_{i\in I} S_i$ are maps $f : I \to \bigcup_{i\in I} S_i$ such that $f(i) \in S_i$ for every $i$. Setting $s_i = f(i)$, we can write $f$ as $(s_i)$. If $I$ is finite, say $I = \{1, \dots, n\}$, then we write $(s_i)$ as $(s_1, \dots, s_n)$ and $\Pi_{i\in I} S_i$ as $S_1 \times \cdots \times S_n$. In particular, $S_1 \times S_2$ is the set $\{(s_1, s_2) \mid s_1 \in S_1, s_2 \in S_2\}$.

By $\mathrm{id}_S$ we denote the identity map on the set $S$. Given a map $f : X \to Y$ and sets $S \subseteq X$, $T \subseteq Y$, we write $f(S)$ for $\{f(s) \mid s \in S\}$ and $f^{-1}(T)$ for $\{x \in X \mid f(x) \in T\}$. By $f|_S$ we denote the restriction of $f$ to $S$. If $f$ is bijective, then $f^{-1}$ denotes its inverse map. The function composition will be denoted by $fg$ rather than $f \circ g$. This is because functions occurring in algebra are often considered as elements of algebraic systems in which multiplication is defined as composition.

A relation $\sim$ on a set $S$ is said to be an **equivalence relation** if it is reflexive, symmetric, and transitive. This means that $a \sim a$, if $a \sim b$ then $b \sim a$, and if $a \sim b$ and $b \sim c$, then $a \sim c$. The **equivalence class** of an element $a$ is the set of all $b \in S$ such that $a \sim b$. It is easy to see that $S$ is a disjoint union of its equivalence classes. An equivalence relation on $S$ thus gives rise to a partition of $S$.

A set $S$ together with a relation $\leq$ which is reflexive ($a \leq a$), anti-symmetric ($a \leq b$ and $b \leq a$ implies $a = b$), and transitive ($a \leq b$ and $b \leq c$ implies $a \leq c$) is called a **partially ordered set**. For example, the power set of a given set is partially ordered by inclusion. A subset $C$ of a partially ordered set $S$ is called a **chain** if for each pair $a, b \in C$ we have either $a \leq b$ or $b \leq a$. An element $u \in S$ is called an **upper bound** for $C$ if $c \leq u$ for every $c \in C$. Finally, an element $m \in S$ is called a

**maximal element** of $S$ if for every $a \in S$, $m \leq a$ implies $m = a$. We are now in a position to state the following important result.

**Zorn's Lemma** *Let $S$ be a nonempty partially ordered set. If every chain in $S$ has an upper bound in $S$, then $S$ contains a maximal element.*

Zorn's lemma is one of the most frequently used results in mathematics. It is equivalent to the **axiom of choice** which says the following: If $\{A_i \mid i \in I\}$ is a family of nonempty sets, then there exists a family of elements $\{a_i \mid i \in I\}$ such that $a_i \in A_i$ for every $i \in I$.

## Groups

A **semigroup** is a set $S$ together with an **associative** binary operation. This means that $S$ is endowed with a map $S \times S \to S$, $(x, y) \mapsto xy$, such that $(xy)z = x(yz)$ for all $x, y, z \in S$. We therefore omit parentheses and write this element as $xyz$. It is easy to see that parentheses can be omitted when involving more than three elements, that is, we can write $x_1x_2 \cdots x_n$ with $x_i \in S$ for any $n \in \mathbb{N}$. In particular, we can define the $n$th power $x^n$ of $x \in S$ for every $n \in \mathbb{N}$. We say that elements $x, y \in S$ **commute** if $xy = yx$. If $xy = yx$ holds for all $x, y \in S$, then we say that $S$ is **commutative**.

An element $e \in S$ that satisfies $ex = x$ for every $x \in S$ is called a left identity element. Similarly, a right identity element is an element $e' \in S$ satisfying $xe' = x$ for every $x \in S$. If $S$ has both a left identity element $e$ and a right identity element $e'$, then they are equal; indeed, they both must be equal to $ee'$. In this case we call $e = e'$ an **identity element** of $S$. From now on we will denote an identity element by 1.

A semigroup with a (necessarily unique) identity element 1 is called a **monoid**. An element $a$ in a monoid $M$ is said to be **right invertible** if there exists a $b \in M$ such that $ab = 1$. Every such element $b$ is called a **right inverse** of $a$. Analogously we define left invertibility and left inverses. We say that $a$ is **invertible** if it is both right and left invertible. In this case, any right inverse of $a$ is equal to any left inverse of $a$. Indeed, $ab = ca = 1$ implies $b = (ca)b = c(ab) = c$. Thus, there exists a unique element which is both a left and right inverse of $a$. We call it the **inverse** of $a$ and denote by $a^{-1}$. If $a$ is invertible, then so is $a^n$ for every $n \in \mathbb{N}$. Its inverse is $(a^{-1})^n$ which we denote by $a^{-n}$. For convenience, we set $a^0 = 1$ for every $a \in M$. Further, if $a$ and $b$ are invertible, then so is $ab$, and $(ab)^{-1} = b^{-1}a^{-1}$. The set of all invertible elements of $M$ will be denoted by $M^*$.

A monoid in which every element is invertible is called a **group**. A group $G$ is **finite** if the underlying set $G$ is a finite set. We call $|G|$, the cardinality of the set $G$, the **order** of the group $G$.

In the next paragraphs, we give some basic examples of groups.

If $M$ is a monoid, then $M^*$ is clearly a group. For example, $\mathbb{Z}$ is a monoid under (the usual) multiplication, and hence $\mathbb{Z}^* = \{-1, 1\}$ is a group. Similarly, if $F \in$

$\{\mathbb{Q}, \mathbb{R}, \mathbb{C}\}$, then $F$ is a monoid under multiplication and $F^* = F \setminus \{0\}$. The set $M_n(F)$ of all $n \times n$ matrices with entries in $F$ is a monoid under the usual matrix multiplication; the corresponding group $M_n(F)^*$ of all invertible matrices is called the **general linear group** (over $F$). It is denoted by $\mathrm{GL}_n(F)$.

Let $X$ be a set. A bijective map from $X$ onto $X$ is called a **permutation**. The set of all permutations of $X$ is a group under the composition. Indeed, the composition is associative, $\mathrm{id}_X$ is the identity element, and inverses clearly exist. We call it the **symmetric group** of $X$. If $X = \{1, \ldots, n\}$, then we denote the symmetric group of $X$ by $S_n$. It is easy to see that $|S_n| = n!$. A permutation in $S_n$ is called a **transposition** if it interchanges two of the elements and leaves all the rest fixed. Every permutation $\sigma \in S_n$ can be written as a product of transpositions. If $\sigma = \tau_1 \ldots \tau_p = \rho_1 \ldots \rho_q$, where all $\tau_i$ and $\rho_i$ are transpositions, then either both $p$ and $q$ are even or both $p$ and $q$ are odd. In the first case we say that $\sigma$ is an even permutation and in the second case we say that it is an odd permutation. We define the **sign** of $\sigma$, $\mathrm{sgn}(\sigma)$, by $\mathrm{sgn}(\sigma) = \begin{cases} 1 & \text{if } \sigma \text{ is even} \\ -1 & \text{if } \sigma \text{ is odd} \end{cases}$.

A commutative group is called an **abelian group**. So far we have used only the multiplicative notation for binary operations. For abelian groups we often use the additive notation. This means that the result of the operation is written as $x + y$ instead of $xy$. An abelian group in which this notation is used is called an **additive group**. (The convention that an additive group must necessarily be abelian is not entirely standard, but it is convenient to adopt it in this book.) An identity element in an additive group is written as 0. Further, we write $-x$ instead of $x^{-1}$ and $nx$ instead of $x^n$. A simple example of an additive group is $\mathbb{Z}$ under the usual addition. Another example is $\mathbb{Z}_n = \{0, 1, \ldots, n-1\}$ where addition is taken modulo $n$.

Until further notice we continue using the multiplicative notation. A map $\varphi$ from a group $G$ to a group $G'$ is called a **group homomorphism**, or simply a **homomorphism** when it is clear from the context that groups are the subject of consideration, if $\varphi(xy) = \varphi(x)\varphi(y)$ for all $x, y \in G$. The set $\ker\varphi = \{x \in G \mid \varphi(x) = 1\}$ is called the **kernel** of $\varphi$, and the set $\mathrm{im}\,\varphi = \{\varphi(x) \mid x \in G\}\ (= \varphi(G))$ is called the **image** of $\varphi$. Note that $\varphi$ is injective if and only if $\ker\varphi = \{1\}$.

A bijective homomorphism is called an **isomorphism**. We say that groups $G$ and $G'$ are isomorphic, and write $G \cong G'$, if there exists an isomorphism $\varphi$ from $G$ onto $G'$. In this case its (set-theoretic) inverse $\varphi^{-1}$ is an isomorphism from $G'$ onto $G$. The relation of being isomorphic is an equivalence relation. Isomorphic groups have exactly the same properties; informally we consider them as identical, although they may appear very different at a glance. For example, consider the set $G$ of all matrices of the form $\left[\begin{smallmatrix} \alpha & \beta \\ -\beta & \alpha \end{smallmatrix}\right]$ with $\alpha, \beta \in \mathbb{R}$, $\alpha \neq 0$ or $\beta \neq 0$. One can check that $G$ is a group under multiplication and that the map

$$\begin{bmatrix} \alpha & \beta \\ -\beta & \alpha \end{bmatrix} \mapsto \alpha + \beta i$$

is an isomorphism from $G$ onto $\mathbb{C}^* = \mathbb{C} \setminus \{0\}$. Thus, as long as we are interested only in multiplication, the set of matrices $G$ is just a disguised form of the more familiar set of nonzero complex numbers. Another example: The additive group $\mathbb{R}$ is isomorphic to the multiplicative group $(0, \infty)$ via the isomorphism $x \mapsto e^x$.

A homomorphism from a group to itself is called an **endomorphism**. An **automorphism** is an endomorphism which is also an isomorphism. An injective homomorphism $\varphi : G \to G'$ is sometimes called an **embedding**. This term is usually used when we wish to point out that by identifying $G$ with $\operatorname{im}\varphi$ we may view $G$ inside $G'$. We also say that $G$ embeds in $G'$. For example, the additive group $\mathbb{R}$ can be canonically embedded into the additive group $\mathbb{C}$ by $\alpha \mapsto \alpha + 0i$.

Every group $G$ can be embedded into a symmetric group. This result is known as **Cayley's theorem**, and can be proved as follows. For every $a \in G$, we define $\ell_a : G \to G$ by $\ell_a(g) = ag$. One immediately checks that $\ell_a$ is a permutation of $G$, $\ell_a = \mathrm{id}_G$ if and only if $a = 1$, and $\ell_{ab} = \ell_a \ell_b$ for all $a, b \in G$. Accordingly, $a \mapsto \ell_a$ is an embedding of the group $G$ into the symmetric group of the set $G$.

A homomorphism between additive groups is also called an **additive map**. We will also come across biadditive maps: We say that a map $f : G_1 \times G_2 \to G'$, where $G_1, G_2, G'$ are additive groups, is **biadditive** if it is additive in each variable, i.e.,

$$f(x_1 + y_1, x_2) = f(x_1, x_2) + f(y_1, x_2),$$
$$f(x_1, x_2 + y_2) = f(x_1, x_2) + f(x_1, y_2)$$

for all $x_1, y_1 \in G_1, x_2, y_2 \in G_2$.

A subset $H$ of a group $G$ is called a **subgroup** if it contains 1, is closed under the group operation (i.e., $h, h' \in H$ implies $hh' \in H$), and $h^{-1} \in H$ whenever $h \in H$. Note that $H$ is itself a group. For any $a \in G$, $\langle a \rangle = \{a^m \mid m \in \mathbb{Z}\}$ is a subgroup. It is called the **cyclic subgroup** generated by $a$.

A (left) **coset** of a subgroup $H$ in $G$ is a set $xH = \{xh \mid h \in H\}$, where $x$ is a fixed element in $G$. It is easy to see that $|xH| = |yH|$ for all $x, y \in G$, $xH = yH$ if and only if $x^{-1}y \in H$, and $xH \neq yH$ if and only if $xH \cap yH = \emptyset$. From these observations one easily infers **Lagrange's theorem** which says that $[G : H]$, the number of cosets of a subgroup $H$ in a finite group $G$, is equal to $\frac{|G|}{|H|}$. In particular, $|H|$ divides $|G|$. This is the most basic fact in finite group theory. One of its well-known applications is **Fermat's little theorem** which states that for any prime $p$ and integer $a$, $a^p \equiv a \pmod{p}$.

We say that an element $a$ in a group $G$ has **finite order** if $a^r = 1$ for some $r \in \mathbb{N}$. If $r$ is the smallest natural number with this property, then $r$ is called the **order** of $a$. In this case, $\langle a \rangle = \{1, a, \ldots, a^{r-1}\}$ and $|\langle a \rangle| = r$. If $G$ is finite, then every element in $G$ is of finite order, and hence Lagrange's theorem tells us that $r$ divides $|G|$ for every $a \in G$. This implies that $a^{|G|} = 1$ for every $a \in G$.

Let $a$ be an arbitrary element in a group $G$. Every element of the form $xax^{-1}$, $x \in G$, is called a **conjugate** of $a$. A subgroup $N$ of $G$ is called a **normal** subgroup if it contains every conjugate of each of its elements, i.e., $a \in N$ implies $xax^{-1} \in N$

for every $x \in G$. Every subgroup of an abelian group is normal. The **center** of a group $G$, defined as the set of all elements in $G$ that commute with every other element in $G$, is a normal subgroup of $G$. If $\varphi : G \to G'$ is a homomorphism, then $\ker\varphi$ is a normal subgroup of $G$, while $\operatorname{im}\varphi$ is a subgroup of $G'$ which is not necessarily normal.

The next construction works for normal subgroups of arbitrary groups, but we confine ourselves to additive groups as this is all we need. Thus, let $H$ be a subgroup of an additive group $G$. A coset of $H$ in $G$ is in the additive notation written as $x+H$. Let $G/H$ denote the set of all cosets of $H$ in $G$ endowed with the addition defined by

$$(x + H) + (y + H) = (x + y) + H.$$

It is easy to see that this operation is well defined, i.e., $x + H = x' + H$ and $y + H = y' + H$ implies $(x + y) + H = (x' + y') + H$, and that $G/H$ is a group under this operation. We call it the **factor group** of $G$ by $H$ (another standard name is the **quotient group**). The map $x \mapsto x + H$ is a surjective homomorphism from $G$ onto $G/H$ with the kernel $H$. It is called the **canonical homomorphism**.

Let $G'$ be another additive group. If $\varphi : G \to G'$ is a homomorphism, then

$$G/\ker\varphi \cong \operatorname{im}\varphi.$$

An isomorphism is given by $x + \ker\varphi \mapsto \varphi(x)$. Indeed, this map is well-defined since $x + \ker\varphi = y + \ker\varphi$ implies $x - y \in \ker\varphi$ and hence $\varphi(x) = \varphi(y)$. Clearly, it is additive and a bijection from $G/\ker\varphi$ onto $\operatorname{im}\varphi$.

Given groups $G_1, \ldots, G_n$, we define their **direct product** $G_1 \times \cdots \times G_n$ as the Cartesian product of the sets $G_1, \ldots, G_n$ endowed with the componentwise operation:

$$(x_1, \ldots, x_n)(y_1, \ldots, y_n) = (x_1 y_1, \ldots, x_n y_n).$$

It is immediate that $G_1 \times \cdots \times G_n$ is a group and that each $G_i$ can be naturally (canonically) embedded in $G_1 \times \cdots \times G_n$. The direct product $\Pi_{i\in I} G_i$ of a possibly infinite family of groups $\{G_i \mid i \in I\}$ is defined analogously. The operation is thus given by $(x_i)(y_i) = (x_i y_i)$. If $G_i$ are additive groups, then the operation in $\Pi_{i\in I} G_i$ is also written additively: $(x_i) + (y_i) = (x_i + y_i)$.

Let now $G_1, \ldots, G_n$ be subgroups of an additive group $G$. Then so is their sum

$$G_1 + \cdots + G_n = \{x_1 + \cdots + x_n \mid x_1 \in G_1, \ldots, x_n \in G_n\}.$$

If

$$G_i \cap (G_1 + \cdots + G_{i-1} + G_{i+1} + \cdots + G_n) = \{0\}$$

for every $i$, then $G_1 + \cdots + G_n$ is called the **direct sum** of the subgroups $G_i$, and is denoted by $G_1 \oplus \cdots \oplus G_n$. Every element in $G_1 \oplus \cdots \oplus G_n$ can be written as $x_1 + \cdots + x_n$ with $x_i \in G_i$ in a unique way. The difference between the groups $G_1 \oplus \cdots \oplus G_n$ and $G_1 \times \cdots \times G_n$ is merely formal. In more accurate terms, they are canonically isomorphic via $x_1 + \cdots + x_n \mapsto (x_1, \ldots, x_n)$.

## Rings

A **ring** $R$ is an additive group endowed with another associative binary operation $(x, y) \mapsto xy$, called multiplication, such that the distributive laws hold in $R$, i.e.,

$$(x + y)z = xz + yz \quad \text{and} \quad x(y + z) = xy + xz$$

for all $x, y, z \in R$. In other words, $(x, y) \mapsto xy$ is a biadditive map. The following simple facts follow easily from the definition: $0x = x0 = 0$, $(-x)y = x(-y) = -xy$, $(-x)(-y) = xy$ for all $x, y \in R$.

A ring $R$ is **commutative** if $xy = yx$ for all $x, y \in R$. For example, $\mathbb{Z}$, equipped with the usual addition and multiplication, is a commutative ring. The additive group $\mathbb{Z}_n$ becomes a commutative ring if we equip it with multiplication modulo $n$. The set $C[a, b]$ of continuous functions $f : [a, b] \to \mathbb{R}$ is a commutative ring under the usual addition and multiplication of functions.

Commutative rings play a secondary role in this book. However, we will be often concerned with the "most commutative" part of our rings. The set

$$Z(R) = \{c \in R \mid cx = xc \text{ for all } x \in R\}$$

is called the **center** of the ring $R$ and its elements are called **central elements**. Obviously, $R$ is commutative if and only if $Z(R) = R$. If $R$ is not commutative, then $Z(R)$ is more often than not a relatively small subset of $R$.

We assume the reader is familiar with at least one example of a noncommutative ring, namely with $M_n(\mathbb{R})$, the ring of all $n \times n$ matrices with entries in $\mathbb{R}$. Here one takes the usual matrix addition and multiplication. We will introduce matrix rings in a more accurate and general manner in the next section.

One of the most intriguing properties in rings is that the product of two nonzero elements can be 0. This cannot happen in, say, $\mathbb{Z}$, but can easily occur in $\mathbb{Z}_n$ if $n$ is not prime, or in $M_n(\mathbb{R})$. A nonzero element $x$ in a ring $R$ is said to be a **left zero-divisor** if there exists a nonzero element $y \in R$ such that $xy = 0$. Analogously we define a **right zero-divisor**. An element that is a left or a right zero-divisor is called a **zero-divisor**.

Every ring is a semigroup under multiplication. If it is actually a monoid, then it will be called a **unital ring**. Thus, a ring $R$ is unital if it contains an element 1 such that $1x = x1 = x$ for all $x \in R$. We will call 1 the **unity** of $R$. Most rings that first come to our mind are unital; a simple example of a ring without unity is $2\mathbb{Z}$, the ring

of even integers. If $1 = 0$ in a ring $R$, then $R$ is the **trivial** or **zero ring** whose only element is 0.

In a unital ring we can speak about left invertible, right invertible, and invertible elements. A nonzero unital ring $D$ in which every nonzero element is invertible (i.e., $D^* = D \setminus \{0\}$) is called a **division ring**. Division rings obviously have no zero-divisors. A commutative division ring is called a **field** (another name for a possibly noncommutative division ring is a **skew field**, but we will not use it). Obvious examples of fields are $\mathbb{Q}$, $\mathbb{R}$, and $\mathbb{C}$. It is easy to show that $\mathbb{Z}_p$ with $p$ prime is a field. Conversely, $\mathbb{Z}_n$ is not a field if $n$ is not prime.

We say that a ring $R$ has **characteristic** $n$ if $n$ is the smallest natural number such that $nx = 0$ for every $x \in R$ (note that in a unital ring $R$, the condition $nx = 0$ for every $x \in R$ is equivalent to $n1 = 0$). If no such natural number $n$ exists, then we say that $R$ has **characteristic** 0. The characteristic of $R$ will be denoted by $\operatorname{char}(R)$. For example, $\operatorname{char}(\mathbb{Z}) = 0$ and $\operatorname{char}(\mathbb{Z}_n) = n$. If $R$ is unital and $m \in \mathbb{N}$ is such that $m1 = 0$, then $m = rs$ implies $r1 \cdot s1 = 0$. Hence, either $R$ has zero-divisors or $\operatorname{char}(R)$ is prime. In particular, the characteristic of a field is either 0 or a prime number $p$. Fields of prime characteristic are called fields of **finite characteristic**. The subset $F_0$ of a field $F$ consisting of all elements of the form $(m1)(n1)^{-1}$, $m, n \in \mathbb{Z}$, $n1 \neq 0$, is easily seen to be a field isomorphic to $\mathbb{Q}$ if $\operatorname{char}(F) = 0$, and to $\mathbb{Z}_p$ if $\operatorname{char}(F) = p$. We call $F_0$ the **prime subfield** of $F$.

A map $\varphi$ from a ring $R$ to a ring $R'$ is called a **ring homomorphism** (or simply a **homomorphism**, when the context is clear) if

$$\varphi(x + y) = \varphi(x) + \varphi(y) \quad \text{and} \quad \varphi(xy) = \varphi(x)\varphi(y)$$

for all $x, y \in R$. Thus, $\varphi$ is an additive group homomorphism and simultaneously a multiplicative semigroup homomorphism. Its **kernel** is $\ker \varphi = \{x \in R \mid \varphi(x) = 0\}$, and its **image** is $\operatorname{im} \varphi = \{\varphi(x) \mid x \in R\}$.

The other necessary definitions related to ring homomorphisms are the same as those related to group homomorphisms (as well as to homomorphisms of other algebraic structures). Thus, an **isomorphism** is a bijective homomorphism, an **endomorphism** is a homomorphism from a ring to itself, an **automorphism** is an endomorphism which is also an isomorphism, and an **embedding** is an injective homomorphism. The relation of being isomorphic is an equivalence relation. We write $R \cong R'$ if $R$ and $R'$ are isomorphic. For illustration, we modify our earlier example on group isomorphisms. The set $R$ of all matrices of the form $\left[\begin{smallmatrix} \alpha & \beta \\ -\beta & \alpha \end{smallmatrix}\right]$ with $\alpha, \beta \in \mathbb{R}$, is easily seen to be a ring isomorphic to $\mathbb{C}$. An isomorphism is defined exactly as before, just that now there is no need to exclude the case where $\alpha = \beta = 0$. We often ignore the difference between isomorphic objects, so sometimes we simply say that $R$ *is* the field of complex numbers.

A subset of a ring $R$ which is a subgroup with respect to addition is called an **additive subgroup** of $R$. Let $S$ be an arbitrary nonempty subset of $R$. The set $\overline{S}$ of all elements of the form $m_1 s_1 + \cdots + m_n s_n$ where $m_i \in \mathbb{Z}$ and $s_i \in S$ is an additive subgroup of $R$ which contains $S$. Moreover, $\overline{S}$ is the smallest additive subgroup

containing $S$ in the sense that it is contained in every other additive subgroup which contains $S$. We call $\overline{S}$ the **additive subgroup generated by** $S$. (For $S = \emptyset$ we define $\overline{S} = \{0\}$.)

An additive subgroup of $R$ which is closed under multiplication is called a **subring** of $R$. A subring is itself a ring. For example, the center $Z(R)$ is a subring of $R$ which is commutative as a ring. We remark that $Z(R)$ can contain only 0 even when $R$ is a nonzero ring. If, however, $R$ is unital, then $1 \in Z(R)$. A subring $D$ of a unital ring $R$ that contains 1 and is a division ring is called a **division subring** of $R$. If it is also commutative, then we call it a **subfield** of $R$.

The subring generated by the subset $S$ of $R$ is, of course, defined as the smallest subring of $R$ containing $S$. Note that it is equal to the additive subgroup generated by the set of all possible (finite) products $s_1 \dots s_r$ of elements $s_i \in S$. If the subring generated by $S$ is $R$ itself, then we call $S$ a **generating set** for the ring $R$, and its elements are called **generators** of $R$. We also say that $S$ **generates** $R$. A ring is said to be **finitely generated** if it contains a finite generating set.

The terms "generated by," "generating set," "generators," "finitely generated" are used, in the same sense as above, also for other algebraic structures. We hope that their meaning can be understood without always giving a detailed explanation.

A subring $I$ of $R$ is called a **left ideal** of $R$ if $xu \in I$ for all $x \in R$ and $u \in I$. Analogously, a subring $I$ of $R$ is called a **right ideal** of $R$ if $ux \in I$ for all $x \in R$ and $u \in I$. If $I$ is both a left and right ideal, then $I$ is called an **ideal**; we also say a **two-sided ideal** when we wish to stress that it is not only left or right. Left and right ideals are called **one-sided ideals**.

For example, the ideals of the ring $\mathbb{Z}$ are $n\mathbb{Z} = \{nk \mid k \in \mathbb{Z}\}$, $n \geq 0$. In commutative rings, the adjectives "left" and "right" are of course unnecessary. The set of all matrices of the form $\left[\begin{smallmatrix} \alpha & 0 \\ \beta & 0 \end{smallmatrix}\right]$, $\alpha, \beta \in \mathbb{R}$, is an example of a left ideal but not an ideal of the ring $M_2(\mathbb{R})$. By putting zeros in one of the rows instead of in a column we obtain a right ideal that is not an ideal.

Every ring $R$ has at least two ideals: $\{0\}$ and $R$. From now on we will write $\{0\}$ as 0. The reader should thus be warned that the symbol 0 can mean two different things, either the zero element or the **zero** (or **trivial**) **ideal**. From the context it should always be clear what we have in mind.

A one-sided or two-sided ideal is said to be **nonzero** (or **nontrivial**) if it is different from 0, and is said to be **proper** if it is different from $R$.

If $I$ and $J$ are ideals of $R$, then so is their intersection $I \cap J$, their **sum** $I + J = \{u + v \mid u \in I, v \in J\}$, and their **product** $IJ$ which is defined as the additive subgroup generated by the set $\{uv \mid u \in I, v \in J\}$. The same is true if we replace "ideal" by "left ideal" or "right ideal" (incidentally, the sum and the product of subrings may not be a subring). We remark that, in accordance with the definition of the product, $I^n$ should be understood as the additive subgroup generated by the set $\{u_1 \dots u_n \mid u_i \in I\}$. Observe that $IJ \subseteq I \cap J$ if $I$, $J$ are two-sided ideals.

Let $S$ be a subset of $R$ and let $a, b \in R$. We write

$$aS = \{ax \mid x \in S\}, \quad Sa = \{xa \mid x \in S\}, \quad aSb = \{axb \mid x \in S\}.$$

Note that $Ra$ is a left ideal and $aR$ is a right ideal. By $RaR$ we denote the additive subgroup generated by the set $\{xay \mid x, y \in R\}$, which is clearly an ideal. If $R$ is unital, then $Ra$ is the left ideal generated by $a$, $aR$ is the right ideal generated by $a$, and $RaR$ is the ideal generated by $a$. If $R$ is not unital, then the left ideal generated by $a$ is equal to $Ra + \mathbb{Z}a$ where $\mathbb{Z}a = \{ka \mid k \in \mathbb{Z}\}$, the right ideal generated by $a$ is equal to $aR + \mathbb{Z}a$, and the ideal generated by $a$, which is sometimes denoted by $(a)$, is equal to $RaR + Ra + aR + \mathbb{Z}a$.

We have thus described the (left, right, two-sided) ideals generated by one element. Replacing one element by an arbitrary subset $S$ of $R$ does not make the task of describing much harder. For example, the **ideal generated by** $S$ is the additive subgroup of $R$ generated by the set of elements of the form $xsy$, $xs$, $sy$, $s$ with $x, y \in R$ and $s \in S$.

The **sum of an arbitrary family of ideals** $\{I_i \mid i \in I\}$ is defined as the ideal generated by $\bigcup_{i \in I} I_i$. It is easy to see that it is actually equal to the additive subgroup generated by $\bigcup_{i \in I} I_i$. The same is true for left or right ideals.

Let $I$ be an ideal of a ring $R$. As $I$ is, in particular, an additive subgroup of $R$, we can form the factor additive group $R/I$. Recall that $R/I$ consists of all cosets $x + I$, $x \in R$, which are added according to

$$(x + I) + (y + I) = (x + y) + I.$$

We can make $R/I$ into a ring by defining the multiplication by

$$(x + I)(y + I) = xy + I.$$

Using $RI \subseteq I$ and $IR \subseteq I$ it is easy to see that this multiplication is well-defined. The ring axioms are obviously fulfilled. We call $R/I$ the **factor ring** of $R$ by $I$ (or the **quotient ring**, but we will avoid this term to prevent confusion with the notion of a ring of quotients). Note that the **canonical homomorphism** $x \mapsto x + I$, mentioned in the additive group setting, is now a ring homomorphism.

The basic example of the factor ring is $\mathbb{Z}_n = \mathbb{Z}/n\mathbb{Z}$.

Let $\varphi : R \to R'$ be a ring homomorphism. Note that $\ker \varphi$ is an ideal of $R$, while $\operatorname{im} \varphi$ is, in general, only a subring of $R'$. Just as in the group-theoretic setting we have

$$R/\ker \varphi \cong \operatorname{im} \varphi,$$

with an isomorphism defined by $x + \ker \varphi \mapsto \varphi(x)$. This result is called the **first isomorphism theorem**. The other two **isomorphism theorems** state that if $I$ and $J$ are ideals of a ring $R$, then $J/J \cap I \cong (J + I)/I$ and $(R/I)/(J/I) \cong R/J$ if $I \subseteq J$. Similar theorems hold for groups, but we will not need them.

If ideals $I_1, \ldots, I_n$ of a unital ring $R$ satisfy $I_i + I_j = R$ for all $i \neq j$, then

$$R/(I_1 \cap \cdots \cap I_n) \cong R/I_1 \times \cdots \times R/I_n.$$

This is known as the **Chinese remainder theorem**.

Given rings $R_1, \ldots, R_n$, we define their **direct product** $R_1 \times \cdots \times R_n$ as the Cartesian product endowed with the componentwise operations:

$$(x_1, \ldots, x_n) + (y_1, \ldots, y_n) = (x_1 + y_1, \ldots, x_n + y_n),$$
$$(x_1, \ldots, x_n)(y_1, \ldots, y_n) = (x_1 y_1, \ldots, x_n y_n).$$

Clearly, $R_1 \times \cdots \times R_n$ is a ring. We may view $R_i$ as an ideal of $R_1 \times \cdots \times R_n$ via the canonical embedding $x_i \mapsto (0, \ldots, 0, x_i, 0, \ldots, 0)$.

One analogously defines the direct product $\Pi_{i \in I} R_i$ of a possibly infinite family of rings $\{R_i \mid i \in I\}$. The operations are thus given as follows:

$$(x_i) + (y_i) = (x_i + y_i),$$
$$(x_i)(y_i) = (x_i y_i).$$

The subset of $\Pi_{i \in I} R_i$ consisting of all $(x_i)$ such that all but finitely many $x_i$ are 0 is a subring of $\Pi_{i \in I} R_i$, called the **direct sum** of the family $\{R_i \mid i \in I\}$. The notions "direct product" and "direct sum" thus coincide for finite families of rings. It is therefore common to use the notation $R_1 \oplus \cdots \oplus R_n$ instead of $R_1 \times \cdots \times R_n$. However, we will stick with the latter.

## Matrices

We have already mentioned that the set of all real $n \times n$ matrices, $M_n(\mathbb{R})$, is a ring under the usual matrix operations. This ring is not commutative as long as $n \geq 2$, and has plenty of zero-divisors. One can replace the role of $\mathbb{R}$ in the definition of addition and multiplication not only by another field, but by any other ring. This does not affect the proof that the ring axioms are fulfilled.

Let us be precise. Take an arbitrary ring $R$ and $n \in \mathbb{N}$. Given any $a_{ij} \in R$, $1 \leq i, j \leq n$, we denote by $(a_{ij})$ the $n \times n$ matrix

$$\begin{bmatrix} a_{11} & a_{12} & \ldots & a_{1n} \\ a_{21} & a_{22} & \ldots & a_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ a_{n1} & a_{n2} & \ldots & a_{nn} \end{bmatrix}.$$

Define addition and multiplication of such matrices by

$$(a_{ij}) + (b_{ij}) = (a_{ij} + b_{ij})$$

and

$$(a_{ij})(b_{ij}) = (c_{ij}), \text{ where } c_{ij} = \sum_{k=1}^{n} a_{ik}b_{kj}.$$

One readily checks that, in this way, the set of all $n \times n$ matrices with entries in $R$ becomes a ring. It will be denoted by $M_n(R)$. Note that $M_1(R)$ can be identified with $R$. If $R$ is unital, then so is $M_n(R)$. Its unity is the identity matrix $I_n$, i.e., the matrix whose entries on the main diagonal are 1 and all other entries are 0.

Sometimes we call $M_n(R)$ the **full matrix ring** over $R$, in order to point out that it contains all $n \times n$ matrices with entries in $R$. The term **matrix ring** is often used as a synonym for the full matrix ring, but it can also mean any ring whose elements are matrices.

A special but extremely important case is when $R = F$ is a field. When working with $M_n(F)$, we often rely on linear algebra methods. We assume the reader is familiar with the notions such as eigenvalue, determinant, trace, and characteristic polynomial of a matrix.

## Polynomials

Let $R$ be a ring $R$. We define a **polynomial** over $R$ as a formal sum

$$\sum_{i=0}^{\infty} a_i \omega^i = a_0 + a_1\omega + a_2\omega^2 + \ldots,$$

where $a_i \in R$ and all but finitely many $a_i$ are 0. The elements $a_i$ are called **coefficients**, and $\omega$ is a formal symbol called an **indeterminate**. The coefficient $a_0$ is called the **constant term**. If $n \geq 0$ is such that $a_i = 0$ for all $i > n$ and $a_n \neq 0$, then we say that the polynomial $\sum_{i=0}^{\infty} a_i\omega^i$ has **degree** $n$, we write it as

$$a_0 + a_1\omega + \cdots + a_n\omega^n,$$

and we say that $a_n$ is its **leading coefficient**. Denote this polynomial shortly by $f(\omega)$. If $a_n = 1$, then $f(\omega)$ is said to be **monic**. If $a_i = 0$ for all $i \geq 0$, then we write $f(\omega) = 0$. We say that $f(\omega)$ is a **constant polynomial** if it is 0 or has degree 0. We identify constant polynomials with elements in $R$, so we have $R \subseteq R[\omega]$.

The addition and multiplication of two polynomials are defined by

$$\sum_{n=0}^{\infty} a_n\omega^n + \sum_{n=0}^{\infty} b_n\omega^n = \sum_{n=0}^{\infty} (a_n + b_n)\omega^n,$$

$$\Big(\sum_{n=0}^{\infty} a_n\omega^n\Big)\Big(\sum_{n=0}^{\infty} b_n\omega^n\Big) = \sum_{n=0}^{\infty} c_n\omega^n, \text{ where } c_n = \sum_{i=0}^{n} a_i b_{n-i}.$$

This makes the set of all polynomials over $R$ a ring, which we denote by $R[\omega]$ and call a **polynomial ring**. If $R$ is unital, then so is $R[\omega]$. Elements in $R$ commute with $\omega$, but do not necessarily commute with each other. In fact, $R$ is commutative if and only if $R[\omega]$ is commutative.

We are primarily interested in the situation where $R$ is a field. In this case, we write $F$ for $R$ and thus denote the polynomial ring by $F[\omega]$. The fundamental result on $F[\omega]$, called the **division algorithm for polynomials**, states that if $f(\omega), g(\omega) \in F[\omega]$ and $g(\omega) \neq 0$, then there exist unique $q(\omega), r(\omega) \in F[\omega]$ such that $f(\omega) = q(\omega)g(\omega) + r(\omega)$ and $r(\omega)$ is either 0 or has lower degree than $g(\omega)$. If $r(\omega) = 0$, we say that $g(\omega)$ **divides** $f(\omega)$ (or that $f(\omega)$ is divisible by $g(\omega)$).

Every ideal of $F[\omega]$ is **principal**, that is, of the form

$$(g(\omega)) = \{h(\omega)g(\omega) \mid h(\omega) \in F[\omega]\}$$

for some $g(\omega) \in F[\omega]$. Clearly, $g(\omega)$ divides $f(\omega)$ if and only if $(f(\omega)) \subseteq (g(\omega))$.

A monic polynomial $d(\omega) \in F[\omega]$ is called the **greatest common divisor** of the polynomials $f(\omega), g(\omega) \in F[\omega]$ (that are not both zero) if $d(\omega)$ divides both $f(\omega)$ and $g(\omega)$ and is divisible by any polynomial which also divides $f(\omega)$ and $g(\omega)$. The greatest common divisor exists, is unique and lies in the sum of the principal ideals $(f(\omega))$ and $(g(\omega))$. If $d(\omega) = 1$, we say that $f(\omega))$ and $g(\omega)$ are **relatively prime**. In this case, $(f(\omega)) + (g(\omega)) = F[\omega]$.

A nonconstant polynomial $p(\omega) \in F[\omega]$ is said to be **irreducible** if it cannot be written as a product of two nonconstant polynomials. This is equivalent to the condition that $F[\omega]/(p(\omega))$ is a field. If an irreducible polynomial divides the product of two polynomials, then it divides at least one of them. Every nonconstant monic polynomial is a product of irreducible monic polynomials. This factorization is unique up to the order of the factors.

So far, we have considered polynomials in one indeterminate. The notion of a **polynomial in several indeterminates** $\omega_1, \ldots, \omega_n$ is not conceptually much more demanding, but notationally heavier. It can be defined as an infinite sum

$$\sum_{i_1=0}^{\infty} \cdots \sum_{i_n=0}^{\infty} a_{i_1,\ldots,i_n} \omega_1^{i_1} \ldots \omega_n^{i_n},$$

where $a_{i_1,\ldots,i_n} \in R$ and all but finitely many $a_{i_1,\ldots,i_n}$ are 0. Each term $a_{i_1,\ldots,i_n} \omega_1^{i_1} \ldots \omega_n^{i_n}$ whose coefficient $a_{i_1,\ldots,i_n}$ is not 0 is called a **monomial**. We may neglect the terms with zero coefficient, so every polynomial in several indeterminates is a finite sum of monomials. The addition is defined by adding the coefficients of each $\omega_1^{i_1} \ldots \omega_n^{i_n}$. The multiplication is determined by the rule

$$(a_{i_1,\ldots,i_n} \omega_1^{i_1} \ldots \omega_n^{i_n})(b_{j_1,\ldots,j_n} \omega_1^{j_1} \ldots \omega_n^{j_n}) = (a_{i_1,\ldots,i_n} b_{j_1,\ldots,j_n}) \omega_1^{i_1+j_1} \ldots \omega_n^{i_n+j_n}$$

and the distributive laws. These two operations make the set of all polynomials in $\omega_1, \ldots, \omega_n$ a ring, which we denote by $R[\omega_1, \ldots, \omega_n]$. In a more precise way, we define $R[\omega_1, \ldots, \omega_n]$ recursively as $R[\omega_1, \ldots, \omega_{n-1}][\omega_n]$.

One can also start with a possibly infinite set of indeterminates $\Omega = \{\omega_i \mid i \in I\}$ and introduce the polynomial ring $R[\Omega]$. Each polynomial in $R[\Omega]$ involves only finitely many indeterminates, so the above definitions regarding $R[\omega_1, \ldots, \omega_n]$ still make sense.

We will usually denote a polynomial in $n$ indeterminates by $f(\omega_1, \ldots, \omega_n)$ or simply by $f$.

## Vector Spaces

A **vector space** over a field $F$ is an additive group $V$ endowed with an additional operation $F \times V \to V$, $(\lambda, v) \mapsto \lambda v$, called the **scalar multiplication**, which satisfies

(a) $(\lambda + \mu)v = \lambda v + \mu v$
(b) $\lambda(v + w) = \lambda v + \lambda w$
(c) $\lambda(\mu v) = (\lambda\mu)v$
(d) $1v = v$

for all $\lambda, \mu \in F$, $v, w \in V$. Note that $\lambda v = 0$ if and only if $\lambda = 0$ or $v = 0$, and $(-\lambda)v = -\lambda v$.

Elements in $V$ are called **vectors** and elements in $F$ are called **scalars**. Instead of the term vector space, we often use the term **linear space** or simply space. A vector space over $\mathbb{R}$ is called a **real (vector) space**, and a vector space over $\mathbb{C}$ is called a **complex (vector) space**.

The prototype example of a vector space is $F^n$, the Cartesian product of $n$ copies of $F$, endowed with componentwise operations

$$(\lambda_1, \ldots, \lambda_n) + (\mu_1, \ldots, \mu_n) = (\lambda_1 + \mu_1, \ldots, \lambda_n + \mu_n),$$
$$\lambda(\lambda_1, \ldots, \lambda_n) = (\lambda\lambda_1, \ldots, \lambda\lambda_n).$$

In particular, $F = F^1$ is a vector space. We have introduced $F[\omega]$ and $C[a, b]$ as rings, but they are also vector spaces if we define scalar multiplication in the usual way.

A nonempty subset $W$ of a vector space $V$ over $F$ is a **(linear) subspace** if $\lambda w + \lambda' w' \in W$ for all $\lambda, \lambda' \in F$ and $w, w' \in W$. For example, the set of all polynomials in $F[\omega]$ of degree at most $n$ is a subspace of $F[\omega]$. Clearly, every subspace is itself a vector space.

A **linear combination** of vectors $v_1, \ldots, v_n$ is every vector of the form $\lambda_1 v_1 + \cdots + \lambda_n v_n$ where $\lambda_i \in F$. If $S$ is a nonempty subset of a space $V$, then the subspace generated by $S$ is equal to the set of all linear combinations of vectors in $S$. It is

called the **linear span** of $S$ and denoted span $S$. Note that, by definition, a linear combination involves a finite number of vectors. If $S$ is an infinite set, then span $S$ is the union of the linear spans of all finite subsets of $S$.

Vector space homomorphisms are **linear maps**, also called **linear operators**. These are maps $\varphi : V \to V'$ that satisfy

$$\varphi(\lambda v + \mu w) = \lambda\varphi(v) + \mu\varphi(w)$$

for all $\lambda, \mu \in F$, $v, w \in V$. The notions such as "kernel," "image," "endomorphism," "isomorphism," etc. should be self-explanatory. Sometimes we will denote linear operators with capital letters ($A$, $B$, etc.). A linear map from a vector space over $F$ to the space $F$ is called a **linear functional**.

A map $f : V_1 \times V_2 \to V'$, where $V_1$, $V_2$, $V'$ are vector spaces over $F$, is said to be **bilinear** if it is linear in each variable, i.e.,

$$f(\lambda u_1 + \mu v_1, v_2) = \lambda f(u_1, v_2) + \mu f(v_1, v_2),$$
$$f(v_1, \lambda u_2 + \mu v_2) = \lambda f(v_1, u_2) + \mu f(v_1, v_2)$$

for all $\lambda, \mu \in F$, $u_1, v_1 \in V_1$, $u_2, v_2 \in V_2$.

A subset $S$ of a vector space $V$ over $F$ is said to be **linearly independent** if for all distinct vectors $s_1, \ldots, s_n \in S$ and all scalars $\lambda_1, \ldots, \lambda_n \in F$, $\lambda_1 s_1 + \cdots + \lambda_n s_n = 0$ implies $\lambda_1 = \cdots = \lambda_n = 0$. If $S$ is not linearly independent, then we say that $S$ is **linearly dependent**. Instead of saying that the set $\{s_i \mid i \in I\}$ is linearly (in)dependent, we can also say that the vectors $s_i$, $i \in I$, are linearly (in)dependent. If we wish to emphasize the role of $F$, we speak about the **$F$-linear (in)dependence**.

A linearly independent subset of $V$ that spans $V$ is called a **basis** of $V$. For example, $\{(1, 0), (1, 1)\}$ is a basis of $F^2$. The standard basis of $F^2$ is $\{(1, 0), (0, 1)\}$, and the standard basis of $F[\omega]$ is $\{1, \omega, \omega^2, \ldots\}$. The basis of $\{0\}$ is $\emptyset$.

A vector space $V$ over $F$ is **finite dimensional** if it is finitely generated, i.e., it is spanned by a finite set. Such a space has a finite basis. Moreover, all its bases have the same cardinality, called the **dimension** of $V$ and denoted $\dim_F V$. (The concept of dimension also makes sense for infinite dimensional spaces, but we will not use it for them.) If $V$ is $n$-dimensional, i.e., $\dim_F V = n$, then every linearly independent set of $n$ vectors in $V$ is a basis of $V$, every subset of $V$ having more than $n$ vectors is linearly dependent, and the dimension of every proper subspace of $V$ is less than $n$. An example of such a space is $F^n$, and every space $V$ with $\dim_F V = n$ is isomorphic to $F^n$.

Let $W$ be a subspace of a vector space $V$ over $F$. Since $W$ is, in particular, an additive subgroup of $V$, we can form the factor group $V/W$. By defining

$$\lambda(v + W) = \lambda v + W$$

for all $\lambda \in F$ and $v \in V$, $V/W$ becomes a vector space. If $V$ is finite dimensional, then the dimension of $V/W$ is equal to $\dim_F V - \dim_F W$. We call this number the **codimension** of $W$.

We say that a linear operator $A$ from one space to another has **finite rank** if its image is a finite dimensional space. In this case, we call the dimension of the image the **rank** of $A$.

Every vector space, finite or infinite dimensional, has a basis. We will actually prove this in Sect. 3.3, but in a more general framework.

## Algebras

A vector space $A$ over a field $F$ is an **algebra** over $F$, or an **$F$-algebra**, if it is endowed with associative multiplication $A \times A \to A$, $(x, y) \mapsto xy$, satisfying

$$(\lambda x + \mu y)z = \lambda(xz) + \mu(yz) \quad \text{and} \quad x(\lambda y + \mu z) = \lambda(xy) + \mu(xz)$$

for all $\lambda, \mu \in F$, $x, y, z \in A$. We often say only "algebra" without specifying the associated field.

An $\mathbb{R}$-algebra is also called a **real algebra**, and a $\mathbb{C}$-algebra is called a **complex algebra**. There is also the notion of an algebra over a commutative unital ring. The axioms are the same, just the role of the field is replaced by a commutative unital ring. Rings can be then viewed as algebras over $\mathbb{Z}$, which conveniently puts rings and algebras under one roof. However, we will consider only algebras over fields in this book.

An algebra is a ring which is also a vector space and multiplication is not only biadditive but bilinear. Many rings mentioned earlier are actually algebras. For example, $F[\omega]$ is an $F$-algebra and $C[a, b]$ is a real algebra. If $A$ is an algebra, then the ring $M_n(A)$ is an algebra if we define scalar multiplication by

$$\lambda(a_{ij}) = (\lambda a_{ij}).$$

Similarly, the ring $A[\omega]$ becomes an algebra by defining

$$\lambda\Big(\sum_{i=0}^{\infty} a_i \omega^i\Big) = \sum_{i=0}^{\infty} (\lambda a_i)\omega^i.$$

We say that $A$ is a **finite dimensional algebra** if it is finite dimensional as a vector space. The dimension of $A$ over $F$ will be denoted by

$$[A : F].$$

Thus, $[A : F]$ is nothing but $\dim_F A$. However, this notation will be used exclusively for algebras.

Algebras are, in particular, rings, so all ring-theoretic notions make sense for algebras. Some of them need small adjustments. A **subalgebra** is a subring which is also a subspace. Thus, for example, the real algebra $\mathbb{R}$ has no proper nonzero subalgebras, but the ring $\mathbb{R}$ has plenty of subrings. Similarly, **algebra ideals**, one-sided or two-sided, are ideals in the ring-theoretic sense which are also subspaces. If $I$ is an ideal of an $F$-algebra $A$, then $A/I$ is both a ring (since $I$ is an ideal of the ring $A$) and a vector space (since $I$ is a subspace of a vector space $A$), and is in fact an $F$-algebra. We call it the **factor algebra** of $A$ by $I$.

The **direct product of algebras** is an algebra with operations defined componentwise. An **algebra homomorphism** is a ring homomorphism which is also a linear map. For example, the aforementioned isomorphism between the ring of matrices of the form $\left[\begin{smallmatrix} \alpha & \beta \\ -\beta & \alpha \end{smallmatrix}\right]$, $\alpha, \beta \in \mathbb{R}$, and the ring $\mathbb{C}$ is actually an $\mathbb{R}$-algebra isomorphism.

The **center of an algebra** is defined just as the center of a ring. A **unital algebra** is an algebra which is unital as a ring, and a **division algebra** is an algebra which is a division ring. We define the **characteristic of an $F$-algebra** $A$, $\operatorname{char}(A)$, as the characteristic of $A$ considered as a ring. The axioms readily imply that $\operatorname{char}(A)$ coincides with the characteristic of the associated field $F$ as long as $A$ is nonzero; therefore $\operatorname{char}(A)$ is either 0 or prime. Hence, $\mathbb{Z}_n$ is not an algebra over any field if $n$ is not prime ($\mathbb{Z}_p$, with $p$ prime, is a field and therefore an algebra over itself). It is also easy to see that $\mathbb{Z}$ is not an algebra over any field.

An $F$-algebra is, in particular, a vector space. Let $\{e_i \mid i \in I\}$ be its basis. For each pair $i, j \in I$ there exist $\alpha_{ijk} \in F$, $k \in I$, such that

$$e_i e_j = \sum_{k \in I} \alpha_{ijk} e_k$$

and all but finitely many $\alpha_{ijk}$'s are 0. We call these equations the **multiplication table**.

Two $F$-algebras $A$ and $B$ are isomorphic, $A \cong B$, if and only if they have the same multiplication tables with respect to some bases. By this we mean that there exist a basis $\{e_i \mid i \in I\}$ of $A$, a basis $\{f_i \mid i \in I\}$ of $B$, and scalars $\alpha_{ijk} \in F$, $i, j, k \in I$, such that $e_i e_j = \sum_{k \in I} \alpha_{ijk} e_k$ and $f_i f_j = \sum_{k \in I} \alpha_{ijk} f_k$ for all $i, j \in I$. Indeed, if this holds, then the linear map that sends $e_i$ into $f_i$ is an algebra isomorphism (note that such a map exists and is unique). Conversely, an algebra isomorphism from $A$ onto $B$ clearly maps a basis of $A$ onto a basis of $B$ and preserves the multiplication table. Thus, if $[A : F] = n$, then the multiplication in $A$ is determined by $n^3$ scalars $\alpha_{ijk}$. Note that not every choice of the $\alpha_{ijk}$'s is appropriate since the associativity condition must be fulfilled.

Let $A$ be a nonzero unital algebra. We may consider $F$ as a one-dimensional subalgebra of $A$ via the canonical embedding $\lambda \mapsto \lambda 1$. Accordingly, the element $\lambda 1 \in A$ will be often written simply as $\lambda$. Given $a \in A$ and a polynomial

$$f(\omega) = \lambda_0 + \lambda_1\omega + \cdots + \lambda_n\omega^n \in F[\omega],$$

we define the **evaluation** of $f$ at $a$ as

$$f(a) = \lambda_0 + \lambda_1 a + \cdots + \lambda_n a^n \in A.$$

The map $f \mapsto f(a)$ is a homomorphism from $F[\omega]$ to $A$. If $f$ has constant term 0, then we can define $f(a)$ also when $A$ is not unital.

We conclude this section with a particularly important example of an algebra. Let $V$ be a vector space over $F$. Then $\mathrm{End}_F(V)$, the set of all endomorphisms of $V$ (i.e., linear operators from $V$ to $V$), becomes an $F$-algebra if we define addition, scalar multiplication, and multiplication by

$$\begin{aligned}(A + B)(v) &= A(v) + B(v),\\ (\lambda A)(v) &= \lambda A(v),\\ (AB)(v) &= A(B(v)).\end{aligned}$$

Suppose $\dim_F V = n < \infty$. Take a basis $\{e_1, \ldots, e_n\}$ of $V$. For every $A \in \mathrm{End}_F(V)$, there exist scalars $a_{ij} \in F$ such that

$$A(e_j) = \sum_{i=1}^{n} a_{ij} e_i, \quad j = 1, \ldots, n.$$

A straightforward verification shows that the map $A \mapsto (a_{ij})$ is an algebra isomorphism from $\mathrm{End}_F(V)$ onto the matrix algebra $M_n(F)$ (in Sect. 3.4 we will prove a generalization of this fact). Thus,

$$\mathrm{End}_F(V) \cong M_n(F) \text{ if } \dim_F V = n.$$

In particular, $[\mathrm{End}_F(V) : F] = (\dim_F V)^2$.

## Fields

We start with a field construction which is usually considered a part of commutative ring theory rather than field theory. A commutative ring without zero-divisors is called a **commutative domain** (the reader is probably familiar with a more common term "integral domain," but usually one assumes that integral domains must be unital, which we do not need right now). Typical examples are the ring of integers $\mathbb{Z}$ and the polynomial ring $F[\omega]$ where $F$ is a field. Of course, fields are also commutative domains, but they are too "perfect" to be listed among typical examples.

The field $\mathbb{Q}$ is constructed from the commutative domain $\mathbb{Z}$. Essentially the same construction produces a field from any nonzero commutative domain $Z$. Let us sketch it (for more details the reader can look at Sect. 7.1 where this construction is presented in a slightly more general context). Define the relation $\sim$ on the set $Z \times \big(Z \setminus \{0\}\big)$ by $(r, s) \sim (r', s')$ if $rs' = r's$. It is easy to check that this is an equivalence relation. Denote by $rs^{-1}$ the equivalence class of $(r, s)$. Thus, $rs^{-1} = r's'^{-1}$ if and only if $rs' = r's$. Define addition and multiplication on the set $\widehat{Z}$ of all equivalence classes as follows:

$$rs^{-1} + tu^{-1} = (ru + ts)(su)^{-1},$$
$$rs^{-1} \cdot tu^{-1} = rt(su)^{-1}.$$

These operations are well-defined and $\widehat{Z}$ is a field. It is called the **field of quotients** of $Z$. One can embed $Z$ into $\widehat{Z}$ and accordingly consider $Z$ as a subring of $\widehat{Z}$. We remark that there does not exist a subfield $E$ of $\widehat{Z}$ such that $Z \subseteq E \subsetneq \widehat{Z}$. If $Z = \mathbb{Z}$, then $\widehat{Z} = \mathbb{Q}$. If $Z = F[\omega]$, then $\widehat{Z}$ is called the **field of rational functions over** $F$ **in** $\omega$ and is denoted $F(\omega)$. More generally, if $\Omega$ is an arbitrary set of indeterminates, then the field of quotients of $F[\Omega]$ is called the **field of rational functions over** $F$ **in** $\Omega$ and is denoted $F(\Omega)$.

We proceed to the genuine field theory. A **field extension** of a field $F$ is any field $K$ that contains $F$ as a subfield. For instance, $\mathbb{R}$ is a field extension of $\mathbb{Q}$ and $\mathbb{C}$ is a field extension of $\mathbb{R}$. Given a field $F$, one usually wants to construct field extensions satisfying certain requirements. The original field $F$ is often contained in all other fields that occur in the study. In this case, we call it the **base field**.

If $K$ is a field extension of $F$, then we may view $K$ as an algebra over $F$. Indeed, $K$ is a ring by definition, and the scalar multiplication should be simply understood as the given multiplication in $K$ restricted to the situation where the first factor lies in $F$. We say that $K$ is a **finite extension** of $F$ if $[K : F] < \infty$. For example, $\mathbb{C}$ is a finite extension of $\mathbb{R}$ with $[\mathbb{C} : \mathbb{R}] = 2$, while $\mathbb{R}$ is not a finite extension of $\mathbb{Q}$. If $K$ is a finite extension of $F$ and $L$ is a finite extension of $K$, then $L$ is a finite extension of $F$ and $[L : F] = [L : K] \cdot [K : F]$.

The main point is that $K$ can be viewed as a vector space over $F$. Let us give a simple illustration of the usefulness of this viewpoint. Let $K$ be a **finite field**. Then $\mathrm{char}(K)$ is a prime number $p$. Hence, $K$ can be considered as a finite dimensional space over the prime subfield $F_0 \cong \mathbb{Z}_p$, and as such is isomorphic to $F_0^n$ for some $n \in \mathbb{N}$. Therefore, $|K| = |F_0^n| = p^n$. We have thereby found out that the cardinality of a finite field can only be $p^n$ with $p$ prime and $n \in \mathbb{N}$. This assertion has a sort of converse, which is far less obvious: For every prime $p$ and $n \in \mathbb{N}$, there exists an (up to isomorphism) unique field having $p^n$ elements.

An element $a$ in a field extension $K$ of $F$ is said to be a **root** of the polynomial $f(\omega) \in F[\omega]$ if $f(a) = 0$. For example, $\sqrt{2} \in \mathbb{R}$ is a root of $\omega^2 - 2 \in \mathbb{Q}[\omega]$. Roots of polynomials $\omega^n - 1$, $n \in \mathbb{N}$, are called **roots of unity**. We say that $a \in K$ is **algebraic** over $F$ if $a$ is a root of some nonzero polynomial in $F[\omega]$. If $p(\omega) \in F[\omega]$ is the monic polynomial of the lowest degree such that $p(a) = 0$, we call $p(\omega)$ the

**minimal polynomial** of the algebraic element $a$. Such a polynomial is irreducible over $F$. If the degree of the minimal polynomial $p(\omega)$ of $a$ is $m$, we say that $a$ is **algebraic of degree** $m$.

Let $a_1, \ldots, a_n$ be elements of a field extension $K$ of $F$. By $F(a_1, \ldots, a_n)$ we denote the subfield of $K$ generated by $F$ and all $a_i$. If each $a_i$ is algebraic over $F$, then $F(a_1, \ldots, a_n)$ is a finite extension of $F$. More can be said if $n = 1$: If $a$ is algebraic of degree $m$, then $[F(a) : F] = m$, $\{1, a, \ldots, a^{m-1}\}$ is a basis of the vector space $F(a)$ over $F$, and $F(a) \cong F[\omega]/(p(\omega))$ where $(p(\omega))$ is the principal ideal generated by the minimal polynomial $p(\omega)$ of $a$. For example, $[\mathbb{Q}(\sqrt{2}) : \mathbb{Q}] = 2$ and $\mathbb{Q}(\sqrt{2}) = \{\lambda_0 + \lambda_1\sqrt{2} \mid \lambda_0, \lambda_1 \in \mathbb{Q}\} \cong \mathbb{Q}[\omega]/(\omega^2 - 2)$.

If $a \in K$ is a root of the polynomial $f(\omega) \in F[\omega]$, then there exists a polynomial $q(\omega) \in K[\omega]$ such that $f(\omega) = (\omega - a)q(\omega)$. We say that $a$ is a **simple root** of $f(\omega)$ if $q(a) \neq 0$. If $f(\omega)$ has degree $n$, then it has at most $n$ roots in $K$. Suppose $f(\omega)$ can be written as $f(\omega) = c(\omega - a_1) \ldots (\omega - a_n)$, where $c \in F$ is the leading coefficient of $f(\omega)$ and $a_1, \ldots, a_n$ are not necessarily distinct elements in $K$. If, in this situation, $K = F(a_1, \ldots, a_n)$, then we call $K$ a **splitting field** for $f(\omega)$ over $F$. This is thus the smallest field extension of $F$ that contains all roots of $f(\omega)$. A splitting field for a nonconstant polynomial always exists and is unique up to isomorphism. We say that $f(\omega) \in F[\omega]$ is a **separable polynomial** if all of its roots in a splitting field are simple. That is, $f(\omega) = c(\omega - a_1) \ldots (\omega - a_n)$ and $a_i \neq a_j$ if $i \neq j$.

We say that $K$ is an **algebraic extension** of $F$ if all elements in $K$ are algebraic over $F$. Every finite extension is algebraic. An algebraic extension $K$ of $F$ with the property that the minimal polynomial of every element in $K$ is separable is called a **separable extension** of $F$. The **primitive element theorem** states that for every finite separable extension $K$ of $F$ there exists an $a \in K$ such that $K = F(a)$.

A field $F$ with the property that every finite extension of $F$ is separable is called a **perfect field**. All fields of characteristic 0, all algebraically closed fields, and all finite fields are perfect. A field that is not perfect is called **imperfect**. An example is the field of rational functions $\mathbb{Z}_p(\omega)$ with $p$ a prime.

A field $K$ is said to be **algebraically closed** if every nonconstant polynomial in $K[\omega]$ has a root in $K$. Equivalently, every polynomial $f(\omega) \in K[\omega]$ of degree $n \geq 1$ splits into linear factors: $f(\omega) = c(\omega - a_1) \ldots (\omega - a_n)$ where $c, a_i \in K$ ($a_i$ are not necessarily distinct). The **fundamental theorem of algebra** states that the field $\mathbb{C}$ is algebraically closed. The polynomial $\omega^2 + 1$ does not have a root in $\mathbb{R}$, so $\mathbb{R}$ is not algebraically closed. From the fundamental theorem of algebra it can be easily deduced that every nonconstant polynomial in $\mathbb{R}[\omega]$ splits into linear factors and quadratic factors $a\omega^2 + b\omega + c$ where $a, b, c \in \mathbb{R}$ are such that $b^2 - 4ac < 0$.

Working with algebraically closed fields is often essentially easier than working with general fields. Let us only point out one matter that is of importance in noncommutative algebra. Recall that square matrices $A$ and $B$ are **similar** if there exists an invertible matrix $T$ such that $A = TBT^{-1}$. If $K$ is an algebraically closed field, then every matrix $A \in M_n(K)$ is similar to a block diagonal matrix $J \in M_n(K)$ whose blocks are matrices in $M_{k_i}(K)$, $n = \sum_i k_i$, having a fixed scalar $\lambda_i$ on the main diagonal, the scalar 1 on the superdiagonal, and zeros elsewhere. The

matrix $J$ is called the **Jordan normal form** of $A$. The scalars $\lambda_i$ appearing in $J$ are the eigenvalues of $A$.

A field extension $\overline{F}$ of a field $F$ is said to be an **algebraic closure** of $F$ if $\overline{F}$ is algebraically closed and every element in $\overline{F}$ is algebraic over $F$. For example, $\mathbb{C}$ is an algebraic closure of $\mathbb{R}$. Every field $F$ has an algebraic closure, which, moreover, is unique up to isomorphism. This is one of the deeper results stated in this preparatory chapter.

# Symbols

| | |
|---|---|
| $\mathbb{N}$ | Set of natural numbers |
| $\mathbb{Z}$ | Ring of integers |
| $\mathbb{Z}_n$ | Ring of integers modulo $n$ |
| $\mathbb{Q}$ | Field of rational numbers |
| $\mathbb{R}$ | Field of real numbers |
| $\mathbb{C}$ | Field of complex numbers |
| $\mathbb{H}$ | Division algebra of quaternions |
| $\delta_{ij}$ | Kronecker delta |
| $\lvert S\rvert$ | Cardinality of the set $S$ |
| $S \setminus T$ | Set difference |
| $\mathrm{id}_S$ | Identity map on $S$ |
| $\det(A)$ | Determinant of the matrix $A$ |
| $\mathrm{tr}(A)$ | Trace of the matrix $A$ |
| $M^*$ | Group of invertible elements of the monoid $M$ |
| $S_n$ | Symmetric group on $\{1, \ldots, n\}$ |
| $\mathrm{sgn}(\sigma)$ | Sign of the permutation $\sigma$ |
| $R \cong R'$ | Isomorphic rings (or algebras, modules, etc.) |
| $\ker \varphi$ | Kernel of the homomorphism $\varphi$ |
| $\mathrm{im}\, \varphi$ | Image of the homomorphism $\varphi$ |
| $R/I$ | Factor ring |
| $IJ$ | Product of ideals |
| $[a, b]$ | Commutator of $a$ and $b$ |
| $Z(R)$ | Center of the ring $R$ |
| $\mathrm{char}(R)$ | Characteristic of $R$ |
| $R^{\mathrm{o}}$ | Opposite ring of $R$ |
| $M_n(R)$ | Ring of all $n \times n$ matrices over $R$ |
| $E_{ij}$ | Standard matrix units |
| $e_{ij}$ | Matrix units |
| $\Pi_{i \in I} R_i$ | Direct product of a family of rings |
| $R_1 \times \cdots \times R_n$ | Direct product of a finite family of rings |
| $R[\omega]$ | Ring of polynomials over $R$ in one indeterminate |

| | |
|---|---|
| $R[\Omega]$ | Ring of polynomials over $R$ in indeterminates from the set $\Omega$ |
| $R[[\omega]]$ | Formal power series ring over $R$ |
| $\mathrm{soc}(R)$ | Socle of $R$ |
| $\mathrm{rad}(R)$ | Jacobson radical of $R$ |
| $\dim_F V$ | Dimension of the vector space $V$ over the field $F$ |
| $[A : F]$ | Dimension of the algebra $A$ over $F$ |
| $T_n(F)$ | Algebra of upper triangular $n \times n$ matrices over $F$ |
| $C[a, b]$ | Algebra of continuous functions from $[a, b]$ to $\mathbb{R}$ |
| $\mathscr{A}_n$ | $n$th Weyl algebra |
| $F[G]$ | Group algebra of the group $G$ over $F$ |
| $L_a$ | Left multiplication map $x \mapsto ax$ |
| $R_b$ | Right multiplication map $x \mapsto xb$ |
| $M(A)$ | Multiplication algebra of the algebra $A$ |
| $A^\sharp$ | Unitization of $A$ |
| $A_K$ | Scalar extension of $A$ to the field $K$ |
| $C_A(S)$ | Centralizer of the set $S$ in $A$ |
| $\mathrm{ann}_R(M)$ | Annihilator of the $R$-module $M$ |
| $\mathrm{End}_R(M)$ | Ring of endomorphisms of the $R$-module $M$ |
| $\mathrm{End}_D(V)$ | Ring of linear operators of the vector space $V$ over the division ring $D$ |
| $\mathrm{End}_F(V)$ | Algebra of linear operators of the vector space $V$ over the field $F$ |
| $\oplus_{i\in I} M_i$ | Direct sum of a family of submodules (or additive subgroups, or subspaces) |
| $M_1 \oplus \cdots \oplus M_n$ | Direct sum of a finite family of submodules (or additive subgroups, or subspaces) |
| $M_1 + \cdots + M_n$ | Sum of a finite family of submodules (or additive subgroups, or subspaces) |
| $U \otimes V$ | Tensor product of vector spaces |
| $u \otimes v$ | Simple tensor |
| $\varphi \otimes \psi$ | Tensor product of linear maps |
| $A \otimes B$ | Tensor product of algebras |
| $F\langle\xi, \eta\rangle$ | Free algebra in two indeterminates |
| $F\langle X\rangle$ | Free algebra in indeterminates from the set $X$ |
| $\deg(f)$ | Degree of the polynomial $f$ |
| $s_n$ | Standard polynomial |
| $c_n$ | Capelli polynomial |
| $\mathrm{Id}(A)$ | T-ideal of all polynomial identities of the algebra $A$ |
| $\mathrm{GM}_n(F)$ | Algebra of generic matrices |
| $\widehat{Z}$ | Field of quotients of the commutative domain $Z$ |
| $F(\Omega)$ | Field of rational functions in $\Omega$ |
| $Q_Z(R)$ | Ring of central quotients of the ring $R$ |
| $Q_{rc}(R)$ | Right classical ring of quotients of $R$ |
| $Q_r(R)$ | Right Martindale ring of quotients of $R$ |
| $R_C$ | Central closure of $R$ |
| $\mathrm{UD}_n(F)$ | Universal division algebra |

# Chapter 1
# Finite Dimensional Division Algebras




Finite dimensional division algebras are as simple and "spotless" as a (not necessarily commutative) ring can be. Their definition gathers together all of the most favorable properties. They are also the oldest, the most classical topic of noncommutative algebra. However, the simplest objects are not always the easiest to study, and classical themes are not necessarily less profound than modern ones. In many algebra textbooks, finite dimensional division algebras are treated in the last chapters by using advanced tools. Our goals in this chapter, however, are relatively modest. We will give an elementary, self-contained introduction to finite dimensional division algebras, which in particular includes two beautiful old results: Frobenius' theorem on real division algebras and Wedderburn's theorem on finite division rings.

We shall not strictly confine ourselves neither to division algebras nor to finite dimensional algebras. Presenting some of the results in more general contexts will not change the level of complexity of arguments and shall prove useful later.

## 1.1 After the Complex Numbers: What Comes Next?

A 2-dimensional real vector space can be transformed into the complex numbers by appropriately defining multiplication. Just take two linearly independent vectors, denote one of them by 1 to indicate that it plays the role of unity, require that the square of the other one is $-1$, and extend this multiplication to the whole space by bilinearity. If one is interested in the geometric interpretation of this multiplication, then one will choose these two vectors more carefully. However, from the algebraic point of view it does not matter which two vectors we take, as long as they are linearly independent and therefore form a basis of our space. In any case we get a 2-dimensional $\mathbb{R}$-algebra isomorphic to the complex number field $\mathbb{C}$.

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Why confine ourselves to 2-dimensional spaces? Are there other ways to create some kind of "numbers" from finite dimensional real vector spaces? In any set of "numbers" one should be able to add, subtract, and multiply elements, and multiplication should be associative and bilinear. Thus, our space should in particular be an $\mathbb{R}$-algebra (in the early literature elements from algebras were actually called *hypercomplex numbers*). In a decent set of "numbers" one should also be able to divide elements, with the obvious exclusion of dividing by 0. Recall from Prerequisites that a unital algebra in which every nonzero element is invertible is called a *division algebra*. We can now rephrase our question in a rigorous manner as follows: Is it possible to define multiplication on an $n$-dimensional real space so that it becomes a real division algebra?

For $n = 1$, the question is trivial: Every element is a scalar multiple of unity and therefore, up to isomorphism, $\mathbb{R}$ itself is the only such algebra. For $n = 2$ we know one example, $\mathbb{C}$, but are there any others? This question is easy and the reader can try to answer it immediately. What about $n = 3$? After finding a multiplication in 2-dimensional spaces of such enormous importance in mathematics, should not the next step be finding something similar in 3-dimensional spaces? Furthermore, what about $n = 4, 5, 6, \ldots$ ?

We will give complete answers to the above questions. The only tools we will need are the following standard results:

(A) Every nonconstant polynomial $f(\omega) \in \mathbb{R}[\omega]$ is a product of linear and quadratic polynomials in $\mathbb{R}[\omega]$.
(B) Every linear operator from an odd-dimensional real vector space to itself has a real eigenvalue.
(C) If $U$ is a proper subspace of a finite-dimensional real inner product vector space $V$, then $V$ contains a unit vector $e$ orthogonal to all vectors in $U$.

Throughout the section, we *assume that $D$ is an $n$-dimensional division $\mathbb{R}$-algebra*. Recall the notational convention from Prerequisites: For every $\lambda \in \mathbb{R}$, we write $\lambda 1 \in D$ simply as $\lambda$. In fact, we identify $\mathbb{R}$ with $\mathbb{R}1$ and in this way consider $\mathbb{R}$ as a subalgebra of $D$.

Suppose $n$ is odd. Take $d \in D$ and consider the linear map $x \mapsto dx$ from $D$ to $D$. By (B), there exists a $\lambda \in \mathbb{R}$ and a nonzero $x \in D$ such that $dx = \lambda x$. That is, $(d - \lambda)x = 0$. Since $x \neq 0$, this yields $d = \lambda \in \mathbb{R}$. Thus, $n = 1$ and hence $D \cong \mathbb{R}$.

Although not needed later, this observation gives us an insight into our problem. In particular, we now know that there are no 3-dimensional real division algebras. After $n = 2$, the lowest possible dimension of $D$ is thus $n = 4$.

We now make the first step in solving the problem.

**Lemma 1.1** *For every $x \in D$ there exists a $\beta \in \mathbb{R}$ such that $x^2 + \beta x \in \mathbb{R}$.*

***Proof*** Since the dimension of $D$ is $n$, the elements $1, x, \ldots, x^n$ are linearly dependent. This means that there exists a nonconstant polynomial $f(\omega) \in \mathbb{R}[\omega]$ of degree at most $n$ such that $f(x) = 0$. Thus, in light of (A), there exist $\alpha_i, \beta_i, \gamma_i \in \mathbb{R}$ such that

$$(x-\alpha_1)\cdots(x-\alpha_r)(x^2+\beta_1 x+\gamma_1)\cdots(x^2+\beta_s x+\gamma_s)=0.$$

As $D$ is a division algebra, one of the factors must be 0. That is, $x$ is a root of a linear or quadratic polynomial in $\mathbb{R}[\omega]$. In each case, the desired conclusion holds. □

The 1-dimensional subspace $\mathbb{R}i$ of $\mathbb{C}$ can be described as the set of all $z \in \mathbb{C}$ such that $z^2$ is a nonpositive real number. The analysis of the set of all such elements in our abstract setting will lead us to the answers to our questions. We thus define

$$V = \{v \in D \mid v^2 \leq 0\}.$$

Here, $v^2 \leq 0$ means that $v^2$ is a nonpositive scalar multiple of 1. Note that $V \cap \mathbb{R} = \{0\}$. We now examine other properties of $V$.

**Lemma 1.2** *If $u, v \in V$ are linearly independent, then so are $u$, $v$, and 1.*

***Proof*** If the lemma is not true, there must exist $\beta_1, \beta_2 \in \mathbb{R}$ such that $1+\beta_1 u = \beta_2 v$. However, squaring both sides gives $2\beta_1 u \in \mathbb{R}$ and hence $\beta_1 = 0$, yielding the contradiction that $1 \in V$. □

For any $x, y \in D$ we write

$$x \circ y = xy + yx.$$

We call $x \circ y$ the **Jordan product** of $x$ and $y$. Observe that $x \circ y = y \circ x$ and $x \circ (y+z) = x \circ y + x \circ z$. If $x \circ y = 0$, then we say that $x$ and $y$ **anticommute**.

**Lemma 1.3** *For all $u, v \in V$, $u \circ v \in \mathbb{R}$.*

***Proof*** We may assume that $u$ and $v$ are linearly independent. By Lemma 1.2, $W = \mathrm{span}\{1, u, v\}$ is a 3-dimensional space. Lemma 1.1 implies that $x^2 \in W$ for all $x \in W$, and hence $x \circ y = (x+y)^2 - x^2 - y^2 \in W$ for all $x, y \in W$. The linear operator $x \mapsto x \circ v$ therefore maps $W$ to itself. By (B), there exist a $\lambda \in \mathbb{R}$ and a $w \neq 0$ in $W$ such that $w \circ v = \lambda w$. That is, $wvw^{-1} = \lambda - v$. Squaring both sides and using $v^2 \in \mathbb{R}$ we obtain $-2\lambda v \in \mathbb{R}$ and hence $\lambda = 0$ since $v \notin \mathbb{R}$. Thus, $w \circ v = 0$. Therefore, writing $w$ as $\alpha_0 + \alpha_1 u + \alpha_2 v$ we see that $\alpha_1 \neq 0$, from which it follows that $u \circ v \in \mathrm{span}\{1, v\}$. Since $u$ and $v$ occur symmetrically, we must also have $u \circ v \in \mathrm{span}\{1, u\}$. Now, $u$, $v$, and 1 are linearly independent and so $u \circ v \in \mathbb{R}$ follows. □

**Lemma 1.4** *If $x \in D$ satisfies $x^2 \in \mathbb{R}$, then $x \in \mathbb{R}$ or $x \in V$.*

***Proof*** If $x \notin V$, then $x^2 > 0$ and hence there exists a $\gamma \in \mathbb{R}$ such that $x^2 = \gamma^2$. Consequently, $(x-\gamma)(x+\gamma) = 0$ and therefore $x = \pm\gamma \in \mathbb{R}$. □

**Lemma 1.5** *The set $V$ is a linear subspace and $D = \mathbb{R} \oplus V$.*

***Proof*** Take $x \in D \setminus \mathbb{R}$ and let $\beta$ be as in Lemma 1.1. Then $(x + \frac{\beta}{2})^2 \in \mathbb{R}$ and hence $x + \frac{\beta}{2} \in V$ by Lemma 1.4. This implies that every element in $D$ is a sum

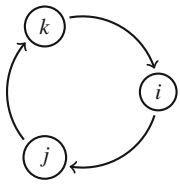



**Fig. 1.1** Multiplication of quaternions

of a real number and an element from $V$. We still have to prove that $V$ is a linear subspace. As it is clearly closed under scalar multiplication, it is enough to show that $u+v \in V$ for each linearly independent pair $u, v \in V$. To this end, observe that $(u+v)^2 = u^2 + u \circ v + v^2 \in \mathbb{R}$ by Lemma 1.3. Since $u+v \notin \mathbb{R}$ by Lemma 1.2, $u+v \in V$ follows from Lemma 1.4. □

Lemma 1.3 implies that

$$\langle u \mid v \rangle = -\frac{1}{2} u \circ v$$

defines an inner product on $V$. The rest of the section is based on this observation.

**Lemma 1.6** *If $n = 1$, then $D \cong \mathbb{R}$. If $n = 2$, then $D \cong \mathbb{C}$. If $n > 2$, then there exist $i, j, k \in D$ such that*

$$\begin{gathered} i^2 = j^2 = k^2 = -1, \\ ij = -ji = k, \quad jk = -kj = i, \quad ki = -ik = j, \end{gathered} \tag{1.1}$$

*and $1, i, j, k$ are linearly independent.*

***Proof*** The case where $n = 1$ is clear, so let $n > 1$. By Lemma 1.5, $V$ has dimension $n-1$. Let $i \in V$ be a unit vector (with respect to the inner product). Then $i^2 = -1$, so $D \cong \mathbb{C}$ if $n = 2$. Let $n > 2$. By (C), there is a unit vector $j \in V$ orthogonal to $i$. Set $k = ij$. It is straightforward to check that (1.1) holds, from which we obtain that

$$(\alpha_0 + \alpha_1 i + \alpha_2 j + \alpha_3 k)(\alpha_0 - \alpha_1 i - \alpha_2 j - \alpha_3 k) = \alpha_0^2 + \alpha_1^2 + \alpha_2^2 + \alpha_3^2 \tag{1.2}$$

is a positive real number for any $\alpha_0, \alpha_1, \alpha_2, \alpha_3 \in \mathbb{R}$ that are not all 0. This proves, in particular, that $1, i, j, k$ are linearly independent. □

The second equation in (1.1) can be easily remembered with the help of Fig. 1.1: going clockwise, the product of two consecutive elements is the third one; going counterclockwise, we obtain the negative of the third one.

If $n = 4$, then $D$ has a basis $1, i, j, k$ with multiplication table (1.1). This is a real finding. Let us change the point of view. Suppose we are given a 4-dimensional real vector space $D$ with basis $\{1, i, j, k\}$ in which we define multiplication by (1.1) and the requirement that 1 is a unity. A simple verification shows that $D$ is indeed an

$\mathbb{R}$-algebra (note that it is enough to check associativity for elements from a basis!). We define the **conjugate** of an element $h = \alpha_0 + \alpha_1 i + \alpha_2 j + \alpha_3 k$ in $D$ by

$$\overline{h} = \alpha_0 - \alpha_1 i - \alpha_2 j - \alpha_3 k.$$

If $h \neq 0$, then $h\overline{h}$ $(= \overline{h}h)$ is a nonzero scalar by (1.2). Therefore, $h$ is invertible with inverse $\frac{1}{h\overline{h}}\overline{h}$. Accordingly, $D$ is a division algebra.

The standard notation for this 4-dimensional noncommutative real division algebra is $\mathbb{H}$, and its elements are called (real) **quaternions**. With $\mathbb{H}$ in hand, we are now in a position to answer our questions from the beginning. The following theorem is named after the German mathematician F. G. Frobenius.

**Theorem 1.7 (Frobenius' Theorem)** *A finite dimensional real division algebra $D$ is isomorphic to $\mathbb{R}$, $\mathbb{C}$, or $\mathbb{H}$.*

***Proof*** In light of Lemma 1.6, it suffices to show that the dimension $n$ of $D$ is at most 4. Suppose this is not true. Then the dimension of $V$ is greater than 3 by Lemma 1.5. Using (C), we see that there is a unit vector $e \in V$ orthogonal to $\text{span}\{i, j, k\}$. This means that each of $i$, $j$, $k$ anticommutes with $e$. However, since $k = ij$ and the product of two elements anticommuting with $e$ obviously commutes with $e$, we arrive at a contradiction that $ek = 0$. □

## 1.2 Beyond Frobenius' Theorem

The notation $\mathbb{H}$ for the algebra of quaternions is in honor of its discoverer, the Irish mathematician W. R. Hamilton. As the first noncommutative algebraic system ever introduced, $\mathbb{H}$ is of great historic value. Before this discovery, Hamilton had been struggling for a long time to extend the multiplication of complex numbers to the 3-dimensional space, that is, he had wanted to solve our problem for $n = 3$. We have seen that this had been hopeless. On October 16, 1843, when walking with his wife along the Royal Canal in Dublin, the idea of 4-dimensional quaternions suddenly came into his mind and he carved the formula $i^2 = j^2 = k^2 = ijk = -1$ in a stone of Broom Bridge. This story may have a flavor of romantic legend, but is documented in Hamilton's letters.

Frobenius established Theorem 1.7 in 1878. This is a classification theorem par excellence. Three important examples of finite dimensional real division algebras were discovered by mathematicians, each of them has turned out to be immensely important, and, up to isomorphism, these three are in fact the only ones that exist.

A famous theorem is always a source of inspiration and often a basis for further development. Some extensions of Frobenius' theorem concern **nonassociative** algebras, i.e., algebras in which multiplication is not necessarily associative. This goes beyond the scope of this book. Still, it seems appropriate to at least mention the algebra of **octonions** $\mathbb{O}$, the natural successor in the sequence $\mathbb{R}, \mathbb{C}, \mathbb{H}$. This

is an 8-dimensional nonassociative real algebra, which contains a copy of $\mathbb{H}$ as a subalgebra. Its standard basis contains a unity 1 and seven elements whose squares equal $-1$.

Let us make this precise. We define $\mathbb{O}$ as the vector space $\mathbb{H} \oplus \mathbb{H}$ endowed with the multiplication

$$(a, b)(c, d) = (ac - d\overline{b}, \overline{a}d + cb) \tag{1.3}$$

for all $a, b, c, d \in \mathbb{H}$. If $a, b, c, d$ are real numbers (and hence $\overline{a} = a$ and $\overline{b} = b$), (1.3) reduces to the definition of the multiplication of complex numbers. The multiplication of quaternions can also be alternatively defined by (1.3), just take $a, b, c, d$ to be complex numbers.

This method of constructing new algebras from old by applying (1.3) is called the **Cayley-Dickson process**. We should add here that at each step we define the new conjugation by

$$\overline{(a, b)} = (\overline{a}, -b).$$

Starting with $\mathbb{R}$ and trivial conjugation $\overline{a} = a$, this process thus yields $\mathbb{C}$ in the first step, $\mathbb{H}$ in the second step, $\mathbb{O}$ in the third step, and can be continued.

Although not associative, $\mathbb{O}$ satisfies weak associativity conditions $x^2y = x(xy)$ and $yx^2 = (yx)x$ for all $x, y \in \mathbb{O}$; nonassociative algebras satisfying these two conditions are called **alternative**. Further, $\mathbb{O}$ is a **nonassociative division algebra**, which, by definition, means that for all $a, b \in \mathbb{O}$ with $a \neq 0$, the equations $ax = b$ and $ya = b$ have unique solutions in $\mathbb{O}$.

Let this be enough in our attempt to make the reader curious and challenged for seeking further information about octonions elsewhere. Two books that include an introduction to nonassociative algebras and can be recommended are [10] and [14].

Let us return to the quaternions. A close inspection of the arguments from the previous section gives further insight. First, if $n = 4$ then $V = \operatorname{span}\{i, j, k\}$, and identifying each $u = \alpha_1 i + \alpha_2 j + \alpha_3 k \in V$ with $\vec{u} = (\alpha_1, \alpha_2, \alpha_3) \in \mathbb{R}^3$ we see that the inner product $(u, v) \mapsto -\frac{1}{2}u \circ v$ is exactly the dot product, $-\frac{1}{2}u \circ v = \vec{u} \cdot \vec{v}$. Another interesting observation is that $\frac{1}{2}(uv - vu) = \vec{u} \times \vec{v}$, where $\times$ is the cross product. Using $uv = \frac{1}{2}u \circ v + \frac{1}{2}(uv - vu)$ one now easily sees that $\mathbb{H}$ can be equivalently introduced as the vector space $\mathbb{R} \times \mathbb{R}^3$ (with componentwise addition and scalar multiplication) endowed with the multiplication

$$(\alpha_0, \vec{u})(\beta_0, \vec{v}) = (\alpha_0\beta_0 - \vec{u} \cdot \vec{v}, \alpha_0\vec{v} + \beta_0\vec{u} + \vec{u} \times \vec{v}).$$

The multiplication of quaternions thus involves all basic operations with vectors.

Let us mention that the quaternions can also be introduced as certain $2 \times 2$ complex matrices. We will arrive at this in Example 2.84.

After describing finite dimensional real division algebras, one may wonder what about the complex ones? The answer is simple and comes easily, not only for $\mathbb{C}$ but any algebraically closed field.

**Proposition 1.8** *If $D$ is a finite dimensional division algebra over an algebraically closed field $F$, then $D = F$.*

***Proof*** Let $x \in D$. Like in the proof of Lemma 1.1, we start by observing that the finite dimensionality of $D$ implies that there exists a nonzero polynomial $f(\omega) \in F[\omega]$ such that $f(x) = 0$. Since $F$ is algebraically closed, this means that there exist $\alpha_i \in F$ such that $(x - \alpha_1) \cdots (x - \alpha_r) = 0$. As $D$ is a division algebra, this gives $x - \alpha_i = 0$ for some $i$, and hence $x = \alpha_i \in F$. □

This proposition may seem disappointing after encountering the quaternions. But later we will see its positive aspects.

Observe that the proofs of both Frobenius' theorem and Proposition 1.8 use the finite dimensionality of $D$ only once, at their very beginnings, for showing that every $x \in D$ is a root of a nonzero polynomial. We can therefore use the latter condition as the assumption that replaces the assumption that $D$ is finite dimensional. Let us introduce some terminology to make this clear.

We say that an element $a$ from an $F$-algebra $A$ is **algebraic** if there exists a nonzero polynomial $f(\omega) \in F[\omega]$ such that $f(a) = 0$ (if $A$ is not unital, then the constant term of $f(\omega)$ is 0). This is, of course, just the repetition of the definition given in Prerequisites for field extensions, and just as therein we define the **degree** and the **minimal polynomial** of an algebraic element. If every element in $A$ is algebraic, then $A$ is said to be an **algebraic algebra**.

The following simple fact was used in the aforementioned proofs.

**Lemma 1.9** *Every finite dimensional algebra is an algebraic algebra.*

***Proof*** The set of all powers of each element is linearly dependent. □

Infinite dimensional algebraic algebras do exist and we will encounter them later (for the first time in Example 2.7). The following are therefore more general versions of Theorem 1.7 and Proposition 1.8:

- An algebraic real division algebra is isomorphic to $\mathbb{R}$, $\mathbb{C}$, or $\mathbb{H}$.
- If $D$ is an algebraic algebra over an algebraically closed field $F$, then $D = F$.

In the next proposition we gather together some elementary properties of algebraic algebras.

**Proposition 1.10** *Let $A$ be a unital algebra over a field $F$. If $A$ is algebraic (in particular, if $A$ is finite dimensional), then the following holds:*

(a) *Every nonzero element in $A$ is either invertible or a zero-divisor.*
(b) *If $a, b \in A$ are such that $ab = 1$, then $ba = 1$.*
(c) *If $A$ is a division algebra, then so is every nonzero subalgebra of $A$.*

***Proof*** Take a nonzero $a \in A$ and let $f(\omega) \in F[\omega]$ be its minimal polynomial. We can write $f(a) = 0$ as $ag(a) = \alpha$ where $\alpha$ is the negative of the constant term of $f$ and $g(\omega) \in F[\omega]$ is a nonzero polynomial with lower degree than $f(\omega)$. The latter implies that $g(a) \neq 0$. If $\alpha \neq 0$, then $a^{-1} = \alpha^{-1}g(a)$; if $\alpha = 0$, then $ag(a) = g(a)a = 0$ and so $a$ is a zero-divisor and cannot have neither a left or right inverse. This proves (a) and (b). If $A$ is a division algebra, then $ag(a) = \alpha \neq 0$. Therefore, every proper subalgebra of $A$ contains the unity of $A$ as well as the inverse of each of its elements. This proves (c). □

The following examples show that Proposition 1.10 does not hold in all algebras.

**Example 1.11** The polynomial algebra $F[\omega]$ contains elements that are neither invertible nor zero-divisors. Moreover, it is contained in the field of rational functions $F(\omega)$. Thus, (a) and (c) do not always hold.

**Example 1.12** From calculus we know that the differentiation and integration are inverse operations. One needs to be a little careful to interpret this statement correctly. If by integration we mean the operator $J$ given by $J(f)(x) = \int_0^x f(t)dt$, then by first integrating and then differentiating $f$ we get $f$, but if we first differentiate and then integrate the constant function 1 we get 0.

Let us translate this phenomenon to an appropriate algebraic setting. Let $D, J$ be elements in the algebra $\mathrm{End}_{\mathbb{R}}(\mathbb{R}[\omega])$, defined by $D(\omega^n) = n\omega^{n-1}$ and $J(\omega^n) = \frac{\omega^{n+1}}{n+1}$, $n \geq 0$ (note that it is enough to specify the action of a linear operator on a basis). Then $DJ = I$, but $JD \neq I$. Thus, $D$ is right invertible but not left invertible, and $J$ is left invertible but not right invertible.

We have chosen this example because of its connection to calculus. Here is a slightly simpler example of two operators $A$ and $B$ that satisfy $AB = I$ and $BA \neq I$: $A(1) = 0$, $A(\omega^n) = \omega^{n-1}$, $n \geq 1$, and $B(\omega^n) = \omega^{n+1}$, $n \geq 0$.

Thus, (b) does not hold in every algebra. We also remark that $ab = 1$ and $ba \neq 1$ implies that $a(1 - ba) = 0$ and $cab = c$ for every element $c$. Therefore, $a$ is a left zero-divisor but not a right zero-divisor. Similarly, $b$ is a right zero-divisor but not a left zero-divisor.

Proposition 1.10 (c) along with Frobenius' theorem gives the following corollary.

**Corollary 1.13** *A proper subalgebra of* $\mathbb{H}$ *of dimension more than* 1 *is isomorphic to* $\mathbb{C}$.

It may be an instructive exercise for the reader to prove Corollary 1.13 directly, without referring to Frobenius' theorem. A standard result, which we present in the next lemma, reduces the problem to subalgebras of dimension two. We first make some preparatory remarks.

A subalgebra $K$ of a unital $F$-algebra $A$ which is a field and contains $F$ (that is, it contains the unity of $A$), is called a **subfield** of $A$. We can consider $A$ as a vector space over a subfield $K$. Indeed, we regard the multiplication of elements from $K$ with elements from $A$ as scalar multiplication $K \times A \to A$. The vector space axioms are readily fulfilled. We remark that this does not yet mean that $A$ is

also a $K$-algebra; this is true only if $K$ is contained in the center $Z(A)$ of $A$. For example, $\mathbb{H}$ is a vector space, but not an algebra, over $\mathbb{R} \oplus \mathbb{R}i \cong \mathbb{C}$. Nevertheless, we will write $[A : K]$ for the dimension of $A$ over $K$ even when $K \not\subseteq Z(A)$.

**Lemma 1.14** *If $A$ is finite dimensional unital $F$-algebra and $K$ is a subfield of $A$, then $[A : F] = [A : K] \cdot [K : F]$.*

***Proof*** In the special case where $A$ itself is a field, this lemma is a fundamental result on field extensions. The same proof works in the general case. Indeed, if $\{k_i \mid i = 1, \ldots, m\}$ is a basis of $K$ over $F$, and $\{a_j \mid j = 1, \ldots, n\}$ is a basis of $A$ over $K$, then $\{k_i a_j \mid i = 1, \ldots, m,\ j = 1, \ldots, n\}$ is easily seen to be a basis of $A$ over $F$. □

## 1.3 Introducing Central Simple Algebras

Even if interested only in division algebras, it is useful, if not necessary, to get familiar with the more general algebras from the title of the section. We start by defining "simple".

**Definition 1.15** A ring $R$ is said to be **simple** if $R^2 \neq 0$ and 0 and $R$ are the only ideals of $R$.

**Remark 1.16** In any ring $R$, the set of all $x \in R$ such that $xR = 0$ is clearly an ideal of $R$. If $R$ is simple, it must be 0 since $R^2 \neq 0$. Similarly we see that $Rx = 0$ implies $x = 0$ in a simple ring $R$.

The condition that $R^2 \neq 0$ is automatically fulfilled if $R$ is a nonzero unital ring. The condition about ideals in Definition 1.15 is, of course, more restrictive. Most rings are not simple, yet some of the most significant are. The first to mention are division rings.

**Lemma 1.17** *A proper one-sided ideal of a unital ring does not contain invertible elements. Therefore, a division ring $D$ has no one-sided ideals different from 0 and $D$; in particular, $D$ is simple.*

***Proof*** Let $L$ be, say, a left nonzero ideal of a unital ring $R$. If $a \in L$ is invertible in $R$, then $x = (xa^{-1})a \in L$ for every $x \in R$. Thus, $L = R$. □

There are many important simple rings that are not division rings. We will now present the most fundamental example. First we introduce some notation and terminology.

Consider the matrix ring $M_n(S)$ where $S$ is a unital ring. Let $E_{ij}$ denote the matrix whose $(i, j)$ entry is 1 and all other entries are 0. We call $E_{ij}$, $1 \leq i, j \leq n$, the **standard matrix units**. By $aE_{ij}$ we denote the matrix whose $(i, j)$ entry is $a \in S$ and other entries are 0. Note that for every $A = (a_{ij}) \in M_n(S)$,

$$E_{ij} A E_{kl} = a_{jk} E_{il} \quad \text{for all } 1 \leq i, j, k, l \leq n. \tag{1.4}$$

This equation considerably simplifies computations in $M_n(S)$. Yet the reader might be more persuaded in the correctness of the arguments based on (1.4) by occasionally writing down matrices and then making the usual matrix calculations. Just do this with $2 \times 2$ matrices, it is enough to feel certain.

**Lemma 1.18** *If $D$ is a division ring, then $M_n(D)$ is a simple ring for every $n \in \mathbb{N}$.*

***Proof*** Let $I$ be a nonzero ideal of $M_n(D)$. Take a nonzero $(a_{ij}) \in I$ and choose $j$ and $k$ such that $a_{jk} \neq 0$. From (1.4) we see that $a_{jk}E_{il} \in I$ for all $i$ and $l$, and hence also $(da_{jk}^{-1})E_{ii} \cdot a_{jk}E_{il} = dE_{il} \in I$ for every $d \in D$. Since every matrix in $M_n(D)$ is a sum of matrices of the form $dE_{il}$, $I = M_n(D)$. □

**Remark 1.19** The ring $M_n(D)$ with $n \geq 2$ has many proper nonzero left and right ideals. In fact, if $R$ is any unital ring and $a \in R$ is not left invertible, then $Ra$ does not contain 1 and is therefore a proper nonzero left ideal. Similarly, $aR$ is proper nonzero right ideal if $a$ is not right invertible. See Exercise 1.1 for a more general result.

We will encounter other examples of simple rings in later chapters. For our goals in this and the next chapter, division rings and matrices over division rings are sufficient. As a matter of fact, rather than in simple rings, in these two chapters we are interested in **simple algebras**, i.e., algebras $A$ such that $A^2 \neq 0$ and 0 and $A$ are the only ideals of $A$.

**Remark 1.20** We can also define a simple algebra as an algebra that is simple as a ring. This is not entirely obvious since an ideal of an algebra $A$ must be a vector subspace, while Definition 1.15 speaks about ideals of a ring which are merely additive subgroups. However, if $A$ is a simple algebra and $I$ is a **ring ideal** of $A$ (i.e., an ideal of the ring $A$) such that $I \neq 0$ and $I \neq A$, then $AI$ is an ideal of the algebra $A$, so it is either 0 or $A$. Since $AI \subseteq I \neq A$, the only possibility is that $AI = 0$. This implies that the algebra ideal $J = \{x \in A \mid Ax = 0\}$ is nonzero, so that $J = A$. That is, $A^2 = 0$, which contradicts the simplicity of the algebra $A$. Thus, an algebra $A$ is simple as an algebra if and only if it is simple as a ring.

**Example 1.21** Without assuming that $A^2 \neq 0$ it is possible that $A$ has many ring ideals, but no algebra ideals different from 0 and $A$. For instance, take the 1-dimensional algebra $A = \mathbb{R}a$ with trivial multiplication, i.e., $a^2 = 0$. Then clearly $A$ has no algebra ideals different from 0 and $A$, but has plenty of ring ideals (say, $\mathbb{Z}a$).

The center of a unital algebra obviously contains scalar multiples of unity. In many important algebras, these are also the only central elements.

**Definition 1.22** A nonzero unital algebra is said to be **central** if scalar multiples of unity are the only elements in its center.

Unless specified otherwise, by an algebra we shall mean an algebra over the field $F$. Recall the convention that in a nonzero unital algebra $A$, we identify $F$ with $F \cdot 1$

and write $\lambda$ instead of $\lambda 1$ for every scalar $\lambda \in F$. Thus, $A$ is central if $Z(A) = F$. In other words, $A$ is central if it is an algebra over its center.

**Lemma 1.23** *Let $A$ be a unital algebra and let $n \in \mathbb{N}$. Then $A$ is central if and only if $M_n(A)$ is central.*

***Proof*** If $c \in Z(A)$, then the diagonal matrix with all diagonal entries equal to $c$ lies in $Z(M_n(A))$. Thus, $A$ is central if $M_n(A)$ is. Now let $A$ be central. Pick $C \in Z(M_n(A))$. Since $C$ commutes with every $E_{kl}$, $k \neq l$, one easily shows that $C$ is a diagonal matrix with all diagonal entries equal. As $C$ also commutes with every $aE_{11}$, $a \in A$, we conclude that this diagonal entry lies in the center of $A$, and hence it is a scalar. Thus, $M_n(A)$ is central. □

**Example 1.24** As the simplest application of Lemma 1.23, we have that $M_n(F)$ is a central algebra.

**Example 1.25** The algebra of quaternions $\mathbb{H}$ is a central $\mathbb{R}$-algebra. This can be easily checked and also follows from Lemma 1.35 in the next section.

**Example 1.26** A field is a central algebra over itself, but not over its proper subfield. Thus, $\mathbb{C}$ is central as a $\mathbb{C}$-algebra, but not as an $\mathbb{R}$-algebra.

Frobenius' theorem therefore implies the following corollary.

**Corollary 1.27** *A finite dimensional central division $\mathbb{R}$-algebra is isomorphic to $\mathbb{R}$ or $\mathbb{H}$.*

The following lemma presents a useful viewpoint for considering simple unital rings.

**Lemma 1.28** *If $A$ is a simple unital ring (in particular, if $A$ is a division ring), then its center is a field. Therefore, $A$ can be viewed as a central algebra over its center.*

***Proof*** If $c$ is a nonzero element from the center $Z(A)$, then $cA$ is a nonzero ideal of $A$ and is hence equal to $A$. Therefore, $c$ is invertible. If we multiply $cx = xc$ from the left and right by $c^{-1}$, we see that $c^{-1}$ is also a central element. This proves that $Z(A)$ is a field. By defining scalar multiplication simply as the ordinary multiplication of elements from $Z(A)$ with elements from $A$, we see that all algebra axioms are fulfilled. □

**Remark 1.29** If $A$ is a finite dimensional simple unital algebra over some field $F$, then $F \cdot 1$ is contained in $Z(A)$, and so $A$ is also finite dimensional over its center $Z(A)$. For example, $A = M_n(\mathbb{C})$ is a simple real $2n^2$-dimensional algebra, but also an $n^2$-dimensional algebra over $Z(A) \cong \mathbb{C}$.

By a **central simple algebra** we simply mean an algebra that is both central and simple. In the literature, these algebras are often assumed to be automatically finite dimensional. As we will also consider infinite dimensional algebras that are central and simple, we will not follow this convention. Still, we will be primarily interested in **finite dimensional central simple algebras**, which form an extremely important class of algebras. As evident from the above examples, the algebras $M_n(F)$ and $\mathbb{H}$

belong to this class. Later we will see that every finite dimensional central simple algebra is built from matrices and division algebras (Corollary 2.77). However, in this chapter we wish to derive some results on these algebras directly from the definition, without relying on the description of their structure.

## 1.4 Quaternion Algebras

Look again at the definition of the algebra $\mathbb{H}$. Observe that the 4-dimensional algebra with basis $\{1, i, j, k\}$ and multiplication determined by (1.1) can be formed over any field $F$, not only over $\mathbb{R}$. However, the $F$-algebra so obtained is not necessarily a division algebra. In fact, from (1.2) we see that the necessary and sufficient condition for this to be true is that for all $\alpha_i \in F$, $\alpha_0^2 + \alpha_1^2 + \alpha_2^2 + \alpha_3^2 = 0$ implies that each $\alpha_i = 0$. This is fulfilled in $\mathbb{R}$ and some other fields (like $\mathbb{Q}$), but not in algebraically closed (and many other) fields.

In this section, we consider 4-dimensional $F$-algebras which are defined by appropriately modifying Eqs. (1.1). The only condition on $F$ that we require is that

$$\operatorname{char}(F) \neq 2.$$

This assumption will be used throughout the section without further mention.

**Definition 1.30** A 4-dimensional unital $F$-algebra is called a **quaternion algebra** if it has a basis $\{1, i, j, k\}$ such that

$$i^2 = \lambda, \quad j^2 = \mu, \quad ij = -ji = k \tag{1.5}$$

for some $\lambda, \mu \in F \setminus \{0\}$. In this case, we denote this algebra by $(\lambda, \mu)_F$.

Equations (1.5) determine the multiplication in $(\lambda, \mu)_F$. For example,

$$k^2 = ij \cdot ij = i(ji)j = -i^2 j^2 = -\lambda\mu.$$

Observe also that

$$jk = -kj = -\mu i, \quad ki = -ik = -\lambda j.$$

**Example 1.31** $\mathbb{H} = (-1, -1)_{\mathbb{R}}$.

**Example 1.32** The algebra $(1, 1)_F$ is isomorphic to $M_2(F)$. Indeed, $A_i = \left[\begin{smallmatrix} 1 & 0 \\ 0 & -1 \end{smallmatrix}\right]$ and $A_j = \left[\begin{smallmatrix} 0 & 1 \\ 1 & 0 \end{smallmatrix}\right]$ are a pair of anticommuting matrices whose squares are equal to the identity matrix $I$ and $\{I, A_i, A_j, A_i A_j\}$ is a basis of $M_2(F)$, so there is an algebra isomorphism from $(1, 1)_F$ to $M_2(F)$ sending $i$ to $A_i$ and $j$ to $A_j$.

For the proof of the next lemma we introduce the following important notion, which arises naturally in the study of noncommutative rings.

**Definition 1.33** The **commutator** of ring elements $a$ and $b$ is the element

$$[a, b] = ab - ba.$$

Obviously, $a$ and $b$ commute if and only if $[a, b] = 0$. Note also that $[a, b] = -[b, a]$.

**Remark 1.34** Sometimes we call $ab - ba$ the **additive commutator** to distinguish it from the **multiplicative commutator** $aba^{-1}b^{-1}$ from group theory, which is also denoted by $[a, b]$.

**Lemma 1.35** *For any $\lambda, \mu \in F \setminus \{0\}$, $(\lambda, \mu)_F$ is a central simple algebra.*

***Proof*** Let $I \neq 0$ be an ideal of $A = (\lambda, \mu)_F$. Take a nonzero element $a = \alpha_0 + \alpha_1 i + \alpha_2 j + \alpha_3 k$ from $I$. If $\alpha_1 = \alpha_2 = \alpha_3 = 0$, then $a = \alpha_0$ is invertible and hence $I = A$. If $\alpha_1 \neq 0$, then $[[a, j], i] = -4\alpha_1 \lambda j$ lies in $I$ and is invertible, so again we have $I = A$. Similarly we see that $I = A$ if $\alpha_2 \neq 0$ or $\alpha_3 \neq 0$. Thus, $A$ is simple. To prove that $A$ is central, observe that for every $c = \beta_0 + \beta_1 i + \beta_2 j + \beta_3 k \in A$ we have $[c, i] = -2\beta_3 \lambda j - 2\beta_2 k$ and $[c, j] = 2\beta_3 \mu i + 2\beta_1 k$. Therefore, $c \in Z(A)$ implies that $\beta_1 = \beta_2 = \beta_3 = 0$ and hence $c = \beta_0 \in F$. □

An element in $\text{span}\{i, j, k\}$ is called a **pure quaternion**. Every $a \in (\lambda, \mu)_F$ can be written as $a = \alpha_0 + v$ where $\alpha_0 \in F$ and $v = \alpha_1 i + \alpha_2 j + \alpha_3 k$ is a pure quaternion. Like in $\mathbb{H}$, we define the **conjugate** $\overline{a}$ of $a$ by

$$\overline{a} = \alpha_0 - v.$$

We call $N(a) = a\overline{a}$ the **reduced norm** of $a$. One easily checks that $a$ and $\overline{a}$ commute and that

$$N(a) = \alpha_0^2 - \lambda\alpha_1^2 - \mu\alpha_2^2 + \lambda\mu\alpha_3^2 \in F.$$

**Lemma 1.36** *An element $a \in (\lambda, \mu)_F$ is invertible if and only if $N(a) \neq 0$.*

***Proof*** If $a \neq 0$ and $N(a) = a\overline{a} = 0$, then $a$ is a zero-divisor and so is not invertible. If $N(a) \neq 0$, then $a^{-1} = \frac{1}{N(a)}\overline{a}$. □

The following definition will be needed in the proof of the next theorem.

**Definition 1.37** Let $R$ be a unital ring (or algebra). A map $\varphi : R \to R$ is said to be an **inner automorphism** if there exists an invertible element $a \in R$ such that $\varphi(x) = axa^{-1}$, $x \in R$.

Every inner automorphism is indeed an automorphism, but not every automorphism is inner. We will say more about this soon, in Sect. 1.6.

**Theorem 1.38** *A 4-dimensional central $F$-algebra $A$ is a division algebra if and only if $A$ is a quaternion algebra in which every nonzero element has nonzero reduced norm.*

***Proof*** In light of Lemma 1.36, all we have to prove is that $A$ is a quaternion algebra if it is a division algebra. Assume, therefore, that $A$ is a division algebra. We point out that $A$ is not commutative since its center is $F$ by assumption.

Take $a \in A \setminus F$. The subalgebra $K$ generated by $a$ is a subfield (see Proposition 1.10 (c)), so it follows from Lemma 1.14 that $[K : F] = 2$. Therefore, $1, a, a^2$ are linearly dependent, so $a^2 + \alpha a \in F$ for some $\alpha \in F$. Hence, $i = a + \frac{\alpha}{2}$ satisfies $i^2 = \lambda \in F$. Let $\varphi : A \to A$ be the inner automorphism $\varphi(x) = ixi^{-1}$ and let

$$A^+ = \{x \in A \mid \varphi(x) = x\}, \quad A^- = \{x \in A \mid \varphi(x) = -x\}.$$

Observe that $\varphi^2 = \mathrm{id}_A$ and that every $x \in A$ can be written as $x = x^+ + x^-$ where

$$x^+ = \frac{x + \varphi(x)}{2} \in A^+, \quad x^- = \frac{x - \varphi(x)}{2} \in A^-.$$

Thus, $A$ is the direct sum of its subspaces $A^+$ and $A^-$. Fix a nonzero $j \in A^-$. From $K \subseteq A^+$ we infer that $Kj \subseteq A^-$. Since $[K : F] = 2$, the dimension argument shows that $K = A^+$ and $Kj = A^-$.

The condition that $j \in A^-$ means that $i$ and $j$ anticommute. The elements $j$ and $k = ij$ form a basis of $A^-$, so $\{1, i, j, k\}$ is a basis of $A$. Repeating the argument concerning $a$ from the beginning, we see that $1, j, j^2$ are linearly dependent. However, $1, j^2 \in A^+$ and $j \in A^-$, so actually $1$ and $j^2$ are linearly dependent. Thus, $j^2 = \mu \in F$. This completes the proof that $A$ is a quaternion algebra. □

It depends on $F$ whether there exist $\lambda, \mu \in F \setminus \{0\}$ such that nonzero elements in $(\lambda, \mu)_F$ have nonzero reduced norms. They certainly do not exist if $F$ is algebraically closed. If $F = \mathbb{R}$, we must require that both $\lambda$ and $\mu$ are negative, but for any such choice of $\lambda$ and $\mu$ we get an algebra isomorphic to $\mathbb{H}$. The case where $F = \mathbb{Q}$ is much more interesting, as the next example shows.

**Example 1.39** We claim that if $p$ is a prime such that $p \equiv 3 \pmod 4$, then $(-1, p)_{\mathbb{Q}}$ is a division algebra. We prove this in two steps.

First, we show that if $a, b \in \mathbb{Z}$ are such that $p \mid a^2 + b^2$, then $p \mid a$ and $p \mid b$. Suppose, by way of obtaining a contradiction, that $p \nmid a$. Then $p$ and $a$ are relatively prime, so there is a $u \in \mathbb{Z}$ such that $au \equiv 1 \pmod p$. Since $p \mid (au)^2 + (bu)^2$ it follows that $c = bu$ satisfies $c^2 \equiv -1 \pmod p$. In particular, $p \nmid c$ and hence $c^{p-1} \equiv 1 \pmod p$ by Fermat's little theorem. Writing $p = 4k + 3$ we thus have $(c^2)^{2k+1} \equiv 1 \pmod p$, which together with $c^2 \equiv -1 \pmod p$ leads to a contradiction that $-1 \equiv 1 \pmod p$.

In the second step, we show that $N(a) = 0$ with $a \in (-1, p)_{\mathbb{Q}}$ implies $a = 0$. Write $a = \alpha_0 + \alpha_1 i + \alpha_2 j + \alpha_3 k$ and observe that $N(a) = 0$ can be written as $\alpha_0^2 + \alpha_1^2 = p(\alpha_2^2 + \alpha_3^2)$. Multiplying by the square of a common denominator of $\alpha_0, \alpha_1, \alpha_2, \alpha_3$, we see that we can assume that each $\alpha_i \in \mathbb{Z}$. By the first step, $p \mid \alpha_0$

and $p \mid \alpha_1$. This implies that $p \mid \alpha_2^2 + \alpha_3^2$ and hence $p \mid \alpha_2$ and $p \mid \alpha_3$. Continuing in this way, we obtain that each $\alpha_i$ is divisible by every power of $p$. Therefore, $\alpha_i = 0$. This proves that $(-1, p)_{\mathbb{Q}}$ is a division algebra.

There are infinitely many primes $p$ such that $p \equiv 3 \pmod 4$, so we have an infinite family of division algebras $(-1, p)_{\mathbb{Q}}$. We leave as an exercise for the interested reader to prove that none of these algebras are isomorphic to each other. Also, none of them is isomorphic to the division algebra $(-1, -1)_{\mathbb{Q}}$. Indeed, $(-1, p)_{\mathbb{Q}}$ contains the pure quaternion $j$ satisfying $j^2 = p$, while in $(-1, -1)_{\mathbb{Q}}$, which is a subring of $\mathbb{H}$, only central elements have positive squares.

In Example 1.32 we saw that $M_2(F)$ is a quaternion algebra. As a byproduct of the general theory developed later, we will see that every quaternion algebra that is not a division algebra is isomorphic to $M_2(F)$ (Corollary 2.82), and that quaternion division algebras over a finite field $F$ do not exist (Corollary 2.83). We will not go deeper than that into the theory of quaternion algebras. Let us just remark that this theory is extremely rich. It is enough to glance at the book [13] to realize this.

## 1.5 Multiplication Algebra

Recall from Lemma 1.28 that every division ring is a central simple algebra over its center. Sometimes it is more natural to study central simple algebras than only division rings. We will focus on them in this and the next section.

Let $A$ be an algebra. For $a, b \in A$, we define maps $L_a, R_b : A \to A$, called the **left multiplication map** and **right multiplication map**, by

$$L_a(x) = ax, \quad R_b(x) = xb.$$

These maps will play important roles in this book. They can be defined if $A$ is merely a ring, but we will usually treat them on algebras. In this case, they can be considered as elements of $\mathrm{End}_F(A)$, the algebra of all linear operators from $A$ to itself.

Note that for all $a, b \in A$, $\lambda, \mu \in F$, we have

- $L_{ab} = L_a L_b$ and $R_{ab} = R_b R_a$,
- $L_a R_b = R_b L_a$,
- $L_{\lambda a + \mu b} = \lambda L_a + \mu L_b$ and $R_{\lambda a + \mu b} = \lambda R_a + \mu R_b$.

These identities imply that

$$M(A) = \{L_{a_1} R_{b_1} + \cdots + L_{a_n} R_{b_n} \mid a_i, b_i \in A, n \in \mathbb{N}\}$$

is a subalgebra of $\mathrm{End}_F(A)$.

**Definition 1.40** The algebra $M(A)$ is called the **multiplication algebra of** $A$.

If $A$ is unital, then $L_a = L_a R_1$ and $R_b = L_1 R_b$ lie in $M(A)$, and so in this case we can describe $M(A)$ as the subalgebra of $\mathrm{End}_F(A)$ generated by all left and right multiplication maps. We also remark that $a = L_a(1) = R_a(1)$, and so the conditions $a = 0$, $L_a = 0$, and $R_a = 0$ are equivalent if $A$ is unital.

**Remark 1.41** Take $f \in M(A)$. Then there exist $a_i, b_i \in A$ such that $f = \sum_{i=1}^{n} L_{a_i} R_{b_i}$, i.e.,

$$f(x) = \sum_{i=1}^{n} a_i x b_i, \ \ x \in A.$$

These elements are not unique, $f$ can be presented as a sum of operators of the form $L_a R_b$ in many ways. If $f \neq 0$, then $a_i, b_i$ can be chosen so that both the set of all $a_i$'s and the set of all $b_i$'s are linearly independent. For example, this is fulfilled if we require that $n$ is minimal. Indeed, if, under this assumption, one of the elements, say $b_n$, was a linear combination of the others, $b_n = \sum_{i=1}^{n-1} \lambda_i b_i$, then $f$ would be equal to $\sum_{i=1}^{n-1} L_{a_i + \lambda_i a_n} R_{b_i}$, contradicting the minimality of $n$.

We also remark that if $\{u_i \mid i \in I\}$ is a basis of $A$, then we can express each $R_b$ as a linear combination of operators $R_{u_i}$, and hence every $f \in M(A)$ is a sum of operators of the form $L_{a_i} R_{u_i}$. Similarly, $f$ can be written as a sum of operators of the form $L_{u_i} R_{b_i}$, and, moreover, as a linear combination of operators $L_{u_i} R_{u_j}$.

The fact that the elements from $M(A)$ can be expressed in different ways through left and right multiplication maps may cause difficulties. The following lemma shows that in central simple algebras these difficulties can be controlled.

**Lemma 1.42** *Let $A$ be a central simple algebra, and let $a_i, b_i \in A$ be such that $\sum_{i=1}^{n} L_{a_i} R_{b_i} = 0$. If the $a_i$'s are linearly independent, then each $b_i = 0$. Similarly, if the $b_i$'s are linearly independent, then each $a_i = 0$.*

***Proof*** The two assertions of the lemma are analogous, so we consider only the case where the $a_i$'s are independent. Suppose $b_n \neq 0$. Since $A$ is simple, the ideal generated by $b_n$ is equal to $A$. That is, $\sum_{j=1}^{m} w_j b_n z_j = 1$ for some $w_j, z_j \in A$. Hence,

$$0 = \sum_{j=1}^{m} R_{z_j} \Big( \sum_{i=1}^{n} L_{a_i} R_{b_i} \Big) R_{w_j} = \sum_{i=1}^{n} L_{a_i} \Big( \sum_{j=1}^{m} R_{w_j b_i z_j} \Big) = \sum_{i=1}^{n} L_{a_i} R_{c_i}$$

where $c_i = \sum_{j=1}^{m} w_j b_i z_j$; thus, $c_n = 1$. This in particular shows that $n > 1$. We may assume that $n$ is the smallest natural number for which the lemma does not hold. Consequently,

$$0 = \Big( \sum_{i=1}^{n} L_{a_i} R_{c_i} \Big) R_x - R_x \Big( \sum_{i=1}^{n} L_{a_i} R_{c_i} \Big) = \sum_{i=1}^{n-1} L_{a_i} R_{x c_i - c_i x}$$

implies that $xc_i - c_i x = 0$ for every $x \in A$; that is, $c_i \in F$. But then

$$0 = \sum_{i=1}^{n} L_{a_i} R_{c_i} = L_{c_1 a_1 + \cdots + c_n a_n},$$

which contradicts the linear independence of the $a_i$'s. □

In the next theorem we consider the finite dimensional situation. Recall that if $[A : F] = d$, then $[\mathrm{End}_F(A) : F] = d^2$.

**Theorem 1.43** *If $A$ is a finite dimensional central simple algebra, then $M(A) = \mathrm{End}_F(A)$.*

***Proof*** Let $\{u_1, \ldots, u_d\}$ be a basis of $A$. Lemma 1.42 implies that the operators $L_{u_i} R_{u_j}$, $1 \leq i, j \leq d$, are linearly independent; indeed, this becomes clear if we rewrite $\sum_{i,j=1}^{d} \lambda_{ij} L_{u_i} R_{u_j}$ as $\sum_{i=1}^{d} L_{u_i} R_{b_i}$ where $b_i = \sum_{j=1}^{d} \lambda_{ij} u_j$. Therefore,

$$[M(A) : F] \geq d^2 = [\mathrm{End}_F(A) : F],$$

which gives $M(A) = \mathrm{End}_F(A)$. □

It may be instructive for the reader to give an independent, direct proof of this theorem for the special case where $A = M_n(F)$ (what is the natural basis of $\mathrm{End}_F(A)$?).

Lemma 1.42 and Theorem 1.43 will be used at different places throughout the book. In the next sections we will see that three important results can be derived from them.

## 1.6 Automorphisms of Central Simple Algebras

The first result that we will derive from Lemma 1.42 and Theorem 1.43 is an important special case of a famous theorem proved by T. Skolem in 1927 and somewhat later rediscovered by E. Noether. We will establish the general Skolem-Noether theorem later, in Sect. 5.11. Treating a special case at this early stage of development of the theory gives us an opportunity to demonstrate a simple, but efficient method of proof.

We introduced inner automorphisms in Sect. 1.4. An automorphism that is not inner is called **outer**. Here are a few examples of such automorphisms.

**Example 1.44** The conjugation $z \mapsto \overline{z}$ is an automorphism of the $\mathbb{R}$-algebra $\mathbb{C}$. It is outer since the only inner automorphism of a commutative ring is the identity map. We also remark that it is an element of $\mathrm{End}_{\mathbb{R}}(\mathbb{C})$, but not of $M(\mathbb{C})$. The assumption that $A$ is central therefore cannot be omitted in Theorem 1.43.

**Example 1.45** For every $\alpha \in F \setminus \{0\}$, $f(\omega) \mapsto f(\omega+\alpha)$ is an outer automorphism of the algebra $F[\omega]$.

**Example 1.46** Let $S$ be a ring and let $R = S \times S$ be the direct product of two copies of $S$. Then $(s,t) \mapsto (t,s)$ is an outer automorphism of $R$.

In view of these examples, we can appreciate the next theorem.

**Theorem 1.47 (Skolem-Noether Theorem (Special Case))** *Every automorphism of a finite dimensional central simple algebra $A$ is inner.*

***Proof*** Let $\varphi$ be an automorphism of $A$. Theorem 1.43 shows that $\varphi = \sum_{i=1}^{n} L_{a_i} R_{b_i}$ for some $a_i, b_i \in A$. We may assume that $a_1 \neq 0$ and the $b_i$'s are linearly independent (see Remark 1.41). Since $\varphi$ preserves multiplication, we have $L_{\varphi(x)}\varphi = \varphi L_x$ for every $x \in A$. Writing $\sum_{i=1}^{n} L_{a_i} R_{b_i}$ for $\varphi$ we obtain

$$\sum_{i=1}^{n} L_{\varphi(x)a_i - a_i x} R_{b_i} = 0.$$

Lemma 1.42 tells us that, in particular,

$$\varphi(x)a_1 - a_1 x = 0, \ \ x \in A. \tag{1.6}$$

It remains to show that $a_1$ is invertible since this then gives $\varphi(x) = a_1 x a_1^{-1}$. As $A$ is simple, there are $w_j, z_j \in A$ such that $\sum_{j=1}^{m} w_j a_1 z_j = 1$. In view of (1.6), this can be written as

$$\Big(\sum_{j=1}^{m} w_j \varphi(z_j)\Big) a_1 = 1.$$

This shows that $a_1$ has a left inverse and so is invertible by Proposition 1.10 (b). □

**Remark 1.48** Example 1.44 shows that the assumption that $A$ is central is indispensable in Theorem 1.47.

## 1.7 Maximal Subfields

Recall that by a subfield of a unital $F$-algebra $A$ we mean a subalgebra that is a field and contains $F$.

**Definition 1.49** A subfield of $A$ that is not properly contained in a larger subfield of $A$ is called a **maximal subfield** of $A$.

**Example 1.50** From Corollary 1.13 we know that every subalgebra $K$ of $\mathbb{H}$ such that $\mathbb{R} \subsetneq K \subsetneq \mathbb{H}$ is a maximal subfield of $\mathbb{H}$, isomorphic to $\mathbb{C}$. Observe that every such subalgebra $K$ can be written as $\mathbb{R} \oplus \mathbb{R}v$ where $v$ is a nonzero pure quaternion.

As noticed at the end of Sect. 1.2, we may consider an algebra $A$ as a vector space over any of its subfields $K$. Observe that $R_u \in \mathrm{End}_K(A)$ for every $u \in A$. Indeed, $R_u(kx) = kxu = kR_u(x)$ for all $k \in K$, $x \in A$. We also remark that given $f \in \mathrm{End}_K(A)$, its scalar multiple $kf$, where $k \in K$, is the operator $L_k f$. This is a trivial observation, but should be kept in mind when reading the next proof.

**Theorem 1.51** *Let $D$ be a finite dimensional central division $F$-algebra. If $K$ is a maximal subfield of $D$, then $[D : F] = [K : F]^2$.*

***Proof*** Pick a basis $\{u_1, \ldots, u_r\}$ of the $F$-algebra $D$. We claim that $\{R_{u_1}, \ldots, R_{u_r}\}$ is a basis of the $K$-algebra $\mathrm{End}_K(D)$. Lemma 1.42 implies that this set is linearly independent, so it suffices to show that it spans $\mathrm{End}_K(D)$. Let $f \in \mathrm{End}_K(D)$. In particular, $f \in \mathrm{End}_F(D)$, so it follows from Theorem 1.43 (and Remark 1.41) that $f = \sum_{i=1}^{r} L_{a_i} R_{u_i}$ for some $a_i \in D$. Since $f$ is $K$-linear, which means that $fL_k = L_k f$ for every $k \in K$, we have

$$\sum_{i=1}^{r} L_{a_i k - k a_i} R_{u_i} = 0.$$

Lemma 1.42 tells us that $a_i k - k a_i = 0$ for every $i$ and every $k \in K$. This implies that $a_i \in K$, since otherwise the subalgebra generated by $a_i$ and $K$ would be, by Proposition 1.10 (c), a field larger than $K$. Thus, $f$ is a $K$-linear combination of the $R_{u_i}$'s, proving our claim.

Accordingly,

$$[D : F] = r = [\mathrm{End}_K(D) : K] = [D : K]^2.$$

On the other hand,

$$[D : F] = [D : K] \cdot [K : F]$$

by Lemma 1.14, and the result follows. □

**Corollary 1.52** *The dimension of a finite dimensional central division algebra $D$ is a perfect square.*

***Proof*** We only need to know that $D$ contains maximal subfields. Since $D$ is finite dimensional, these are exactly the subfields of maximal dimensions. □

**Remark 1.53** The converse to Theorem 1.51 is also true. That is, if $K$ is a subfield of $D$ such that $[D : F] = [K : F]^2$, then $K$ is maximal. Indeed, if this was not true, then $K$ would be contained in a subfield $K_1$ that has higher dimension and satisfies $[D : F] = [K_1 : F]^2$.

We can now better understand why the simplest noncommutative division algebras, studied in the early sections, have dimension 4. Let us also mention that later we will extend Corollary 1.52 to general finite dimensional central simple algebras (see Corollary 2.78).

## 1.8 Wedderburn's Theorem on Finite Division Rings

Finite fields exist and their structure is well known. Do there exist finite division rings that are not commutative? The nonzero elements of such a ring would form a nonabelian finite group under the ring multiplication. Understanding this group should be sufficient for understanding the ring. As finite groups are one of the most studied mathematical objects, it seems natural to involve group-theoretic techniques when addressing the above question.

We will need the **class formula** from elementary group theory. Let us recall it along with its proof. Take a finite group $G$ with center $Z(G)$. For every $a \in G$, we set

$$C_G(a) = \{g \in G \mid ag = ga\}.$$

This is a subgroup of $G$, called the **centralizer** of $a$. The set $\{xax^{-1} \mid x \in G\}$, called the **conjugacy class** of $a$, has the same cardinality as the set of all (left) cosets of $C_G(a)$ in $G$. Indeed, $xax^{-1} \mapsto xC_G(a)$ is easily seen to be a well-defined bijection. Conjugacy is an equivalence relation on $G$. Partitioning $G$ into its disjoint conjugacy classes and using Lagrange's theorem telling us that the number of cosets is $\frac{|G|}{|C_G(a)|}$, we obtain the class formula

$$|G| = |Z(G)| + \sum \frac{|G|}{|C_G(a)|}. \tag{1.7}$$

Here, the sum is taken over representatives of nontrivial conjugacy classes.

We can now show that the answer to our question from the first paragraph is negative. This was established by J. H. M. Wedderburn in 1905.

**Theorem 1.54 (Wedderburn)** *A finite division ring is commutative.*

***Proof*** Suppose this is not true. Then there exists a noncommutative finite division ring $D$ of minimal cardinality; all its proper division subrings are therefore commutative. Consider $D$ as a central division algebra over its center $F$.

The **centralizer** of an element $a$ in $D$ is defined just as in groups, i.e.,

$$C_D(a) = \{x \in D \mid ax = xa\}.$$

Clearly, $C_D(a)$ is a division subring, and hence a subfield of $D$ if $a \notin F$. Since elements in any subfield that contains $a$ commute with $a$, $C_D(a)$ is a maximal

subfield. Let $q = |F|$. (As we know, $q$ is a prime power, but we will not need this.) Observe that an $m$-dimensional vector space over $F$ has $q^m$ elements. Accordingly, by Theorem 1.51, there exists a $d \geq 2$ such that $|D| = q^{d^2}$ and $|C_D(a)| = q^d$ for every $a \in D \setminus F$. Now apply (1.7) to $G = D^* = D \setminus \{0\}$. Since $Z(G) = F \setminus \{0\}$ and $C_G(a) = C_D(a) \setminus \{0\}$, it follows that there exists an $s \in \mathbb{N}$ such that

$$q^{d^2} - 1 = q - 1 + s\frac{q^{d^2} - 1}{q^d - 1}.$$

Since $q^{d^2} - 1$ is a multiple of $\frac{q^{d^2}-1}{q^d-1}$, this leads to a contradiction that

$$\frac{q^{d^2} - 1}{q^d - 1} = \frac{(q^d)^d - 1}{q^d - 1} = 1 + q^d + \cdots + q^{d(d-2)} + q^{d(d-1)}$$

is a divisor of $q - 1$. □

This theorem is incredibly simple. It provides a beautiful connection between three fundamental concepts: invertibility, finiteness, and commutativity. Such simple, basic results often have equally simple, few line proofs. But no such proof of Wedderburn's theorem is known (nor expected). There are, however, numerous different proofs, both elementary and advanced. Probably the most famous one is a short proof found by E. Witt in 1931. Although Witt's proof is a rather standard choice for a textbook (for example, it is included in [6]), we have demonstrated the proof by T. Nagahara and H. Tominaga from 1974. It is also short and, hopefully the reader will agree, beautiful. Besides, we will need the auxiliary results that led to it later.

## 1.9 Division Rings of Laurent Series

The only examples of division rings that are not fields we know so far are quaternion division algebras. It is time to provide new examples.

A class of rings is often closed under some constructions. Is it possible to form new division rings from a given division ring $D$? The "usual suspects", direct products and matrix ring construction, obviously fail because of abundance of zero-divisors. What about the polynomial ring $D[\omega]$? It is even farther from a division ring as polynomials of positive degree cannot be invertible in $D[\omega]$. Let us consider a larger **ring** $D[[\omega]]$ **of formal power series** over $D$. Its elements are written as formal sums $\sum_{n=0}^{\infty} a_n\omega^n$, $a_n \in D$. Unlike for polynomials, we impose no restriction on the $a_i$'s. The addition and multiplication are defined in the same way as for polynomials:

$$\sum_{n=0}^{\infty} a_n\omega^n + \sum_{n=0}^{\infty} b_n\omega^n = \sum_{n=0}^{\infty}(a_n + b_n)\omega^n,$$

$$\Big(\sum_{n=0}^{\infty} a_n\omega^n\Big)\Big(\sum_{n=0}^{\infty} b_n\omega^n\Big) = \sum_{n=0}^{\infty} c_n\omega^n, \text{ where } c_n = \sum_{i=0}^{n} a_i b_{n-i}.$$

This definition obviously makes sense if we replace $D$ by an arbitrary ring $R$. Thus, for every ring $R$ we can construct the **formal power series ring** $R[[\omega]]$ which contains the polynomial ring $R[\omega]$ as a subring. However, we will be interested in the case where $R = D$.

**Lemma 1.55** *Let $D$ be a division ring. Then $\sum_{n=0}^{\infty} a_n\omega^n$ is invertible in $D[[\omega]]$ if and only if $a_0 \neq 0$.*

***Proof*** The right invertibility of $\sum_{n=0}^{\infty} a_n\omega^n$ is equivalent to the existence of elements $b_n \in D$, $n \geq 0$, satisfying

$$a_0b_0 = 1, \quad a_0b_1 + a_1b_0 = 0, \quad a_0b_2 + a_1b_1 + a_2b_0 = 0, \text{ etc.}$$

If $a_0 \neq 0$, then these equations can be solved. Indeed, the first equation gives $b_0 = a_0^{-1}$, the second one gives $b_1 = -a_0^{-1}a_1b_0$, etc. Thus, in this case, $\sum_{n=0}^{\infty} a_n\omega^n$ is right invertible. Similarly we see that it is left invertible, and hence invertible. Conversely, if $\sum_{n=0}^{\infty} a_n\omega^n$ is invertible, then $a_0 \neq 0$ since $a_0b_0 = 1$ holds for some $b_0 \in R$. □

Thus, although much closer than $D[\omega]$, $D[[\omega]]$ is still far from being a division ring. If only $\omega$ was invertible, then the invertibility of elements with zero constant term would not be a problem. Why not make it invertible?

This is actually easy. For any ring $R$, we define the **Laurent series ring** $R((\omega))$ as the set of all formal series $\sum_{n=-\infty}^{\infty} a_n\omega^n$ where only finitely many $a_n$'s with $n < 0$ can be nonzero, and with operations defined in the obvious way by formally adding and multiplying series. This multiplication is possible since, by assumption, for every $f \neq 0$ in $R((\omega))$ there exists an $n_0 \in \mathbb{Z}$ such that

$$f = \sum_{n=n_0}^{\infty} a_n\omega^n = a_{n_0}\omega^{n_0} + a_{n_0+1}\omega^{n_0+1} + a_{n_0+2}\omega^{n_0+2} + \dots$$

and $a_{n_0} \neq 0$. It is easy to check that $R((\omega))$ is a ring. Thus, we have the following chain of subrings:

$$R \subsetneq R[\omega] \subsetneq R[[\omega]] \subsetneq R((\omega)). \tag{1.8}$$

As before, we are interested in the case where $R = D$ is a division ring.

**Proposition 1.56** *If $D$ is a division ring, then so is $\Delta = D((\omega))$. If $D$ is finite dimensional over its center, then so is $\Delta$ and $[\Delta : Z(\Delta)] = [D : Z(D)]$.*

***Proof*** Take $f = \sum_{n=n_0}^{\infty} a_n \omega^n \in \Delta$ where $a_{n_0} \neq 0$. Then $f\omega^{-n_0}$ lies in $D[[\omega]]$ and is invertible by Lemma 1.55, implying that $f$ is invertible. Thus, $\Delta$ is a division ring.

One readily checks that $Z(\Delta) = Z(D)((\omega))$. If $D$ is finite dimensional over $Z(D)$ and $\{d_1, \ldots, d_m\}$ is a basis of $D$ over $Z(D)$, then $\{d_1, \ldots, d_m\}$, which can be viewed as a subset of $\Delta$ in light of (1.8), is clearly also a basis of $\Delta$ over $Z(\Delta)$. □

**Remark 1.57** The final part of the proof shows that if $D$ is a finite dimensional over any of its subfields $L$, then $[D((\omega)) : L((\omega))] = [D : L]$.

The equation $[\Delta : Z(\Delta)] = [D : Z(D)]$ is in some sense disappointing. It shows that we did not arrive at essentially new examples of division rings. If $D$ is a quaternion division algebra—and at present we do not know any other noncommutative division ring—then $\Delta$ is also a quaternion division algebra by Theorem 1.38. For example, if $D = \mathbb{H}$, then we see from the proof that $\Delta$ is an algebra with the same basis $\{1, i, j, k\}$ satisfying the standard rule $i^2 = j^2 = k^2 = ijk = -1$, but over the field $\mathbb{R}((\omega))$ rather than $\mathbb{R}$; thus, $\Delta = \mathbb{H}((\omega)) \cong (-1, -1)_{\mathbb{R}((\omega))}$. Indeed $\Delta$ is infinite dimensional over $\mathbb{R}$, but not over $Z(\Delta) = \mathbb{R}((\omega))$. This illustrates why it is natural to study division rings as algebras over their centers (that is, as central division algebras).

## 1.10 Division Rings of Skew Laurent Series

To obtain more satisfactory examples, we will modify the multiplication of Laurent series. The idea is to make it "more noncommutative". Note that in the rings $R[\omega]$, $R[[\omega]]$, and $R((\omega))$, $\omega$ commutes with elements in $R$ and is therefore a central element. We will change this as follows. Let $\sigma$ be a ring endomorphism of $R$. Instead of $\omega a = a\omega$ we require that

$$\omega a = \sigma(a)\omega$$

for every $a \in R$. Accordingly,

$$(a\omega^i)(b\omega^j) = a\sigma^i(b)\omega^{i+j}. \tag{1.9}$$

Observe that each of the expressions $\big((a\omega^i)(b\omega^j)\big)(c\omega^k)$ and $(a\omega^i)\big((b\omega^j)(c\omega^k)\big)$ is equal to $a\sigma^i(b)\sigma^{i+j}(c)\omega^{i+j+k}$, so this new multiplication rule is associative.

It is straightforward to check that the set of polynomials $\sum_{i=0}^{n} a_i\omega^i$, endowed with the usual addition and multiplication determined by (1.9), is a ring. We denote it by $R[\omega; \sigma]$ and call it the **skew polynomial ring**. By naturally extending these

operations to the set of all formal power series $\sum_{n=0}^{\infty} a_n\omega^n$ we obtain a ring, denoted $R[[\omega;\sigma]]$ and called the **skew power series ring**.

Now assume that $\sigma$ is an automorphism. Then (1.9) makes sense also for the negative integers $i$ and $j$, which enables us to introduce the **skew Laurent series ring** $R((\omega;\sigma))$. The elements in this ring are multiplied as follows:

$$\Big(\sum_{n=n_0}^{\infty} a_n\omega^n\Big)\Big(\sum_{n=r_0}^{\infty} b_n\omega^n\Big) = \sum_{n=n_0+r_0}^{\infty} c_n\omega^n,$$

where $n_0, r_0 \in \mathbb{Z}$ and

$$c_{n_0+r_0} = a_{n_0}\sigma^{n_0}(b_{r_0}), \quad c_1 = a_{n_0}\sigma^{n_0}(b_{r_0+1}) + a_{n_0+1}\sigma^{n_0+1}(b_{r_0}), \quad \text{etc.}$$

By making obvious modifications in the proof of the first statement of Proposition 1.56, we see that the following is true.

**Proposition 1.58** *If $\sigma$ is an automorphism of a division ring $D$, then $D((\omega;\sigma))$ is a division ring.*

The question now is what is the dimension of this division ring over its center. We will answer it for the case where $D$ is a field. This is enough for our purposes since $D((\omega;\sigma))$ is noncommutative as long as $\sigma$ is not the identity.

A small piece of Galois theory is needed to solve our problem. We will state and prove a version of what is known as **Artin's lemma**. This lemma is linear algebraic in nature and all we need to know from Galois theory is the following definition: the **fixed field of a group** $G$ **of automorphisms** of the field $E$ is the subfield $L = \{x \in E \mid \sigma(x) = x \text{ for all } \sigma \in G\}$.

**Lemma 1.59** *Let $E$ be a field, let $G$ be a finite group of automorphisms of $E$, and let $L$ be the fixed field of $G$. Then $[E : L] = |G|$.*

***Proof*** Let $n = |G|$ and write $G = \{\sigma_1, \ldots, \sigma_n\}$.

We will first prove that $[E : L] \leq n$ (this will in particular show that $[E : L]$ is finite). Suppose this is not true. Then $E$ contains $L$-linearly independent elements $b_1, \ldots, b_{n+1}$. The $n+1$ vectors

$$v_j = \big(\sigma_1(b_j), \ldots, \sigma_n(b_j)\big), \quad j = 1, \ldots, n+1,$$

in the $n$-dimensional vector space $E^n$ must be $E$-linearly dependent. Each $v_j \neq 0$, so there is no loss of generality in assuming that there exists an $r \leq n$ such that the vectors $v_1, \ldots, v_r$ are $E$-linearly independent and $v_{r+1} \in \operatorname{span}\{v_1, \ldots, v_r\}$. Let $a_1, \ldots, a_r \in E$ be such that $v_{r+1} = \sum_{j=1}^{r} a_j v_j$, that is,

$$\sigma_i(b_{r+1}) = \sum_{j=1}^{r} a_j\sigma_i(b_j), \quad i = 1, \ldots, n. \tag{1.10}$$

If each $a_j \in L$, then $a_j\sigma_i(b_j) = \sigma_i(a_jb_j)$ and so (1.10) implies that $b_{r+1} = \sum_{j=1}^{r} a_jb_j$, contradicting the $L$-linear independence of the $b_j$'s. We may therefore assume that $a_1 \notin L$. Let $t$ be such that $\sigma_t(a_1) \neq a_1$. Applying $\sigma_t$ to (1.10) and using that $G = \{\sigma_t\sigma_1, \ldots, \sigma_t\sigma_n\}$ (since $G$ is a group), we obtain

$$\sigma_i(b_{r+1}) = \sum_{j=1}^{r} \sigma_t(a_j)\sigma_i(b_j), \quad i = 1, \ldots, n. \tag{1.11}$$

Set $a'_j = a_j - \sigma_t(a_j)$ and note that $a'_1 \neq 0$. Subtracting (1.11) from (1.10), we obtain $\sum_{j=1}^{r} a'_j\sigma_i(b_j) = 0$ for all $i = 1, \ldots, n$, that is, $\sum_{j=1}^{r} a'_jv_j = 0$. This contradicts our assumption that $v_1, \ldots, v_r$ are $E$-linearly independent. Therefore, $[E : L] \leq n$.

Suppose $m = [E : L] < n$. Let $\{b_1, \ldots, b_m\}$ be a basis of $E$ over $L$. The $n$ vectors

$$w_i = (\sigma_i(b_1), \ldots, \sigma_i(b_m)), \quad i = 1, \ldots, n,$$

in the $m$-dimensional vector space $E^m$ must be $E$-linearly dependent. Since each $w_i \neq 0$, we may assume that there exists an $s < n$ such that $w_1, \ldots, w_s$ are $E$-linearly independent and $\sum_{i=1}^{s+1} c_iw_i = 0$ for some $c_1, \ldots, c_{s+1} \in E$ with $c_{s+1} \neq 0$. Not every $c_i$, $i = 1, \ldots, s$, is 0, so we may also assume that $c_1 \neq 0$. From the definition of $w_i$ we see that $\sum_{i=1}^{s+1} c_i\sigma_i(b_j) = 0$ for all $j = 1, \ldots, m$. Observe that each $\sigma_i$ is $L$-linear. Since $\{b_1, \ldots, b_m\}$ is a basis of $E$ over $L$, this implies that $\sum_{i=1}^{s+1} c_i\sigma_i(b) = 0$ for every $b \in E$. Replacing $ab$ for $b$ in this equation we obtain

$$c_1\sigma_1(a)\sigma_1(b) + c_2\sigma_2(a)\sigma_2(b) + \cdots + c_{s+1}\sigma_{s+1}(a)\sigma_{s+1}(b) = 0. \tag{1.12}$$

On the other hand, multiplying the same equation by $\sigma_{s+1}(a)$, we obtain

$$c_1\sigma_{s+1}(a)\sigma_1(b) + c_2\sigma_{s+1}(a)\sigma_2(b) + \cdots + c_{s+1}\sigma_{s+1}(a)\sigma_{s+1}(b) = 0. \tag{1.13}$$

Fix $a \in E$ such that $\sigma_1(a) \neq \sigma_{s+1}(a)$. Set $c'_i = c_i(\sigma_i(a) - \sigma_{s+1}(a))$ and note that $c'_1 \neq 0$. Subtracting (1.13) from (1.12) we obtain $\sum_{i=1}^{s} c'_i\sigma_i(b) = 0$ for every $b \in E$. However, this gives $\sum_{=1}^{s} c'_iw_i = 0$, which contradicts our assumption that $w_1, \ldots, w_s$ are $E$-linearly independent. Therefore, $[E : L] \geq n$. □

Now we can address the question about the dimension over the center we posed before the lemma. The answer depends on whether the automorphism $\sigma$ has finite order $d$ (i.e., $\sigma^d$ is the identity while $\sigma^j$ for $1 \leq j < n$ is not) or has infinite order. In the first case, the cyclic group $\langle\sigma\rangle$ has order $d$ and the corresponding fixed field is simply $\{x \in E \mid \sigma(x) = x\}$. We refer to it as the **fixed field of** $\sigma$.

**Theorem 1.60** *Let $\sigma$ be an automorphism of a field $E$. The division ring $\Delta = E((\omega; \sigma))$ is finite dimensional over its center $Z(\Delta)$ if and only if $\sigma$ has finite order $d$. In this case, $[\Delta : Z(\Delta)] = d^2$.*

***Proof*** To determine the center, take $f = \sum_{n=n_0}^{\infty} a_n \omega^n \in Z(\Delta)$. From $f\omega = \omega f$ we obtain $(a_n\omega^n)\omega = \omega(a_n\omega^n)$, and hence $\sigma(a_n) = a_n$. Thus, each $a_n$ lies in the fixed field $L$ of $\sigma$. Next, from $fb = bf$, $b \in E$, we obtain $(a_n\omega^n)b = b(a_n\omega^n)$, and hence $a_n(\sigma^n(b) - b) = 0$. Therefore, $\sigma^n \neq \mathrm{id}_E$ implies $a_n = 0$.

Suppose $\sigma$ has infinite order. Then $\sigma^n = \mathrm{id}_E$ holds only for $n = 0$, so $f = a_0 \in L$. As elements from $L$ indeed lie in $Z(\Delta)$, we thus have $Z(\Delta) = L$ and hence $\Delta$ is obviously infinite dimensional over $Z(\Delta)$.

Assume now that $\sigma$ has finite order $d$. Then $\sigma^n = \mathrm{id}_E$ if and only if $d \mid n$. Therefore, every element in $Z(\Delta)$ can be written as $\sum_{j=j_0}^{\infty} \ell_j(\omega^d)^j$ with $\ell_j \in L$. Conversely, an element of such a form belongs to $Z(\Delta)$. Thus, $Z(\Delta) = L((\omega^d; \sigma))$. Note that $L((\omega^d; \sigma))$ is actually equal to $L((\omega^d))$, and, more generally, $E((\omega^d; \sigma))$ is equal to $E((\omega^d))$, since the elements $a(\omega^d)^j$ commute with each other. Clearly, $\{1, \omega, \ldots, \omega^{d-1}\}$ is a basis of $\Delta$ over its subfield $E((\omega^d))$. Lemma 1.14 therefore implies that

$$[\Delta : Z(\Delta)] = [\Delta : E((\omega^d))] \cdot [E((\omega^d)) : Z(\Delta)] = d \cdot [E((\omega^d)) : Z(\Delta)].$$

Now it suffices to show that $[E((\omega^d)) : Z(\Delta)] = d$. As $Z(\Delta) = L((\omega^d))$ and $[E((\omega^d)) : L((\omega^d))] = [E : L]$ by Remark 1.57, this follows from Lemma 1.59. □

**Remark 1.61**

(a) In the proof we saw that $Z(\Delta)$ is equal to the fixed field $L$ of $\sigma$ if $\sigma$ has infinite order, and to $L((\omega^d))$ if $L$ has finite order $d$. In each case, $L$ is contained in $Z(\Delta)$.

(b) We also saw in the proof that if $\sigma$ has finite order $d$, then $[\Delta : Z(\Delta)] = [E((\omega^d)) : Z(\Delta)]^2$. Therefore, $E((\omega^d))$ is a maximal subfield of $\Delta$ by Remark 1.53. Of course, one can verify this directly.

Recall from Corollary 1.52 that the dimension of a division ring over its center is either infinite or a perfect square. Theorem 1.60 therefore covers all possible dimensions. The following corollary gives a more clear picture of its meaning.

**Corollary 1.62** *Let $F$ be a field. For every $d \in \mathbb{N}$, there exists a division algebra $\Delta_d$ over $F$ such that $[\Delta_d : Z(\Delta_d)] = d^2$. Moreover, there exists a division algebra $\Delta_\infty$ over $F$ that is infinite dimensional over $Z(\Delta_\infty)$.*

***Proof*** Let $d \in \mathbb{N} \cup \{\infty\}$. Theorem 1.60, together with Remark 1.61 (a), tells us that it is enough to find a field extension $E$ of $F$ having an automorphism $\sigma$ of order $d$ and such that $F$ is contained in its fixed field. We can take the field of rational functions $E = F(\Omega)$ with $\Omega = \{\omega_n \mid n \in \mathbb{N}\}$, choose a permutation $\sigma_0$ of the set $\Omega$ having degree $d$, and define $\sigma$ to be the (unique) automorphism of $E$ that sends $\omega_n$ to $\omega_{\sigma_0(n)}$, $n \in \mathbb{N}$, and every element from $F$ to itself. Observe that the order of the automorphism $\sigma$ is equal to the order of the permutation $\sigma_0$. (Note that in the case of finite order we can take $\Omega$ to be a finite set.) □

Establishing this corollary was actually our main goal in this section. This is because, on the one hand, we will need it in the final part of the book (for the proof of Theorem 7.63), and, on the other hand, it indicates that the study of division algebras is not vacuous. We will see later that these algebras naturally appear in the structure theory, and now we are convinced that they exist; for some further examples, see Sects. 7.3 and 7.10.

We will not study division algebras in much greater detail in this book. The next step for the reader who would like to learn more about them can be consulting the books [6, 11], and [12]. One soon realizes that the study of division algebras may be quite difficult. In particular, some old and easily stated problems about them are still open. Here is an example.

**Problem 1.63** Suppose a division algebra $D$ is algebraic and finitely generated (as an algebra). Is $D$ finite dimensional?

This question was posed for general (not necessarily division) algebras by A. G. Kurosh in the early 1940s. In 1964, E. S. Golod and I. R. Shafarevitch showed that in this generality the answer is negative. It is thus necessary to impose some restrictions on the algebras in question. Problem 1.63 is often referred to as the "Kurosh problem for division algebras".

## Exercises

**1.1** Prove that the following statements are equivalent for a nonzero ring $D$:

(i) $D$ is a division ring.
(ii) For all $a, b \in D$ with $a \neq 0$, the equations $ax = b$ and $ya = b$ have unique solutions in $D$.
(iii) For all $a, b \in D$ with $a \neq 0$, the equation $ax = b$ has a solution in $D$.
(iv) $D^2 \neq 0$ and $D$ has no right ideals other than 0 and $D$.

**Remark** As mentioned in Sect. 1.2, nonassociative division algebras are defined through condition (ii). The equivalence of (i) and (ii) thus means that in associative algebras this definition coincides with the standard one. In nonassociative algebras, however, the condition of being a division algebra is independent of the condition that every nonzero element $a$ has a right and left inverse. The latter only means that the equations $ax = b$ and $ya = b$ can be solved for $b = 1$, while the former does not even imply the existence of unity. For example, take an associative division algebra (possibly a field) $D$ having an automorphism $\varphi \neq \mathrm{id}_D$. The vector space $D$ endowed with the new product $x \cdot y = \varphi(xy)$ is then a nonassociative division algebra without unity.

**1.2** Let $x, y$ be elements in a division ring $D$ such that $x \neq 0$, $y \neq 0$, and $x \neq y^{-1}$. Prove **Hua's identity** $xyx = x - \left(x^{-1} + (y^{-1} - x)^{-1}\right)^{-1}$.

**1.3** Show that a division ring $D$ satisfying $(xy)^2 = (yx)^2$ for all $x, y \in D$ is commutative.
*Hint*: When considering an identity in a ring that involves arbitrary elements it is often useful to replace these elements by the sum (or product) of two elements.

**1.4** Suppose a finite dimensional real unital algebra $A$ contains a subfield isomorphic to $\mathbb{C}$. Show that every left ideal of $A$ has even dimension.
*Hint*: Consider the map $x \mapsto ix$ from a left ideal $L$ to itself, where $i \in A$ is such that $i^2 = -1$.

**Remark** The converse is also true, see Exercise 5.21.

**1.5** Let $D$ be a unital $\mathbb{R}$-algebra. Suppose there exists a linear map $x \mapsto \tilde{x}$ from $D$ to $D$ such that $x\tilde{x} \in \mathbb{R}$ for every $x \in D$, and $x\tilde{x} \neq 0$ if $x \neq 0$. Show that $D$ is isomorphic to $\mathbb{R}$, $\mathbb{C}$, or $\mathbb{H}$.
*Hint*: See the comment following Lemma 1.9 and the hint from Exercise 1.3.

**1.6** Let $\alpha \in \mathbb{R}$. Find all solutions of the equation $x^2 = \alpha$ in $\mathbb{H}$.

**1.7** Show that $(\lambda, \mu)_{\mathbb{R}} \cong \mathbb{H}$ if both $\lambda$ and $\mu$ are negative and $(\lambda, \mu)_{\mathbb{R}} \cong M_2(\mathbb{R})$ if one of them is positive.

**1.8** Show that $(\lambda, \mu)_F \cong (\mu, \lambda)_F \cong (\lambda, -\lambda\mu)_F \cong (\mu, -\lambda\mu)_F$.

**1.9** A linear map $x \mapsto x^*$ from an $F$-algebra $A$ to itself is called an **involution** if $(ab)^* = b^*a^*$ and $(a^*)^* = a$ for all $a, b \in A$. The familiar examples from linear algebra are the transpose involution $M \mapsto M^t$ on $M_n(F)$ and the conjugate transpose involution $M \mapsto M^*$ on $M_n(\mathbb{C})$ (this map is $\mathbb{R}$-linear but not $\mathbb{C}$-linear, so here we consider $M_n(\mathbb{C})$ as a real algebra). Show that the conjugation $a \mapsto \overline{a}$ is an involution on every quaternion algebra, and hence derive that the reduced norm is multiplicative (i.e., $N(ab) = N(a)N(b)$).

**1.10** Let $A$ be a quaternion algebra and let $a \in A$. The element $\mathrm{Tr}(a) = a + \overline{a}$ is called the **reduced trace** of $a$. Show that $a \mapsto \mathrm{Tr}(a)$ is a linear functional on $A$ satisfying $\mathrm{Tr}(ab) = \mathrm{Tr}(ba)$ for all $a, b \in A$. Observe also that $a^2 - \mathrm{Tr}(a)a + N(a) = a^2 - (a + \overline{a})a + a\overline{a} = 0$ for every $a \in A$.

**1.11** Identify the matrix algebra $M_2(F)$ with the quaternion algebra $(1, 1)_F$ via the isomorphism given in Example 1.32. Show that the conjugate of the matrix $M = \left[\begin{smallmatrix} a & b \\ c & d \end{smallmatrix}\right]$ is the adjugate matrix $\left[\begin{smallmatrix} d & -b \\ -c & a \end{smallmatrix}\right]$, the reduced norm of $M$ is the determinant of $M$, the reduced trace of $M$ is the usual trace of the matrix $M$, and $\omega^2 - \mathrm{Tr}(M)\omega + N(M)$ is the characteristic polynomial of $M$ (the last observation from the preceding exercise therefore becomes the Cayley-Hamilton Theorem for $2 \times 2$ matrices).

**1.12** Show that a (not necessarily unital) ring $R$ is simple if and only if $M_n(R)$ is simple.

**1.13** Show that the center of a simple ring is either 0 or a field. In particular, a commutative simple ring is thus a field.

**1.14** Give an alternative proof of Proposition 1.8 by considering the eigenvalues of left multiplication maps.

**1.15** Show that a ring $R$ has a left unity (i.e., a left identity element with respect to multiplication) if and only if the left multiplication maps $L_a$, $a \in R$, are the only maps $f : R \to R$ satisfying $f(xy) = f(x)y$, $x, y \in R$. Describe all such maps $f$ in the case where $R = M_n(2\mathbb{Z})$.

**1.16** Let $A$ be a unital $F$-algebra with $[A : F] = n$. Show that $[M(A) : F] \geq n$ and find an example where $[M(A) : F] = n > 1$.

**Remark** The other extreme where $[M(A) : F] = n^2$ is treated in Theorem 1.43.

**1.17** Find an element in $M(\mathbb{H})$ that cannot be written as $L_a R_b$, $a, b \in \mathbb{H}$.

**1.18** Let $A$ be a finite dimensional central simple algebra. Suppose that a linear map $d : A \to A$ satisfies $d(xy) = d(x)y + xd(y)$, $x, y \in A$. Show that there exists an $a \in A$ such that $d(x) = [a, x]$, $x \in A$.

**Remark** In terminology introduced in Definition 2.92 and Example 2.94 below, this can be stated as that every derivation of $A$ is inner.

**1.19** Check that if $\varphi$ is an automorphism of the algebra $A$, then $(a_{ij}) \mapsto (\varphi(a_{ij}))$ is an automorphism of the algebra $M_n(A)$. Use this to find an outer automorphism of the $\mathbb{R}$-algebra $M_n(\mathbb{C})$.

**1.20** Show that a nonzero algebra endomorphism of a finite dimensional simple algebra is an automorphism.

**1.21** Let $A$ be a finite dimensional central simple algebra. Describe all algebra endomorphisms of $A \times A$.

**1.22** A bijective linear map $\theta$ from an $F$-algebra $A$ to itself is called an **anti-automorphism** if $\theta(xy) = \theta(y)\theta(x)$ for all $x, y \in A$ (involutions are thus antiautomorphisms of order 2). Describe all antiautomorphisms of $\mathbb{R}$-algebras $M_n(\mathbb{R})$ and $M_n(\mathbb{H})$.
*Hint*: What is the product of two antiautomorphisms?

**1.23** Find a maximal subfield of the $F$-algebra $M_2(F)$ for $F = \mathbb{R}$ and $F = \mathbb{C}$.

**1.24** Let $A$ be a quaternion division $F$-algebra. Show that $F \oplus Fa$ is a maximal subfield of $A$ for every $a \in A \setminus F$, and conversely, every maximal subfield is of such a form. If $F = \mathbb{R}$ and hence $A = \mathbb{H}$, show further that for any two maximal subfields $K$ and $K'$ there exists a $b \in A$ such that $K' = bKb^{-1}$. Is this true for every field $F$?
*Hint*: The statement about $\mathbb{H}$ can be proved directly, but it may be instructive to give a more conceptual proof as follows: assume without loss of generality that $K = \mathbb{R} \oplus \mathbb{R}i$, use Corollary 1.13 to obtain an isomorphism from $K$ to $K'$, extend it to an automorphism of $\mathbb{H}$, and then apply Theorem 1.47. (In this way, one proves "by hand" a special case of Theorem 5.60 below).

**1.25** Prove that a finite dimensional division algebra $D$ is a finite field if and only if there exists an integer $n \geq 2$ such that $x^n = x$ for all $x \in D$.

**Remark** In Sect. 4.10, we will establish an incomparably deeper result stating that only a commutative ring $R$ can satisfy the condition that for every $x \in R$ there exists an integer $n(x) \geq 2$ such $x^{n(x)} = x$. This exercise is meant just as a small illustration of the applicability of rudimentary tools that are at our disposal now.

**1.26** Let $D$ be a division ring. Show that every nonzero ideal of $D[[\omega]]$ is generated by $\omega^n$ for some $n \geq 0$.

**1.27** Let $\mathbb{E}$ be a field, let $\mathfrak{S}$ be an automorphism of $\mathbb{E}$ of order $d$, let $\ell$ be a nonzero element from the fixed field $\mathbb{L}$ of $\mathfrak{S}$, and let $I$ be the ideal of $\mathbb{E}[\omega; \mathfrak{S}]$ generated by $\omega^d - \ell$. We call the ring $\mathbb{E}[\omega; \mathfrak{S}]/I$ the **cyclic algebra** $(\mathbb{E}, \mathfrak{S}, \ell)$. Show that:

(a) $(\mathbb{E}, \mathfrak{S}, \ell)$ is a simple ring.
(b) The center of $(\mathbb{E}, \mathfrak{S}, \ell)$ is $\mathbb{L}$.
(c) $[(\mathbb{E}, \mathfrak{S}, \ell) : \mathbb{L}] = d^2$.
(d) $\mathbb{E}$ is a maximal subfield of $(\mathbb{E}, \mathfrak{S}, \ell)$.
(e) If $\mathfrak{S}$ is the conjugation of complex numbers, then $(\mathbb{C}, \mathfrak{S}, -1) \cong \mathbb{H}$.
(f) If $\sigma$ is an automorphism of a field $E$ of order $d$ and $\mathfrak{S}$ is its extension to $E((\omega^d))$ satisfying $\mathfrak{S}(\omega^d) = \omega^d$, then $(E((\omega^d)), \mathfrak{S}, \omega^d) \cong E((\omega; \sigma))$.

**Remark** Our main examples of division algebras thus belong to a more general class of cyclic division algebras. These algebras play a crucial role in the theory of finite dimensional division algebras. Not every cyclic algebra is a division algebra, however.

# Chapter 2
# Structure of Finite Dimensional Algebras




This chapter is centered around Wedderburn's structure theory, which under rather mild and inevitable restrictions describes the form of a finite dimensional algebra. In some sense, it reduces the problem of understanding finite dimensional algebras to the problem of understanding finite dimensional division algebras. The latter can be difficult, but at least for division algebras over some fields definitive answers were given in Chap. 1.

Like in the preceding chapter, the conceptual simplicity will be our basic principle. Rather than building powerful general machinery and then derive classical theorems as its byproducts, we will give direct proofs that only illustrate more general phenomena. There will be enough opportunities for demonstrating more profound theories later. That said, we will usually say a bit more than needed for achieving our main goals, and make occasional digressions from the main themes. Partially this will be done in order to make the exposition less condensed and more illuminating, and partially because of future goals. Thus, we shall give a survey of the very basic theory, but with an eye on more general topics.

## 2.1 Nilpotent Ideals

So far we have treated rings without proper nontrivial ideals. Such rings can be relatively easily handled, they are considered nice from the structural point of view. We will now briefly consider rings that are in complete contrast; they contain nontrivial ideals with a particularly "bad" property.

**Definition 2.1** An ideal $I$ of a ring $R$ is said to be a **nilpotent ideal** if there exists an $n \in \mathbb{N}$ such that $I^n = 0$.

The condition $I^n = 0$ means that $u_1 u_2 \dots u_n = 0$ for all $u_i \in I$. Nilpotent ideals should not be confused with the following two notions.

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M. Brešar, *Introduction to Noncommutative Algebra*, Universitext,
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**Definition 2.2** An element $a$ in a ring $R$ is said to be a **nilpotent element** if there exists an $n \in \mathbb{N}$ such that $a^n = 0$.

**Definition 2.3** An ideal $I$ of a ring $R$ is said to be a **nil ideal** if all elements in $I$ are nilpotent.

Obviously, a nilpotent ideal is nil. Example 2.7 below shows that the converse is not true.

A trivial example of a nilpotent ideal is $I = 0$, but of course we are interested in nontrivial examples. An extreme case is when the ring $R$ itself is nilpotent.

**Example 2.4** Take any additive group $R$, and equip it with trivial product: $xy = 0$ for all $x, y \in R$. Then $R^2 = 0$.

**Example 2.5** A nilpotent element lying in the center $Z(R)$ of the ring $R$ clearly generates a nilpotent ideal. Taking a commutative ring $C$ with nonzero nilpotent elements, such as the ring $K[\omega]/(\omega^n)$ where $K$ is a commutative ring, we can therefore use standard ring constructions (like direct products, matrix rings, polynomials rings, etc.) to obtain rings having $C$ in their centers and hence containing nonzero nilpotent ideals.

**Example 2.6** Let $A = T_n(F)$ be the algebra of all *upper triangular* $n \times n$ *matrices* over a field $F$, i.e., matrices that have zeros below the main diagonal. Let $N$ be the set of all *strictly upper triangular matrices*, i.e., matrices in $A$ that have zeros also on the diagonal. Then $N$ is an ideal of $A$ such that $N^n = 0$ and $N^{n-1} \neq 0$. One can find a variety of other ideals of $A$ that are contained in $N$, and are therefore nilpotent. For instance, the set $I$ of all matrices whose $(1, n)$ entry is arbitrary and all other entries are 0 is an ideal of $A$ satisfying $I^2 = 0$.

**Example 2.7** Let $A$ be the set of all infinite, $\mathbb{N} \times \mathbb{N}$ matrices over $F$ that are upper triangular and have only finitely many nonzero entries. Note that $A$ is an algebra under the standard matrix operations. Let $I$ be the set of all matrices in $A$ that are strictly upper triangular. Then $I$ is a nil, but not nilpotent ideal of $A$.

**Remark 2.8** **Nilpotent one-sided ideals** are defined in the same way as nilpotent ideals. If $L$ is a nilpotent left ideal of $R$, then $L$ is contained in a nilpotent (two-sided) ideal $L + LR$. Indeed, it is easy to see that $L^n = 0$ implies $(L + LR)^n = 0$. Similarly, every nilpotent right ideal is contained in a nilpotent ideal.

At this point we can mention a celebrated open problem. From Remark 2.8 we see that a ring with a nonzero nilpotent one-sided ideal also contains a nonzero nilpotent ideal. In 1930, G. Köthe conjectured that an analogous statement holds for nil ideals. The following question is therefore known as **Köthe's problem**.

**Problem 2.9** If a ring $R$ has a nonzero nil one-sided ideal, does $R$ also have a nonzero nil ideal?

This is known to be true in various special situations. In general, however, the problem is still open.

Let us return to nilpotent ideals.

**Lemma 2.10** *The sum of two nilpotent ideals is a nilpotent ideal.*

***Proof*** Let $I, J$ be ideals such that $I^n = 0$ and $J^m = 0$. We claim that $(I + J)^{n+m-1} = 0$. That is, the product of $n + m - 1$ elements of the form $u + v$, $u \in I$, $v \in J$, is 0. Such a product can be written as a sum of products $w = w_1 w_2 \dots w_{n+m-1}$ where each $w_i \in I \cup J$. If at least $n$ of these $w_i's$ are in $I$, then $w = 0$ as $I^n = 0$. If the number of the $w_i$'s belonging to $I$ is smaller than $n$, then at least $m$ of them lie in $J$, and hence $w = 0$ since $J^m = 0$. □

By a **maximal nilpotent ideal** we mean a nilpotent ideal that is not properly contained in a larger nilpotent ideal.

**Lemma 2.11** *If a ring $R$ has a maximal nilpotent ideal $N$, then $N$ contains all nilpotent ideals of $R$.*

***Proof*** If $I$ is another nilpotent ideal, then $I + N$ is again a nilpotent ideal by Lemma 2.10. The maximality of $N$ implies $I + N = N$, and hence $I \subseteq N$. □

Thus, a maximal nilpotent ideal, if it exists, is unique and is equal to the sum of all nilpotent ideals. The next example shows that not every ring has such an ideal. That is to say, the sum of all nilpotent ideals of a ring is not always nilpotent (although it is nil because each of its elements is contained in a nilpotent ideal by Lemma 2.10).

**Example 2.12** Let $A$ and $I$ be the algebra and its nil ideal from Example 2.7. For every $k \in \mathbb{N}$, let $I_k$ denote the set of all matrices in $I$ with the property that their nonzero entries appear only in the first $k$ rows. It is easy to check that $I_k$ is a nilpotent ideal of $A$; in fact, $I_k^{k+1} = 0$. If $A$ had a maximal nilpotent ideal $N$, then $N$ would contain each $I_k$ by Lemma 2.11, and hence also $\bigcup_{k=1}^{\infty} I_k = I$. However, $I$ is not a nilpotent ideal.

Every finite dimensional algebra, however, has a maximal nilpotent ideal. Indeed, it contains at least one nilpotent ideal, namely 0, and therefore the (nonempty) set of all of its nilpotent ideals contains an element of maximal dimension. This is obviously a maximal nilpotent ideal.

**Definition 2.13** The maximal nilpotent ideal of a finite dimensional algebra $A$ is called the **radical** of $A$.

This definition is adjusted to the finite dimensional context. There are several nonequivalent definitions of radicals of general rings, which agree with Definition 2.13 in the case of finite dimensional algebras.

**Example 2.14** The radical of the algebra $A = T_n(F)$ from Example 2.6 is $N$, the set of all strictly upper triangular matrices. Indeed, $N$ is a nilpotent ideal of $A$, and any ideal of $A$ that properly contains $N$ cannot be nilpotent since it necessarily contains a nonzero diagonal matrix.

**Example 2.15** The radical of a finite dimensional simple algebra is 0. This is because a simple algebra $A$ cannot be nilpotent. Indeed, by Remark 1.16, $A^n = 0$ implies $A^{n-1} = 0$.

From now on in this chapter, we will be mostly interested in rings that have no nonzero nilpotent ideals. Our central goal is to describe the structure of all finite dimensional algebras with this property.

## 2.2 Prime and Semiprime Rings

The goal of this section is to introduce two important classes of rings, the prime rings and the semiprime rings. Let us start, however, with another class with which the reader is presumably already familiar, at least in the commutative context.

**Definition 2.16** A ring $R$ is said to be a **domain** if for all $a, b \in R$, $ab = 0$ implies $a = 0$ or $b = 0$.

In other words, $R$ is a domain if it has no zero-divisors. Equivalently, $R$ is a domain if it has the *cancellation property*: If $a \neq 0$, then each of $ab = ac$ and $ba = ca$ implies $b = c$.

Commutative domains are of utmost importance in algebra. Noncommutative domains also form a notable class of rings, but their role in noncommutative algebra is not entirely parallel to the role of commutative domains in commutative algebra. The most basic examples of commutative rings are domains, while the matrix ring $M_n(F)$, which may be considered as the prototype of a noncommutative ring, is not a domain. The next lemma introduces a wider class of rings, which can be regarded as a noncommutative counterpart of the class of commutative domains.

**Lemma 2.17** *Let $R$ be a ring. The following conditions are equivalent:*

(i) *For all $a, b \in R$, $aRb = 0$ implies $a = 0$ or $b = 0$.*
(ii) *For all left ideals $I$ and $J$ of $R$, $IJ = 0$ implies $I = 0$ or $J = 0$.*
(iii) *For all right ideals $I$ and $J$ of $R$, $IJ = 0$ implies $I = 0$ or $J = 0$.*
(iv) *For all ideals $I$ and $J$ of $R$, $IJ = 0$ implies $I = 0$ or $J = 0$.*

***Proof*** If $I$ and $J$ are left ideals satisfying $IJ = 0$, then $IRJ = 0$ since $RJ \subseteq J$. Hence we see that (i) implies (ii). Similarly, (i) implies (iii), and each of (ii) and (iii) trivially implies (iv).

Assume that (iv) holds and that $a, b \in R$ satisfy $aRb = 0$. The product of the ideals $RaR$ and $RbR$ is then 0. Hence one of them, say $RaR$, is 0. This implies that $Ra$ and $aR$ are two-sided ideals such that $Ra \cdot R = R \cdot aR = 0$. By (iv), $Ra = aR = 0$. But then $\mathbb{Z}a$ is an ideal of $R$ satisfying $\mathbb{Z}a \cdot R = 0$, and so (iv) now yields $a = 0$. Thus, (iv) implies (i). □

**Definition 2.18** A ring $R$ is said to be **prime** if it satisfies one, and hence all, of the conditions of Lemma 2.17.

**Lemma 2.19** *A commutative ring is prime if and only if it is a domain.*

***Proof*** It is enough to observe that $ab = 0$ implies $aRb = 0$ if $R$ is commutative. □

**Remark 2.20** Let $R$ be a prime ring with $\mathrm{char}(R) \neq 0$. First observe that if $0 \neq a \in R$ and $n \in \mathbb{N}$ are such that $na = 0$, then $aR(nb) = 0$ for every $b \in R$, and hence $nR = 0$. Secondly, if $nR = 0$ and $n = rs$ for some $r, s \in \mathbb{N}$, then $rR \cdot sR = 0$. Since $rR$ and $sR$ are ideals of $R$, we must have $rR = 0$ or $sR = 0$. This implies that $\mathrm{char}(R)$ is a prime number $p$. We can therefore consider $R$ as an algebra over $\mathbb{Z}_p$ in the natural way. That is, for $k \in \mathbb{Z}_p = \{0, 1, \ldots, p-1\}$ and $x \in R$, we define $kx$ as $x + \cdots + x$ ($k$ times).

We proceed by introducing a yet wider class of rings.

**Lemma 2.21** *Let $R$ be a ring. The following conditions are equivalent:*

(i) *For all $a \in R$, $aRa = 0$ implies $a = 0$.*
(ii) *For all left ideals $I$ of $R$, $I^2 = 0$ implies $I = 0$.*
(iii) *For all right ideals $I$ of $R$, $I^2 = 0$ implies $I = 0$.*
(iv) *For all ideals $I$ of $R$, $I^2 = 0$ implies $I = 0$.*
(v) *$R$ has no nonzero nilpotent ideals.*

***Proof*** The proof that (i)–(iv) are equivalent is similar to the proof of Lemma 2.17, so we omit it. It is trivial that (v) implies (iv). Conversely, if (iv) holds and $I$ is an ideal such that $I^n = 0$, then $(I^{n-1})^2 = 0$ and hence $I^{n-1} = 0$. Inductively, we obtain $I = 0$. □

**Definition 2.22** A ring $R$ is said to be **semiprime** if it satisfies one, and hence all, of the conditions of Lemma 2.21.

**Remark 2.23** As a special case of (i) we have that each of the conditions $aR = 0$ and $Ra = 0$ implies $a = 0$ if $R$ is semiprime.

**Remark 2.24** If $A$ is an algebra and $I$, $J$ are ring ideals of $A$ such that $IJ = 0$, then the linear spans of $I$ and $J$ are algebra ideals of $A$ whose product is 0. Hence it follows that a **prime algebra** can be equivalently defined as an algebra which is prime as a ring, or as an algebra in which the product of any two of its nonzero algebra ideals is nonzero. A similar remark holds for **semiprime algebras** (as well as for simple algebras, cf. Remark 1.20).

**Remark 2.25** Let $I$ be a nonzero ideal of a prime ring $R$. If $a, b \in R$ are such that $aIb = 0$, then $aRuRb = 0$ for every $u \in I$. Using (i) from Lemma 2.17 twice it follows that $a = 0$ or $b = 0$. This in particular shows that an ideal of a prime ring is again a prime ring. Similarly, an ideal of a semiprime ring is a semiprime ring.

A ring having no nonzero nilpotent elements is called a **reduced ring**. The following is an analog of Lemma 2.19.

**Lemma 2.26** *A commutative ring is semiprime if and only if it is reduced.*

Another useful observation is the following.

**Lemma 2.27** *A ring is a domain if and only if it is prime and reduced.*

***Proof*** It is enough to prove the "if" part. Let $R$ be a prime reduced ring, and assume that $a, b \in R$ satisfy $ab = 0$. Then every element in $bRa$ has square zero, so it must be zero since $R$ is reduced. But then $b = 0$ or $a = 0$ since $R$ is prime. □

We know that a maximal nilpotent ideal contains all nilpotent ideals (Lemma 2.11). The following can be added.

**Lemma 2.28** *If $N$ is a maximal nilpotent ideal of a ring $R$, then the factor ring $R/N$ is semiprime.*

***Proof*** Let $K$ be an ideal of $R/N$ such that $K^2 = 0$. Then $J = \{x \in R \mid x + N \in K\}$ is an ideal of $R$ such that $J^2 \subseteq N$. As $N$ is nilpotent, it follows that $J$ is nilpotent as well. Lemma 2.11 therefore implies that $J \subseteq N$, and hence $K = 0$. □

**Example 2.29** Let $A$ and $N$ be as in Example 2.14. Then $A/N$ is isomorphic to $F^n$, the direct product of $n$ copies of $F$. Indeed, the map $(a_{ij}) \mapsto (a_{11}, a_{22}, \ldots, a_{nn})$ from $A$ to $F^n$ is easily seen to be a surjective algebra homomorphism with kernel $N$.

Let us clarify the relationships between the classes of rings introduced so far. The following implications are obvious from definitions:

$$\text{division ring} \implies \text{simple and domain} \implies \text{simple}$$
$$\implies \text{simple or domain} \implies \text{prime} \implies \text{semiprime}.$$

None of them can be reversed. A standard example of a simple domain that is not a division ring is the Weyl algebra $\mathscr{A}_1$, which will be introduced in Sect. 3.11. Providing counterexamples to other reverse implications is easy.

**Example 2.30** Let $F$ be a field. The matrix ring $M_n(F)$ is simple (see Lemma 1.18), but not a domain if $n \geq 2$.

**Example 2.31** The ring $\mathbb{Z}$ is a domain, but is not simple.

**Example 2.32** The ring $M_n(\mathbb{Z})$, $n \geq 2$, is easily seen to be prime, but it is neither a domain nor a simple ring. The obvious, and in fact the only examples of its ideals are $M_n(k\mathbb{Z})$, $k \geq 0$.

The next example points out an important difference between the classes of prime and semiprime rings, and also indicates a slight analogy between prime rings and prime numbers.

**Example 2.33** Let $R_1$ and $R_2$ be nonzero rings. Their direct product $R = R_1 \times R_2$ is not a prime ring, as $R_1 \times 0$ and $0 \times R_2$ are nonzero ideals of $R$ whose product is 0. On the other hand, if both $R_1$ and $R_2$ are semiprime, then $R$ is also semiprime.

The following implications are also obvious:

$$\text{domain} \implies \text{reduced} \implies \text{semiprime}.$$

Again, none of them can be reversed. Indeed, the direct product $R_1 \times R_2$ of nonzero reduced rings $R_1$ and $R_2$ is a reduced ring that is not a domain (and neither prime), and the matrix ring $M_n(F)$ is semiprime but not reduced if $n \geq 2$.

Our main results in this chapter concern finite dimensional prime and semiprime algebras. We need some preliminaries first. In the next few sections, we will consider some notions of general interest in ring theory.

## 2.3 Unitization

Let $A$ be an $F$-algebra. The set $F \times A$ becomes an $F$-algebra if we define addition, scalar multiplication, and product as follows:

$$\begin{aligned}(\lambda, x) + (\mu, y) &= (\lambda + \mu, x + y),\\ \mu(\lambda, x) &= (\mu\lambda, \mu x),\\ (\lambda, x)(\mu, y) &= (\lambda\mu, \mu x + \lambda y + xy).\end{aligned}$$

Denote this algebra by $A^\sharp$. We consider $A$ as a subalgebra of $A^\sharp$ via the embedding $x \mapsto (0, x)$. Note that $A$ is actually an ideal of $A^\sharp$. A crucial observation for us is that $A^\sharp$ is a unital algebra, with unity $(1, 0)$. We call $A^\sharp$ the **unitization of the algebra** $A$.

Replacing the role of $F$ by $\mathbb{Z}$ one defines the **unitization** $R^\sharp$ **of the ring** $R$ in exactly the same way (just ignore the scalar multiplication).

The unitization is primarily intended for algebras (and rings) without unity. The idea behind is to reduce problems on general algebras to unital algebras; this does not always work, but sometimes it does. In principle, one can form $A^\sharp$ even when $A$ is unital. This may seem artificial, especially since the unity of $A$ is then different from the unity of $A^\sharp$. However, constructions of new rings and algebras from old sometimes turn out to be unexpectedly useful. In the proof of Theorem 2.71, we will make use of $A^\sharp$ to show that the original algebra $A$ is necessarily unital.

The unitization of $A$ does not preserve all properties of $A$. For instance, the simplicity is definitely not preserved since $A$ is an ideal of $A^\sharp$. If $A$ is a domain with unity $1_A$, then $A^\sharp$ is not a domain since $(1, -1_A)(0, 1_A) = 0$. On the other hand, the property of being semiprime is preserved, as we now show.

**Lemma 2.34** *If $A$ is a semiprime algebra, then so is $A^\sharp$.*

***Proof*** Let $(\lambda, a) \in A^\sharp$ be such that $(\lambda, a)A^\sharp(\lambda, a) = 0$. Then $(\lambda, a)(1, 0)(\lambda, a) = 0$, which gives $\lambda^2 = 0$ and hence $\lambda = 0$. From $(0, a)(0, x)(0, a) = 0$ we now obtain $aAa = 0$ and so $a = 0$ because $A$ is semiprime. This proves that $A^\sharp$ is semiprime. □

## 2.4 Idempotents

Elements equal to their squares play a special role in ring theory. In this section, we will survey some of their basic properties. We start by giving them a name.

**Definition 2.35** A ring element $e$ satisfying $e^2 = e$ is called an **idempotent**.

**Example 2.36** Obvious examples of idempotents are 0 and 1. An idempotent different from 0 and 1 is called a **nontrivial idempotent**.

**Remark 2.37** A domain has no nontrivial idempotents. Indeed, if $R$ is unital, then $e^2 = e$ implies $e(1 - e) = 0$ and hence $e = 0$ or $e = 1$. If $R$ is not unital, then $e^2 = e$ implies $e(ex - x) = (xe - x)e = 0$ and hence, since $e$ is not a unity, $e = 0$.

**Example 2.38** Let $R = M_n(S)$ where $S$ is a unital ring. Every diagonal matrix in $R$ with 1's and 0's on the diagonal (i.e., a matrix which is a sum of some of the standard matrix units $E_{ii}$) is an idempotent. If both 0 and 1 appear on the diagonal, then it is nontrivial.

**Remark 2.39** The product $ef$ of idempotents $e$ and $f$ is not always an idempotent. However, if $e$ and $f$ commute, then it is. For example, the idempotents from Example 2.38 commute with each other, and their products are again idempotents of the same type, that is, diagonal matrices with 1's and 0's on the diagonal. Of course, these products can be zero.

**Definition 2.40** Idempotents $e$ and $f$ are **orthogonal** if $ef = fe = 0$.

**Remark 2.41**

(a) If $e$ is an idempotent in a unital ring $R$, then $1 - e$ is also an idempotent, and $e$ and $1 - e$ are orthogonal.
(b) The sum of orthogonal idempotents is an idempotent.
(c) If $e$ and $f$ are orthogonal idempotents in a ring $R$, then $e + exf$ and $e + fxe$ are idempotents for every $x \in R$. In particular, if $e$ is an idempotent in a unital ring $R$, then $e + ex(1 - e)$ and $e + (1 - e)xe$ are idempotents for every $x \in R$.

Remark 2.41 indicates that the existence of one nontrivial idempotent implies the existence of many. We also remark that if $a$ is an invertible element in a unital ring $R$, then $aea^{-1}$ is an idempotent whenever $e$ is an idempotent. However, if $e$ is a **central idempotent**, i.e., an element from the center of the ring, then these observations do not tell us much.

**Example 2.42** Let $R_1$ and $R_2$ be nonzero unital rings, and let $R = R_1 \times R_2$. Then $e = (1, 0)$ and $f = (0, 1)$ are nontrivial, central, and orthogonal idempotents. If $R_1$ and $R_2$ are domains, then $e$ and $f$ are the only nontrivial idempotents in $R$.

If $e$ is a central idempotent of a ring $R$, then $I = eR$ is an ideal of $R$. Considering $I$ as a ring, we see that $e$ is its unity. Ideals are rarely unital rings. In fact, the following lemma shows that only those generated by central idempotents are.

**Lemma 2.43** *Let $I$ be an ideal of a ring $R$. If $I$ is a unital ring, then:*

(a) *The unity $e$ of $I$ is a central idempotent in $R$ and $I = eR$.*
(b) *There exists an ideal $J$ of $R$ such that $R = I \oplus J$. Moreover, $R \cong I \times J$.*
(c) *Every ideal of $R$ is of the form $I' \oplus J'$ where $I'$ is an ideal of $I$ and $J'$ is an ideal of $J$.*

***Proof***

(a) From $e \in I$ it follows that $eR \subseteq I$. Conversely, $I = eI \subseteq eR$. Thus, $I = eR$. Since $ex, xe \in I$ for every $x \in R$, we have $ex = (ex)e$ and $xe = e(xe)$. Therefore, $ex = xe$, proving that $e$ is a central idempotent.
(b) The set $J = \{y - ey \mid y \in R\}$ ( $= (1 - e)R$ if $R$ is unital) is clearly an ideal of $R$. If $ex = y - ey$ for some $x, y \in R$, then by multiplying this equation by $e$ we see that both sides are zero. Thus, $I \cap J = 0$ and so it follows from $x = ex + (x - ex)$ that $R = I \oplus J$. An isomorphism from $R$ to $I \times J$ is given by $x \mapsto (ex, x - ex)$. To prove that it is surjective, observe that it maps $ex + y - ey$ to $(ex, y - ey)$.
(c) Let $K$ be an ideal of $R$. Write $I'$ for the set of all $u \in I$ such that $u + v \in K$ for some $v \in J$, and similarly, $J'$ for the set of all $v \in J$ such that $u + v \in K$ for some $u \in I$. Clearly, $I'$ is an ideal of $I$, $J'$ is an ideal of $J$, and $K \subseteq I' \oplus J'$. Now take $u \in I'$. By definition, $u + v \in K$ for some $v \in J$ and hence $u = eu = e(u + v) \in K$. Similarly, if $v \in J'$, then $u + v \in K$ for some $u \in I$ and hence $v = v - ev = (u + v) - e(u + v) \in K$. Thus, $I', J' \subseteq K$ and so $I' \oplus J' \subseteq K$. Consequently, $K = I' \oplus J'$. □

The next lemma is a small variation of (c).

**Lemma 2.44** *If $R_1, \ldots, R_n$ are unital rings, then every ideal of $R = R_1 \times \cdots \times R_n$ is of the form $I_1 \times \cdots \times I_n$ where $I_j$ is an ideal of $R_j$. Similarly, every left ideal of $R$ is of the form $L_1 \times \cdots \times L_n$ where $L_j$ is a left ideal of $R_j$.*

***Proof*** In the case where $n = 2$, the first statement is obtained by applying (c) to $I = R_1 \times 0$. The general case follows trivially by induction on $n$. The proof of the second statement is essentially the same. □

**Remark 2.45** If each $R_j$ is a simple ring, then $I_j = 0$ or $I_j = R_j$. Therefore, in this case, every ideal of $R$ is a unital ring.

**Remark 2.46** Let $e$ now be arbitrary, not necessarily central idempotent of a ring $R$. We call the subring

$$eRe = \{exe \mid x \in R\}$$

the **corner ring** corresponding to $e$. This term obviously arises from the matrix ring example (what is $eRe$ if $R = M_n(S)$ and $e = E_{11} + \cdots + E_{kk}$ where $E_{ii}$ are standard matrix units?). The corner ring $eRe$ is unital, even if $R$ is not; its unity is $e$.

For simplicity of exposition, assume that $R$ is unital and write $f = 1-e$. Suppose $ex_1e + ex_2f + fx_3e + fx_4f = 0$ for some $x_i \in R$. Multiplying from the left and right by $e$ we obtain $ex_1e = 0$. Similarly we see that all other terms are 0. Since every $x \in R$ can be written as $x = exe + exf + fxe + fxf$, $R$ can therefore be decomposed into the following additive group direct sum:

$$R = eRe \oplus eRf \oplus fRe \oplus fRf. \tag{2.1}$$

We call (2.1) the **Peirce decomposition** of $R$ (with respect to $e$). If $e$ is central, then $eRf = fRe = 0$ and the corner rings $eRe$ and $fRf$ are equal to $eR$ and $fR$, respectively. Thus, in this case (2.1) reduces to the decomposition from Lemma 2.43.

The assumption that $R$ is unital was obviously necessary for introducing the idempotent $f = 1 - e$. However, the presence of 1 can actually be avoided. We did not really deal with $f$ itself, but with products of $f$ with elements from $R$. If one does not assume that $R$ is unital and then writes $x - ex$ instead of $fx$, $x - ex - xe + exe$ instead of $fxf$, etc., then we still have the Peirce decomposition, but the price we have to pay for the absence of 1 are lengthy equations.

We will say more about the Peirce decomposition in Remark 2.54.

Let $I$ be an ideal of a ring $R$. If $e$ is an idempotent in $R$, then $e+I$ is an idempotent in $R/I$. If every idempotent in $R/I$ is of such a form—that is, for every idempotent $\varepsilon \in R/I$ there exists an idempotent $e \in R$ such that $\varepsilon = e + I$—then we say that **idempotents in $R/I$ can be lifted to $R$**. This is not always true. For example, if $1 < k < n$ are natural numbers such that $n \mid k(k-1)$, then $k + n\mathbb{Z}$ is an idempotent in $\mathbb{Z}/n\mathbb{Z}$ which cannot be lifted to $\mathbb{Z}$.

**Lemma 2.47** *If an ideal $I$ of a ring $R$ is nil, then idempotents in $R/I$ can be lifted to $R$.*

***Proof*** We first prove the lemma for the case where $R$ is unital. Let $\varepsilon$ be an idempotent in $R/I$. If $a \in R$ is such that $\varepsilon = a + I$, then $a^2 - a \in I$ and is therefore a nilpotent element. We want to show that it is actually 0 for some $a$. Suppose this is not true. Let $n \geq 2$ be the smallest natural number with the property that $(a^2 - a)^n = 0$ for some $a \in R$ such that $\varepsilon = a + I$. Let $b = 3a^2 - 2a^3$. We have $\varepsilon = b + I$ since $b - a = (1 - 2a)(a^2 - a) \in I$. Further, since

$$b^2 - b = (4a^2 - 4a - 3)(a^2 - a)^2$$

and $((a^2 - a)^2)^{n-1} = 0$, $(b^2 - b)^{n-1} = 0$. This contradicts the choice of $n$.

Assume now that $R$ is not unital. Take $a \in R$ such that $a + I$ is an idempotent in $R/I$. Let $R^\sharp$ be the unitization of $R$. Then $I$ is also an ideal of $R^\sharp$, and so, by what we proved above, there exists an idempotent $e \in R^\sharp$ such that $e + I = a + I$. However, since $e - a \in I \subseteq R$, $e$ actually lies in $R$. □

The proof may have seemed mysterious. How did we come up with introducing $b = 3a^2 - 2a^3$? The idea behind was to find a polynomial $f(\omega) \in \mathbb{Z}[\omega]$ such that

$b = a + f(a)(a^2 - a)$ (to guarantee that $\varepsilon = b + I$) and $b^2 - b = g(a)(a^2 - a)^2$ for some polynomial $g(\omega) \in \mathbb{Z}[\omega]$ (to guarantee that $(b^2 - b)^{n-1} = 0$). A short calculation shows that $f(\omega) = 1 - 2\omega$ is a right choice.

We will say that **orthogonal idempotents in $R/I$ can be lifted to $R$** if for any finite set of pairwise orthogonal idempotents $\{\varepsilon_1, \ldots, \varepsilon_n\}$ in $R/I$ there exists a set of pairwise orthogonal idempotents $\{e_1, \ldots, e_n\}$ in $R$ such that $\varepsilon_i = e_i + I$, $i = 1, \ldots, n$.

**Lemma 2.48** *If an ideal $I$ of a ring $R$ is nil, then orthogonal idempotents in $R/I$ can be lifted to $R$.*

***Proof*** Essentially the same argument as in the preceding proof shows that we may assume without loss of generality that $R$ is unital.

We proceed by induction on $n$. Lemma 2.47 covers the base case. We may therefore assume that there exist pairwise orthogonal idempotents $e_1, \ldots, e_{n-1}$ in $R$ such that $\varepsilon_i = e_i + I$, $i = 1, \ldots, n-1$. Again referring to Lemma 2.47, we also have an idempotent $f \in R$ such that $\varepsilon_n = f + I$. Observe that $e = e_1 + \cdots + e_{n-1}$ is an idempotent and that $e + I$ and $f + I$ are orthogonal, so $ef, fe \in I$. Since $I$ is nil, $(fe)^n = 0$ for some $n \geq 1$. Hence, $1 - fe$ is invertible and has inverse $1 + fe + \cdots + (fe)^{n-1}$. Note that

$$g = (1 - fe)^{-1} f (1 - fe)$$

is also an idempotent and that $fe \in I$ implies $g + I = f + I = \varepsilon_n$. Finally, let

$$e_n = (1 - e)g.$$

Since $ef \in I$ and $g - f \in I$, $eg = e(g - f) + ef \in I$ and hence $\varepsilon_n = e_n + I$. Next, observe that $f(1 - fe)e = 0$ and therefore $ge = 0$. Consequently,

$$e_n^2 = (1 - e)g(1 - e)g = (1 - e)g^2 = (1 - e)g = e_n.$$

It remains to prove that $e_n e_i = 0$ and $e_i e_n = 0$ for $i = 1, \ldots, n-1$. Since $e_i = ee_i = e_i e$, it is enough to show that $e_n e = 0$ and $ee_n = 0$. The latter is obvious and the former follows from $ge = 0$. □

## 2.5 Matrix Units

We are familiar with the *standard matrix units* $E_{ij}$. Let us introduce their abstract generalization.

**Definition 2.49** Let $R$ be a unital ring and let $n \in \mathbb{N}$. We call $\{e_{ij} \in R \mid 1 \leq i, j \leq n\}$ a **set of $n \times n$ matrix units** if

$$e_{11} + e_{22} + \cdots + e_{nn} = 1$$

and

$$e_{ij}e_{kl} = \delta_{jk}e_{il}$$

for all $1 \leq i, j, k, l \leq n$. Here, $\delta_{jk}$ stands for the *Kronecker delta*: $\delta_{jk} = \begin{cases} 1 \text{ if } j = k \\ 0 \text{ if } j \neq \end{cases}$.

**Example 2.50** Let $S$ be a unital ring and let $n \in \mathbb{N}$. Note that the standard matrix units $E_{ij}$ in $R = M_n(S)$ indeed form a set of matrix units in $R$. There are others. For example, if $e_{ij}$ are some matrix units (say the standard ones) in $R$ and $a \in R$ is invertible, then $ae_{ij}a^{-1}$ are also matrix units. More generally, automorphisms send matrix units to matrix units.

**Remark 2.51** The matrix units $e_{ij}$ with $i \neq j$ are nilpotent, in fact square-zero elements, while $e_{ii}$ are pairwise orthogonal idempotents whose sum is 1. Every $e_{ii}$ gives rise to the corner subring $e_{ii}Re_{ii}$, and all of these rings are isomorphic:

$$e_{ii}Re_{ii} \cong e_{jj}Re_{jj}.$$

Indeed, it is an easy exercise to check that

$$e_{ii}ae_{ii} \mapsto e_{ji}(e_{ii}ae_{ii})e_{ij} = e_{jj}(e_{ji}ae_{ij})e_{jj}$$

is an isomorphism. This explains why we will usually deal only with $e_{11}Re_{11}$.

We have defined matrix units in an arbitrary unital ring. However, we will now show that a ring with matrix units is in fact a full matrix ring.

**Lemma 2.52** *If a unital ring $R$ contains a set of $n \times n$ matrix units $e_{ij}$, then $R \cong M_n(S)$ where $S = e_{11}Re_{11}$.*

***Proof*** Define $\varphi : R \to M_n(S)$ by

$$\varphi(a) = (a_{ij}), \text{ where } a_{ij} = e_{1i}ae_{j1}.$$

Note that $a_{ij} = e_{11}a_{ij}e_{11}$ and so $a_{ij}$ indeed lies in $S$. The additivity of $\varphi$ is clear. The $(i, j)$ entry of $\varphi(a)\varphi(b)$ is equal to

$$\sum_{k=1}^{n} e_{1i}ae_{k1}e_{1k}be_{j1} = e_{1i}a\Big(\sum_{k=1}^{n} e_{kk}\Big)be_{j1} = e_{1i}abe_{j1},$$

which is the $(i, j)$ entry of $\varphi(ab)$. Therefore, $\varphi(ab) = \varphi(a)\varphi(b)$. If $a_{ij} = 0$ for all $i, j$, then $e_{ii}ae_{jj} = e_{i1}a_{ij}e_{1j} = 0$, and so $a = 0$ since the sum of the $e_{ii}$'s is 1. Thus, $\varphi$ is injective. Finally, observe that $\varphi(e_{k1}ae_{1l})$ is the matrix whose $(k, l)$ entry

is $e_{11}ae_{11}$ and all other entries are 0. Since every matrix in $M_n(S)$ is a sum of such matrices, $\varphi$ is surjective. □

**Remark 2.53** If $R$ is an algebra, then the isomorphism from Lemma 2.52 is an algebra isomorphism.

**Remark 2.54** A pair of idempotents $e$ and $f$ in a unital ring $R$ such that $e + f = 1$ gives rise to the Peirce decomposition (2.1). Its meaning can be better understood after familiarity with matrix units. Indeed, introducing the notation

$$R_{11} = eRe, \quad R_{12} = eRf, \quad R_{21} = fRe, \quad R_{22} = fRf,$$

we can write (2.1) as $R = R_{11} \oplus R_{12} \oplus R_{21} \oplus R_{22}$. We remark that $R_{11}$ and $R_{22}$ are corner rings, and $R_{12}$ and $R_{21}$ are subrings with zero multiplication. Moreover,

$$R_{ij}R_{kl} \subseteq \delta_{jk}R_{il}$$

for all $1 \leq i, j, k, l \leq 2$. This resembles calculations with matrix units. The point we wish to make is that when having a nontrivial idempotent $e$ in our ring, we can, with the help of its indispensable companion $f = 1 - e$, mimic the $2 \times 2$ matrix setup. This can be productive. We did not say, however, that a ring $R$ with a nontrivial idempotent $e$ is necessarily isomorphic to a $2 \times 2$ matrix ring (over some ring). This holds under the additional assumption that the equations $exfye = e$ and $fyexf = f$ are solvable in $R$. Indeed, the elements $e_{11} = e, e_{12} = exf, e_{21} = fye$, $e_{22} = f$ then form a set of $2 \times 2$ matrix units of $R$, and so $R \cong M_2(eRe)$ by Lemma 2.52. We will arrive at such a situation in Lemma 2.65 below.

Lemma 2.52 tells us that in order to show that a ring $R$ is a full matrix ring, it is enough to find matrix units in $R$. The next lemma simplifies this task.

**Lemma 2.55** *If a unital ring $R$ contains pairwise orthogonal idempotents $e_{ii}$, $i = 1, \ldots, n$, whose sum is 1, and elements $e_{1i}, e_{i1}$, $i = 2, \ldots, n$, satisfying*

$$\text{(a) } e_{1i}e_{i1} = e_{11}, \quad \text{(b) } e_{i1}e_{1i} = e_{ii}, \quad \text{(c) } e_{1i} \in e_{11}Re_{ii}, \quad \text{(d) } e_{i1} \in e_{ii}Re_{11},$$

*then the set $\{e_{ii}, e_{1i}, e_{i1} \mid i = 1, \ldots, n\}$ can be extended to a set of $n \times n$ matrix units.*

***Proof*** For $i \neq j$ and $i, j \neq 1$, define $e_{ij} = e_{i1}e_{1j}$. Observe that (b), (c), and (d) show that $e_{ij} = e_{i1}e_{1j}$ holds in any case, that is, also if $i = j$ or one of $i, j$ is 1. Further, (a), (c), and (d) imply that $e_{1j}e_{k1} = \delta_{jk}e_{11}$ for all $j$ and $k$. Consequently, for all $i, j, k, l$ we have

$$e_{ij}e_{kl} = e_{i1}e_{1j}e_{k1}e_{1l} = \delta_{jk}e_{i1}e_{11}e_{1l} = \delta_{jk}e_{i1}e_{1l} = \delta_{jk}e_{il},$$

proving that $e_{ij}$ are matrix units. □

We now continue from where we left off at the end of the preceding section. The next lemma considers the **lifting of matrix units**. Let us first point out Exercise 2.19 which shows that factor rings of (unital) full matrix rings $M_n(S)$ are again full matrix rings. We will be interested in the converse of this elementary fact. That is, if a unital ring $R$ has an ideal $I$ such that $R/I$ is a full matrix ring, does it follow that $R$ is a full matrix ring? In general, the answer is certainly negative, just consider the direct product $R = M_n(S) \times I$ where $I$ is any unital ring. If, however, $I$ is a nil ideal, the answer is positive.

**Lemma 2.56** *Let $I$ be a nil ideal of a unital ring $R$. If $R/I \cong M_n(T)$ for some unital ring $T$ and $n \in \mathbb{N}$, then there exists an idempotent $e \in R$ such that $R \cong M_n(eRe)$ and $T \cong (e+I)(R/I)(e+I)$.*

***Proof*** Let $\varepsilon_{ij}$, $1 \leq i, j \leq n$, be matrix units in $R/I$ (so, $T \cong \varepsilon_{11}(R/I)\varepsilon_{11}$). By Lemma 2.48, there exist pairwise orthogonal idempotents $e_{ii}$, $i = 1, \ldots, n$, in $R$ such that $\varepsilon_{ii} = e_{ii} + I$ (so, $e = e_{11}$ satisfies $T \cong (e+I)(R/I)(e+I)$). Observe that $1 - (e_{11} + \cdots + e_{nn})$ is an idempotent lying in $I$. Since $I$ is nil and a nonzero idempotent cannot be nilpotent, it follows that $e_{11} + \cdots + e_{nn} = 1$.

In light of Lemma 2.52, it suffices to find elements $e_{1i}, e_{i1} \in R$, $i = 2, \ldots, n$, that satisfy conditions (a)–(d) from Lemma 2.55. To this end, we first choose $u_{1i} \in R$ such that $\varepsilon_{1i} = u_{1i} + I$. As $\varepsilon_{1i} = \varepsilon_{11}\varepsilon_{1i}\varepsilon_{ii}$, $e_{1i} = e_{11}u_{1i}e_{ii}$ also satisfies $\varepsilon_{1i} = e_{1i} + I$, and moreover, it satisfies condition (c). The definition of $e_{i1}$ will be less straightforward. We start by choosing $u_{i1} \in R$ such that $\varepsilon_{i1} = u_{i1} + I$. Then $a_{i1} = e_{ii}u_{i1}e_{11}$ satisfies $\varepsilon_{i1} = a_{i1} + I$ and $a_{i1} = e_{ii}a_{i1}e_{11}$. From $\varepsilon_{11} = \varepsilon_{1i}\varepsilon_{i1}$ we infer that $e_{11} - e_{1i}a_{i1}$ lies in $I$, and hence $(e_{11} - e_{1i}a_{i1})^n = 0$ for some $n \geq 1$. Therefore, $1 - (e_{11} - e_{1i}a_{i1})$ is invertible, with inverse

$$1 + (e_{11} - e_{1i}a_{i1}) + \cdots + (e_{11} - e_{1i}a_{i1})^{n-1}.$$

We now define

$$e_{i1} = a_{i1}(1 - (e_{11} - e_{1i}a_{i1}))^{-1}.$$

From $a_{i1} = e_{ii}a_{i1}e_{11}$ we infer that condition (d) is fulfilled. Observe also that $e_{11} - e_{1i}a_{i1} \in I$ yields $e_{i1} - a_{i1} \in I$, and consequently, $e_{ii} - e_{i1}e_{1i} \in I$. Next,

$$\begin{aligned} e_{1i}e_{i1} &= e_{1i}a_{i1}(1 - (e_{11} - e_{1i}a_{i1}))^{-1} \\ &= e_{11}(1 - (e_{11} - e_{1i}a_{i1}))(1 - (e_{11} - e_{1i}a_{i1}))^{-1} \\ &= e_{11}, \end{aligned}$$

so (a) holds. This implies that $e_{ii} - e_{i1}e_{1i}$ is an idempotent. As an element of a nil ideal $I$, it must be zero. Thus, (b) holds too. □

Before stating our final lemma, we mention that a ring $R$ does not necessarily contain subrings isomorphic to the factor rings $R/I$. Just think of the ring of integers and its ideals.

**Lemma 2.57** *Let $I$ be a nil ideal of an algebra $A$ over a field $F$. Suppose that $A/I \cong M_{n_1}(F) \times \cdots \times M_{n_r}(F)$ for some $n_1, \ldots, n_r \in \mathbb{N}$. Then $A$ contains a subalgebra $A_0$ such that $A_0 \cong A/I$.*

***Proof*** By Lemma 2.48, $A$ contains pairwise orthogonal idempotents $e_1, \ldots, e_n$ such that $e_i + I$ is the unity of the subalgebra (in fact, the ideal) of $A/I$ isomorphic to $M_{n_i}(F)$. Observe that $I \cap e_i A e_i$ is a nil ideal of the corner algebra $e_i A e_i$ and that $e_i A e_i / I \cap e_i A e_i \cong M_{n_i}(F)$. Lemma 2.56 implies that $e_i A e_i$ contains a set of $n_i \times n_i$ matrix units. Their linear span, which we denote by $A_i$, is a subalgebra of $e_i A e_i$ isomorphic to $M_{n_i}(F)$. If $a_1 + \cdots + a_n = 0$ where $a_i \in A_i$, then by multiplying from the left and right by $e_i$ we see that each $a_i = 0$. Let $A_0 = A_1 \oplus \cdots \oplus A_r$. Since $A_i A_j = 0$ if $i \neq j$, $A_0$ is a subalgebra of $A$, and clearly, $A_0 \cong A_1 \times \cdots \times A_r \cong A/I$. □

We will need Lemma 2.57 for the proof of an important result, called Wedderburn's principal theorem. Lemmas 2.47, 2.48, and 2.56 that led to it are of independent interest in ring theory.

## 2.6 Minimal Left Ideals

When considering one-sided ideals, we will usually give preference to the left ones.

**Definition 2.58** A left ideal $L$ of a ring $R$ is called a **minimal left ideal** if $L \neq 0$ and $L$ does not properly contain a nonzero left ideal of $R$.

The existence of minimal left ideals in rings is an exception rather than a rule. For example, the ring $\mathbb{Z}$ does not have them since, say, $2n\mathbb{Z} \subsetneq n\mathbb{Z}$ for every $n \in \mathbb{N}$. The simplest example of a minimal left ideal is the following.

**Example 2.59** The unique minimal left ideal of a division ring $D$ is $D$ itself.

The next example presents a model of a minimal left ideal.

**Example 2.60** Let $R = M_n(D)$, $D$ a division ring, and let $L$ be the set of matrices in $R$ that have arbitrary entries in the $i$th column and zeros in all other columns. It is an easy exercise to show that $L$ is a minimal left ideal of $R$. We remark that $L = RE_{ii}$ where $E_{ii}$ is the standard matrix unit, and that $E_{ii} R E_{ii} = \{dE_{ii} \mid d \in D\}$ is a division ring isomorphic to $D$.

We will give further examples in Sect. 4.4 where we will continue the study of minimal left ideals. The example just given is sufficient for comprehending the following fundamental lemma.

**Lemma 2.61** *Let $L$ be a minimal left ideal of a ring $R$. If $L^2 \neq 0$, then there exists an idempotent $e \in L$ such that $L = Re$ and $eRe$ is a division ring.*

***Proof*** Take $x, y \in L$ such that $xy \neq 0$. In particular, $Ly \neq 0$. Since $Ly$ is a left ideal of $R$ contained in $L$ and $L$ is minimal, $Ly = L$. Therefore, there exists an $e \in L$ such that $ey = y$. Hence, $e^2 y = ey$, meaning that $e^2 - e$ lies in the set $J = \{z \in L \mid zy = 0\}$. Observe that $J$ is a left ideal of $R$ contained in $L$. Since $x \in L \setminus J$, it follows that $J = 0$. In particular, $e^2 = e$. As $e \in L$, we have $Re \subseteq L$, and since $0 \neq e \in Re$ it follows from the minimality assumption that $L = Re$.

Now consider the corner ring $eRe$. Let $a \in R$ be such that $eae \neq 0$. We must prove that $eae$ is invertible in $eRe$. We have $0 \neq Reae \subseteq Re = L$, and so $Reae = L$. Therefore, $beae = e$ for some $b \in R$, which gives $(ebe)(eae) = e$. Since $ebe$ is a nonzero element in $eRe$, by the same argument there exists a $c \in R$ such that $(ece)(ebe) = e$. However, a left inverse coincides with a right inverse, so $eae = ece$ is invertible in $eRe$, with inverse $ebe$. □

**Remark 2.62** A similar statement holds for analogously defined **minimal right ideals**. Also, without any change in the proof, we see that the lemma holds for algebras. Minimal left ideals of algebras are of course defined in the same way as minimal left ideals of rings.

The condition that $L^2 \neq 0$ is automatically fulfilled in semiprime rings, in which the connection between minimal left ideals $Re$ and division rings $eRe$ is even tighter.

**Proposition 2.63** *The following statements are equivalent for an idempotent $e$ in a semiprime ring $R$:*

(i) *$eRe$ is a division ring.*
(ii) *$Re$ is a minimal left ideal of $R$.*
(iii) *$eR$ is a minimal right ideal of $R$.*

***Proof*** Since (i) is, unlike (ii) and (iii), a left-right symmetric condition, it suffices to prove that it is equivalent to (ii). As we know that (ii) implies (i) by (the proof of) Lemma 2.61, we must only show that (i) implies (ii).

Thus, assume that $eRe$ is a division ring and take a left ideal $I$ of $R$ such that $0 \neq I \subseteq Re$. It is enough to show that $e \in I$, as this clearly yields $I = Re$. Pick $0 \neq u \in I$. By the semiprimeness of $R$, $uru \neq 0$ for some $r \in R$. Note that $u = ue$ since $u \in I \subseteq Re$. Consequently, $eru = erue$ is a nonzero element of $eRe$, and hence there exists a $v \in R$ such that $(eve)(eru) = e$. Therefore, $e \in Ru \subseteq I$. □

**Definition 2.64** An idempotent $e$ in a semiprime ring $R$ that satisfies one, and hence all, of the conditions of Proposition 2.63 is called a **minimal idempotent**.

**Lemma 2.65** *Let $e$, $f$ be a pair of orthogonal minimal idempotents in a semiprime ring $R$. If $eRf \neq 0$, then there exist a $u \in eRf$ and a $v \in fRe$ such that $uv = e$ and $vu = f$.*

***Proof*** Let $a \in R$ be such that $eaf \neq 0$. Since $R$ is semiprime, there exists a $b \in R$ such that $(eaf)b(eaf) \neq 0$. In particular, $eafbeaf \neq 0$. As $eRe$ is a division ring with

unity $e$, there is a $c \in R$ such that $(eafbe)(ece) = e$. Thus, $u = eaf \in eRf$ and $v = fbece \in fRe$ satisfy $uv = e$. Suppose $vu \neq f$. Then $vu - f$ is a nonzero, and hence invertible element of the division ring $fRf$. Observe that $vuv = ve = v$ and therefore $(vu - f)v = 0$. Multiplying this equation from the left by the inverse of $vu - f$, we arrive at a contradiction $v = fv = 0$. Thus, $vu = f$. □

**Remark 2.66** A nonzero idempotent $f$ in a ring $R$ is called a **primitive idempotent** if the corner ring $fRf$ has no nontrivial idempotents. Obviously, a minimal idempotent is primitive. The converse is not true. For example, 1 is obviously a primitive, but not a minimal idempotent of the ring $\mathbb{Z}$. It is therefore interesting that in Lemma 2.65 we can replace the assumption that $f$ is minimal by a milder assumption that $f$ is primitive. This is because $uv = e$ with $u \in eRf$ and $v \in fRe$ implies $(vu)^2 = v(uv)u = veu = vu$. Similarly, in the next three results it is enough to assume that only one idempotent is minimal and the others are primitive.

**Lemma 2.67** *Let $e_1, \ldots, e_n$ be pairwise orthogonal minimal idempotents in a semiprime ring $R$. Write $e = e_1 + \cdots + e_n$ and $D = e_{11}Re_{11}$. If $e_iRe_j \neq 0$ for all $i, j$, then $eRe \cong M_n(D)$.*

***Proof*** Write $e_{ii}$ for $e_i$. Applying Lemma 2.65 to the idempotents $e_{11}$ and $e_{ii}$, $i \geq 2$, we see that there exist elements $e_{1i} \in e_{11}Re_{ii}$ and $e_{i1} \in e_{ii}Re_{11}$ satisfying $e_{1i}e_{i1} = e_{11}$ and $e_{i1}e_{1i} = e_{ii}$. Lemma 2.55 now tells us that the set $\{e_{ii}, e_{1i}, e_{i1} \mid, i = 1, \ldots, n\}$ can be extended to a set of $n \times n$ matrix units, and hence $R \cong M_n(e_{11}Re_{11})$ by Lemma 2.52. □

We are now ready to prove the cornerstone theorem from which Wedderburn's structure theorems will be easily derived.

**Theorem 2.68** *A unital ring $R$ is semiprime and contains $n$ pairwise orthogonal minimal idempotents whose sum is 1 if and only if there exist $n_1, \ldots, n_r \in \mathbb{N}$ and division rings $D_1, \ldots, D_r$ such that $n_1 + \cdots + n_r = n$ and $R \cong M_{n_1}(D_1) \times \cdots \times M_{n_r}(D_r)$.*

***Proof*** We start with the easy "if" part. If $R = M_n(D)$, then $R$ is simple (Lemma 1.18) and hence semiprime, and the obvious choice for idempotents having the desired properties are the matrix units $E_{ii}, i = 1, \ldots, n$. The general case readily follows from this particular case.

Now to the "only if" part. Let $E = \{e_1, \ldots, e_n\}$ be a set of pairwise orthogonal minimal idempotents in $R$ whose sum is 1. Define a relation $\sim$ on $E$ as follows: $e_i \sim e_j$ if $e_iRe_j \neq 0$. It is clear that $\sim$ is reflexive. If $e_jRe_i = 0$, then $(e_iRe_j)R(e_iRe_j) = 0$, and therefore $e_iRe_j = 0$ since $R$ is semiprime. This shows that $\sim$ is symmetric. Assume that $e_i \sim e_j$ and $e_j \sim e_k$. Lemma 2.65 shows that there exist $u \in e_iRe_j$ and $v \in e_jRe_i$ such that $vu = e_j$ and hence $ve_iu = e_j$. Therefore, $e_iRe_k = 0$ leads to a contradiction $e_jRe_k = ve_iuRe_k = 0$. Thus, $\sim$ is transitive and is an equivalence relation.

Let $E_1, \ldots, E_r$ be the equivalence classes. Write $f_i$ for the sum of all idempotents in $E_i$. Note that $f_i$ are pairwise orthogonal idempotents with sum 1, $f_iRf_j = 0$

for $i \neq j$, and, by Lemma 2.67, $f_i R f_i \cong M_{n_i}(D_i)$ where $n_i = |E_i|$ and $D_i$ is a division ring. For every $x \in R$, $f_i x = f_i x(f_1 + \cdots + f_r) = f_i x f_i$. Similarly we see that $x f_i = f_i x f_i$, meaning that each $f_i$ is a central idempotent. Therefore, $f_i R = f_i R f_i$ is an ideal of $R$ and $R = f_1 R \oplus \cdots \oplus f_r R$, from which $R \cong M_{n_1}(D_1) \times \cdots \times M_{n_r}(D_r)$ follows. Finally, $n_1 + \cdots + n_r = |E_1| + \cdots + |E_r| = n$. □

**Remark 2.69** If $R$ is an algebra, so are $D_1, \ldots, D_r$ and the isomorphism from Theorem 2.68 is an algebra isomorphism (compare Remark 2.53).

**Corollary 2.70** *A unital ring $R$ is prime and contains $n$ pairwise orthogonal minimal idempotents whose sum is 1 if and only if there exists a division ring $D$ such that $R \cong M_n(D)$.*

## 2.7 Wedderburn's Structure Theorems

In 1907, Wedderburn laid down the foundations of the structure theory of noncommutative algebras. Equipped with the knowledge from the preceding section, we are now in a position to prove Wedderburn's theorems.

Here is the fundamental theorem.

**Theorem 2.71 (Wedderburn)** *A nonzero finite dimensional algebra $A$ is semiprime if and only if there exist $n_1, \ldots, n_r \in \mathbb{N}$ and division algebras $D_1, \ldots, D_r$ such that $A \cong M_{n_1}(D_1) \times \cdots \times M_{n_r}(D_r)$.*

***Proof*** The "if" part is a special case of the "if" part of Theorem 2.68.

We first prove the "only if" part under the additional assumption that $A$ is unital. The proof is by induction on $d = [A : F]$. The case where $d = 1$ is trivial (since then $A \cong F$), so let $d > 1$. Take a nonzero left ideal $L$ of $A$ of minimal dimension. Obviously, $L$ is a minimal left ideal and $L^2 \neq 0$ as $A$ is semiprime. By Lemma 2.61, $L$ is generated by a minimal idempotent $e$. We may assume that $e \neq 1$, since otherwise the desired result holds for $r = 1$ and $n_1 = 1$. Set $f = 1 - e$. Then $fAf$ is a nonzero algebra with unity $f$ and $[fAf : F] < d$ as $e \notin fAf$. Moreover, $fAf$ is semiprime since $(faf)fAf(faf) = 0$ implies $faf = 0$ by the semiprimeness of $A$. By the induction hypothesis, $fAf$ is isomorphic to the direct product of algebras $M_{m_i}(D_i)$, and hence, by the "if" part of Theorem 2.68, the algebra $fAf$ contains pairwise orthogonal minimal idempotents $e_1, \ldots, e_m$ whose sum is $f$. As $e_i f = f e_i = e_i$, $e_i A e_i$ is equal to $e_i fAf e_i$ and is therefore a division algebra. Consequently, $e_i$ is a minimal idempotent of $A$ (not only of $f_i A f_i$). Now, by the "only if" part of Theorem 2.68 (together with Remark 2.69) applied to the idempotents $e, e_1, \ldots, e_m$, there exist division algebras $D_i$ and natural numbers $n_i$ (whose sum is $m + 1$) such that $A \cong M_{n_1}(D_1) \times \cdots \times M_{n_r}(D_r)$.

It remains to show that a finite dimensional semiprime algebra $A$ is automatically unital. To this end, recall from Lemma 2.34 that the unitization $A^\sharp$ is a semiprime

unital algebra. By what was proved in the preceding paragraph, $A^\sharp$ is isomorphic to the direct product of (unital) simple algebras $M_{n_i}(D_i)$. Remark 2.45 tells us that every ideal of $A^\sharp$ is unital when considered as an algebra. This applies to $A$ which is therefore unital. □

The observation from the final paragraph of the proof deserves a separate statement.

**Corollary 2.72** *Every finite dimensional semiprime algebra is unital.*

A more common name for a finite dimensional semiprime algebra is a (finite dimensional) **semisimple algebra**. Accordingly, a standard wording of Theorem 2.71 is

$$A \text{ is semisimple} \iff A \cong M_{n_1}(D_1) \times \cdots \times M_{n_r}(D_r).$$

A semisimple ring can be defined as a ring satisfying the conditions of Theorem 2.68. However, this is not a standard definition and we postpone the study of semisimple rings to the next chapter.

The following special case of Theorem 2.71 is of crucial importance.

**Theorem 2.73 (Wedderburn)** *Let $A$ be a nonzero finite dimensional algebra. The following statements are equivalent:*

(i) *$A$ is simple.*
(ii) *$A$ is prime.*
(iii) *There exist an $n \in \mathbb{N}$ and a division algebra $D$ such that $A \cong M_n(D)$.*

***Proof*** The implication (i) $\Longrightarrow$ (ii) trivial, (ii) $\Longrightarrow$ (iii) follows from Theorem 2.71, and (iii) $\Longrightarrow$ (i) from Lemma 1.18. □

We can add that $D$ is, of course, finite dimensional, and that

$$[A : F] = n^2[D : F].$$

Let us also mention that the classical version of Theorem 2.73 does not involve prime algebras. Usually it is stated as

$$A \text{ is simple} \iff A \cong M_n(D).$$

Nevertheless, it is interesting that finite dimensional prime algebras coincide with simple ones. In general, the class of prime algebras is much larger than the class of simple algebras.

If a finite dimensional algebra $A$ has nonzero nilpotent ideals, then we can factor out its radical (i.e., the maximal nilpotent ideal) and thereby obtain a semiprime algebra (Lemma 2.28). This yields the following theorem which summarizes our findings on finite dimensional algebras.

**Theorem 2.74** *Let $A$ be a finite dimensional algebra with radical $N$. Assume that $A \neq N$ (i.e., $A$ is not a nilpotent algebra). Then $A/N \cong M_{n_1}(D_1) \times \cdots \times M_{n_r}(D_r)$ for some $n_1, \ldots, n_r \in \mathbb{N}$ and division algebras $D_1, \ldots, D_r$.*

We will give additional information on Wedderburn's structure theorems in Sects. 3.10, 4.3, and 4.4.

## 2.8 Corollaries of Wedderburn's Theorems

In this section, we examine several situations in which Wedderburn's theorems yield more definitive conclusions.

The following result was proved (for $F = \mathbb{C}$) already in 1892 by T. Molien.

**Corollary 2.75 (Molien)** *Let $A$ be a nonzero finite dimensional semiprime algebra over an algebraically closed field $F$. Then $A \cong M_{n_1}(F) \times \cdots \times M_{n_r}(F)$ for some $n_1, \ldots, n_r \in \mathbb{N}$. Moreover, if $A$ is prime, then $r = 1$.*

***Proof*** Apply Theorems 2.71 and 2.73, along with Proposition 1.8. □

The next corollary, which is also due to Wedderburn, is deeper.

**Corollary 2.76 (Wedderburn's Principal Theorem (Special Case))** *Let $F$ be an algebraically closed field and let $A$ be a finite dimensional algebra over $F$ with radical $N$. Then $A$ contains a subalgebra $A_0$ such that $A_0 \cong A/N$.*

***Proof*** We may assume that $A \neq N$. Theorem 2.74 and Proposition 1.8 show that $A/N \cong M_{n_1}(F) \times \cdots \times M_{n_r}(F)$ for some $n_i \in \mathbb{N}$. The desired conclusion therefore follows from Lemma 2.57. □

We will derive the general Wedderburn's principal theorem in Sect. 5.10, where we will also discuss its meaning. Corollary 2.76 will be used in the proof.

In the rest of the section, we confine ourselves to simple algebras. The next two corollaries treat central simple algebras over an arbitrary field $F$.

**Corollary 2.77** *A finite dimensional algebra $A$ is a central simple algebra if and only if there exist an $n \in \mathbb{N}$ and a central division algebra $D$ such that $A \cong M_n(D)$.*

***Proof*** Apply Theorem 2.73 and Lemma 1.23. □

**Corollary 2.78** *The dimension of a finite dimensional central simple algebra is a perfect square.*

***Proof*** We have $[M_n(D) : F] = n^2[D : F]$. Now use Corollary 1.52. □

If a central simple algebra $A$ has dimension $n^2$, then we say that $A$ has **degree** $n$. However, we will not use this term in what follows.

The proof of Corollary 2.78 also shows that the following is true.

**Corollary 2.79** *Let $A$ be a central simple algebra over a field $F$. If $[A : F] = p^2$, where $p$ is a prime, then either $A$ is a division algebra or $A \cong M_p(F)$.*

Wedderburn's theorems reduce the problem of classifying finite dimensional (semi)prime $F$-algebras to classifying finite dimensional division $F$-algebras. This can be extremely difficult, but at least the cases where $F$ is algebraically closed, $F = \mathbb{R}$, or $F$ is a finite field have been handled in the preceding chapter. The algebraically closed case was considered in Corollaries 2.75 and 2.76, so let us look at the other two cases.

**Corollary 2.80** *Let $A$ be a finite dimensional $\mathbb{R}$-algebra.*

(a) *If $A$ is simple, then it is isomorphic to $M_n(\mathbb{R})$, $M_n(\mathbb{C})$, or $M_n(\mathbb{H})$ for some $n \in \mathbb{N}$.*
(b) *If $A$ is central simple, then it is isomorphic to $M_n(\mathbb{R})$ or $M_n(\mathbb{H})$ for some $n \in \mathbb{N}$.*

***Proof*** Theorems 2.73 and 1.7 imply (a), while (b) follows from (a) and Corollary 2.77. □

**Corollary 2.81** *A finite simple ring $R$ is isomorphic to $M_n(F)$ for some $n \in \mathbb{N}$ and some finite field $F$.*

***Proof*** As noticed in Remark 2.20, $\mathrm{char}(R)$ is a prime number $p$ and we can consider $R$ as a $\mathbb{Z}_p$-algebra. Theorem 2.73 therefore tells us that $R \cong M_n(D)$ where $D$ is a finite dimensional division algebra over $\mathbb{Z}_p$. By Theorem 1.54, $D$ is a field. □

Wedderburn's theorem on simple algebras also enables us to obtain a more complete picture of quaternion algebras studied in Sect. 1.4.

**Corollary 2.82** *Let $F$ be a field with $\mathrm{char}(F) \neq 2$. Every quaternion algebra $(\lambda, \mu)_F$ is either a division algebra or is isomorphic to the algebra $M_2(F)$.*

***Proof*** Lemma 1.35 tells us that $(\lambda, \mu)_F$ is a 4-dimensional central simple algebra, so we can apply Corollary 2.79. □

**Corollary 2.83** *Let $F$ be a finite field with $\mathrm{char}(F) \neq 2$. Every quaternion algebra $(\lambda, \mu)_F$ is isomorphic to the algebra $M_2(F)$.*

***Proof*** The algebra $(\lambda, \mu)_F$ is noncommutative and, as a finite dimensional algebra over a finite field, finite. By Theorem 1.54, it cannot be a division algebra. □

## 2.9 Scalar Extension (A Naive Approach)

In most sections of this chapter, the field $F$ has played a formal role in our study of $F$-algebras. In the preceding section, however, we saw that $F$ is of essential importance if we wish to describe algebras precisely. We now aim to present a

simple construction which enables one to replace the given field $F$ by a "better" field for which the structure theory yields definitive results.

The standard way for introducing this construction requires the concept of the tensor product. In accordance with the down-to-earth style of this chapter, we will use an alternative way which has many disadvantages and one advantage: it is very simple and intuitively clear. It is our hope that a naive approach used here will help the reader to easily understand (and appreciate) the "correct" definition given later, in Sect. 5.7.

Let us begin with an illuminating example.

**Example 2.84** Consider the algebra $\mathbb{H}_{\mathbb{C}}$ of "complex quaternions", i.e., $\mathbb{H}_{\mathbb{C}}$ is the 4-dimensional $\mathbb{C}$-algebra with basis $\{1, i, j, k\}$ satisfying the same multiplication rules $i^2 = j^2 = k^2 = ijk = -1$ as the real quaternions $\mathbb{H}$. That is to say, $\mathbb{H}_{\mathbb{C}}$ is the quaternion algebra $(-1,-1)_{\mathbb{C}}$. Since the field $\mathbb{C}$ is algebraically closed, and hence there are no 4-dimensional division algebras over $\mathbb{C}$ (by Proposition 1.8), Corollary 2.82 shows that

$$\mathbb{H}_{\mathbb{C}} \cong M_2(\mathbb{C}).$$

Let us also give a more direct proof. Write i for the complex imaginary unit (to distinguish it from $i \in \mathbb{H}_{\mathbb{C}}$). It is straightforward to check that the elements

$$e_{11} = \frac{1 - \mathrm{i}i}{2},\ e_{12} = \frac{j - \mathrm{i}k}{2},\ e_{21} = \frac{-j - \mathrm{i}k}{2},\ e_{22} = \frac{1 + \mathrm{i}i}{2},$$

form a set of $2 \times 2$ matrix units of $\mathbb{H}_{\mathbb{C}}$, and that $e_{11}\mathbb{H}_{\mathbb{C}}e_{11} = \mathbb{C}e_{11} \cong \mathbb{C}$. This induces an explicit isomorphism from $\mathbb{H}_{\mathbb{C}}$ to $M_2(\mathbb{C})$. Considering its restriction to $\mathbb{H}$, we see that $\mathbb{H}$ is isomorphic to the real algebra of all complex matrices of the form $\left[\begin{smallmatrix} z & w \\ -\overline{w} & \overline{z} \end{smallmatrix}\right]$.

One might object that by changing the real scalars to the complex ones we did not make the quaternions nicer. On the contrary, we have spoiled the property of being a division algebra. However, our aim was to indicate a general principle that is hidden behind this example. Let us consider another, admittedly less appealing, in fact quite trivial example.

**Example 2.85** The standard matrix units $E_{11}$, $E_{12}$, $E_{21}$, $E_{22}$ form the standard basis of $M_2(\mathbb{R})$. Again substituting the scalars we can consider the $\mathbb{C}$-algebra $M_2(\mathbb{R})_{\mathbb{C}}$ consisting of elements $\sum_{i,j=1}^{2} \lambda_{ij} E_{ij}$, $\lambda_{ij} \in \mathbb{C}$, whose multiplication is determined by $E_{ij}E_{kl} = \delta_{jk}E_{il}$. It is obvious that

$$M_2(\mathbb{R})_{\mathbb{C}} \cong M_2(\mathbb{C}).$$

Let us summarize what has been inferred in these two examples. Corollary 2.80 shows that, up to isomorphism, there are exactly two 4-dimensional simple real algebras: $\mathbb{H}$ and $M_2(\mathbb{R})$ (they both happen to be central). If we take the standard basis of any of them and introduce the algebra that has the same basis and the same

multiplication table, but the scalars are changed from $\mathbb{R}$ to $\mathbb{C}$, then this new algebra is isomorphic to $M_2(\mathbb{C})$.

Now we proceed to the abstract situation. Take an algebra $A$ over $F$. Assume, for simplicity, that it is finite dimensional. Choose a basis $\{e_1, \ldots, e_d\}$ of $A$ with multiplication table

$$e_i e_j = \sum_{k=1}^{d} \alpha_{ijk} e_k. \tag{2.2}$$

The constants $\alpha_{ijk} \in F$ determine the multiplication in $A$. Let $K$ be a field extension of $F$. We now introduce the $K$-algebra $A_K$ which has the same basis $\{e_1, \ldots, e_d\}$ and whose multiplication is determined by the same formula (2.2) (this makes sense since $\alpha_{ijk} \in F \subseteq K$). Thus, we only change the scalars but keep the operations. The associativity of the multiplication in $A_K$ is a consequence of the associativity of the products of the $e_i$'s. We call $A_K$ the **scalar extension** of $A$ to $K$.

Since $F$ is a subfield of $K$, we can consider $A_K$ as an $F$-algebra and $A$ as its $F$-subalgebra. It is evident from the construction of $A_K$ that

$$[A_K : K] = [A : F]. \tag{2.3}$$

Our definition of scalar extension depends on a basis choice. It would be formally more correct to denote it by something like $A_{K,\{e_1,\ldots,e_d\}}$ instead of just $A_K$. However, let us put aside a natural question of what happens if we take another basis, and stick with the simplified notation $A_K$. The "correct" definition that will be given in Sect. 5.7 is independent of basis, and when we reach that point, everything will be clear. Let us focus on generalizing the above observations on 4-dimensional simple real algebras to general finite dimensional central simple algebras.

**Theorem 2.86** *Let $A$ be a finite dimensional central simple $F$-algebra. If $\overline{F}$ is the algebraic closure of $F$, then $A_{\overline{F}} \cong M_n(\overline{F})$ where $n = \sqrt{[A : F]}$.*

***Proof*** Let us show that $A_{\overline{F}}$ is simple. Take a nonzero ideal $I$ of $A_{\overline{F}}$, and pick a nonzero $u = \lambda_1 e_1 + \cdots + \lambda_d e_d \in I$. We may assume that $\lambda_1 \neq 0$. Let $\varphi \in \mathrm{End}_F(A)$ be such that $\varphi(e_1) = 1$ and $\varphi(e_i) = 0$, $i \geq 2$. By Theorem 1.43, there exist $a_i, b_i \in A$ such that $\varphi = \sum_{i=1}^{r} L_{a_i} R_{b_i}$. Therefore,

$$I \ni \sum_{i=1}^{r} a_i u b_i = \sum_{j=1}^{d} \lambda_j \Big( \sum_{i=1}^{r} a_i e_j b_i \Big) = \sum_{j=1}^{d} \lambda_j \varphi(e_j) = \lambda_1.$$

The ideal $I$ thus contains an invertible element, and so it is equal to $A_{\overline{F}}$. This proves that $A_{\overline{F}}$ is simple. Corollary 2.75 shows that $A_{\overline{F}} \cong M_n(\overline{F})$ for some $n \in \mathbb{N}$, and (2.3) shows that $[A : F] = n^2$. □

Theorem 2.86 can be used to reduce various problems on finite dimensional central simple algebras to matrix algebras over fields (even better, over algebraically

closed fields). A quick illustration: Theorem 2.86 immediately yields a new proof of Corollary 2.78.

## 2.10 The Regular Representation

The purpose of this section is to introduce a useful and natural notion that appears throughout ring theory. Its main application will be given in the next section.

Let us begin with a question. Given a unital algebra $A$, is it possible that

$$[a, b] = 1 \tag{2.4}$$

for some $a, b \in A$? It is not our intention to discuss the reasons for the relevance of this equation. This problem was primarily chosen in order to illustrate the usefulness of the notion that we are going to introduce.

The reader's answer to our question might be "why not?", and it is difficult to argue against it. Let us look at the following example.

**Example 2.87** Let $F$ be a field, let $\mathrm{End}_F(F[\omega])$ be the algebra of all linear operators from the vector space $F[\omega]$ to itself, and let $D, L \in \mathrm{End}_F(F[\omega])$ be given by

$$D\big(f(\omega)\big) = f'(\omega), \quad L\big(f(\omega)\big) = \omega f(\omega).$$

Here, $f'(\omega)$ is the derivative of $f(\omega)$ (we remark that $D$ was already treated in Example 1.12). It is straightforward to check that $[D, L] = I$, the identity operator.

We will return to Example 2.87 in Sect. 3.11. What is important for us now is that it shows that the equation (2.4) has a solution in $A = \mathrm{End}_F(F[\omega])$. This algebra is infinite dimensional. Let us therefore modify our question: Can (2.4) occur in a finite dimensional algebra?

The answer is closer as it may seem. A shortcut to it is based on left multiplication maps $L_a : x \mapsto ax$, which we introduced in Sect. 1.5. Here, $a$ is an element from an arbitrary algebra $A$. Recalling the formulas $L_{\lambda a+\mu b} = \lambda L_a + \mu L_b$ and $L_{ab} = L_a L_b$, we see that $a \mapsto L_a$ is an algebra homomorphism from $A$ to $\mathrm{End}_F(A)$, the algebra of all linear operators from $A$ to $A$.

**Definition 2.88** The homomorphism $a \mapsto L_a$ from $A$ to $\mathrm{End}_F(A)$ is called the **regular representation** of $A$.

The regular representation is injective, unless $aA = 0$ for some nonzero $a \in A$. The latter condition is quite special, in particular it cannot occur if $A$ is unital. Thus, under a mild assumption, $a \mapsto L_a$ is an embedding of $A$ into $\mathrm{End}_F(A)$, which allows us to identify $a \in A$ with $L_a \in \mathrm{End}_F(A)$. When trying to deal with an element $a$ from an abstract algebra one might feel barehanded. The advantage of

considering $L_a$ is that it is a linear operator and therefore we can rely on linear algebra methods.

The answer to our question is now within reach, but let us first record the following proposition giving a firm basis for using the approach just indicated. The reader will notice the analogy with Cayley's theorem from group theory.

**Proposition 2.89** *Every $F$-algebra $A$ can be embedded into the algebra* $\mathrm{End}_F(V)$ *for some vector space $V$. If $A$ is finite dimensional, then $V$ can be chosen to be finite dimensional, and hence $A$ can be embedded into $M_n(F)$ for some $n \in \mathbb{N}$.*

***Proof*** If $A$ is unital, then we can take $V = A$ and apply the regular representation. If it is not unital, then we can embed $A$ to its unitization $A^\sharp$, and accordingly take $V = A^\sharp$. In any case, $V$ is finite dimensional if $A$ is; if $V$ is $n$-dimensional, then $\mathrm{End}_F(V) \cong M_n(F)$. □

This proposition enables us to consider elements from a finite dimensional algebra as matrices. The proof was easy, and this observation is often useless. But sometimes it can be very helpful, as we are about to demonstrate.

Let us return to our question. Thus, assume that a finite dimensional algebra contains elements $a, b$ satisfying (2.4). Then, by Proposition 2.89, there exist matrices $A, B \in M_n(F)$ such that $[A, B] = I$ (this is because the regular representation maps 1 into the identity). Now, the trace of the matrix $[A, B] = AB - BA$ is 0, while the trace of the matrix $I$ is $n$. If $\mathrm{char}(F) = 0$, then this is impossible, and hence (2.4) cannot occur.

Observe that essentially the same argument yields the following more general result.

**Proposition 2.90** *Let $A$ be a nonzero finite dimensional unital algebra over a field $F$ with* $\mathrm{char}(F) = 0$. *Then* 1 *cannot be written as a sum of commutators in $A$.*

**Example 2.91** Let us show that Proposition 2.90 does not hold if $F$ has prime characteristic $p$. Let $D, L \in \mathrm{End}_F(F[\omega])$ be as in Example 2.87, let $I = (\omega^p)$ be the ideal of $F[\omega]$ generated by $\omega^p$, let $V = F[\omega]/I$, and let $\phi : F[\omega] \to V$ be the canonical homomorphism. We will actually consider $V$ only as a vector space; note that it is $p$-dimensional. Since $D(\omega^p) = 0$, $D(I) \subseteq I$. We also have $L(I) \subseteq I$. These inclusions show that the maps $d, \ell : V \to V$ given by $d\phi = \phi D$ and $\ell\phi = \phi L$ are well-defined. Clearly $d, \ell \in \mathrm{End}_F(V) \cong M_p(F)$, and $[d, \ell] = 1$, the unity of $\mathrm{End}_F(V)$. The reader may wish to write down the matrix representations of $d$ and $\ell$ to obtain a more explicit interpretation of this example.

One might wonder whether there is really no other way to handle (2.4) than reducing the problem to matrices. This does seem to be the most natural and simple way, but there are others. Let us allow ourselves a digression, and prove a lovely result by N. Jacobson from 1935, dealing with a more general condition $[[a, b], a] = 0$. This can easily occur in finite dimensional algebras even when $a$ and $b$ do not commute. For example, the standard matrix units $E_{11}, E_{12}$ satisfy $[E_{12}, E_{11}] = -E_{12}$, and hence $[[E_{12}, E_{11}], E_{12}] = 0$. The most convenient way to consider this condition is through the following concept.

**Definition 2.92** Let $A$ be an algebra (resp. ring). A linear (resp. additive) map $d : A \to A$ is called a **derivation** if $d(xy) = d(x)y + xd(y)$ for all $x, y \in A$.

It is easy to guess where this name comes from.

**Example 2.93** The operator of differentiation $D$ on $F[\omega]$ from Example 2.87 is a derivation.

The basic example in noncommutative rings looks quite different.

**Example 2.94** For every $a$ in an algebra (or ring) $A$, the map $d : A \to A$ given by $d(x) = [a, x]$ is a derivation. Such a derivation is said to be an **inner derivation**.

**Proposition 2.95 (Jacobson)** *Let $A$ be a finite dimensional algebra over a field $F$ with $\mathrm{char}(F) = 0$. If $a, b \in A$ satisfy $[[a, b], a] = 0$, then $[a, b]$ is a nilpotent element.*

***Proof*** Let $d$ be the inner derivation $d(x) = [a, x]$. Our condition can be written as $d^2(b) = 0$. A simple induction argument shows that

$$d(d(b)^k) = 0 \quad \text{for every } k \in \mathbb{N}. \tag{2.5}$$

We claim that

$$d^n(b^n) = n!d(b)^n \quad \text{for every } n \in \mathbb{N}. \tag{2.6}$$

This is trivial for $n = 1$, so let $n > 1$ and assume that (2.6) holds for $n - 1$. From the definition of a derivation one easily infers that $d^n$ satisfies the Leibniz rule that we know from calculus. Thus,

$$\begin{aligned} d^n(b^n) = d^n(bb^{n-1}) &= \sum_{i=0}^{n} \binom{n}{i} d^i(b) d^{n-i}(b^{n-1}) \\ &= bd(d^{n-1}(b^{n-1})) + nd(b)d^{n-1}(b^{n-1}). \end{aligned}$$

Now use the induction hypothesis together with (2.5) and (2.6) follows.

From (2.5) and (2.6) we see that $d^m(b^k) = 0$ whenever $k < m$. By Lemma 1.9, there exists an $m \in \mathbb{N}$ such that $b^m$ is a linear combination of $b^k$ with $k < m$. Hence, $d^m(b^m) = 0$, and so $m!d(b)^m = 0$ by (2.6); that is, $[a, b]^m = 0$. □

The proof just given was actually discovered by D. Kleinecke, but is similar to Jacobson's original proof.

**Remark 2.96** From the proof we see that a slightly more general result than stated is true: If $A$ is as in Proposition 2.95, $d$ is a derivation of $A$, and $d^2(b) = 0$ for some $b \in A$, then $d(b)$ is nilpotent.

**Remark 2.97** We can also reword Proposition 2.95 in terms of derivations as follows: If $\delta$ is an inner derivation of $A$ (induced by $b$), and $a \in A$ is such that

$[\delta(a), a] = 0$, then $\delta(a)$ is nilpotent. We will now give another illustration of the usefulness of the regular representation by showing that this is true for any (not necessarily inner) derivation $\delta$. We start by observing that $[\delta(a), a] = 0$ implies $[L_{\delta(a)}, L_a] = 0$. The condition that $\delta$ is a derivation can be expressed as $L_{\delta(a)} = [\delta, L_a]$. Therefore, $[[\delta, L_a], L_a] = 0$. Proposition 2.95 applied to the (also finite dimensional) algebra $\mathrm{End}_F(A)$ tells us that $L_{\delta(a)} = [\delta, L_a]$ is nilpotent. Hence, $\delta(a)$ is nilpotent too.

## 2.11 Group Algebras

Group theory and ring theory are two branches of algebra. In a first course in abstract algebra one might get an impression that, although sometimes similar, they are essentially unconnected. However, there are many links between groups and rings. The concept that we are about to introduce gives rise to striking interactions.

Assume temporarily that $G$ is an arbitrary set. If $F$ is, as always, a field, then we can form the vector space over $F$ whose basis is $G$. Its elements are formal sums $\sum_{g\in G} \lambda_g g$ where $\lambda_g \in F$ and all but finitely many $\lambda_g$ are zero. The definitions of addition and scalar multiplication are self-explanatory.

Assume now that $G$ is a group. Then this vector space becomes an algebra, denoted $F[G]$, if we define multiplication by simply extending the group multiplication on $G$ to the whole space; taking into account the algebra axioms, this can be done in a unique way. That is,

$$\Big(\sum_{g\in G} \lambda_g g\Big)\Big(\sum_{h\in G} \mu_h h\Big) = \sum_{k\in G} \nu_k k, \quad \text{where } \nu_k = \sum_{gh=k} \lambda_g \mu_h.$$

**Definition 2.98** The algebra $F[G]$ is called the **group algebra** of $G$ over $F$.

It is actually enough to assume that $G$ is merely a semigroup to construct $F[G]$. In this case we call $F[G]$ a **semigroup algebra**. Similarly, we can speak about a **monoid algebra** if $G$ is a monoid. Furthermore, we can replace the role of the field $F$ by any ring $R$, and define addition and multiplication in formally the same way. Everything still makes sense, but of course the resulting object is a ring rather than an algebra. It is called a **(semi)group ring** and is denoted by $R[G]$. For example, considering $\mathbb{N}_0 = \mathbb{N} \cup \{0\}$ as a monoid under addition, we readily see that $R[\mathbb{N}_0]$ is isomorphic to the polynomial ring $R[\omega]$.

If an element $g$ of a group $G$ satisfies $g^n = 1$, then $(1-g)(1+g+\cdots+g^{n-1}) = 0$. This shows that if $G$ has elements different from 1 of finite order, then $F[G]$ is not a domain. A group $G$ is said to be **torsion-free** if 1 is the only element in $G$ of finite order. The following question has been open for a long time.

**Problem 2.99** Is $F[G]$ a domain if $G$ is torsion-free?

This problem is attributed to I. Kaplansky. It is often mentioned together with another Kaplansky's problem which asks whether the only invertible elements of $F[G]$, where $G$ is torsion-free, are nonzero scalar multiples of group elements. It is known that a positive answer to this second problem would imply a positive answer to Problem 2.99. However, very recently, in 2021, G. Gardam constructed an example showing that the second problem has a negative answer. To the best of the author's knowledge, Problem 2.99 is still open at present.

If $G$ is a nontrivial group, then the set of all $\sum_{g\in G} \lambda_g g$ such that $\sum_{g\in G} \lambda_g = 0$ is a proper nonzero ideal of $F[G]$. It is called the **augmentation ideal** of $F[G]$. The algebra $F[G]$ therefore is not simple. The main issue of this section is the question whether $F[G]$ is semiprime for a finite group $G$. We begin with the simplest example.

**Example 2.100** Let $G = \mathbb{Z}_2$. It is more convenient to use the multiplicative notation, so we write $G = \{1, g\}$ where $g^2 = 1$. The group algebra $F[G]$ thus consists of elements of the form $\lambda 1+\mu g$, $\lambda, \mu \in F$, which are multiplied as follows:

$$(\lambda 1 + \mu g)(\lambda' 1 + \mu' g) = (\lambda\lambda' + \mu\mu')1 + (\lambda\mu' + \mu\lambda')g.$$

It is easy to verify that $\lambda 1 + \mu g \mapsto (\lambda + \mu, \lambda - \mu)$ is a homomorphism from $F[G]$ to $F \times F$. If $\mathrm{char}(F) \neq 2$, then it is bijective; thus, $F[G] \cong F \times F$ in this case. In particular, $F[G]$ is semiprime. This is no longer true if $\mathrm{char}(F) = 2$ since $F(1 + g)$ is then a nilpotent ideal of $F[G]$. Note that, in this case, $F[G]$ is isomorphic to the subalgebra of $M_2(F)$ consisting of matrices of the form $\left[\begin{smallmatrix} \alpha & \beta \\ 0 & \alpha \end{smallmatrix}\right]$. An isomorphism is given by $\lambda 1 + \mu g \mapsto \left[\begin{smallmatrix} \lambda+\mu & \mu \\ 0 & \lambda+\mu \end{smallmatrix}\right]$.

Thus, the group algebra $F[\mathbb{Z}_2]$ is semiprime if and only if $\mathrm{char}(F) \neq 2$. We will generalize this observation to group algebras over arbitrary finite groups.

First we recall some facts from linear algebra. Let $V$ be a finite dimensional vector space, and let $T : V \to V$ be a linear operator. Choosing a basis in $V$, we can represent $T$ as a matrix relative to this basis. Different bases give rise to similar matrices. Therefore, no matter which basis we choose, the trace of a matrix representing $T$ is always the same. We can therefore define $\mathrm{tr}(T)$, the **trace of** $T$, as the trace of any matrix representation of $T$. If $T$ is nilpotent, then $\mathrm{tr}(T) = 0$. This can be deduced, for example, from the Jordan normal form of a matrix representation of $T$, as 0 is clearly the only eigenvalue of $T$. (Note that here we have to use the fact that every field $F$ has an algebraic closure $\overline{F}$, and then consider the matrix representation of $T$ as a matrix in $M_n(\overline{F})$. More elementary proofs are also possible.)

The following theorem was proved in 1898 by H. Maschke.

**Theorem 2.101 (Maschke's Theorem)** *Let $G$ be a finite group. Then the group algebra $F[G]$ is semiprime if and only if either* $\mathrm{char}(F) = 0$ *or* $\mathrm{char}(F)$ *is a prime number $p$ that does not divide $|G|$.*

***Proof*** Since $F[G]$ is a finite dimensional vector space over $F$, we can define a linear functional $\rho : F[G] \to F$ by

$$\rho(a) = \mathrm{tr}(L_a)$$

(here, $L_a$ is the left multiplication map, see the preceding section). Let $n = |G|$ and denote the elements in $G$ by $g_1, g_2, \ldots, g_n$ where $g_1 = 1$. Obviously, $\rho(g_1) = n$. If $i \geq 2$, then

$$L_{g_i}(g_j) = g_i g_j \in G \setminus \{g_j\}$$

for every $j$. The matrix representation of $L_{g_i}$ with respect to the basis $\{g_1, g_2, \ldots, g_n\}$ has therefore zeros on the diagonal. Consequently, $\rho(g_i) = 0$. This is a crucial observation upon which our proof is based.

Suppose $F[G]$ has a nonzero nilpotent ideal $I$. We want to show that $F$ then has finite characteristic $p$ which divides $n = |G|$. Take a nonzero $a = \sum_{i=1}^{n} \lambda_i g_i \in I$. By replacing $a$ by $g_i^{-1} a \in I$, if necessary, we see that we may assume that $\lambda_1 \neq 0$. We have

$$\rho(a) = \lambda_1 \rho(g_1) + \lambda_2 \rho(g_2) + \cdots + \lambda_n \rho(g_n) = n\lambda_1.$$

As an element of a nilpotent ideal, $a$ is a nilpotent element. Hence, $L_a$ is a nilpotent linear operator, and so $\rho(a) = 0$. That is, $n\lambda_1 = 0$. Since $\lambda_1 \neq 0$, this is possible only when $p = \mathrm{char}(F)$ divides $n$.

Conversely, assume that $p = \mathrm{char}(F)$ divides $|G|$. Set $r = \sum_{i=1}^{n} g_i$. Since $r g_j = g_j r = r$ for every $j$, we see that the 1-dimensional space $Fr$ is an ideal of $F[G]$. As $r^2 = |G| r = 0$, $Fr$ is a nilpotent ideal. □

Theorem 2.101 becomes useful when combined with Wedderburn's structure theory. If we are interested primarily in the group $G$, we have freedom in selecting the field $F$. By requiring that $\mathrm{char}(F) = 0$, we will be on the safe side. In view of Corollary 2.75, choosing $F$ to be algebraically closed seems a good idea. An obvious choice now is $F = \mathbb{C}$.

**Corollary 2.102** *Let $G$ be a finite group. Then there exist $n_1, \ldots, n_r \in \mathbb{N}$ such that $\mathbb{C}[G] \cong M_{n_1}(\mathbb{C}) \times \cdots \times M_{n_r}(\mathbb{C})$.*

**Example 2.103** Consider the group algebra $\mathbb{C}[S_3]$, where $S_3$ is the symmetric group on three elements. As $[\mathbb{C}[S_3] : \mathbb{C}] = |S_3| = 6$, representing $\mathbb{C}[S_3]$ as $M_{n_1}(\mathbb{C}) \times \cdots \times M_{n_r}(\mathbb{C})$, we must have $\sum_{i=1}^{r} n_i^2 = 6$. Moreover, at least one $n_i$ must be different from 1 since $S_3$ is noncommutative. The only possibility is that $\mathbb{C}[S_3] \cong M_2(\mathbb{C}) \times \mathbb{C} \times \mathbb{C}$.

Thus, starting with an abstract finite group, we are able to represent it through its group algebra as a "concrete" object, the direct product of full matrix algebras. This is of fundamental importance in the **representation theory** of finite groups.

The basic concept of this theory is a **group representation**, which is defined as a group homomorphism from $G$ to the general linear group $\mathrm{GL}_n(F)$, i.e., the group of all invertible $n \times n$ matrices over the field $F$.

## Exercises

**2.1** Determine the radical of the subalgebra of $M_n(F)$ consisting of all matrices whose last $k$ rows are zero.

**2.2** Find an ideal of the algebra $\Pi_{n\geq 2} T_n(F)$ that is nil but not nilpotent.

**2.3** Let $R$ be an arbitrary ring. Show that:

(a) The left ideal generated by $a \in R$ is nil if and only if the right ideal generated by $a$ is nil.
(b) The sum of all nil left ideals of $R$ is equal to the sum of all nil right ideals of $R$, and is therefore an ideal of $R$.
(c) The sum of a nil left ideal of $R$ and a nil ideal of $R$ is nil.
(d) The sum of all nil ideals of $R$, $\mathrm{Nil}(R)$, is a nil ideal of $R$, and the ring $R/\mathrm{Nil}(R)$ has no nonzero nil ideals.
(e) If $R/\mathrm{Nil}(R)$ is commutative (in particular, if $R$ itself is commutative), then $\mathrm{Nil}(R)$ is equal to the set of all nilpotent elements in $R$.

**Remark** We call $\mathrm{Nil}(R)$ the **upper nilradical** of $R$. There is also the notion of the **lower nilradical** (also called the **prime radical**) of $R$ which is defined as the intersection of all **prime ideals** of $R$, i.e., ideals with the property that the corresponding factor rings are prime. It can be shown that the lower nilradical is a nil ideal, and hence contained in $\mathrm{Nil}(R)$, and that a ring is semiprime if and only if its lower nilradical is zero.

**2.4** Prove that the following statements are equivalent:

(i) Köthe's problem has a positive answer.
(ii) The upper nilradical of an arbitrary ring contains all nil one-sided ideals of this ring.
(iii) The sum of two nil left ideals of an arbitrary ring is nil.

**2.5** Show that a semiprime ring $R$ is prime if and only if the intersection of any two nonzero ideals of $R$ is nonzero. Give an example showing that this is not always true if $R$ is not semiprime.

**2.6** Show that a ring $R$ is prime (resp. semiprime) if and only if $M_n(R)$ is prime (resp. semiprime).

**2.7** Let $a, b$ be elements in a ring $R$. Note that $aRb = 0$ implies $bRa = 0$ in case $R$ is semiprime, and give an example showing that this is not true in every ring $R$.

**2.8** Let $R$ be a prime ring. Show that if $a \in R$ commutes with every element from a nonzero left ideal $L$ of $R$, then $a \in Z(R)$. Is this still true if $R$ is semiprime?

**2.9** Let $R$ be a semiprime ring, and let $a \in R$ be such that $axa = a^2x$ for every $x \in R$. Show that $a \in Z(R)$.

**2.10** Let $A$ be a unital $F$-algebra. Show that its unitization $A^\sharp$ is isomorphic to the direct product $F \times A$.

**2.11** Let $A$ be a prime $F$-algebra. Show that its unitization $A^\sharp$ is prime if and only if $A$ is not unital. Show also that this statement does not hold for rings. More precisely, find an example of a (possibly commutative) prime ring $R$ without unity such that $R^\sharp$ is not prime.

**2.12** Show that a unital $F$-algebra $B$ is isomorphic to the unitization $A^\sharp$ of some algebra $A$ if and only if there exists a nonzero algebra homomorphism from $B$ to $F$. Note that every group algebra $F[G]$ is an example of such an algebra.

**2.13** Show that if $I$ and $J$ are ideals of a unital ring $R$ such that $R = I \oplus J$, then there exists a central idempotent $e \in R$ such that $I = eR$ and $J = (1 - e)R$.

**Remark** This is a kind of converse to Lemma 2.43.

**2.14** Show that the subring generated by all idempotents in a ring $R$ contains the ideal generated by all commutators $[e, x]$, where $e$ is an idempotent and $x$ is an arbitrary element in $R$. Hence conclude that a simple ring containing at least one nontrivial idempotent is generated (as a ring) by its idempotents.
*Hint:* Express $[e, x]$ as a difference of two idempotents (have a look at Remark 2.41 (c)), and use the identity $yu = [e, y[e, u]] - [e, y][e, u]$ where $u = [e, x]$.

**2.15** Show that a unital ring $R$ contains an element which is right invertible but not left invertible if and only if $M_2(R)$ contains an invertible upper triangular matrix whose inverse is not upper triangular.
*Hint*: If $ab = 1$, then $e = 1 - ba$ is an idempotent and $ae = eb = 0$.

**2.16** Show that the following conditions are equivalent for a nonzero idempotent $e$ in a ring $R$:

(i) $e$ is primitive.
(ii) $e$ is not a sum of two nonzero orthogonal idempotents in $R$.
(iii) The left ideal $Re$ is not a direct sum of two nonzero left ideals.

**2.17** Let $I$ be a nil ideal of a ring $R$. Show that an idempotent $e \in R$ is primitive if and only if $e + I \in R/I$ is primitive.
*Hint*: Apply Lemma 2.48 to the ring $eRe$.

**2.18** Show that an idempotent in a finite dimensional semiprime algebra is primitive if and only if it is minimal.

**2.19** Let $S$ be a unital ring. Show that every ideal of $M_n(S)$ is of the form $M_n(I)$ for some ideal $I$ of $S$, and that $M_n(S)/M_n(I) \cong M_n(S/I)$. Show also that the ideal generated by any matrix unit $e_{ij}$ in $M_n(S)$ is the whole ring.

**2.20** Let $A$ be a nonzero finite dimensional algebra. Show that:

(a) $A$ is nil if and only if $A$ is nilpotent.
(b) $A$ is a domain if and only if $A$ is a division algebra.
(c) $A$ is reduced if and only if $A$ is isomorphic to a direct product of division algebras.
(d) There exists an $n \in \mathbb{N}$ such that $x^n = x$ for every $x \in A$ if and only if $A$ is isomorphic to a direct product of finite fields.

**Remark** The statement (d) extends the statement of Exercise 1.25. In Sect. 4.10, we will consider the condition $x^n = x$ in arbitrary rings and, moreover, under the assumption that $n$ depends on $x$.

**2.21** Let $A$ be a finite dimensional central simple $F$-algebra such that $[x, y][z, w] + [z, w][x, y] \in F$ for all $x, y, z, w \in A$. Prove that either $A = F$ or $[A : F] = 4$.
*Hint:* Scalar extension.

**2.22** Let $A$ be a nonzero finite dimensional unital algebra over a field $F$ with $\mathrm{char}(F) = 0$, and let $\varphi$ be an automorphism of $A$. Show that $\varphi(a) \neq a + 1$ for every $a \in A$.
*Hint:* Express $L_{\varphi(a)}$ through $\varphi$, $\varphi^{-1}$, and $L_a$.

**2.23** Let $e_1, e_2$ be idempotents in an $F$-algebra $A$. Observe that if $\mathrm{char}(F) \neq 2$, then $e_1 + e_2$ is an idempotent if and only if $e_1$ and $e_2$ are orthogonal. Show that this can be extended to any number of idempotents in case $A$ is finite dimensional and $\mathrm{char}(F) = 0$; that is, the sum of idempotents $e_1, \ldots, e_n \in A$ is an idempotent if and only if $e_1, \ldots, e_n$ are pairwise orthogonal.
*Hint:* The trace of an idempotent matrix is equal to its rank.

**2.24** Let $A$ be a unital algebra and let $G$ be a subgroup of the group $A^*$ of invertible elements in $A$. Explain why, in general, the group algebra of $G$ is not isomorphic to the subalgebra of $A$ generated by $G$ (which is just the linear span of $G$, so the elements in both algebras look the same), and find an example (with $G \neq \{1\}$) where these two algebras are isomorphic.

**2.25** Let $G$ be a group, let $F$ be a field, let $B$ be a unital $F$-algebra, and let $B^*$ be the group of invertible elements of $B$.

(a) Let $\varphi : F[G] \to B$ be an algebra homomorphism sending 1 to 1. Show that the restriction of $\varphi$ to $G$ is a group homomorphism from $G$ to $B^*$.
(b) Conversely, show that a group homomorphism $\varphi_0 : G \to B^*$ can be uniquely extended to an algebra homomorphism $\varphi : F[G] \to B$.

Derive the conclusions from Example 2.100 by first embedding $\mathbb{Z}_2$ to $\mathrm{GL}_2(F)$ and then extend it to an embedding of $F[\mathbb{Z}_2]$ to $M_2(F)$.

**2.26** Show that every group algebra $F[G]$ has an involution.

**2.27** Let $G = \{g_1, \ldots, g_n\}$ be group and let $F$ be a field such that $\mathrm{char}(F) = 0$ or $\mathrm{char}(F)$ does not divide $|G|$. Show that $e = \frac{1}{|G|}(g_1 + \cdots + g_n)$ is a central idempotent in $F[G]$, $[eF[G] : F] = 1$, and $(1 - e)F[G]$ is the augmentation ideal.

**2.28** Show that $F[\mathbb{Z}_n] \cong F[\omega]/(\omega^n - 1)$ for every field $F$. Use this to find the Wedderburn decomposition (in the sense of Theorem 2.71) of $\mathbb{R}[\mathbb{Z}_3]$.

**Remark** This can also be used for describing $\mathbb{C}[\mathbb{Z}_3]$; however, Corollary 2.102 readily implies that $\mathbb{C}[G] \cong \mathbb{C} \times \cdots \times \mathbb{C}$ for every finite abelian group $G$.

**2.29** The subset $Q = \{1, -1, i, -i, j, -j, k, -k\}$ of $\mathbb{H}$ is clearly a group under multiplication. It is called the **quaternion group**. Show that for any field $F$ with $\mathrm{char}(F) \neq 2$, $F[Q] \cong F \times F \times F \times F \times A$, where $A$ is a quaternion $F$-algebra. What is $A$ if $F = \mathbb{C}$? What is $A$ if $F = \mathbb{R}$?
*Hint*: Is $\mathbb{R}[Q]$ reduced?

# Chapter 3
# Modules and Semisimple Rings




Modules at last! In most books on a similar subject, modules appear on the very first pages. And there are good reasons for this. Ring theory and module theory are symbiotic. Many questions about rings can be handled most effectively by using modules. In fact, modules often seem inevitable. In spite of that, we have decided to postpone introducing modules up until now. Our viewpoint has been that an intrinsic approach from the first chapters, where results on rings and algebras were rather quickly derived from basic definitions, should be easier and more appealing for a newcomer to the subject. We have now reached the point where continuing without modules would be inefficient.

A large part of the chapter is devoted to the introduction of various concepts, without immediately demonstrating their significance. At the end, however, we will present a generalization and supplement to Wedderburn's structure theory, centered around the notion of a semisimple ring.

## 3.1 The Concept of a Module

Being familiar with vector spaces, the definition of a module can be easily memorized. One just replaces the role of a field by an arbitrary ring.

**Definition 3.1** Let $R$ be a ring. A **left $R$-module** (or left module over $R$) is an additive group $M$ together with a map from $R \times M$ to $M$, $(r, m) \mapsto rm$, such that for all $r, s \in R$, $m, n \in M$,

(a) $(r + s)m = rm + sm$,
(b) $r(m + n) = rm + rn$,
(c) $r(sm) = (rs)m$.

If, additionally, $R$ is a unital ring and

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(d) $1m = m$

holds for every $m \in M$, then $M$ is said to be a **unital left $R$-module**.

A **(unital) right $R$-module** is defined similarly, only the order of elements from $R$ and $M$ is interchanged. That is, one requires the following obvious analogues of (a)–(d) for a map $(m, r) \mapsto mr$ where $m \in M$ and $r \in R$: (a') $m(r+s) = mr+ms$, (b') $(m+n)r = mr+nr$, (c') $(mr)s = m(rs)$, and (d') $m1 = m$.

**Remark 3.2** If $R$ is commutative, then every left $R$-module $M$ becomes a right $R$-module by defining $mr = rm$ for all $m \in M$, $r \in R$.

Distinguishing between left and right $R$-modules is therefore more substantial when $R$ is noncommutative. This does not mean, of course, that the theory of left $R$-modules is any different from the theory of right $R$-modules. Results on left modules have obvious analogues for right modules, and vice versa. Among two equivalent choices, let us decide to work with the left version. Thus, unless specified otherwise, we will henceforth adopt the following convention:

$$R\text{-module} = \text{left } R\text{-module}.$$

When it will be clear which ring $R$ we have in mind, or when $R$ will play just a formal role in our discussion, we will simply use the term "module".

Modules appear throughout mathematics; finding examples is easy, making a selection is harder.

**Example 3.3** A vector space over a field $F$ is a unital $F$-module.

**Example 3.4** Every additive group $M$ is a unital $\mathbb{Z}$-module, with $nm$, where $n \in \mathbb{Z}$ and $m \in M$, having the usual meaning.

**Example 3.5** Let $M$ be an additive group. The set $R = \mathrm{End}(M)$ of all endomorphisms of $M$ (i.e., all additive maps from $M$ into $M$) becomes a ring if we define the sum $f + g$ and the product $fg$ of $f, g \in R$ as follows:

$$(f+g)(m) = f(m) + g(m), \quad (fg)(m) = f(g(m)). \tag{3.1}$$

We can view $M$ as a unital $R$-module via $fm = f(m)$, $f \in R, m \in M$.

**Example 3.6** A left ideal $L$ of a ring $R$ is a left $R$-module with $r\ell$, where $r \in R$ and $\ell \in L$, being the ordinary product in $R$. In particular, $R$ itself is an $R$-module.

Similarly, a right ideal of $R$ can be considered as a right $R$-module. An ideal of $R$ (in particular, $R$ itself) is thus both a left and right $R$-module; moreover, it satisfies the conditions of the following definition.

**Definition 3.7** Let $R$ and $S$ be rings. If $M$ is both a left $R$-module and a right $S$-module, and also

$$(rm)s = r(ms) \quad \text{for all } r \in R, m \in M, s \in S,$$

then $M$ is said to be an $(R, S)$-**bimodule**. An $(R, R)$-bimodule is also called an $R$-**bimodule**.

**Example 3.8** An ideal of a ring $R$ is an $R$-bimodule.

**Example 3.9** If $R$ is a subring of a ring $R'$, then $R'$ is an $R$-bimodule (via the ordinary product in $R'$).

**Example 3.10** Let $M = M_{m,n}(S)$ be the set of all $m \times n$ matrices with entries from a ring $S$. Under the usual matrix operations, $M$ is an $(M_m(S), M_n(S))$-bimodule.

Replacing rings by algebras, one has to make some obvious modifications in the above definitions (and examples). For instance, a **(left)** $A$**-module**, where $A$ is an algebra over a field $F$, is defined as a vector space $V$ over $F$ together with a map $A \times V \to V$, $(a, v) \mapsto av$, that satisfies (a), (b), (c), and

$$(\lambda a)v = a(\lambda v) = \lambda(av) \quad \text{for all } \lambda \in F, a \in A, v \in V.$$

In other words, this map must be bilinear, not only biadditive.

**Example 3.11** The $n$-dimensional space $F^n$ can be viewed as a unital $M_n(F)$-module if we write vectors in $F^n$ as columns and define the module operation as the standard product of a matrix with a column vector.

**Example 3.12** A vector space $V$ over $F$ is a unital $\mathrm{End}_F(V)$-module.

Note that Example 3.12 is a refined version of Example 3.5 which also involves the scalar multiplication, and Example 3.11 is just a disguised form of its finite dimensional version. The importance of these examples will become evident from the following discussion.

Let $A$ be an algebra over $F$ and let $V$ be a vector space over $F$. An algebra homomorphism from $A$ to $\mathrm{End}_F(V)$ is called a **representation** of $A$. If $\varphi : A \to \mathrm{End}_F(V)$ is a representation, then $V$ becomes an $A$-module by defining $av = \varphi(a)(v)$. Conversely, every $A$-module $V$ gives rise to a representation $\varphi : A \to \mathrm{End}_F(V)$ given by $\varphi(a)(v) = av$.

We may therefore regard representations and modules as equivalent concepts. For example, instead of with the regular representation we can deal with the (left) $A$-module $A$; thus, in a hidden manner modules were already treated before.

We will usually present our results in the language of modules rather than in that of representations. Still, being aware of a possible interpretation in terms of representations is important for understanding the philosophy of the subject. Considering a module over an algebra is essentially the same as representing algebra elements as linear operators of a vector space. This indicates why modules are useful for the structure theory. Namely, they can be used to identify abstract algebras with algebras of linear operators. In the previous chapter, we have shown that some abstract algebras can be identified with algebras of matrices. From linear algebra we know that the two seemingly different notions, matrices and linear operators, are essentially equivalent. It therefore should not be surprising that constructing matrix

units, the approach taken in the previous chapter, is not the only way to describe algebra elements as matrices—one can rely on modules instead. This may sound vague at this point, especially since the matrices from the previous chapter were not the "ordinary" matrices we know from linear algebra, but matrices with entries from a division algebra. However, as will become apparent in Sect. 3.4, this is not really a problem. In the present chapter, as well as in Chap. 4, we will show that the module-theoretic approach to the structure theory yields profound generalizations of Wedderburn's theory.

Similar observations as for modules over algebras can be made for modules over rings: an $R$-module $M$ can be identified with a ring homomorphism from $R$ to $\mathrm{End}(M)$, the ring of endomorphisms of the additive group of $M$. Its kernel will later turn out to be important, so we give it a name.

**Definition 3.13** The **annihilator** of an $R$-module $M$ is the set

$$\mathrm{ann}_R(M) = \{r \in R \mid rM = 0\}.$$

If $\mathrm{ann}_R(M) = 0$, then $M$ is said to be a **faithful** module (here, 0 stands for $\{0\}$ and $rM$ for $\{rm \mid m \in M\}$).

Observe that $\mathrm{ann}_R(M)$ is a (two-sided) ideal.

**Example 3.14** Consider the abelian group $\mathbb{Z}_n$ as a $\mathbb{Z}$-module. Then $\mathrm{ann}_{\mathbb{Z}}(\mathbb{Z}_n) = n\mathbb{Z}$.

**Example 3.15** The modules from Examples 3.11 and 3.12 are faithful.

## 3.2 Basic Module-Theoretic Notions

Even the readers that have never met modules before might experience a sense of déjà vu when reading this section. Basic notions of module theory are very similar to the familiar notions from the theories of vector spaces, groups, and rings. We will therefore introduce them in a rather quick and perhaps slightly dry manner. The reader is advised to translate these notions into the (presumably more familiar) language of vector spaces.

For simplicity of exposition, we will consider only modules over rings. It should be obvious how to make the necessary adjustments for modules over algebras. One just has to require the natural scalar multiplication properties at appropriate places.

Throughout this section, $R$ stands for a ring and $M$ for an $R$-module. Recall our convention that a module means a left module.

**Definition 3.16** A subset $L$ of $M$ is called a **submodule** of $M$ if $L$ is an additive subgroup of $M$ and $r\ell \in L$ for all $r \in R$, $\ell \in L$.

Of course, a submodule is itself a module.

**Example 3.17** Every module $M$ has at least two submodules: 0 and $M$. Submodules different from $M$ are said to be *proper*, and submodules different from 0 are said to be *nonzero* or *nontrivial*.

**Example 3.18** If we consider an additive group as a $\mathbb{Z}$-module, then its submodules are exactly its subgroups. Similarly, if viewing a vector space over $F$ as an $F$-module, its submodules are its subspaces.

**Example 3.19** The submodules of the $R$-module $R$ are the left ideals of $R$. This simple fact will be frequently used.

**Example 3.20** Let $J$ be a left ideal of $R$. The additive subgroup of $M$ generated by the elements of the form $um$, where $u \in J$ and $m \in M$, is a submodule. We denote it by $JM$.

The intersection of submodules is obviously a submodule. Given a subset $X$ of $M$, the intersection of all submodules containing $X$ is therefore the **submodule generated by** $X$. Explicitly, it can be described as the additive subgroup generated by the elements $rx + nx$, $r \in R$, $n \in \mathbb{Z}$, $x \in X$ (there is no need to involve $nx$ if $M$ is unital).

A (sub)module generated by some finite set is said to be **finitely generated**. A module generated by a single element is called **cyclic**. A unital cyclic module can be written as

$$Rm = \{rm \mid r \in R\}$$

where $m$ is its generator.

As in any algebraic structure, homomorphisms play an important role.

**Definition 3.21** Let $N$ be another $R$-module. A map $f : M \to N$ is an **$R$-module homomorphism** if

$$f(m + m') = f(m) + f(m') \quad \text{and} \quad f(rm) = rf(m)$$

for all $r \in R$ and $m, m' \in M$.

The **kernel** of $f$, $\ker f = \{m \in M \mid f(m) = 0\}$, is clearly a submodule of $M$, and the **image** of $f$, $\operatorname{im} f = \{f(m) \mid m \in M\}$, is a submodule of $N$.

An **$R$-module isomorphism** is, of course, a bijective $R$-module homomorphism. We say that $M$ and $N$ are **isomorphic modules**, and write $M \cong N$, if there exists an isomorphism $f : M \to N$. It is immediate to check that in this case its inverse $f^{-1} : N \to M$ is also an isomorphism. An **$R$-module endomorphism** is an $R$-module homomorphism from $M$ to itself, and an **$R$-module automorphism** is an endomorphism which is also an isomorphism. We also speak about **$R$-module embeddings** which are of course injective $R$-module homomorphisms.

If $f : M \to N$ and $g : N \to P$ are $R$-module homomorphisms, then so is their composition $g \circ f : M \to P$. We will usually write $g \circ f$ as $gf$.

We define the **endomorphism ring**

$$\mathrm{End}_R(M)$$

as the set of all $R$-module endomorphisms of $M$ endowed with operations defined as in (3.1), i.e., pointwise addition and composition as multiplication. It is immediate to check that $\mathrm{End}_R(M)$ is indeed a (unital) ring. Observe that invertible elements of $\mathrm{End}_R(M)$ are exactly the automorphisms of $M$.

After being familiar with factor groups and factor rings, the following notion should be easy to grasp. Take a submodule $L$ of $M$. As $L$ is, in particular, a subgroup of the additive group of $M$, one can form the factor group $M/L$. Recall that its elements are cosets $m + L = \{m + \ell \mid \ell \in L\}$, $m \in M$, which are added according to

$$(m + L) + (m' + L) = (m + m') + L.$$

We claim that $M/L$ becomes an $R$-module if we set

$$r(m + L) = rm + L, \quad r \in R,\ m \in M.$$

Indeed, this operation is easily seen to be well-defined, and the module axioms are clearly fulfilled.

**Definition 3.22** The $R$-module $M/L$ is called the **factor module**.

The map $m \mapsto m+L$ is an $R$-module homomorphism from $M$ onto $M/L$, called the **canonical homomorphism**. Note that its kernel is equal to $L$. Another fundamental fact connecting homomorphisms and submodules is the following: If $f : M \to N$ is an $R$-module homomorphism, then

$$M/\ker f \cong \mathrm{im}\, f.$$

Indeed, the map $m + \ker f \mapsto f(m)$ is well-defined and is an isomorphism.

Our last topic concerns sums and products of (sub)modules.

**Definition 3.23** If $\{M_i \mid i \in I\}$ is a family of submodules of $M$, then the submodule generated by $\bigcup_{i\in I} M_i$ is called the **sum of the submodules** $\{M_i \mid i \in I\}$. We write it as $\sum_{i\in I} M_i$. If every $M_j$ has trivial intersection with the sum of the other submodules, i.e., $M_j \bigcap \left(\sum_{i\in I\setminus\{j\}} M_i\right) = 0$, then we say that this sum is a **direct sum**, and write it as $\oplus_{i\in I} M_i$.

Note that $\sum_{i\in I} M_i$ consists of all (finite) sums of elements from $\bigcup_{i\in I} M_i$. Hence we see that the sum $\sum_{i\in I} M_i$ is direct if and only if for all distinct $i_1, \ldots, i_n \in I$ and all $m_{i_1} \in M_{i_1}, \ldots, m_{i_n} \in M_{i_n}$, $m_{i_1} + \cdots + m_{i_n} = 0$ implies $m_{i_1} = \cdots = m_{i_n} = 0$. If the index set $I$ is finite, say $I = \{1, \ldots, n\}$, then we can write the sum $\sum_{i\in I} M_i$ as $M_1 + \cdots + M_n$, or $M_1 \oplus \cdots \oplus M_n$ if it is a direct sum.

A formally more adequate term for the above defined direct sum is the **internal direct sum**. One can, on the other hand, start with an arbitrary family of $R$-modules $\{M_i \mid i \in I\}$, not necessarily submodules of a given module, and then form their **direct product** $\Pi_{i\in I} M_i$: this is the Cartesian product of the $M_i$'s endowed with the componentwise operations

$$(m_i) + (m_i') = (m_i + m_i') \quad \text{and} \quad r(m_i) = (rm_i)$$

for all $r \in R, m_i, m_i' \in M_i, i \in I$. Its submodule consisting of all elements $(m_i)$ for which all but finitely many $m_i$ are 0 is called the **external direct sum** of the $M_i$'s. The external direct sum obviously contains isomorphic copies of the modules $M_i$ as its submodules, and, moreover, it is equal to their internal direct sum. Therefore we often neglect the difference between internal and external direct sums, and use the same notation for both. Thus, $M_1 \oplus \cdots \oplus M_n$ may mean an internal or an external direct sum. Since the index set is finite here, this sum coincides with the direct product $\Pi_{i=1}^n M_i$ (written also as $M_1 \times \cdots \times M_n$).

Given a submodule $L$ of $M$, does there exist another submodule $K$ such that $M = L \oplus K$? It is easy to see that the answer is negative in general. Say, the proper nonzero submodules of the $\mathbb{Z}$-module $\mathbb{Z}$ are $n\mathbb{Z}$, $n \geq 2$, and any two of them intersect nontrivially. Submodules for which the answer to our question is positive therefore deserve attention.

**Definition 3.24** A submodule $L$ of $M$ is called a **direct summand** of $M$ if there exists a submodule $K$ of $M$ such that $M = L \oplus K$.

**Example 3.25** Let $L$ be a left ideal of a unital ring $R$. Consider $L$ as a submodule of the $R$-module $R$. We claim that $L$ is a direct summand if and only if there exists an idempotent $e \in R$ such that $L = Re$. The "if" part is obvious: if $L = Re$ then one easily checks that $K = R(1 - e)$ satisfies $R = L \oplus K$. To prove the converse, suppose there exists a left ideal $K$ such that $R = L \oplus K$. Then there exist an $e \in L$ and an $f \in K$ such that $1 = e + f$. For every $\ell \in L$ we thus have $\ell - \ell e = \ell f \in L \cap K = 0$, so $\ell = \ell e$. In particular, $e = e^2$ and $L \subseteq Re$. Since $e \in L$, it follows that $L = Re$.

## 3.3 Vector Spaces Over Division Rings

The following familiar notion from linear algebra can be also considered in modules.

**Definition 3.26** A subset $B$ of an $R$-module $M$ is said to be **linearly independent** if for all distinct $b_1, \ldots, b_n \in B$ and all $r_1, \ldots, r_n \in R$, $r_1 b_1 + \cdots + r_n b_n = 0$ implies $r_i = 0$ for every $i$. If $B$ is not linearly independent, then we say that it is **linearly dependent**.

One should not be misled by the experience with this concept in vector spaces. For instance, if $M$ is not faithful, then even singletons $\{x\}$ are never linearly independent.

**Definition 3.27** A linearly independent subset $B$ of an $R$-module $M$ is called a **basis** of $M$ if $M$ is generated by $B$.

Assume that $R$ is unital. A unital $R$-module is said to be a **free $R$-module** if it has a basis. (A remark for the readers familiar with categories: free $R$-modules are free objects in the category of unital $R$-modules.) Thus, a unital $R$-module $M$ is free if it contains a linearly independent subset $B$ with the following property: for every $m \in M$ there exist $b_1, \ldots, b_n \in B$ and $r_1, \ldots, r_n \in R$ such that $m = r_1 b_1 + \cdots + r_n b_n$.

Being free is a rather exceptional property for an $R$-module, but examples can be easily found. The simplest one is the $R$-module $R$. Indeed, $\{1\}$ is its basis. It is easy to see that every free $R$-module $M$ is isomorphic to a direct sum of copies of the $R$-module $R$.

The structure of free modules is thus quite simple, apparently just as that of vector spaces over fields. Yet a closer look shows that not everything we know about vector spaces holds for free modules. For example, there are rings whose free modules may have finite bases of different cardinalities (see Exercises 3.8 and 3.9).

From now on we will consider only free modules that are relevant for our goals. Before introducing them in the next definition, let us point out that up until now it went without saying that "vector space" means a vector space over a field. The following more general conception turns out to be very useful.

**Definition 3.28** Let $D$ be a division ring. A unital (left) $D$-module is called a **(left) vector space** over $D$.

A right vector space is of course defined as a unital right $D$-module. Vector spaces over fields are traditionally left vector spaces, but this is just a matter of notational convention (see Remark 3.2). One must be more careful about "left" and "right" when considering vector spaces over noncommutative division rings. However, from now on we assume that our vector spaces are left vector spaces.

We believe that the notions such as **subspace**, **linear combination**, **linear span**, etc., need no explanation. They are defined just as for vector spaces over fields. Fortunately, studying vector spaces over division rings at a basic level is no harder than studying vector spaces over fields. Every vector space is a free module over the corresponding division ring, i.e., the following theorem holds.

**Theorem 3.29** *Every vector space has a basis.*

***Proof*** The proof is a typical application of Zorn's lemma. Let $\mathscr{S}$ be the set of all linearly independent subsets of $V$. Since $\emptyset$ is vacuously a linearly independent set, $\mathscr{S} \neq \emptyset$. Partially order $\mathscr{S}$ by inclusion, i.e., define that $A \leq B$ if $A \subseteq B$. If $\{A_i \mid i \in I\}$ is a chain in $\mathscr{S}$, then it is easy to see that $U = \bigcup_{i \in I} A_i$ is a linearly independent set. Thus, $U \in \mathscr{S}$ and is obviously an upper bound for the chain $\{A_i \mid i \in I\}$. By Zorn's lemma, $\mathscr{S}$ contains a maximal element $B$.

Let us show that $B$ is a basis of $V$. Since $B \in \mathscr{S}$, it is enough to show that $B$ generates $V$. If $V = \{0\}$, then $B = \emptyset$ and this is trivially true. Thus, let $V \neq \{0\}$. Pick a nonzero $v \in V$. Our goal is to prove that $v$ is a linear combination of elements from $B$. We may assume that $v \notin B$. Then $B \cup \{v\}$ contains $B$ as its proper subset, and is therefore a linearly dependent set. This means that there exists a finite linearly dependent subset of $B \cup \{v\}$. Since subsets of $B$ are linearly independent, this subset must contain $v$. Thus, there exist $b_1, \ldots, b_n \in B$ and $d_0, d_1, \ldots, d_n \in D$, not all zero, such that

$$d_0 v + d_1 b_1 + \cdots + d_n b_n = 0.$$

The linear independence of $b_1, \ldots, b_n$ implies that $d_0 \neq 0$. Therefore,

$$v = (-d_0^{-1} d_1) b_1 + \cdots + (-d_0^{-1} d_n) b_n$$

is indeed a linear combination of elements from $B$. □

If we introduce $\mathscr{S}$ as the set of all linearly independent subsets of $V$ that contain a given linearly independent set, the same proof gives the following.

**Theorem 3.30** *Every linearly independent subset of a vector space is contained in a basis.*

A vector space $V$ is said to be **finite dimensional** if it has a finite basis. Suppose $V$ contains a basis with $n$ elements. Then:

(a) Every basis of $V$ has $n$ elements.
(b) If $U$ is a subspace of $V$, then its bases have $\leq n$ elements.
(c) If a subspace $U$ of $V$ has a basis with $n$ elements, then $U = V$.

For vector spaces over a field these are the most basic facts that one learns in a standard linear algebra course. The reader must have seen the proofs. Glancing through them one notices that they still work if a field is replaced by a noncommutative division ring. Therefore, we feel that we can omit giving these proofs without harming the reader's understanding of the subject (the unconvinced reader can, for example, take a look at [5, pp. 185–187]). In view of (a), we can define the **dimension** of $V$, $\dim_D V$, to be $n$ if $V$ has a basis with $n$ elements.

## 3.4 Endomorphisms and Matrices

Let $V$ be a vector space over a division ring $D$. The endomorphism ring $\mathrm{End}_D(V)$ will often play an important role in our discussions. Its elements, the endomorphisms, are also called **linear maps** or **linear operators**. Each linear map is uniquely determined by its action on a basis. More precisely, the following holds.

**Remark 3.31** Let $B = \{b_i \mid i \in I\}$ be a basis of $V$. For each family of vectors $\{w_i \mid i \in I\}$ in $V$ there exists a unique $f \in \mathrm{End}_D(V)$ such that $f(b_i) = w_i$ for every $i \in I$. Indeed, the uniqueness is clear since $B$ is a basis, and the existence is also clear as we may extend $f$ from $B$ to $V$ by linearity, i.e., we define

$$f(d_{i_1}b_{i_1} + \cdots + d_{i_n}b_{i_n}) = d_{i_1}w_{i_1} + \cdots + d_{i_n}w_{i_n}$$

for every finite subset $\{i_1, \ldots, i_n\}$ of $I$ and arbitrary $d_{i_1}, \ldots, d_{i_n} \in D$.

Assume now that $V$ is $n$-dimensional. If $D = F$ is a field, then we know that the ring $\mathrm{End}_F(V)$ is isomorphic to the matrix ring $M_n(F)$. Representing every endomorphism by a matrix with respect to a fixed basis gives rise to a standard isomorphism. Let us see if this still works if $D$ is a noncommutative division ring.

Choose and fix a basis $\{b_1, b_2, \ldots, b_n\}$ of $V$. Given $f \in \mathrm{End}_D(V)$, we have

$$f(b_j) = \sum_{i=1}^{n} f_{ij}b_i, \quad j = 1, \ldots, n,$$

for some $f_{ij} \in D$. We attach the $n \times n$ matrix $\varphi(f)$ to $f$ so that $f_{ij}$ is its $(i, j)$ entry, i.e.,

$$\varphi(f) = \begin{bmatrix} f_{11} & f_{12} & \cdots & f_{1n} \\ f_{21} & f_{22} & \cdots & f_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ f_{n1} & f_{n2} & \cdots & f_{nn} \end{bmatrix}.$$

Let $g$ be another element from $\mathrm{End}_D(V)$, and let $g_{ij} \in D$ be such that

$$g(b_j) = \sum_{i=1}^{n} g_{ij}b_i, \quad j = 1, \ldots, n.$$

Then

$$(f + g)(b_j) = \sum_{i=1}^{n} (f_{ij} + g_{ij})b_i,$$

and hence $\varphi(f + g) = \varphi(f) + \varphi(g)$. If $\varphi(f) = 0$, then $f$ vanishes on a basis, and so $f = 0$. This proves the injectivity of $\varphi$. The surjectivity is evident from Remark 3.31. Indeed, given arbitrary $f_{ij} \in D$, we can define $f \in \mathrm{End}_D(V)$ according to $f(b_j) = \sum_{i=1}^{n} f_{ij}b_i$, so that $\varphi(f) = (f_{ij})$.

Thus, $\varphi$ is an isomorphism of additive groups. This is hardly surprising. Let us examine the action of $\varphi$ on products. To determine $\varphi(fg)$, we compute

$$(fg)(b_j) = f(g(b_j)) = \sum_{k=1}^{n} g_{kj} f(b_k) = \sum_{k=1}^{n} g_{kj} \Big( \sum_{i=1}^{n} f_{ik} b_i \Big) = \sum_{i=1}^{n} \Big( \sum_{k=1}^{n} g_{kj} f_{ik} \Big) b_i .$$

Thus, the $(i, j)$ entry of $\varphi(fg)$ is

$$\sum_{k=1}^{n} g_{kj} f_{ik}. \tag{3.2}$$

The $(i, j)$ entry of $\varphi(f)\varphi(g)$, however, is equal to

$$\sum_{k=1}^{n} f_{ik} g_{kj}. \tag{3.3}$$

As $D$ may be noncommutative, there is, unfortunately, no reason to believe that (3.2) and (3.3) coincide. But let us look on the bright side: The summations (3.2) and (3.3) are very similar, the terms differ only in the order of factors. This can be remedied. We need the following definition.

**Definition 3.32** Let $R$ be a ring. The **opposite ring** of $R$, denoted by $R^{\mathrm{o}}$, is the ring consisting of the same elements as $R$, having the same addition as $R$, and multiplication $\cdot$ given by

$$x \cdot y = yx,$$

where $yx$ is the product in $R$.

Note that $R^{\mathrm{o}}$ is indeed a ring; in particular, the product $\cdot$ is associative. This ring often naturally appears in the study of modules. For example, an obvious adaptation of Remark 3.2 shows that there is a canonical correspondence between left $R$-modules and right $R^{\mathrm{o}}$-modules.

The rings $R$ and $R^{\mathrm{o}}$ are not always isomorphic, but they are **antiisomorphic**. We say that a bijective additive map $\theta$ from a ring $R$ onto a ring $S$ is an **antiisomorphism** if $\theta(xy) = \theta(y)\theta(x)$ for all $x, y \in R$. Obviously, the map $x \mapsto x$ is an antiisomorphism from $R$ onto $R^{\mathrm{o}}$. Antiisomorphic rings are simultaneously division rings; thus, $D$ is a division ring if and only if $D^{\mathrm{o}}$ is.

Let us now consider $\varphi(f)$ as an element of $M_n(D^{\mathrm{o}})$ (rather than $M_n(D)$). This point of view does not affect the above discussion up until (3.3). In $M_n(D^{\mathrm{o}})$, the $(i, j)$ entry of $\varphi(f)\varphi(g)$ is equal to

$$\sum_{k=1}^{n} f_{ik} \cdot g_{kj} = \sum_{j=1}^{n} g_{kj} f_{ik}$$

which does coincide with (3.2), the $(i, j)$ entry of $\varphi(fg)$. Thus, $\varphi$ is an isomorphism from $\mathrm{End}_D(V)$ onto $M_n(D^{\mathrm{o}})$. We have thus proved the following.

**Theorem 3.33** *Let $V$ be an $n$-dimensional vector space over a division ring $D$. Then the rings* $\mathrm{End}_D(V)$ *and* $M_n(D^{\mathrm{o}})$ *are isomorphic.*

**Corollary 3.34** *Let $V$ be a vector space over a division ring $D$. The ring* $\mathrm{End}_D(V)$ *is simple if and only if $V$ is finite dimensional.*

***Proof*** If $V$ is finite dimensional, then $\mathrm{End}_D(V)$ is simple by Theorem 3.33 and Lemma 1.18. Assume that $V$ is infinite dimensional. Let $K$ be the set of all **finite rank operators** in $\mathrm{End}_D(V)$. These are operators whose **rank**, i.e., the dimension of their image, is finite. It is easy to check that $K$ is an ideal of $\mathrm{End}_D(V)$. Moreover, $K$ is proper since it does not contain the identity operator, and it is certainly nonzero; in fact, from Remark 3.31 we deduce that for every subspace $W$ of $V$ there is a linear operator having $W$ as its image. Therefore, $\mathrm{End}_D(V)$ is not simple. □

**Remark 3.35** It is clear from the proof that Theorem 3.33 also holds for free modules. That is, $\mathrm{End}_R(M) \cong M_n(R^{\mathrm{o}})$ if $M$ is a free $R$-module having a basis with $n$ elements.

**Remark 3.36** Theorem 3.33 yields an alternative aspect of Wedderburn's structure theory. We now see that we can add another equivalent condition to (i)–(iii) of Theorem 2.73: (iv) $A \cong \mathrm{End}_\Delta(V)$ where $\Delta$ is a division algebra (in fact, $\Delta \cong D^{\mathrm{o}}$) and $V$ is an $n$-dimensional vector space over $\Delta$. Let us remark that $V$ is automatically also a vector space over the base field $F$, since $F(\ = F \cdot 1)$ is contained in $\Delta$. The elements in $\mathrm{End}_\Delta(V)$ are, in particular, $F$-linear maps.

## 3.5 Simple Modules

In Sect. 3.3 we saw that the structure of a (unital) $R$-module is very simple if $R = D$ is a division ring. We shall now consider another class of easily approachable (although quite different) modules, which play a crucial role in the structure theory of rings.

**Definition 3.37** Let $R$ be a ring. An $R$-module $M$ is said to be **simple** if $RM \neq 0$ and its only submodules are 0 and $M$.

If $M$ is unital, then the condition $RM \neq 0$ reduces to $M \neq 0$.

Another name for a simple module is an **irreducible module**. An **irreducible representation** of an algebra $A$ is defined as a representation $\varphi : A \to \mathrm{End}_F(V)$ for which there does not exist a proper nonzero subspace $W$ of $V$ that is invariant under every $\varphi(a)$, $a \in A$. It is easy to check that this is equivalent to the condition that the associated $A$-module is irreducible.

The following lemma shows that simple modules are cyclic modules generated by *every* nonzero element.

**Lemma 3.38** *An $R$-module $M \neq 0$ is simple if and only if $M = Rm$ for every $0 \neq m \in M$.*

***Proof*** Suppose $M$ is simple. For any $m \in M$, $Rm$ is a submodule of $M$, so either $Rm = 0$ or $Rm = M$. Since $N = \{m \in M \mid Rm = 0\}$ is also a submodule of $M$ and since $RM \neq 0$, $N = 0$. Therefore, $Rm = M$ whenever $m \neq 0$.

Conversely, if $M = Rm$ for every $0 \neq m \in M$ and $L$ is a nonzero submodule of $M$, then $M = R\ell \subseteq L$ for every $0 \neq \ell \in L$. Thus, $L = M$. □

Viewing modules as generalizations of vector spaces, one might first wonder what does the simplicity mean for the latter. The answer is simple.

**Example 3.39** A vector space $V$ over a division ring $D$ is simple (as a $D$-module) if and only if $\dim_D V = 1$.

This example, however, is not really illuminating. The next ones better reflect the meaning of simple modules in the theory that we shall develop.

**Example 3.40** Let $V$ be a nonzero vector space over a division ring $D$. As in Example 3.12, we may regard $V$ as a module over $R = \mathrm{End}_D(V)$ (via $fv = f(v)$). We claim that $V$ is a simple $R$-module. In view of Lemma 3.38, it suffices to show that for each pair $v, w \in V$ with $v \neq 0$ there exists an $f \in R$ such that $f(v) = w$. And this is true since, by Theorem 3.30, we may choose a basis of $V$ containing $v$, and then use Remark 3.31 to find such an $f$.

**Example 3.41** Let $L$ be a left ideal of a ring $R$ such that $RL \neq 0$. Then $L$ is simple as an $R$-module if and only if $L$ is a minimal left ideal.

The following lemma indicates the significance of this example.

**Lemma 3.42** *Let $R$ be a simple ring having a minimal left ideal $L$. Then every simple $R$-module is isomorphic to $L$.*

***Proof*** As $R$ is simple and $LR \neq 0$ (see Lemma 2.17), $LR = R$. Hence, for every simple $R$-module $M$ we have $LM \supseteq L(RM) = (LR)M = RM \neq 0$. Choose $m \in M$ such that $Lm \neq 0$. Since $Lm$ is a submodule of $M$ it follows that $Lm = M$. The map $\varphi : L \to M$ given by $\varphi(\ell) = \ell m$ is therefore a surjective $R$-module homomorphism. Its kernel is a left ideal of $R$ properly contained in $L$, so it can only be 0. Therefore, $\varphi$ is an isomorphism. □

**Example 3.43** Let $R = M_n(D)$, $D$ a division ring. Lemma 1.18 tells us that $R$ is a simple ring. A typical example of a minimal left ideal of $R$ is $RE_{11}$, the set of all matrices that have zeros in all columns except in the first one (Example 2.60). Therefore, every simple $R$-module is isomorphic to $RE_{11}$. Neglecting the zero columns, we may identify this module with $D^n$, the abelian group of all $n$-tuple columns on which $R$ acts via the usual matrix multiplication. (A side remark, related to Theorem 3.33: We can consider $D^n$ as a left vector space over $D$, but then the matrices from $R$ do not act on $D^n$ as $D$-linear operators; if, however, we consider $D^n$ as a right vector space over $D$, then they do.)

Minimal left ideals are prototypes of simple modules. However, they only rarely exist in rings. Say, even such a simple and basic example of a ring as $\mathbb{Z}$ does not have them (why?). Let us now consider left ideals having the opposite property. As it turns out, they also have a tight connection with simple modules.

**Definition 3.44** A left ideal $U$ of a ring $R$ is called a **maximal left ideal** if $U \neq R$ and $U$ is not properly contained in a proper left ideal.

Maximal right ideals and maximal ideals are defined analogously.

**Example 3.45** Let $R = M_n(D)$, with $D$ a division ring, and let

$$U = RE_{11} \oplus RE_{22} \oplus \cdots \oplus RE_{n-1,n-1}.$$

That is, $U$ is the set of all matrices that have zeros in the last column. It is easy to check that $U$ is a maximal left ideal of $R$. We remark that $R = U \oplus L$ where $L$ is the minimal left ideal $RE_{nn}$.

**Example 3.46** If $R$ is a commutative unital ring, then an ideal $U$ of $R$ is maximal if and only if the factor ring $R/U$ is a field. This is well known, and the proof is easy (see Lemma 4.40 below, which considers a more general situation). Hence:

(a) $p\mathbb{Z}$ is a maximal ideal of $\mathbb{Z}$ for every prime number $p$.
(b) $U_c = \{f \in C[a,b] \mid f(c) = 0\}$, where $c$ is a fixed real number from $[a,b]$, is a maximal ideal of the ring of continuous functions $C[a,b]$. Indeed, $f \mapsto f(c)$ is a homomorphism from $C[a,b]$ onto $\mathbb{R}$ with kernel $U_c$, so $C[a,b]/U_c \cong \mathbb{R}$ is a field.

It turns out that the $U_c$'s are actually the only maximal ideals of $C[a,b]$. Of course, the $p\mathbb{Z}$'s are the only maximal ideals of $\mathbb{Z}$.

Most rings have plenty of maximal left ideals.

**Lemma 3.47** *If $R$ is a unital ring, then every proper left ideal $L$ of $R$ is contained in a maximal left ideal.*

***Proof*** Let $\mathscr{S}$ be the set of all proper left ideals of $R$ that contain $L$. Since $L \in \mathscr{S}$, $\mathscr{S}$ is nonempty. Partially order $\mathscr{S}$ by inclusion. If $\{K_i \mid i \in I\}$ is a chain in $\mathscr{S}$, then $K = \bigcup_{i\in I} K_i$ is its upper bound. Indeed, $K$ is easily seen to be a left ideal, and $K \neq R$ since $1 \notin K$. By Zorn's lemma, $\mathscr{S}$ contains a maximal element, which is obviously a maximal left ideal of $R$ containing $L$. □

It is natural to expect that a maximal left ideal $U$ of a ring $R$ is "too big" to be simple as an $R$-module. However, we will be interested in the factor module $R/U$ which may be expected to be "small". Here we regard $R$ as an $R$-module and $U$ as its submodule, so the operations in $R/U$ are given by

$$(r+U)+(s+U) = (r+s)+U \quad \text{and} \quad r(s+U) = rs+U.$$

**Lemma 3.48** *Let $R$ be a ring. Every simple $R$-module is isomorphic to $R/U$ for some maximal left ideal $U$ of $R$. Conversely, if $U$ is a maximal left ideal of $R$ such that $R^2 \not\subseteq U$ (this is automatically fulfilled if $R$ is unital), then $R/U$ is a simple $R$-module.*

***Proof*** Let $M$ be a simple $R$-module. Choose $0 \neq m \in M$. Then $Rm = M$ and hence $\varphi : R \to M, \varphi(r) = rm$, is a surjective $R$-module homomorphism. Its kernel $U$ is a left ideal of $R$, and $R/U \cong M$. We have to show that $U$ is maximal. Let $L$ be a left ideal of $R$ such that $U \subseteq L \subseteq R$. Then $L/U$ is a submodule of $R/U$. Since $R/U$, being isomorphic to $M$, is a simple module, it follows that $L/U = 0$ or $L/U = R/U$. Accordingly, $L = U$ or $L = R$, showing that $U$ is maximal.

To establish the converse, take a maximal left ideal $U$ of $R$, and consider a submodule $N$ of $R/U$. Clearly, $L = \{\ell \in R \mid \ell + U \in N\}$ is a left ideal containing $U$. Thus $L = U$ or $L = R$, yielding $N = 0$ or $N = R/U$. Therefore, $R/U$ is simple provided that $R \cdot R/U \neq 0$, i.e., $R^2 \not\subseteq U$. □

**Example 3.49** Let $R$, $U$, and $L$ be as in Example 3.45. Note that $R/U \cong L$.

**Example 3.50** From Example 3.46 it is now clear what are the simple $\mathbb{Z}$-modules and simple $C[a, b]$-modules.

Lemmas 3.47 and 3.48 yield the following.

**Corollary 3.51** *Every nonzero unital ring has a simple module.*

## 3.6 Semisimple Modules

Theorem 3.29 saying that every vector space has a basis can be equivalently stated as follows: every vector space is a direct sum of a family of 1-dimensional subspaces. In view of Example 3.39, this can be further rephrased as follows: If $D$ is a division ring, then every unital $D$-module is a direct sum of a (possibly infinite) family of simple submodules. We will now consider modules with this property over arbitrary rings.

**Definition 3.52** Let $R$ be a ring. An $R$-module $M$ is said to be **semisimple** if $M$ is a direct sum of a family of simple submodules.

The sum of an empty family of submodules should be understood as 0. The zero module is therefore semisimple. As pointed out above, unital modules over division rings are semisimple. Trivially, simple modules are semisimple. A more interesting example, related to Examples 3.41 and 3.45, is the following.

**Example 3.53** Let $R = M_n(D)$, $D$ a division ring. Pick $1 \leq k \leq n$, and set $M = RE_{11} \oplus \cdots \oplus RE_{kk}$. Thus, $M$ is a left ideal of $R$ consisting of all matrices that have zeros in the last $n - k$ columns. Each $RE_{ii}$ is a minimal left ideal of $R$, and hence a simple $R$-module. Therefore, $M$ is a semisimple $R$-module. In particular, $R = RE_{11} \oplus \cdots \oplus RE_{nn}$ is semisimple as an $R$-module.

Recall a familiar fact from linear algebra: Every generating set of a vector space $V$ contains a basis of $V$. In other words, if $V$ is equal to the sum of 1-dimensional subspaces, then $V$ is equal to the *direct* sum of some of them. The next lemma extends this fact to modules over arbitrary rings.

**Lemma 3.54** *If a module $M$ is the sum of a family of simple submodules $\{M_i \mid i \in I\}$, then there exists a subset $K$ of $I$ such that $M$ is the direct sum of the family $\{M_k \mid k \in K\}$.*

***Proof*** Let $\mathscr{S}$ be the set of all $J \subseteq I$ such that the sum of the family $\{M_j \mid j \in J\}$ is direct. Clearly, $\mathscr{S}$ can be partially ordered by inclusion and $\mathscr{S} \neq \emptyset$ since $\{i\} \in \mathscr{S}$ for every $i \in I$. The existence of a maximal element in $\mathscr{S}$ is a typical application of Zorn's lemma. Indeed, just note that the upper bound of every chain is its union. Denote a maximal element by $K$, and set $M_0 = \oplus_{k\in K} M_k$. Suppose there exists an $i \in I$ such that $M_i \not\subseteq M_0$. Then $M_i \cap M_0$ is a proper submodule of $M_i$, and hence $M_i \cap M_0 = 0$ since $M_i$ is simple. Accordingly, the sum of the family $\{M_j \mid j \in K \cup \{i\}\}$ is direct, which contradicts the maximality of $K$. Hence $M_i \subseteq M_0$ for each $i \in I$, proving that $M_0 = M$. □

**Example 3.55** Let $R$ be as in Example 3.53. Then $R$ has plenty of minimal left ideals and each of them is a simple $R$-module. Obviously, their sum is equal to $R$. Example 3.53 shows that $R$ is the direct sum of some of them.

Lemma 3.54 implies that one can equivalently define semisimple modules by omitting the word "direct" in Definition 3.52.

**Corollary 3.56** *If a module is the sum of a family of simple submodules, then it is semisimple.*

Every subspace of a vector space is a direct summand. The reader probably knows this fact at least for finite dimensional vector spaces over fields. For general vector spaces it can be derived from Theorem 3.30. Let us show that it holds more generally for semisimple modules.

**Lemma 3.57** *The following statements are equivalent for a nonzero module $M$:*

(i) *$M$ is semisimple.*
(ii) *Every submodule of $M$ is a direct summand and $Rm \neq 0$ for every nonzero $m \in M$.*

***Proof*** (i) $\Longrightarrow$ (ii). Our assumption is that $M = \oplus_{i\in I} M_i$ where $M_i$ are simple submodules.

Take a submodule $L$ of $M$. We have to prove that $L$ is a direct summand. By simplicity of $M_i$, either $L \cap M_i = M_i$ or $L \cap M_i = 0$, i.e., $L$ either contains $M_i$ or has trivial intersection with it. If the former holds for every $i \in I$, then $L = M$ and the desired conclusion trivially holds. We may therefore assume that $L \cap M_i = 0$ for some $i \in I$. Consequently, the set $\mathscr{S}$ of all $J \subseteq I$ such that $L \bigcap \left(\oplus_{j\in J} M_j\right) = 0$ is nonempty. A standard application of Zorn's lemma shows that $\mathscr{S}$ has a maximal

element $K$ with respect to inclusion. We claim that $L \oplus \left(\oplus_{k\in K} M_k\right) = M$. If this was not true, there would exist an $i \in I$ such that $M_i \not\subseteq L \oplus \left(\oplus_{k\in K} M_k\right)$, and hence, by simplicity of $M_i$, $M_i \bigcap \left(L \oplus \left(\oplus_{k\in K} M_k\right)\right) = 0$. But then $L \bigcap \left(\oplus_{j\in K\cup\{i\}} M_j\right) = 0$, contradicting the maximality of $K$.

Take a nonzero $m \in M$. Then $m = m_{i_1} + \cdots + m_{i_n}$ for some nonzero $m_{i_j} \in M_{i_j}$, $j = 1, \ldots, n$. If $Rm = 0$, then $Rm_{i_j} = 0$ for every $j$. However, $Rm_{i_j} = M_{i_j}$ by Lemma 3.38, so this cannot occur.

(ii) $\Longrightarrow$ (i). Let $N$ be a submodule of $M$. We claim that $N$ also satisfies the condition that every submodule of $N$ is a direct summand. Indeed, if $L$ is a submodule of $N$, then, by assumption, there exists a submodule $K$ of $M$ such that $M = L \oplus K$, which readily implies that $N = L \oplus (K \cap N)$.

By Corollary 3.56, it is enough to show that the sum of all simple submodules of $M$, which we denote by $M_0$, is equal to $M$. Suppose this is not true. Then there exists a nonzero submodule $T$ of $M$ such that $M = M_0 \oplus T$. Observe that it suffices to show that $T$ contains a simple submodule. To this end, take a nonzero $t \in T$. Using Zorn's lemma, we can find a submodule $W$ of $T$ maximal with respect to the property that $t \notin W$. From the preceding paragraph we see that there exists a submodule $Z$ of $T$ such that $T = W \oplus Z$. The proof will be completed by showing that $Z$ is simple.

Since $t \notin W$, $Z \neq 0$. Therefore, $RZ \neq 0$ by the second condition of (ii). It is thus enough to show that $Z$ has no proper nonzero submodules. Suppose $Z_1$ was such a submodule. Again referring to the beginning of the proof, then there would exist a submodule $Z_2$ of $Z$ such that $Z = Z_1 \oplus Z_2$. Since $W \cap Z = 0$ and $Z_i \neq 0$ for $i = 1, 2$, $t \in W \oplus Z_1$ and $t \in W \oplus Z_2$. That is, $t = w + z_1 = w' + z_2$ for some $w, w' \in W$, $z_1 \in Z_1$, $z_2 \in Z_2$. Therefore, $z_1 - z_2 = w' - w \in W \cap Z = 0$. Thus, $z_1 = z_2 \in Z_1 \cap Z_2 = 0$, which yields a contradiction that $t = w \in W$. □

We add a little example preventing possible misinterpretations of the proof of (i) $\Longrightarrow$ (ii). The point we wish to make is that the set $K$ does not necessarily consist of all $k$ such that $K \cap M_k = 0$, and, moreover, is not always unique.

**Example 3.58** From Example 3.53, we know that $R = M_2(D)$ is a semisimple $R$-module, in fact, $R$ is the direct sum of simple submodules (i.e., minimal left ideals) $M_1 = RE_{11}$ and $M_2 = RE_{22}$. Observe that $L = R(E_{11} + E_{12})$ has trivial intersection with each $M_i$, and that $R = L \oplus M_1 = L \oplus M_2$.

**Corollary 3.59** *A unital module $M$ is semisimple if and only if every submodule of $M$ is a direct summand.*

Do not be misled with subspaces of vector spaces—submodules are only seldom direct summands. For instance, proper nonzero submodules of the $\mathbb{Z}$-module $\mathbb{Z}$ are $n\mathbb{Z}$, $n \geq 2$, and none of them is a direct summand. Thus, $\mathbb{Z}$ is not a semisimple $\mathbb{Z}$-module. We will soon see that the semisimplicity of the $R$-module $R$ is in fact an extremely restrictive condition.

## 3.7 Endomorphism Rings of Semisimple Modules

In the preceding chapter, more precisely in Lemma 2.61, we saw that minimal left ideals give rise to division rings. This was our basis for developing the structure theory of finite dimensional algebras. Simple modules, which can be regarded as generalizations of minimal left ideals (see Example 3.41), have a similar role in the structure theory of more general rings and algebras.

As the first step towards this more general theory, we will show in the next lemma that simple modules give rise to division rings. This lemma may be just a straightforward observation, but an extremely useful one. It is named after I. Schur who used it in the representation theory of groups.

**Lemma 3.60 (Schur's Lemma)** *If $M$ is a simple $R$-module, then the endomorphism ring $\mathrm{End}_R(M)$ is a division ring.*

***Proof*** Let $f \in \mathrm{End}_R(M)$, $f \neq 0$. Then $\ker f$ is a submodule of $M$ different from $M$, and $\mathrm{im}\, f$ is a submodule of $M$ different from 0. Since $M$ is simple, the only possibility is that $\ker f = 0$ and $\mathrm{im}\, f = M$. Thus $f$ is an automorphism, and therefore an invertible element of $\mathrm{End}_R(M)$. □

A slightly more general version of Schur's lemma states that a nonzero homomorphism from an arbitrary module to a simple module is surjective, and a nonzero homomorphism from a simple module to an arbitrary module is injective. The following therefore holds.

**Lemma 3.61** *If simple $R$-modules $M$ and $N$ are not isomorphic, then there are no nonzero homomorphisms from one to another.*

Lemma 3.42 implies that if $R$ is a simple ring with a minimal left ideal, then all simple modules are isomorphic. Let us give a very simple example showing that this is not true for every ring $R$.

**Example 3.62** Let $D$ be a division ring and let $R = D \times D$. Then $M = D \times 0$ and $N = 0 \times D$ are minimal ideals of $R$, and hence simple $R$-modules. Observe that there are no nonzero $R$-module homomorphisms from $M$ to $N$ (just consider $f(rm) = rf(m)$ for $r = (1, 0)$ and $m = (d, 0)$).

A minimal left ideal $L$ of a semiprime ring $R$ gives rise to two division rings: the first one is described in Lemma 2.61 and the other one arises from Schur's lemma when regarding $L$ as a simple $R$-module. What is the connection between them? The answer will be given in the corollary to the next lemma.

**Lemma 3.63** *Let $e$ be an idempotent in a ring $R$. Then* $\mathrm{End}_R(Re) \cong (eRe)^{\mathrm{o}}$.

***Proof*** Define $\varphi : \mathrm{End}_R(Re) \to eRe$ by

$$\varphi(f) = f(e).$$

Since $f$ is an endomorphism, we have $f(e) = f(ee) = ef(e)$, and since $f$ maps to $Re$, we have $f(e) = f(e)e$. Accordingly, $f(e) = ef(e)e$ indeed belongs to $eRe$. It is clear that $\varphi$ is additive. Since $f(e) = 0$ implies $f(re) = rf(e) = 0$ for every $r \in R$, $\varphi$ is injective. Pick $a \in eRe$ and define $f : Re \to Re$ by $f(u) = ua$. Then $f \in \mathrm{End}_R(Re)$ and $\varphi(f) = f(e) = a$, which proves that $\varphi$ is surjective. Next,

$$\varphi(fg) = f(g(e)) = f(g(e)e) = g(e)f(e) = \varphi(g)\varphi(f).$$

This means that $\varphi$ is an antiisomorphism. The composition of the canonical antiisomorphism $x \mapsto x$ from $eRe$ onto $(eRe)^{\mathrm{o}}$ and $\varphi$ is thus an isomorphism from $\mathrm{End}_R(Re)$ onto $(eRe)^{\mathrm{o}}$. □

**Corollary 3.64** *Let $L$ be a minimal left ideal of a semiprime ring $R$. Then there exists an idempotent $e \in R$ such that $L = Re$, $D = eRe$ is a division ring, and $\mathrm{End}_R(L) \cong D^{\mathrm{o}}$.*

***Proof*** Combine Lemmas 2.61 and 3.63. □

Our next goal is to describe endomorphism rings of those semisimple modules that are direct sums of finitely many simple modules. To this end, we need some preliminary results. First we record a simple observation.

**Remark 3.65** If $M$ and $M'$ are isomorphic $R$-modules, then $\mathrm{End}_R(M)$ and $\mathrm{End}_R(M')$ are isomorphic rings. Indeed, if $h : M \to M'$ is an $R$-module isomorphism, then $\varphi : \mathrm{End}_R(M) \to \mathrm{End}_R(M')$, $\varphi(f) = hfh^{-1}$, is a ring isomorphism.

We continue with a crucial lemma.

**Lemma 3.66** *Let $M_1, \ldots, M_r$ be $R$-modules and let $\overline{M} = M_1 \oplus \cdots \oplus M_r$.*

(a) *If the modules $M_i$ are pairwise isomorphic, then the ring $\mathrm{End}_R(\overline{M})$ is isomorphic to the matrix ring $M_r(\mathrm{End}_R(M))$ where $M$ is any of $M_i$.*
(b) *If there are no nonzero homomorphisms from $M_i$ to $M_j$ whenever $i \neq j$, then the ring $\mathrm{End}_R(\overline{M})$ is isomorphic to the ring $\mathrm{End}_R(M_1) \times \cdots \times \mathrm{End}_R(M_r)$.*

***Proof*** Write $\pi_i$ for the canonical projection of $\overline{M}$ onto $M_i$ and $\iota_j$ for the canonical embedding of $M_j$ into $\overline{M}$, i.e.,

$$\pi_i(m_1, \ldots, m_r) = m_i \quad \text{and} \quad \iota_j(m_j) = (0, \ldots, 0, m_j, 0, \ldots, 0).$$

Observe that

$$\sum_{k=1}^{r} \iota_k \pi_k = \mathrm{id}_{\overline{M}}. \tag{3.4}$$

(a) Since the modules $M_i$ are pairwise isomorphic, and hence $\overline{M}$ is isomorphic to the direct sum of $r$ copies of $M_1$, we may, in light of Remark 3.65, assume that they are all equal: $M_1 = \cdots = M_r$. Write $M$ instead of $M_i$.

For any $f \in \mathrm{End}_R(\overline{M})$, set

$$f_{ij} = \pi_i f \iota_j \in \mathrm{End}_R(M)$$

and define

$$\varphi : \mathrm{End}_R(\overline{M}) \to M_r(\mathrm{End}_R(M)), \quad \varphi(f) = (f_{ij}).$$

It is immediate that $\varphi$ is additive and has trivial kernel. To show that it is multiplicative, observe that (3.4) yields

$$(fg)_{ij} = \pi_i fg\iota_j = \pi_i f\Big(\sum_{k=1}^{r} \iota_k\pi_k\Big)g\iota_j = \sum_{k=1}^{r} \pi_i f\iota_k\pi_k g\iota_j = \sum_{k=1}^{r} f_{ik}g_{kj}.$$

Finally, $\varphi$ is surjective, since for any $(f_{ij}) \in M_r(\mathrm{End}_R(M))$,

$$f = \sum_{i,j=1}^{r} \iota_i f_{ij}\pi_j \in \mathrm{End}_R(\overline{M})$$

satisfies $\varphi(f) = (f_{ij})$.

(b) We claim that

$$\varphi : \mathrm{End}_R(\overline{M}) \to \mathrm{End}_R(M_1)\times\cdots\times\mathrm{End}_R(M_r), \quad \varphi(f) = (\pi_1 f\iota_1, \ldots, \pi_r f\iota_r),$$

is a ring isomorphism. Indeed, our assumption implies that

$$\pi_i f\iota_j = 0 \quad \text{whenever } i \neq j, \tag{3.5}$$

and hence by using (3.4) we see that

$$\pi_i fg\iota_i = \pi_i f\left(\sum_{k=1}^{r} \iota_k\pi_k\right) g\iota_i = (\pi_i f\iota_i)(\pi_i g\iota_i),$$

showing that $\varphi$ is a ring homomorphism. Note that (3.5) also shows that $\varphi$ has trivial kernel. Given $f_i \in \mathrm{End}_R(M_i)$, $i = 1, \ldots, r$, we have $\varphi(f) = (f_1, \ldots, f_r)$ where $f = \sum_{k=1}^{r} \iota_k f_k\pi_k \in \mathrm{End}_R(\overline{M})$. Thus, $\varphi$ is surjective. □

**Remark 3.67** An essentially equivalent, but neater formulation of (a) is

$$\mathrm{End}_R(M^r) \cong M_r(\mathrm{End}_R(M)).$$

We need one final lemma.

**Lemma 3.68** *If modules $N_1, \ldots, N_n$ and $M_1, \ldots, M_m$ are such that there are no nonzero homomorphisms from $N_j$ to $M_i$ for all $j$ and $i$, then there are no nonzero homomorphisms from $N_1 \oplus \cdots \oplus N_n$ to $M_1 \oplus \cdots \oplus M_m$.*

***Proof*** Let $\pi_i : M_1 \oplus \cdots \oplus M_m \to M_i$ and $\iota_j : N_j \to N_1 \oplus \cdots \oplus N_n$ be the canonical projection and the canonical embedding, respectively. If $f$ is a homomorphism from $N_1 \oplus \cdots \oplus N_n$ to $M_1 \oplus \cdots \oplus M_m$, then $\pi_i f \iota_j$ is a homomorphism from $N_j$ to $M_i$. Therefore, $\pi_i f \iota_j = 0$ for all $j$ and $i$, giving $f = 0$. □

We are now in a position to accomplish our goal.

**Theorem 3.69** *If a nonzero semisimple $R$-module $M$ is a finite direct sum of simple modules, then*

$$\mathrm{End}_R(M) \cong M_{n_1}(\Delta_1) \times \cdots \times M_{n_r}(\Delta_r)$$

*for some $n_i \in \mathbb{N}$ and division rings $\Delta_i$.*

***Proof*** Recall from Lemma 3.61 that two simple modules are either isomorphic or there are no nonzero homomorphisms from one to another. We may therefore assume that $M = M_1 \oplus \cdots \oplus M_r$ where each $M_i$ is a direct sum of $n_i$ pairwise isomorphic simple $R$-modules, and, for $i \neq j$, there are no nonzero homomorphisms between any of the summands of $M_i$ to any of the summands of $M_j$. Lemma 3.68 tells us that there are also no nonzero homomorphisms from $M_i$ to $M_j$. By Lemma 3.66 (b),

$$\mathrm{End}_R(M) \cong \mathrm{End}_R(M_1) \times \cdots \times \mathrm{End}_R(M_r).$$

Further, Lemma 3.66 (a) shows that

$$\mathrm{End}_R(M_i) \cong M_{n_i}(\mathrm{End}_R(N_i))$$

for some simple $R$-module $N_i$. Finally, $\Delta_i = \mathrm{End}_R(N_i)$ is a division ring by Schur's lemma. □

## 3.8 Semisimple Rings

In Theorem 3.69 we reached essentially the same conclusion as in Wedderburn's structure theorem. Thus, if we wish to show that a ring $R$ has the Wedderburn type structure, it is enough to find a module $M$ such that $R \cong \mathrm{End}_R(M)$ and $M$ is a finite direct sum of simple modules. How to find such rings and corresponding modules? Observe that Lemma 3.63 implies that every unital ring $R$ satisfies $\mathrm{End}_R(R) \cong R^{\mathrm{o}}$, and hence $\mathrm{End}_R(R)^{\mathrm{o}} \cong R$. As the rings $\mathrm{End}_R(R)^{\mathrm{o}}$ and $\mathrm{End}_R(R)$ are not that different, the natural candidate for the module we are looking for is the $R$-module $R$. Hence the following definition.

**Definition 3.70** A unital ring $R$ is said to be **semisimple** if $R$ is semisimple as an $R$-module.

Theorem 3.69 speaks about semisimple modules that are finite direct sums of simple modules, while Definition 3.70 requires only the semisimplicity of the module $R$. However, the next theorem shows that the finiteness condition follows from the definition.

**Theorem 3.71** *Let $R$ be a nonzero unital ring. The following statements are equivalent:*

(i) *$R$ is semisimple.*
(ii) *Every unital $R$-module is semisimple.*
(iii) *$R$ is a finite direct sum of minimal left ideals.*
(iv) *Every left ideal of $R$ is of the form $Re$ where $e \in R$ is an idempotent.*
(v) *$R \cong M_{n_1}(D_1) \times \cdots \times M_{n_r}(D_r)$ for some $n_i \in \mathbb{N}$ and division rings $D_i$.*

***Proof*** (i) $\iff$ (ii). We only have to show that (i) implies (ii). Condition (i) means that $R$ is a (direct) sum of a family $\{L_i \mid i \in I\}$ of minimal left ideals (see Examples 3.19 and 3.41). Let $M$ be a unital $R$-module. For every $m \in M$ and every $L_i$, the map $\ell_i \to \ell_i m$ is an $R$-module homomorphism from $L_i$ onto the submodule $L_i m$ of $M$. Since $L_i$ is a simple $R$-module, either $L_i m = 0$ or this map is an isomorphism, in which case $L_i m$ is simple. Writing 1 as a (finite) sum of elements from $L_i$, we see that $m = 1m$ lies in the sum of some simple submodules $L_i m$. This shows that $M$ is the sum of a family of simple submodules, and is hence semisimple (by Corollary 3.56).

(i) $\iff$ (iii). We have to show that if $R$ is a direct sum of a family of minimal left ideals, then it is a direct sum of some *finite* family of minimal left ideals. By assumption, there exist minimal left ideals $L_1, \ldots, L_s$ such that $1 \in L_1 \oplus \cdots \oplus L_s$. Let $\ell_i \in L_i$ be such that $1 = \ell_1 + \cdots + \ell_s$. Then $x = x\ell_1 + \cdots + x\ell_s$ for every $x \in R$, showing that $R = L_1 \oplus \cdots \oplus L_s$.

(i) $\iff$ (iv). By Corollary 3.59, $R$ is semisimple if and only if every left ideal of $R$ is a direct summand. From Example 3.25 we see that this is further equivalent to the condition that left ideals of $R$ are of the form $Re$ with $e$ idempotent.

(iii) $\iff$ (v). Lemma 3.63 implies that $R \cong \mathrm{End}_R(R)^{\mathrm{o}}$. Assuming (iii), it thus follows from Theorem 3.69 that $R \cong M_{n_1}(\Delta_1)^{\mathrm{o}} \times \cdots \times M_{n_r}(\Delta_r)^{\mathrm{o}}$ for some $n_i \in \mathbb{N}$ and division rings $\Delta_i$. Observe that $M_{n_i}(\Delta_i)^{\mathrm{o}} \cong M_{n_i}(\Delta_i^{\mathrm{o}})$ via the isomorphism $(d_{ij}) \mapsto (d_{ji})$. Therefore, $R \cong M_{n_1}(D_1) \times \cdots \times M_{n_r}(D_r)$ where $D_i = \Delta_i^{\mathrm{o}}$.

It remains to show that (v) implies (iii). If $R = M_n(D)$, then $R$ is the direct sum of minimal left ideals $RE_{11}, \ldots, RE_{nn}$ (see Example 3.53). The general case readily follows from this special case. □

**Remark 3.72** Unlike conditions (i)–(iv), condition (v) is left-right symmetric. Therefore, the right versions of (i)–(iv) are also equivalent to (v). In particular, $R$ is semisimple as a left $R$-module if and only if $R$ is semisimple as a right $R$-module.

We can add the uniqueness statement to condition (v).

**Theorem 3.73** *Let $D_1, \ldots, D_r$ and $D'_1, \ldots, D'_{r'}$ be division rings and let $n_1, \ldots, n_r$ and $n'_1, \ldots, n'_{r'}$ be natural numbers. If the rings*

$$R = M_{n_1}(D_1) \times \cdots \times M_{n_r}(D_r) \quad \text{and} \quad R' = M_{n'_1}(D'_1) \times \cdots \times M_{n'_{r'}}(D'_{r'})$$

*are isomorphic, then $r = r'$ and, with a suitable permutation of direct factors if necessary, $n_i = n'_i$ and $D_i \cong D'_i$, $i = 1, \ldots, r$.*

***Proof*** Let $I_n$ denote the identity matrix of the $n \times n$ matrix ring. Write $e_1 = (I_{n_1}, 0, \ldots, 0)$, $e_2 = (0, I_{n_2}, 0, \ldots, 0)$, etc., and similarly define $e'_1$, $e'_2$, etc. in $R'$. Let $\varphi : R \to R'$ be an isomorphism. Since $e_1$ is a central idempotent in $R$, $\varphi(e_1)$ is a central idempotent in $R'$. As $M_{n'_i}(D'_i)$ is a unital simple ring (Lemma 1.18), its center is a field and so 0 and $I_{n'_i}$ are its only central idempotents. Therefore, $\varphi(e_1) = e'_{i_1} + \cdots + e'_{i_k}$ for some distinct $i_1, \ldots, i_k$, giving

$$\varphi(e_1 R) = \varphi(e_1)R' = e'_{i_1} R' \oplus \cdots \oplus e'_{i_k} R'.$$

Since $e_1 R \cong M_{n_1}(D_1)$ is a simple ring, $k$ must be equal to 1. We may assume that $i_1 = 1$, so $\varphi(e_1 R) = e'_1 R'$. Similarly, $\varphi(e_2 R) = e'_2 R'$, etc. Hence, $r = r'$ and $M_{n_i}(D_i) \cong M_{n'_i}(D'_i)$, $i = 1, \ldots, r$.

We are left with the $r = 1$ case. Write $R = M_n(D)$ and $R' = M_{n'}(D')$. Our goal is to show that the existence of an isomorphism $\varphi : R \to R'$ implies $n = n'$ and $D \cong D'$.

We first prove the latter. Write $e$ for the matrix unit $E_{11}$ in $R$ and set $e' = \varphi(e) \in R'$. By Lemma 3.63, $\mathrm{End}_{R'}(R'e') \cong (e'R'e')^{\mathrm{o}}$. On the other hand, $e'R'e' = \varphi(eRe) \cong eRe \cong D$. Therefore,

$$\mathrm{End}_{R'}(R'e') \cong D^{\mathrm{o}}.$$

Taking $f'$ to be the matrix unit $E_{11}$ in $R'$, we have $f'R'f' \cong D'$ and hence

$$\mathrm{End}_{R'}(R'f') \cong D'^{\mathrm{o}}$$

by Lemma 3.63. Either directly or using Proposition 2.63, we see that $R'e'$ and $R'f'$ are minimal left ideals of $R'$. Lemma 3.42 therefore tells us that they are isomorphic as $R'$-modules. This implies that the rings $\mathrm{End}_{R'}(R'e')$ and $\mathrm{End}_{R'}(R'f')$ are isomorphic (Remark 3.65). Thus, $D^{\mathrm{o}} \cong D'^{\mathrm{o}}$, and hence $D \cong D'$.

It remains to show that $n = n'$. Suppose this is not true, say $n < n'$. The matrix units $E_{11}, E_{22}, \ldots, E_{n'n'}$ in $R'$ are nonzero pairwise orthogonal idempotents. Theorem 3.33 tells us that $R$, and hence also $R'$, is isomorphic to $\mathrm{End}_{\Delta}(V)$ where $V$ is an $n$-dimensional vector space over $\Delta = D^{\mathrm{o}}$. Therefore, $\mathrm{End}_{\Delta}(V)$ also contains nonzero pairwise orthogonal idempotents $\varepsilon_1, \ldots, \varepsilon_{n'}$. Choose $v_j \in V$ such that $w_j = \varepsilon_j(v_j) \neq 0$. Observe that $\varepsilon_i(w_j) = \delta_{ij} w_j$ for all $i, j = 1, \ldots, n'$. However, this is impossible since $w_1, \ldots, w_{n'}$ are linearly dependent due to $n < n'$. □

Recall that a module over an $F$-algebra $A$ is defined as a module over the ring $A$ which is also a vector space over $F$ and the module operation is $F$-bilinear. Extending our results on ring modules to algebra modules is straightforward. Of course, all definitions must be appropriately adjusted. For example, a simple module over an algebra should have no proper nonzero algebra submodules, algebra module homomorphisms must be $F$-linear maps, etc. These changes do not affect the proofs. Thus, for example, we can substitute "algebra" for "ring" in Lemma 3.42, which in particular implies the following: *Every simple module of a finite dimensional simple algebra $A$ is isomorphic to a minimal left ideal $L$ of $A$.* Together with the algebra version of Theorem 3.71, this enables us to prove the following corollary.

**Corollary 3.74** *If $A$ is a finite dimensional simple algebra, then any two finite dimensional unital $A$-modules of the same dimension are isomorphic.*

***Proof*** Let $M$ be a unital $A$-module. Since $A \cong M_n(D)$ by Theorem 2.73, we may use Theorem 3.71 to conclude that $M$ is semisimple. If $M$ is also finite dimensional, then it is obviously a finite direct sum of simple modules: $M = M_1 \oplus \cdots \oplus M_s$, $M_i$ simple. By Lemma 3.42, each $M_i$ is isomorphic to a minimal left ideal $L$ of $A$. Therefore, $M$ is isomorphic to the direct sum of $s$ copies of $L$. The number $s$ thus determines $M$ up to isomorphism. On the other hand, $s$ is determined by the dimension of $M$. Therefore, if $N$ is a unital $A$-module of the same dimension as $M$, then $M \cong N$. □

## 3.9 Chain Conditions

The theory presented in the preceding section is obviously similar to Wedderburn's structure theory of finite dimensional algebras exposed in Sect. 2.7. The relation between the two theories will become more clear in the next section. The aim of this one is to introduce two module-theoretic substitutes for finite dimensionality.

Observe first that the following three statements are equivalent for a vector space $V$:

(i) $V$ is infinite dimensional.
(ii) there exists an infinite descending chain $V_1 \supsetneq V_2 \supsetneq \dots$ of subspaces of $V$.
(iii) there exists an infinite ascending chain $W_1 \subsetneq W_2 \subsetneq \dots$ of subspaces of $V$.

Indeed, each of (ii) and (iii) obviously implies (i), and the converses follow by taking a linearly independent subset $\{v_1, v_2, \dots\}$ of $V$ and defining $V_n$ to be the linear span of $\{v_n, v_{n+1}, \dots\}$ and $W_n$ to be the linear span of $\{v_1, \dots, v_n\}$. The following two notions can be therefore viewed as generalizations of finite dimensionality.

**Definition 3.75** A module $M$ is said to be **artinian** (or is said to satisfy the **descending chain condition** on submodules) if for every chain $M_1 \supseteq M_2 \supseteq \dots$ of submodules of $M$, there exists an $m \in \mathbb{N}$ such that $M_m = M_{m+1} = \dots$

**Definition 3.76** A module $N$ is said to be **noetherian** (or is said to satisfy the **ascending chain condition** on submodules) if for every chain $N_1 \subseteq N_2 \subseteq \dots$ of submodules of $N$, there exists an $n \in \mathbb{N}$ such that $N_n = N_{n+1} = \dots$

Let $X$ be a set and $\mathscr{S}$ be a set of subsets of $X$. We say that $A \in \mathscr{S}$ is a **minimal element** of $\mathscr{S}$ (with respect to set inclusion) if there does not exist a $Y \in \mathscr{S}$ such that $A \supsetneq Y$. Similarly, we say that $B$ is a **maximal element** of $\mathscr{S}$ if there does not exist a $Z \in \mathscr{S}$ such that $B \subsetneq Z$.

**Lemma 3.77** *The following statements are equivalent for a module $M$:*

(i) *$M$ is artinian.*
(ii) *Every nonempty set of submodules of $M$ has a minimal element.*

***Proof*** (i) $\Longrightarrow$ (ii). Let $\mathscr{S}$ be a nonempty set of submodules of $M$. Choose $M_1 \in \mathscr{S}$. If $M_1$ is not minimal, there exists an $M_2 \in \mathscr{S}$ such that $M_1 \supsetneq M_2$. If $M_2$ is not minimal, then $M_2 \supsetneq M_3$ for some $M_3 \in \mathscr{S}$. Continuing this process we arrive in a finite number of steps at a minimal element in $\mathscr{S}$; for if not, there would exist an infinite chain $M_1 \supsetneq M_2 \supsetneq \dots$ (Note that the axiom of choice was used in this argument.)

(ii) $\Longrightarrow$ (i). Let $M_1 \supseteq M_2 \supseteq \dots$ be a descending chain of submodules of $M$. If $M_m$ is a minimal element of the set $\{M_1, M_2, \dots\}$, then $M_m = M_{m+1} = \dots$ □

**Lemma 3.78** *The following statements are equivalent for a module $N$:*

(i) *$N$ is noetherian.*
(ii) *Every nonempty set of submodules of $N$ has a maximal element.*
(iii) *Every submodule of $N$ is finitely generated.*

***Proof*** (i) $\Longleftrightarrow$ (ii). This can be proved in the same manner as Lemma 3.77, only the inclusions must be reversed.

(ii) $\Longrightarrow$ (iii). Let $P$ be a submodule of $N$. Denote by $\mathscr{S}$ the set of all finitely generated submodules of $P$. Since it contains 0, $\mathscr{S}$ is nonempty. Thus there exists a maximal element $B$ in $\mathscr{S}$. We have to show that $B = P$. Let $b_1, \dots, b_r$ be generators of $B$. Take an arbitrary $p \in P$. The submodule generated by $p, b_1, \dots, b_r$ lies in $\mathscr{S}$ and contains $B$, and therefore it must be equal to $B$. In particular, $p \in B$.

(iii) $\Longrightarrow$ (i). Let $N_1 \subseteq N_2 \subseteq \dots$ be a chain of submodules of $N$. Note that $\bigcup_{i=1}^{\infty} N_i$ is also a submodule of $N$, and is therefore generated by finitely many elements $c_1, \dots, c_q$. Each $c_i$ lies in one of the $N_j$'s, so there exists an $n \in \mathbb{N}$ such that $N_n$ contains all $c_1, \dots, c_q$. Hence $N_n = \bigcup_{i=1}^{\infty} N_i$, i.e., $N_n = N_{n+1} = \dots$ □

We will not go further into the theory of general artinian and noetherian modules. Rather, we will restrict our attention to the following special cases.

**Definition 3.79** A ring $R$ is said to be **left (resp. right) artinian** if $R$ is an artinian left (resp. right) $R$-module.

**Definition 3.80** A ring $R$ is said to be **left (resp. right) noetherian** if $R$ is a noetherian left (resp. right) $R$-module.

A left artinian (resp. noetherian) ring is not necessarily right artinian (resp. noetherian), and vice versa. For simplicity we will consider only left artinian and left noetherian rings. Of course, all statements that we shall make have analogous right versions. If $R$ is both left and right artinian (resp. noetherian), then we say that $R$ is **artinian (resp. noetherian)**. A left artinian (resp. noetherian) commutative ring is obviously automatically artinian (resp. noetherian).

Since submodules of the left $R$-module $R$ are the left ideals of $R$, $R$ is left artinian if and only if for every chain $L_1 \supseteq L_2 \supseteq \dots$ of left ideals of $R$ there exists an $m \in \mathbb{N}$ such that $L_m = L_{m+1} = \dots$ Analogously, $R$ is left noetherian if it satisfies the ascending chain condition on left ideals.

There is a non-obvious relation between these two classes of rings: A left artinian unital ring is left noetherian. We will not prove this here. On the other hand, the ring $\mathbb{Z}$ is noetherian but not artinian; this follows easily from the fact that the nonzero ideals of $\mathbb{Z}$ are $n\mathbb{Z}$, $n \in \mathbb{N}$, and that $n\mathbb{Z} \subseteq m\mathbb{Z}$ if and only if $m$ divides $n$.

Noetherian modules and rings are named after E. Noether, and Artinian modules and rings are named after E. Artin who developed, some twenty years after Wedderburn's seminal work on finite dimensional algebras, the structure theory we are presenting in these last sections of the chapter. Its usual name is therefore the **Wedderburn-Artin theory**.

## 3.10 Artinian Rings

The artinian property turns out to be a perfect ring-theoretic substitute for finite dimensionality. The basic structure theory of finite dimensional algebras can be generalized to artinian rings, sometimes with identical proofs and sometimes with extra effort.

We restrict our attention to extending Wedderburn's fundamental structure theorems to left artinian rings. We start with two simple lemmas.

**Lemma 3.81** *If $e$ and $f \neq 0$ are orthogonal idempotents in a unital ring $R$, then $R(1-e) \supsetneq R(1-e-f)$.*

***Proof*** Since $e$ and $f$ are orthogonal, $(1-e-f)(1-e) = 1-e-f$. This gives $R(1-e) \supseteq R(1-e-f)$. To show that $R(1-e) \neq R(1-e-f)$, observe that $f = f(1-e) \in R(1-e)$, while $f = x(1-e-f)$ with $x \in R$ implies $f = f^2 = x(1-e-f)f = 0$, a contradiction. □

**Lemma 3.82** *If $R$ is a left artinian ring, then so is $eRe$ for every idempotent $e \in R$.*

***Proof*** Let $L_1 \supseteq L_2 \supseteq \dots$ be a chain of left ideals of $eRe$. Then $RL_1 \supseteq RL_2 \supseteq \dots$ is a chain of left ideals of $R$. Since $R$ is left artinian, there exists an $m \in \mathbb{N}$ such that $RL_m = RL_{m+1} = \dots$, and hence $eRL_m = eRL_{m+1} = \dots$ However, since

$L_n$ is a left ideal of $eRe$, $eRL_n = L_n$ for each $n$. Therefore, $L_m = L_{m+1} = \ldots$, as desired. □

The next theorem is one of the cornerstones of ring theory.

**Theorem 3.83 (Wedderburn-Artin)** *A nonzero unital ring $R$ is semiprime and left artinian if and only if $R \cong M_{n_1}(D_1) \times \cdots \times M_{n_r}(D_r)$ for some $n_i \in \mathbb{N}$ and division rings $D_i$.*

***Proof*** The "if" part is easy. Recall from Theorem 2.68 that the ring $M_{n_1}(D_1) \times \cdots \times M_{n_r}(D_r)$ is semiprime. To prove that it is left artinian, first observe that the ring $M_n(D)$, where $D$ is a division ring, has a natural structure of a finite dimensional vector space over $D$ (of dimension $n^2$, with standard matrix units forming a basis). Since left ideals of $M_n(D)$ are subspaces, every descending chain $L_1 \supseteq L_2 \supseteq \ldots$ terminates. This proves the desired conclusion for the case where $r = 1$. The general case follows easily from Lemma 2.44.

We start the proof of the "only if" part by applying Lemma 3.77 to the set of all nonzero left ideals. This gives the existence of a minimal left ideal $L$ of $R$. Since $R$ is semiprime, $L^2 \neq 0$. Lemma 2.61 therefore shows that $R$ contains a minimal idempotent $e_1$. By Lemma 3.81, $R \supsetneq R(1-e_1)$. If $e_1 \neq 1$, then $(1-e_1)R(1-e_1)$ is a nonzero ring which is easily seen to be semiprime. By Lemma 3.82, it is also left artinian. We can thus repeat the argument from the beginning to conclude that $(1-e_1)R(1-e_1)$ contains a minimal idempotent $e_2$. Observe that $e_2 \in (1-e_1)R(1-e_1)$ implies that $e_1$ and $e_2$ are orthogonal. Therefore, the division ring $e_2(1-e_1)R(1-e_1)e_2$ is equal to $e_2Re_2$, so $e_2$ is also a minimal idempotent of $R$. By Lemma 3.81, $R(1-e_1) \supsetneq R(1-e_1-e_2)$. If $e_1 + e_2 \neq 1$, the same argument shows that there exists an idempotent $e_3$ lying in $(1-e_1-e_2)R(1-e_1-e_2)$ and hence orthogonal to $e_1$ and $e_2$, which is a minimal idempotent of $(1-e_1-e_2)R(1-e_1-e_2)$ and therefore also of $R$, and satisfies $R(1-e_1-e_2) \supsetneq R(1-e_1-e_2-e_3)$. We continue this process, which must stop after finitely many steps since $R$ is left artinian. Therefore, there exist pairwise orthogonal minimal idempotents $e_1, \ldots, e_n \in R$ whose sum is 1.

We can now complete the proof in two ways, either by referring to Theorem 2.68 or by observing that $R$ is the direct sum of minimal left ideals $Re_1, \ldots, Re_n$ and then referring to the implication (iii) $\Longrightarrow$ (v) of Theorem 3.71. □

**Corollary 3.84** *The following condition can be added to the list of equivalent conditions in Theorem 3.71:* (vi) *$R$ is semiprime and left artinian.*

We can of course replace "left" by "right" in Theorem 3.83 and Corollary 3.84.

A unital finite dimensional algebra is obviously an artinian ring. Theorem 3.83 is therefore a generalization of Wedderburn's theorem on the structure of finite dimensional semiprime algebras (Theorem 2.71). To be precise, this is true only under the unitality assumption, since Theorem 2.71 in particular shows that finite dimensional semiprime algebras are automatically unital. The same turns out to be true for semiprime left artinian rings, i.e., the assumption that $R$ is unital is redundant in Theorem 3.83. However, we omit the proof.

Our next goal is to obtain various characterizations of rings isomorphic to $M_n(D)$. We need a lemma first.

**Lemma 3.85** *If a ring $R$ has minimal left ideals, then the sum of all minimal left ideals of $R$ is a two-sided ideal of $R$.*

***Proof*** We have to prove that the sum of all minimal left ideals of $R$ is a right ideal. Note that it is enough to show the following: If $L$ is a minimal left ideal of $R$ and $a \in R$ is such that $La \neq 0$, then $La$ is also a minimal left ideal. To this end, consider an $R$-module homomorphism $L \to La$, $x \mapsto xa$. The minimality of $L$ is equivalent to the condition that $L$, regarded as an $R$-module, has no proper nonzero submodules. Our map has therefore trivial kernel. Since it is obviously surjective, $L$ and $La$ are isomorphic as $R$-modules. Hence $La$ does not have proper nonzero submodules, meaning that $La$ is a minimal left ideal. □

**Theorem 3.86 (Wedderburn-Artin)** *Let $R$ be a nonzero unital ring. The following statements are equivalent:*

(i) *$R$ is simple and has a minimal left ideal.*
(ii) *$R$ is simple and left artinian.*
(iii) *$R$ is prime and left artinian.*
(iv) *$R$ is simple and semisimple.*
(v) *There exist an $n \in \mathbb{N}$ and a division ring $D$ such that $R \cong M_n(D)$.*

***Proof*** (iii) $\Longleftrightarrow$ (v). Use Theorem 3.83.
(iv) $\Longleftrightarrow$ (v). Use Theorem 3.71.
(iii) & (iv) $\Longrightarrow$ (ii). Trivial.
(ii) $\Longrightarrow$ (i). Use Lemma 3.77.
(i) $\Longrightarrow$ (iv). The simplicity of $R$ together with Lemma 3.85 shows that $R$ is a sum of minimal left ideals, and is thus semisimple (by Corollary 3.56). □

The statement (iii) may sound strange, but not every simple ring is semisimple; the terminology becomes a bit awkward here.

Again we can replace "left" by "right" at all places. In (ii) and (iii), we can therefore simply omit "left". The standard name for a ring from Theorem 3.86 is actually a **simple artinian ring**.

The theory we have developed in this chapter generalizes and supplements Wedderburn's structure theory. Indeed, it covers a larger class of rings which in particular includes infinite dimensional division algebras, and, on the other hand, gives additional information by providing new characteristic conditions (Theorems 3.71 and 3.86) and establishing the uniqueness of $n_i$ and $D_i$ (Theorem 3.73).

Still, the methods of Chap. 2 are of independent interest. At the end of the proof of Theorem 3.83 we saw that they yield an alternative proof of the central result of this chapter. It should be said that, in essence, these methods are not that different from those developed in the present chapter. The ones and the others are based on the notion of a minimal left ideal. In Chap. 2 we study them via idempotents, and in the present chapter via modules. It is good to be familiar with both approaches.

## 3.11 Noetherian Rings

The noetherian property gives rise to a considerably richer class of rings. Lemma 3.78 implies that a ring $R$ is left noetherian if and only if every left ideal of $R$ is finitely generated. In particular, if every left ideal of $R$ is principal, i.e., generated by a single element, then $R$ is left noetherian. Besides $\mathbb{Z}$, the basic example of such a ring is $F[\omega]$ where $F$ is a field. We assume the reader knows this from a basic abstract algebra course. A deeper result, known as Hilbert's basis theorem, says that if $T$ is a left noetherian unital ring, then so is $T[\omega]$. We will now prove a slightly more general version which is not limited to polynomials. However, the reader will notice that polynomials are imitated in the proof.

**Theorem 3.87** *Let $R$ be a unital ring, $T$ be a subring of $R$ that contains the unity of $R$, and $d \in R$. If $T$ is left noetherian, $R$ is generated by $T \cup \{d\}$, and $T + dT = T + Td$, then $R$ is left noetherian.*

***Proof*** From $Td \subseteq T + dT$ it follows easily that $Td^s \subseteq T + dT + \cdots + d^s T$ for every $s \geq 1$. Similarly, $dT \subseteq T + Td$ yields $d^s T \subseteq T + Td + \cdots + Td^s$. Consequently,

$$T + dT + \cdots + d^s T = T + Td + \cdots + Td^s \quad \text{for every } s \geq 1. \tag{3.6}$$

We denote the set in (3.6) by $R_s$; additionally we define $R_0 = T$. Clearly, $TR_s = R_s T = R_s$ for every $s \geq 0$.

Take a left ideal $L$ of $R$. Our goal is to prove that $L$ is finitely generated. Set

$$W_0 = L \cap T \quad \text{and} \quad W_k = \{w \in T \mid d^k w \in R_{k-1} + L\}, \quad k \geq 1.$$

Using (3.6), one easily verifies that $W_k$ is a left ideal of $T$. Since $T$ is left noetherian, $W_k$ is finitely generated: $W_k = \sum_{j=1}^{r_k} T w_{jk}$ for some $w_{jk} \in W_k$. Next, using $dL \subseteq L$ and $dR_{k-1} \subseteq R_k$ we infer that $W_k \subseteq W_{k+1}$ for every $k \geq 0$. Therefore, $W_n = W_{n+1} = \ldots$ for some $n \geq 0$.

By definition of $W_k$, $k \geq 1$, for each $w_{jk}$ there is an $\ell_{jk} \in L$ such that $\ell_{jk} - d^k w_{jk} \in R_{k-1}$; for $k = 0$ we simply take $\ell_{j0} = w_{j0} \in L$. Let

$$M = \sum_{k=0}^{n} \sum_{j=1}^{r_k} R \ell_{jk}.$$

We want to show that $L$ is equal to $M$ and is therefore finitely generated. Obviously, $M \subseteq L$. Since $R$ is generated by $T \cup \{d\}$, it can be deduced from (3.6) that $R = \bigcup_{s=0}^{\infty} R_s$. Therefore, it suffices to show that $L \cap R_s \subseteq M$ for every $s \geq 0$.

We proceed by induction on $s$. For $s = 0$ we have $L \cap R_0 = W_0 \subseteq M$, so let $s \geq 1$. Take $\ell \in L \cap R_s$ and let $w_s \in T$ be such that $\ell - d^s w_s \in R_{s-1}$.

First assume $s \leq n$. Since, by the very definition, $w_s \in W_s$, we have $w_s = \sum_{j=1}^{r_s} t_j w_{js}$ for some $t_j \in T$. By (3.6), there exist elements $t'_j \in T$ such that $d^s t_j - t'_j d^s \in R_{s-1}$. A short calculation, based on $\ell_{js} - d^s w_{js} \in R_{s-1}$, shows that $\ell - \sum_{j=1}^{r_s} t'_j \ell_{js} \in R_{s-1}$. Since this is an element from $L$, we may use the induction hypothesis to conclude that it lies in $M$. As $\sum_{j=1}^{r_s} t'_j \ell_{js} \in M$ by definition, it follows that $\ell \in M$.

If $s > n$, then $w_s \in W_s = W_n$. As in the preceding paragraph we can find $t_j, t'_j \in T$ such that $w_s = \sum_{j=1}^{r_n} t_j w_{jn}$ and $d^n t_j - t'_j d^n \in R_{n-1}$. One can verify that $\ell - \sum_{j=1}^{r_n} d^{s-n} t'_j \ell_{jn} \in R_{s-1}$. Therefore, this element lies in $M$ by induction hypothesis, yielding $\ell \in M$. □

**Corollary 3.88 (Hilbert's Basis Theorem)** *If $T$ is a left noetherian unital ring, then the polynomial ring $T[\omega]$ is also left noetherian.*

***Proof*** Use Theorem 3.87 for $d = \omega$. □

We close this section and also this chapter by introducing the Weyl algebra $\mathscr{A}_1$. There are three good reasons for this. The first reason is our lack of examples of simple rings—the only ones we have encountered so far are the simple artinian rings $M_n(D)$. The second reason is that $\mathscr{A}_1$ is important, not only in algebra. Its origin is actually in quantum mechanics. The third reason is that $\mathscr{A}_1$ is mathematically appealing.

To introduce $\mathscr{A}_1$, we return to Example 2.87. Thus, let $F$ be a field and let $\mathrm{End}_F(F[\omega])$ be the algebra of all linear operators of the vector space $F[\omega]$. Define $D, L \in \mathrm{End}_F(F[\omega])$ by

$$D\big(f(\omega)\big) = f'(\omega), \qquad L\big(f(\omega)\big) = \omega f(\omega).$$

Assume further that $\mathrm{char}(F) = 0$ (unlike in Example 2.91). The subalgebra of $\mathrm{End}_F(F[\omega])$ generated by $D$ and $L$ is denoted by $\mathscr{A}_1$ and called the **Weyl algebra**. A more accurate name is **the first Weyl algebra**. We will define the $n$th Weyl algebras $\mathscr{A}_n$ at the end of the section.

The fundamental relation on which the study of $\mathscr{A}_1$ is based is the one already pointed out in Example 2.87:

$$[D, L] = I. \tag{3.7}$$

This in particular implies that $\mathscr{A}_1$ is a unital algebra. The following theorem summarizes other basic properties of $\mathscr{A}_1$.

**Theorem 3.89** *The Weyl algebra $\mathscr{A}_1$ is an (infinite dimensional) central simple noetherian domain, which is not a division algebra.*

***Proof*** Starting with (3.7), one easily shows by induction on $n$ that

$$[D, L^n] = [D, L^{n-1}]L + L^{n-1}[D, L] = nL^{n-1} \tag{3.8}$$

for every $n \in \mathbb{N}$ (here we follow the convention that $T^0 = I$ for every operator $T$). Similarly,

$$[D^n, L] = nD^{n-1}. \tag{3.9}$$

Let $\mathscr{L}$ be the subalgebra of $\mathscr{A}_1$ generated by $L$ and $I$. From (3.8) we see that $[D, \mathscr{L}] \subseteq \mathscr{L}$; in particular,

$$D\mathscr{L} \subseteq \mathscr{L} + \mathscr{L}D. \tag{3.10}$$

Using $[D^m, T] = D[D^{m-1}, T] + [D, T]D^{m-1}$, we obtain inductively that

$$[D^m, \mathscr{L}] \subseteq \mathscr{L} + \mathscr{L}D + \cdots + \mathscr{L}D^{m-1} \quad \text{for every } m \in \mathbb{N}. \tag{3.11}$$

This readily implies that the linear span of $\{TD^m \mid T \in \mathscr{L}, m \geq 0\}$ is a subalgebra of $\mathscr{A}_1$, and therefore, since it contains $L$ and $D$, it is equal to $\mathscr{A}_1$. The following thus holds:

(a) Every element in $\mathscr{A}_1$ can be written as $T_0 + T_1 D + \cdots + T_n D^n$, where $n \geq 0$ and $T_0, T_1, \ldots, T_n \in \mathscr{L}$.

We claim that the elements $T_i$ are uniquely determined. That is, we claim that

$$T_0 + T_1 D + \cdots + T_n D^n = 0 \implies T_i = 0 \text{ for each } i.$$

The proof is easy. Considering the action of the left-hand side operator on 1 we obtain $T_0(1) = 0$, and hence $T_0 = 0$. Similarly, by considering the action on $\omega$ we obtain $T_1 = 0$. We continue with $\omega^2$, $\omega^3$, etc. Note that the assumption that $\operatorname{char}(F) = 0$ is used when deriving $T_2 = 0$, $T_3 = 0$, etc.

Since $\{L^m \mid m \geq 0\}$ is obviously a basis of $\mathscr{L}$, we have thereby proved the following:

(b) The set $\{L^m D^n \mid m, n \geq 0\}$ is a basis of $\mathscr{A}_1$.

(Similarly one shows that so is the set $\{D^n L^m \mid m, n \geq 0\}$.)

Let us prove that $\mathscr{A}_1$ is simple. Take a nonzero ideal $\mathscr{I}$ of $\mathscr{A}_1$. By (a), we can take $S = \sum_{k=0}^{n} T_k D^k \in \mathscr{I}$, $T_k \in \mathscr{L}$, such that $T_n \neq 0$ and $n$ is minimal. Suppose $n > 0$. Since $L$ commutes with each $T_i$, it follows by using (3.9) that

$$[S, L] = \sum_{k=0}^{n} T_k [D^k, L] = \sum_{k=1}^{n} k T_k D^{k-1}. \tag{3.12}$$

However, since $nT_n \neq 0$ and $[S, L] \in \mathscr{I}$, this contradicts the minimality of $n$. Thus $n = 0$, i.e., $\mathscr{I} \cap \mathscr{L} \neq 0$. Now take $T = \sum_{j=0}^{m} \alpha_j L^j \in \mathscr{I} \cap \mathscr{L}$, $\alpha_j \in F$, such that $\alpha_m \neq 0$ and $m$ is minimal. Suppose $m > 0$. Using (3.8), we obtain

$$[D, T] = \sum_{j=1}^{m} j\alpha_j L^{j-1}, \tag{3.13}$$

contradicting the minimality of $m$. Therefore $m = 0$, i.e., $I \in \mathscr{I}$, and so $\mathscr{I} = \mathscr{A}_1$. This proves the simplicity of $\mathscr{A}_1$.

To show that $\mathscr{A}_1$ is central, take $S = \sum_{k=0}^{n} T_k D^k \in Z(\mathscr{A}_1)$. As $[S, L] = 0$, it follows from (3.12) (and (b)) that $T_i = 0$ for $i \geq 1$. Therefore, $S = \sum_{j=0}^{m} \alpha_j L^j$. Using $[D, S] = 0$ we see from (3.13) that $\alpha_j = 0$ for $j \geq 1$. Thus, $S = \alpha_0 \in F$.

Take $S, T \in \mathscr{L}$ and $r, s \geq 0$. Applying (3.11) it follows that

$$(SD^r)(TD^s) = STD^{r+s} + S[D^r, T]D^s = STD^{r+s} + \sum_{i=s}^{r+s-1} S_i D^i$$

for some $S_i \in \mathscr{L}$. Accordingly, for all $S_r, T_s \in \mathscr{L}$ we have

$$\Big(\sum_{r=0}^{m} S_r D^r\Big)\Big(\sum_{s=0}^{n} T_s D^s\Big) = S_m T_n D^{m+n} + \sum_{j=0}^{m+n-1} S'_j D^j$$

where $S'_j \in \mathscr{L}$. Since $\mathscr{L}$ is obviously a domain (isomorphic to $F[\omega]$), it follows from (b) that $\mathscr{A}_1$ is a domain, and also that the only invertible elements in $\mathscr{A}_1$ are the nonzero elements from $F$. In particular, $\mathscr{A}_1$ is not a division algebra.

We claim that $\mathscr{A}_1$ satisfies conditions of Theorem 3.87 with $\mathscr{L}$ playing the role of $T$ and $D$ playing the role of $d$. Indeed, $\mathscr{L}$ is easily seen to be isomorphic to $F[\omega]$ and is therefore (both left and right) noetherian. From (3.10) it follows that $\mathscr{L} + D\mathscr{L} \subseteq \mathscr{L} + \mathscr{L}D$. Analogously we see that $\mathscr{L} + \mathscr{L}D \subseteq \mathscr{L} + D\mathscr{L}$. Of course, $\mathscr{A}_1$ is generated by $\mathscr{L} \cup \{D\}$. Therefore, $\mathscr{A}_1$ is left noetherian. It can be similarly shown that $\mathscr{A}_1$ is right noetherian. □

**Remark 3.90** Theorem 3.89 implies that $\mathscr{A}_1$ is not artinian. Indeed, a simple artinian ring $M_n(D)$ is a domain only for $n = 1$, but $\mathscr{A}_1$ is not a division ring.

For any $n \in \mathbb{N}$, the **$n$th Weyl algebra**, $\mathscr{A}_n$, is defined as the subalgebra of $\mathrm{End}_F(F[\omega_1, \ldots, \omega_n])$, the algebra of all linear operators of $F[\omega_1, \ldots, \omega_n]$, generated by the operators $D_i, L_i$, $i = 1, \ldots, n$, given by

$$D_i\big(f(\omega_1, \ldots, \omega_n)\big) = \frac{\partial f}{\partial \omega_i}(\omega_1, \ldots, \omega_n), \quad L_i\big(f(\omega_1, \ldots, \omega_n)\big) = \omega_i f(\omega_1, \ldots, \omega_n).$$

The fundamental relations in $\mathscr{A}_n$ are

$$[D_i, L_j] = \delta_{ij} I, \quad [D_i, D_j] = 0, \quad [L_i, L_j] = 0.$$

Theorem 3.89 holds for $\mathscr{A}_n$ for every $n$. The proof is not much more difficult than the proof we gave. We restricted ourselves to the case where $n = 1$ primarily because of the simplicity of notation and transparency of calculations.

To provoke the reader's curiosity, we mention a problem by J. Dixmier from 1968.

**Problem 3.91** Is every nonzero algebra endomorphism of $\mathscr{A}_n$ surjective?

It has turned out that this problem, usually called **Dixmier's conjecture**, is equivalent to the famous **Jacobian conjecture** from algebraic geometry. We omit stating this conjecture here; the interested reader will easily find information about it at many places.

## Exercises

**3.1** Let $R$ be a unital ring and $M$ be a unital $R$-module. Show that $M$ is cyclic if and only if $M \cong R/L$ for some left ideal $L$ of $R$.

**3.2** Let $L$ and $N$ be submodules of a module $M$. Show that:

(a) $N/(N \cap L) \cong (N + L)/L$.
(b) If $L \subseteq N$, then $N/L$ is a submodule of $M/L$ and $(M/L)/(N/L) \cong M/N$.

**Remark** These two results, together with the standard fact that $M/\ker\varphi \cong N$ if $\varphi : M \to N$ is a surjective homomorphism, are referred to as the **isomorphism theorems** for modules. The isomorphism theorems for rings were stated in Prerequisites.

**3.3** Let $L$ be a submodule of a module $M$. Show that every submodule of $M/L$ is of the form $N/L$, where $N$ is a submodule of $M$ which contains $L$. Similarly, if $J$ is an ideal of a ring $R$, then every ideal of $R/I$ is of the form $J/I$, where $J$ is an ideal of $R$ which contains $I$.

**3.4** Let $e$ and $f$ be idempotents in a ring $R$. Show that the left ideals $Re$ and $Rf$ are isomorphic as $R$-modules if and only if there exist $u, v \in R$ such that $e = uv$ and $f = vu$. Moreover, if $R = M_n(F)$ with $F$ a field, this is further equivalent to the condition that $e$ and $f$ are similar matrices, i.e., there exists an invertible $t \in R$ such that $f = tet^{-1}$.

**3.5** Let $R$ be a unital ring and let $M$ be a free $R$-module with basis $B$. Note that every function from $B$ to a unital $R$-module $N$ can be (uniquely) extended to a homomorphism from $M$ to $N$. Hence deduce that every unital $R$-module is the homomorphic image of a free $R$-module.

**3.6** Is a submodule of a free module free?

**3.7** A module $P$ is said to be **projective** if for every surjective homomorphism $f : M \to N$ and every homomorphism $g : P \to N$ there exists a homomorphism $h : P \to M$ such that $fh = g$. Show that every free module is projective.

**3.8** Let $R$ be a unital ring. By $R^n$ we denote the direct sum of $n$ copies of the $R$-module $R$. We say that $R$ has the **invariant basis number** property ($R$ has IBN for short) if for all $m, n \in \mathbb{N}$, $R^m \cong R^n$ implies $m = n$. Note that this is equivalent to the condition that all finite bases of a free $R$-module $M$ have the same cardinality. Division rings thus have IBN. Show that if $R$ has a proper ideal $I$ such that $R/I$ has IBN, then $R$ has IBN too. Observe that this implies that commutative unital rings have IBN (just take any maximal ideal $I$).

**Remark** It turns out that if a free $R$-module has an infinite basis, then all of its bases have the same cardinality. This is true for any unital ring $R$. Therefore, $R$ has IBN if and only if all bases of any free $R$-module $M$ have the same cardinality. The term "dimension" for this cardinal number is usually used only when $R$ is a division ring, while for other rings $R$ we call it the **rank** of a free $R$-module $M$.

**3.9** Keep the notation from the preceding exercise. Show that:

(a) $R \cong R^2$ implies $R^m \cong R^n$ for all $m, n \in \mathbb{N}$.
(b) $R \cong R^2$ if and only if there exist $a_1, a_2, b_1, b_2 \in R$ such that $a_1b_1 + a_2b_2 = 1$, $b_1a_1 = b_2a_2 = 1$, and $b_1a_2 = b_2a_1 = 0$.
(c) Let $V$ be a vector space over a division ring $D$ with an infinite countable basis. Show that the ring $R = \mathrm{End}_D(V)$ contains such elements $a_1, a_2, b_1, b_2$ and thus satisfies $R^m \cong R^n$ for all $m, n \in \mathbb{N}$.

**Remark** There also exist rings such that $R^m \cong R^n$ for some, but not all, different $m$ and $n$. The interested reader should search for literature on **Leavitt algebras**, which are basic examples of such rings.

**3.10** Show that the ring $R$ from Exercise 3.9 (c) is isomorphic to the ring of all column finite $\mathbb{N} \times \mathbb{N}$ matrices over $D^{\mathrm{o}}$, i.e., matrices with the property that each of their columns has only finitely many nonzero entries.

**Remark** The column finite matrices indeed form a ring under the standard matrix operations. So do the row finite matrices. On the other hand, the set of all $\mathbb{N} \times \mathbb{N}$ matrices is merely an additive group but not a ring since infinite summations occur when performing multiplication.

**3.11** Note that a ring $R$ is isomorphic to $R^{\mathrm{o}}$ if and only if $R$ has an antiautomorphism. Show that if $R$ is such a ring, then so is $M_n(R)$. Find an example of a ring without antiautomorphisms.
*Hint:* Where does an antiautomorphism send a left unity?

**3.12** Describe the center of the ring $\mathrm{End}_D(V)$.

**3.13** Let $V$ be a finite dimensional vector space over a division ring $D$. Let $\mathrm{rank}(f)$ denote the **rank** of $f \in \mathrm{End}_D(V)$, i.e., the dimension of $\mathrm{im}\, f$. Show that $\mathrm{rank}(f) +$

$\dim_D \ker f = \dim_D V$. Further, show that $d(f, g) = \operatorname{rank}(f - g)$ defines a metric on $\operatorname{End}_D(V)$.

**3.14** Let $D$ be a division ring and let $A = (a_{ij}) \in M_n(D)$. Which of the following conditions imply that $A$ is invertible?

(a) $A$ is left invertible.
(b) $A$ is not a left zero-divisor.
(c) $\sum_{\sigma \in S_n} \operatorname{sgn}(\sigma) a_{1\sigma(1)} \dots a_{n\sigma(n)} \neq 0$.

**3.15** Show that every simple module over a unital ring is unital.

**3.16** Let $V$ be a nonzero vector space over a division ring $D$, and let $R = \operatorname{End}_D(V)$. As in Example 3.40, consider $V$ as a (simple) $R$-module. Show that $\operatorname{End}_R(V) \cong D$.

**3.17** Let $M$ be a simple module over a commutative unital ring $R$. Show that the division ring $\operatorname{End}_R(M)$ from Schur's lemma is a field isomorphic to the factor ring $R/U$ where $U$ is a maximal ideal of $R$.

**3.18** Show that a homomorphic image of a semisimple module is semisimple.

**3.19** A submodule $E$ of a module $M$ is said to be **essential** if $E$ has nontrivial intersection with every nonzero submodule of $M$. Applying Zorn's lemma, show that for every submodule $L$ of $M$ there exists a submodule $K$ such that $L \cap K = 0$ and $L \oplus K$ is essential. Use this to show that a unital module $M$ is semisimple if and only if $M$ has no proper essential submodules.

**3.20** The **left annihilator** of a subset $I$ of a ring $R$ is the left ideal $\operatorname{lann}_R(I) = \{x \in R \mid xI = 0\}$, and the **right annihilator** of $I$ is the right ideal $\operatorname{rann}_R(I) = \{x \in R \mid Ix = 0\}$. Show that if $R$ is semiprime and $I$ is an ideal, then $\operatorname{lann}_R(I) = \operatorname{rann}_R(I)$ is an ideal, and if $R$ is a semisimple, then $\operatorname{lann}_R(\operatorname{lann}_R(I)) = I$. Note that this formula is not, in general, valid if $I$ is merely a left ideal. However, a similar formula that also involves the right annihilator does hold in this case. Find it!

**3.21** Let $L$ be a submodule of a module $M$. Show that $M$ is artinian (resp. noetherian) if and only if both $L$ and $M/L$ are artinian (resp. noetherian).

**3.22** Show by an example that a finitely generated module is not necessarily noetherian (i.e., it may have submodules that are not finitely generated).
*Hint:* A unital ring $R$ is finitely (in fact, singly) generated as an $R$-module.

**3.23** Show that an injective endomorphism of an artinian module is automatically surjective. Similarly, a surjective endomorphism of a noetherian module is injective.

**3.24** Suppose a module $M$ is both artinian and noetherian. Show that for every endomorphism $\varphi$ of $M$ there exists an $n \in \mathbb{N}$ such that $M = \operatorname{im} \varphi^n \oplus \ker \varphi^n$.

**Remark** This result is known as **Fitting's lemma**.

**3.25** Let $R$ be the ring of matrices of the form $\left[\begin{smallmatrix} x & y \\ 0 & q \end{smallmatrix}\right]$ where $x, y \in \mathbb{R}$ and $q \in \mathbb{Q}$. Show that $R$ is left artinian and left noetherian, but neither right artinian nor right noetherian.

**3.26** Show that an artinian domain is a division ring.

**3.27** Show that a left artinian ring contains no infinite sequence of pairwise orthogonal nonzero idempotents.
*Hint*: By interpreting $R(1-a)$ as $\{x - xa \mid x \in R\}$ we see that Lemma 3.81 holds without the assumption that $R$ is unital.

**3.28** Let $V$ be a vector space over a division ring $D$. Show that $R = \mathrm{End}_D(V)$ is a prime ring and that the following statements are equivalent:

(i) $V$ is finite dimensional.
(ii) $R$ is simple.
(iii) $R$ is left artinian.
(iv) $R$ is left noetherian.

**3.29** Show that simple artinian rings $R$ and $S$ are isomorphic if (and only if) there exists an $n \in \mathbb{N}$ such that the rings $M_n(R)$ and $M_n(S)$ are isomorphic.

**3.30** Show that the polynomial ring $F[\Omega]$, where $F$ is a field, is noetherian if and only if the set $\Omega$ is finite. Is the ring of continuous functions $C[a, b]$ noetherian?

**3.31** Show that the only inner automorphism of the Weyl algebra $\mathscr{A}_1$ is $\mathrm{id}_{\mathscr{A}_1}$, and find an example of an outer automorphism of $\mathscr{A}_1$.
*Hint*: What restrictions on the action of an automorphism on $D$ and $L$ does $[D, L] = I$ bring?

**3.32** Show that every derivation $\delta$ of the Weyl algebra $\mathscr{A}_1$ is inner.
*Hint*: Observe that it is enough to show that there exists an $A \in \mathscr{A}_1$ such that $\delta(D) = [A, D]$ and $\delta(L) = [A, L]$. Use $[\delta(D), L] + [D, \delta(L)] = \delta([D, L]) = \delta(I) = 0$ to find such an $A$.

# Chapter 4
# Structure of Rings




Although most of the concepts introduced in the previous chapters involve general rings and algebras, the theory was centered around finite dimensional algebras and (related but more general) rings satisfying chain conditions. The theme of this chapter is the structure theory of general rings.

The topics will be considered in a different order than in Chap. 2 where we presented Wedderburn's structure theory of finite dimensional algebras. In the first part, we will treat primitive rings, which play a similar role among general rings as simple algebras among finite dimensional algebras. The second part will be primarily devoted to the Jacobson radical, a generalization of the radical of a finite dimensional algebra.

Most of the theory that we are about to expose was developed around 1945 by N. Jacobson, a student of Wedderburn.

## 4.1 Primitive Rings

Wedderburn's basic structure theorem states that a finite dimensional prime algebra can be represented as the algebra of all $n \times n$ matrices over a division algebra $D$, or, equivalently, the algebra of all linear operators on an $n$-dimensional vector space over a division algebra $\Delta = D^{\mathrm{o}}$ (see Remark 3.36). We will now introduce a class of prime rings for which a more general structure theorem holds.

Recall that, by convention, our modules are left modules. Let $R$ be a ring and let $V$ be a simple $R$-module. Schur's lemma (Lemma 3.60) tells us that

$$\Delta = \mathrm{End}_R(V)$$

is a division ring. Let $\delta \in \Delta$ and $v \in V$. We simplify the notation by writing

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M. Brešar, *Introduction to Noncommutative Algebra*, Universitext,
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$$\delta v = \delta(v) \in V. \tag{4.1}$$

Obviously, we have

$$(\delta_1 + \delta_2)v = \delta_1 v + \delta_2 v, \quad \delta(v_1 + v_2) = \delta v_1 + \delta v_2,$$
$$(\delta_1 \delta_2)v = \delta_1(\delta_2 v), \quad 1v = v$$

for all $\delta_1, \delta_2, \delta \in \Delta$ and $v_1, v_2, v \in V$. This means that $V$ is a vector space over $\Delta$; more precisely, the additive group of $V$ becomes a vector space over $\Delta$ with respect to the operation $\Delta \times V \to V$ defined by (4.1) (which is why we denoted our simple module by $V$ rather than, as before, by $M$).

From now on we will therefore consider $V$ simultaneously as a module over $R$ and a vector space over $\Delta$. Since $\delta \in \Delta$ is an endomorphism of the $R$-module $V$, the following fundamental equation

$$\delta(rv) = r(\delta v) \quad \text{for all } \delta \in \Delta, r \in R, v \in V, \tag{4.2}$$

connects both structures. Note that $\delta$ and $r$ appear symmetrically in (4.2). Accordingly, why not interpret (4.2) as that $r$ is an endomorphism of the vector space $V$ over $\Delta$? This idea will be formalized in the next section, but let us point out already now that an extra condition that $V$ is faithful is needed for its full realization. This is because if $r$ lies in the annihilator of $V$, then it gives rise to the zero endomorphism of $V$ over $\Delta$.

**Definition 4.1** A ring $R$ is said to be **primitive** if it has a faithful simple module $V$.

Recall from Definition 3.13 and Lemma 3.38 that this means that $V \neq 0$ and the following two conditions are fulfilled:

- $rV \neq 0$ for every $0 \neq r \in R$.
- $Rv = V$ for every $0 \neq v \in V$.

Since by a module we mean a left module, a more adequate name for such a ring is a **left primitive ring**. Analogously one defines a **right primitive ring**, and it turns out that these two classes of rings do not coincide. This should be kept in mind. Nevertheless, we will omit writing "left" before "primitive ring".

**Remark 4.2** Let $R$ be a primitive ring. If a ring $S$ is isomorphic to $R$, then $S$ is primitive too. Indeed, if $V$ is a faithful simple $R$-module, then the additive group of $V$ becomes a faithful simple $S$-module by defining $sv = \varphi(s)v$, where $\varphi$ is an isomorphism from $S$ onto $R$.

The discussion before Definition 4.1 indicates a simple example, in fact the prototype, of a primitive ring.

**Example 4.3** Let $V$ be a nonzero vector space over a division ring $\Delta$. Then the ring $R = \mathrm{End}_\Delta(V)$ is primitive. Indeed, $V$ is a faithful simple $R$-module (via $fv = f(v)$). The condition that $Rv = V$ if $v \neq 0$ follows from Remark 3.31 and the

fact that $v$ is contained in a basis (see Theorem 3.30). Let us emphasize that $V$ is of arbitrary dimension, finite or infinite.

More concrete examples of primitive rings will be given later. Let us now compare the class of primitive rings to other classes of rings introduced earlier.

**Lemma 4.4** *A primitive ring $R$ is prime.*

***Proof*** Let $a, b \in R$ be nonzero elements. Take a faithful simple $R$-module $V$. By faithfulness, there are $u, v \in V$ such that $au \neq 0$ and $bv \neq 0$, and by simplicity there is an $r \in R$ such that $r(bv) = u$. Accordingly, $(arb)v \neq 0$, and so $aRb \neq 0$. □

**Lemma 4.5** *A prime ring $R$ with a minimal left ideal is primitive.*

***Proof*** Let $L$ be a minimal left ideal of $R$. It is clear that $L$ is simple as an $R$-module. The primeness of $R$ implies that it is also faithful. Indeed, $rL = 0$, where $r \in R$, implies $rRL \subseteq rL = 0$, and hence $r = 0$. □

**Lemma 4.6** *A simple unital ring $R$ is primitive.*

***Proof*** Since $R$ is unital, it has a simple module by Corollary 3.51. The annihilator of every module is an ideal, so simple modules over simple rings are automatically faithful. □

A non-unital simple ring is not always primitive. Such rings may have surprising properties. In 2002, A. Smoktunowicz constructed a simple nil ring. Primitive rings, however, cannot be nil. In fact, if $R$ is a nil ring and $M$ is an $R$-module, then we clearly have $rm \neq m$ for every $r \in R$ and $0 \neq m \in M$. Therefore, in view of Lemma 3.38, nil rings do not even have simple modules.

**Lemma 4.7** *A commutative primitive ring $R$ is a field.*

***Proof*** Let $V$ be a faithful simple $R$-module. Pick $0 \neq v \in V$. We claim that for every $r \in R$, $rv = 0$ forces $r = 0$. Indeed, $Rv = V$ since $V$ is simple, hence $rV = rRv = Rrv = 0$, and so $r = 0$ by the faithfulness of $V$. Now take $0 \neq a \in R$. Then $av \neq 0$, and therefore, by the simplicity of $V$, there exists a $b \in R$ such that $b(av) = v$. Consequently, $(xba - x)v = 0$ for every $x \in R$. Therefore, $bax = xba = x$, which shows that $R$ is unital and $b = a^{-1}$. □

**Example 4.8** The ring $\mathbb{Z}$ is the simplest example of a prime ring that is not primitive.

**Example 4.9** The Weyl algebra $\mathscr{A}_1$ is a simple unital ring (see Theorem 3.89), and hence a primitive ring.

**Example 4.10** If $V$ is an infinite dimensional vector space over a division ring $\Delta$, then the ring $\mathrm{End}_\Delta(V)$ is primitive, but not simple since the set of all finite rank operators is its proper nonzero ideal (see Corollary 3.34 and its proof). In Sect. 4.4, we will see that $\mathrm{End}_\Delta(V)$ is a model of a primitive ring having minimal left ideals.

**Remark 4.11** Every $F$-algebra can be embedded into a primitive ring, namely, into $\mathrm{End}_F(V)$ for some vector space $V$ over $F$ (see Proposition 2.89). This is not always true for rings. For example, take a ring $R$ that contains nonzero elements $u$ and $v$ satisfying $2u = 3v = 0$ (e.g., $\mathbb{Z}_2 \times \mathbb{Z}_3$). In light of Remark 2.20, $R$ cannot be embedded into a prime ring, and hence, by Lemma 4.4, neither into a primitive ring.

## 4.2 The Jacobson Density Theorem

The following notion is intimately tied to the notion of a primitive ring.

**Definition 4.12** Let $V$ be a vector space over a division ring $\Delta$. A subring $R$ of the endomorphism ring $\mathrm{End}_\Delta(V)$ is said to be a **dense ring of linear operators** of $V$ if for every $n \in \mathbb{N}$, every linearly independent subset $\{u_1, u_2, \ldots, u_n\}$ of $V$, and every (not necessarily linearly independent) subset $\{v_1, v_2, \ldots, v_n\}$ of $V$, there exists an $f \in R$ such that

$$f(u_1) = v_1,\ f(u_2) = v_2, \ldots, f(u_n) = v_n.$$

The term "dense" comes from a topological characterization, which we will not need and so will not discuss here. Still, the name suggests that a dense ring of linear operators of $V$ is, in some sense, a large subring of $\mathrm{End}_\Delta(V)$, and indeed it will turn out to be large enough for some of our purposes.

**Lemma 4.13** *A ring $R$ is a dense ring of linear operators of the vector space $V$ over $\Delta$ if and only if for every $g \in \mathrm{End}_\Delta(V)$ and every finite dimensional subspace $U$ of $V$ there exists an $f \in R$ such that $f|_U = g|_U$.*

***Proof*** Take $g \in \mathrm{End}_\Delta(V)$ and a finite dimensional subspace $U$ of $V$. Pick a basis $\{u_1, u_2, \ldots, u_n\}$ of $U$. If $R$ is a dense ring, then there exists an $f \in R$ such that $f(u_i) = g(u_i)$, $i = 1, \ldots, n$. Consequently, $f|_U = g|_U$.

Proving the converse is just as easy, only standard facts about endomorphisms and bases (see Remark 3.31) have to be used. □

**Corollary 4.14** *If $V$ is a finite dimensional vector space over $\Delta$, then $\mathrm{End}_\Delta(V)$ is the only dense ring of linear operators of $V$.*

If, however, $V$ is infinite dimensional, then we have a variety of dense rings of operators on $V$. The next lemma gives a partial evidence of this.

**Lemma 4.15** *If $R$ is a dense ring of linear operators of a vector space $V$, then so are its nonzero ideals.*

***Proof*** Let $I$ be a nonzero ideal of $R$. Choose a nonzero $h \in I$, and let $u \in V$ be such that $v = h(u) \neq 0$. Now take linearly independent vectors $u_1, \ldots, u_n \in V$ and arbitrary vectors $v_1, \ldots, v_n \in V$. Since $R$ is dense, there exist $g_1, \ldots, g_n \in R$ such

that $g_i(u_i) = u$ and $g_i(u_j) = 0$ if $j \neq i$. Further, there are $f_1, \ldots, f_n \in R$ such that $f_i(v) = v_i$. Consequently,

$$k = f_1 h g_1 + \cdots + f_n h g_n \in I$$

satisfies $k(u_i) = v_i$, $i = 1, \ldots, n$. □

**Example 4.16** The ring of all finite rank operators on an infinite dimensional vector space $V$ over $\Delta$ is thus a proper dense subring of $\mathrm{End}_\Delta(V)$ (see Example 4.10).

The next theorem is one of the cornerstones of noncommutative algebra.

**Theorem 4.17 (Jacobson Density Theorem)** *A ring $R$ is primitive if and only if it is isomorphic to a dense ring of linear operators of a vector space over a division ring.*

***Proof*** Let us first resolve the easy "if" part. By Remark 4.2, it is enough to show that a dense ring of linear operators of a vector space $V$ is primitive, and this is clear since $V$ is its faithful simple module (via $fv = f(v)$), which follows already from the $n = 1$ case of Definition 4.12.

We proceed to the nontrivial "only if" part. Let $R$ be a primitive ring and let $V$ be its faithful simple module. Then $\Delta = \mathrm{End}_R(V)$ is a division ring by Schur's lemma. Recall from the preceding section that the additive group of $V$ becomes a vector space over $\Delta$ with respect to the operation defined by (4.1) (i.e., $\delta v = \delta(v)$), and that (4.2) (i.e., $\delta(rv) = r(\delta v)$) holds. Now, for each $r \in R$ we define $\overline{r} : V \to V$ by

$$\overline{r}(v) = rv. \tag{4.3}$$

It is clear that $\overline{r}$ is an additive map. Moreover, by (4.2) we have $\overline{r} \in \mathrm{End}_\Delta(V)$. Note that

$$R \to \mathrm{End}_\Delta(V), \quad r \mapsto \overline{r},$$

is a ring homomorphism; the faithfulness of $V$ implies that it is injective. Thus, $R$ is isomorphic to

$$\overline{R} = \{\overline{r} \mid r \in R\} \subseteq \mathrm{End}_\Delta(V).$$

Our goal is to prove that $\overline{R}$ is dense. Observe first that $Rv = V$ for every $v \neq 0$ in $V$ shows that the condition from Definition 4.12 holds for $n = 1$.

Let us pause for a moment. So far we have shown that, up to isomorphism, primitive rings can be characterized as rings satisfying the $n = 1$ case of Definition 4.12. This was obtained almost for free; more precisely, the arguments were straightforward, but they rested upon an ingenious recasting of the definition of primitivity. The second part of the proof is more tricky.

We claim that it is enough to prove the following:

(∗) Let $U$ be a finite dimensional subspace of $V$. For every $w \in V \setminus U$ there exists an $r \in R$ such that $\overline{r}(U) = 0$ and $\overline{r}(w) \neq 0$.

Indeed, assume (∗) and take linearly independent vectors $u_1, \ldots, u_n \in V$ and arbitrary vectors $v_1, \ldots, v_n \in V$. Given $1 \leq i \leq n$, use (∗) with the linear span of $u_1, \ldots, u_{i-1}, u_{i+1}, \ldots, u_n$ playing the role of $U$ and $u_i$ playing the role of $w$. Thus, there exists an $r_i \in R$ such that $\overline{r_i}(u_j) = 0$ if $j \neq i$ and $\overline{r_i}(u_i) \neq 0$. As we know, we can find an $s_i \in R$ such that $\overline{s_i}(\overline{r_i}(u_i)) = v_i$. Note that

$$r = s_1 r_1 + s_2 r_2 + \cdots + s_n r_n \in R$$

satisfies $\overline{r}(u_i) = v_i$, $i = 1, \ldots, n$.

Suppose (∗) is not true. Without loss of generality, we may assume that $U$ is a space of the smallest dimension for which (∗) does not hold. Note that $\dim_\Delta U \geq 1$, since (∗) is obviously true for the zero space. Let $U_0$ be any subspace of $U$ with $\dim_\Delta U_0 = \dim_\Delta U - 1$. We set

$$L = \{\ell \in R \mid \overline{\ell}(U_0) = 0\}.$$

Take a $z \in V \setminus U_0$. Since (∗) does hold for $U_0$ by our assumption, there exists an $\ell \in L$ such that $\overline{\ell}(z) \neq 0$. In view of (4.3), this means that $Lz \neq 0$. As $L$ is clearly a left ideal of $R$, $Lz$ is a submodule of the $R$-module $V$. The simplicity of $V$ therefore yields

$$Lz = V \quad \text{for all } z \in V \setminus U_0. \tag{4.4}$$

Let $w \in V \setminus U$ be a vector for which (∗) does not hold, and choose a $u \in U \setminus U_0$. We claim that the map $\delta : V \to V$ given by

$$\delta(\ell u) = \ell w \quad \text{for all } \ell \in L \tag{4.5}$$

is well-defined. First of all, since $Lu = V$ by (4.4), $\delta$ is defined on the whole space $V$. Suppose that $\ell \in L$ is such that $\ell u = 0$, i.e., $\overline{\ell}(u) = 0$. Then $\overline{\ell}$ vanishes on both $U_0$ and $\Delta u$, so $\overline{\ell}(U) = 0$. Our assumption on $w$ implies that $\overline{\ell}(w) = 0$, i.e., $\ell w = 0$. Therefore $\delta$ is indeed well-defined. One easily checks that $\delta$ is an $R$-module endomorphism, so $\delta \in \Delta$. According to (4.2) we can therefore rewrite (4.5) as

$$\ell(\delta u) = \ell w \quad \text{for all } \ell \in L,$$

that is,

$$L(\delta u - w) = 0.$$

From (4.4) it follows that $\delta u - w \in U_0$. However, this yields

$$w = \delta u - (\delta u - w) \in \Delta u + U_0 = U,$$

contradicting the choice of $w$. This proves $(*)$. □

## 4.3 Alternative Versions and Applications

We have stated the Jacobson density theorem in a condensed manner. For applications, it is sometimes useful to know that the vector space from the theorem is tightly connected with a faithful simple $R$-module $V$. Actually, the faithfulness of $V$ was needed only to assure the injectivity of the canonical homomorphism $r \mapsto \overline{r}$. If $V$ is not faithful, then this homomorphism has a nontrivial kernel, namely $\mathrm{ann}_R(V)$. In this case $R/\mathrm{ann}_R(V)$ is isomorphic to $\overline{R}$. The following *expanded version of the Jacobson density theorem* thus holds.

**Theorem 4.18** *Let $R$ be a ring and let $V$ be a simple $R$-module. Consider $V$ as a vector space over the division ring $\Delta = \mathrm{End}_R(V)$. Then the ring $R/\mathrm{ann}_R(V)$ is isomorphic to a dense ring of linear operators of $V$.*

Our next objective is the *Jacobson density theorem for algebras*. We define a **primitive algebra** as an algebra $A$ over a field $F$ that has a faithful simple $A$-module $V$; here, of course, a module means an algebra module. Therefore, $\Delta = \mathrm{End}_A(V)$ is a division $F$-algebra rather than merely a division ring. Likewise, $\mathrm{End}_\Delta(V)$ is an $F$-algebra, and $r \mapsto \overline{r}$ is an algebra homomorphism. Accordingly, Theorem 4.17 gets the following form.

**Theorem 4.19** *An $F$-algebra $A$ is primitive if and only if it is isomorphic to a dense algebra of linear operators of a vector space over a division $F$-algebra.*

We are now in a position to give yet another, more precisely the third, proof of the fundamental Wedderburn's structure theorem (Theorem 2.73). The previous two proofs were given in Sects. 2.7 and 3.10.

**Corollary 4.20** *Let $A$ be a nonzero finite dimensional algebra. The following statements are equivalent:*

(i) *$A$ is primitive.*
(ii) *$A$ is prime.*
(iii) *$A$ is simple.*
(iv) *There exist a finite dimensional division algebra $\Delta$ and a finite dimensional vector space $V$ over $\Delta$ such that $A \cong \mathrm{End}_\Delta(V)$ (and hence $A \cong M_n(D)$, where $D = \Delta^{\mathrm{o}}$ and $n = \dim_\Delta V$).*

***Proof*** We know that (iv) implies (iii) (Lemma 1.18), and trivially (iii) implies (ii). Since $A$ is finite dimensional, it has a minimal left ideal $L$. Lemma 4.5, which was stated for rings but obviously also holds for algebras, therefore tells us that (ii) implies (i). It remains to show that (i) implies (iv).

Let $A$ be primitive. From Theorem 4.19 we know that $A$ is isomorphic to a dense algebra $\overline{A}$ of linear operators of a vector space $V$ over a division algebra $\Delta$. Let $u_1, \ldots, u_m \in V$ be linearly independent over $\Delta$. Then there exist $f_1, \ldots, f_m \in \overline{A}$ such that $f_i(u_i) \neq 0$ and $f_i(u_j) = 0$ if $j \neq i$. Note that $f_1, \ldots, f_m$ are linearly independent over the base field $F$. Therefore, $m \leq [\overline{A} : F]$. Since $\overline{A}$ is finite dimensional over $F$, $V$ must be finite dimensional over $\Delta$. Corollary 4.14 shows that $\overline{A} = \mathrm{End}_\Delta(V)$ (and hence $A \cong M_n(D)$, $D = \Delta^o$, by Theorem 3.33). Obviously, $\Delta$ is also finite dimensional over $F$. □

We will see later that the Jacobson density theorem not only covers Theorem 2.73, but provides a basis for a profound generalization of the structure theory of finite dimensional algebras. The other main ingredient of this generalization is the notion of the Jacobson radical, which will be introduced in Sect. 4.6.

Let us now change the perspective. Suppose we are given a nonzero vector space $V$ over a division $F$-algebra $\Delta$ and an $F$-subalgebra $A$ of $\mathrm{End}_\Delta(V)$. When is $A$ dense? We can view $V$ as a faithful $A$-module (via $fv = f(v)$), which is simple if $Av = \{f(v) \mid f \in A\}$ equals $V$ for every nonzero $v \in V$. Let us therefore assume the latter. One should not hastily conclude that this already means that $A$ is dense. We also have to assume that $\mathrm{End}_A(V) = \Delta$ (in this context we identify $\delta \in \Delta$ with the map $V \to V$, $v \mapsto \delta v$). Obviously, $\Delta \subseteq \mathrm{End}_A(V)$. The converse inclusion does not always hold.

**Example 4.21** Consider $\mathbb{C}$ as a 2-dimensional space over $\mathbb{R}$ ($= \Delta = F$). Let $A$ be the subalgebra of $\mathrm{End}_\mathbb{R}(\mathbb{C})$ consisting of all (left) multiplication maps $L_z$, $z \in \mathbb{C}$. Obviously, $Av = \mathbb{C}$ for every $0 \neq v \in \mathbb{C}$. It is easy to see that $\mathrm{End}_A(\mathbb{C}) \cong \mathbb{C}$, so it properly contains $\mathbb{R}$. Moreover, from Corollary 4.14 we see that $A$ is not dense.

The *Jacobson density theorem for algebras of linear operators* thus reads as follows.

**Theorem 4.22** *Let $\Delta$ be a division algebra over a field $F$, let $V$ be a vector space over $\Delta$, and let $A$ be an $F$-subalgebra of $\mathrm{End}_\Delta(V)$. If $Av = V$ for every nonzero $v \in V$ and $\mathrm{End}_A(V) = \Delta$, then $A$ is a dense algebra of linear operators of $V$.*

**Remark 4.23** The condition "$Av = V$ for every nonzero $v \in V$" (i.e., the condition that $V$ is simple as an $A$-module) is equivalent to the condition that there does not exist a proper nonzero subspace $W$ of $V$ that is invariant under every $f \in A$. Indeed, just observe that $Av$ is a subspace invariant under every $f \in A$, and the statement follows.

The next two corollaries are concerned with the simplest situation where $\Delta = F$. The first one is a classical result which was, in some form, obtained by W. Burnside in 1905.

**Corollary 4.24** ***(Burnside)*** *Let $V$ be a finite dimensional vector space over an algebraically closed field $F$. If $A$ is a subalgebra of $\mathrm{End}_F(V)$ such that $Av = V$ for every nonzero $v \in V$, then $A = \mathrm{End}_F(V)$.*

***Proof*** All we have to show is that $\mathrm{End}_A(V) = F$, and this is easy: $\mathrm{End}_A(V)$ is a finite dimensional division algebra over $F$, so it must be equal to $F$ by Proposition 1.8. □

If a central simple $F$-algebra $A$ is finite dimensional, then its multiplication algebra $M(A)$ is equal to the whole algebra $\mathrm{End}_F(A)$ (Theorem 1.43). In the next corollary, first proved by E. Artin and G. Whaples a couple of years before Jacobson's discovery of the density theorem, we extend this extremely useful result to infinite dimensional algebras.

**Corollary 4.25** ***(Artin-Whaples)*** *Let $A$ be a (possibly infinite dimensional) central simple algebra. Then its multiplication algebra $M(A)$ is a dense algebra of linear operators of $A$.*

***Proof*** Take $0 \neq v \in A$. Since $A$ is simple, the ideal of $A$ generated by $v$ is equal to $A$. Thus, for every $w \in A$ there exist $a_i, b_i \in A$ such that $\sum_i a_i v b_i = w$. The condition $M(A)v = A$ is thus fulfilled.

It remains to prove that $\mathrm{End}_{M(A)}(A) = F$. That is, we have to show that each $\delta \in \mathrm{End}_{M(A)}(A)$ is a scalar multiple of $\mathrm{id}_A$. To this end we only have to use

$$\delta(L_a(1)) = L_a(\delta(1)) \quad \text{and} \quad \delta(R_a(1)) = R_a(\delta(1)).$$

Indeed, these two equations can be rewritten as $\delta(a) = a\delta(1) = \delta(1)a$ for every $a \in A$, and therefore $\delta(1) \in F$ since $A$ is central. □

## 4.4 Primitive Rings Having Minimal Left Ideals

If the Jacobson density theorem shows that all primitive rings are, in some sense, close to rings of matrices over division rings, then those with minimal left ideals are closer. This is what we intend to show in this section.

We start by recalling some results from previous chapters. Lemma 3.85 states that if a ring $R$ has minimal left ideals, then their sum is a two-sided ideal of $R$. A rough summary of Lemma 2.61, Remark 2.62, and Proposition 2.63 is that minimal one-sided ideals of a semiprime ring $R$ are precisely the one-sided ideals generated by minimal idempotents. In more precise terms, a left (resp. right) ideal $J$ of $R$ is minimal if and only if $J = Re$ (resp. $J = eR$) where $e$ is a minimal idempotent, i.e., an idempotent such that $eRe$ is a division ring. Moreover, $Re$ is a minimal left ideal of $R$ if and only if $eR$ is a minimal right ideal of $R$. In particular, a semiprime ring $R$ has minimal left ideals if and only if $R$ has minimal right ideals. More can be said.

**Lemma 4.26** *Let $R$ be a semiprime ring having minimal one-sided ideals. Then the sum of all minimal left ideals of $R$ coincides with the sum of all minimal right ideals of $R$.*

***Proof*** By left-right symmetry, it is enough to show that every minimal right ideal $J$ of $R$ is contained in the sum of all minimal left ideals. As we know, there exists an idempotent $e \in R$ such that $J = eR$ and $Re$ is a minimal left ideal of $R$. Since $e = ee \in Re$, $e$ thus lies in a minimal left ideal. As the sum of all minimal left ideals is a two-sided ideal, it contains $J = eR$. □

**Definition 4.27** Let $R$ be a semiprime ring. If $R$ has minimal one-sided ideals, then let $\operatorname{soc}(R)$ denote the sum of all minimal left ideals of $R$ (equivalently, $\operatorname{soc}(R)$ is the sum of all minimal right ideals of $R$). Otherwise we let $\operatorname{soc}(R) = 0$. We call $\operatorname{soc}(R)$ the **socle of** $R$.

**Example 4.28** As $\operatorname{soc}(R)$ is a two-sided ideal of $R$, it is equal to $R$ if $R$ is a simple ring having a minimal one-sided ideal. For example, the ring of matrices over a division ring $R = M_n(D)$ readily coincides with $\operatorname{soc}(R)$ since it is equal to the sum of its minimal left ideals $RE_{ii}$, $i = 1, \ldots, n$ (see Example 3.53).

This example is too plain to reveal the true nature of the socle. The discussion that follows gives a better insight.

From now on $R$ will be a primitive ring. Let us first recall what we already know. The Jacobson density theorem says that there is an isomorphism $r \mapsto \overline{r}$ from $R$ onto a dense ring $\overline{R}$ of linear operators of a vector space $V$ over a division ring $\Delta$. We keep this notation until the end of the section. Assume additionally that $R$ has a minimal left ideal $L$. Then we can consider $L$ as a faithful simple $R$-module (see Lemma 4.5 and its proof). Corollary 3.64 shows that $V$ and $\Delta$ can be chosen so that $V = Re$ and $\Delta \cong (eRe)^{\mathrm{o}}$ where $e$ is a minimal idempotent in $R$. However, we shall not elaborate this view any further. The clue for obtaining additional information about $\overline{R}$ is to consider *all* minimal one-sided ideals of $R$, not only the chosen minimal left ideal $L$. More precisely, the socle of $R$ will play a fundamental role.

The following example serves as a motivation for the next lemma.

**Example 4.29** An example of a minimal left ideal of $R = M_n(D)$ is $L = RE_{11}$, the set of all matrices that have zeros in all columns except in the first one (Example 2.60). By Theorem 3.33, elements in $R$ may be viewed as linear operators. Note that all nonzero elements in $L$ have rank one, and each of them generates the left ideal $L$.

**Lemma 4.30** *Let $a$ be an element of a primitive ring $R$. Then the left ideal $L$ generated by $a$ is minimal if and only if the operator $\overline{a}$ has rank one.*

***Proof*** Suppose $\overline{a}$ does not have rank one. We may assume that $a \neq 0$. Therefore there exist $u, v \in V$ such that $\overline{a}(u)$ and $\overline{a}(v)$ are linearly independent over $\Delta$. By density of $\overline{R}$ there exists an $r \in R$ such that $\overline{r}(\overline{a}(u)) = 0$ and $\overline{r}(\overline{a}(v)) \neq 0$. Consequently, the left ideal

$$I = \{x \in L \mid \overline{x}(u) = 0\}$$

is nonzero (since $0 \neq ra \in I$) and properly contained in $L$ (since $a \notin I$). Therefore, $L$ is not minimal.

Now assume that $\overline{a}$ has rank one. Let $J$ be a left ideal of $R$ such that $0 \neq J \subseteq L$. We have to show that $a \in J$. Take a nonzero $y \in J$, and let $u \in V$ be such that $\overline{y}(u) \neq 0$. By density we can find an $r \in R$ satisfying $\overline{r}(\overline{y}(u)) = \overline{a}(u)$. Note that $\overline{a}(u) \neq 0$ since $y \in L = Ra + \mathbb{Z}a$ and $\overline{y}(u) \neq 0$. Now take an arbitrary $v \in V$. Since $\overline{a}$ has rank one, we have $\overline{a}(v) = \delta\overline{a}(u)$ for some $\delta \in \Delta$. Therefore $v - \delta u \in \ker \overline{a} \subseteq \ker \overline{y}$, so $\overline{y}(v) = \delta\overline{y}(u)$. Hence,

$$\overline{ry}(v) = \overline{r}(\delta\overline{y}(u)) = \delta\overline{r}(\overline{y}(u)) = \delta\overline{a}(u) = \overline{a}(v).$$

Thus, $a = ry \in J$. □

Describing the socle of a primitive ring is now an easy task.

**Theorem 4.31** *Let $a$ be an element of a primitive ring $R$. Then $a \in \operatorname{soc}(R)$ if and only if the operator $\overline{a}$ has finite rank.*

***Proof*** If $a \in \operatorname{soc}(R)$, then $a = a_1 + \cdots + a_n$ where each $a_i$ lies in some minimal left ideal $L_i$. Lemma 4.30 shows that $\overline{a_i}$ has rank one whenever $a_i \neq 0$. Hence, $\overline{a}$ has rank at most $n$.

Conversely, assume that $\overline{a}$ has finite rank. Let $\{v_1, \ldots, v_m\}$ be a basis of the image of $\overline{a}$. Since $\overline{R}$ is dense, there exist $r_1, \ldots, r_m \in R$ such that $\overline{r_i}(v_i) = v_i$ and $\overline{r_i}(v_j) = 0$ if $j \neq i$. Note that $\overline{r_1} + \cdots + \overline{r_m}$ acts as the identity on the image of $\overline{a}$, so $a = r_1 a + \cdots + r_m a$. Since each $\overline{r_i a}$ has rank one, $r_i a$ lies in $\operatorname{soc}(R)$ by Lemma 4.30. Consequently, $a \in \operatorname{soc}(R)$. □

**Example 4.32** If $R = \operatorname{End}_\Delta(V)$, then $\operatorname{soc}(R)$ is the set of all finite rank linear operators of $V$.

We are, of course, interested in the situation where $\operatorname{soc}(R) \neq 0$, i.e., $R$ has minimal left ideals. The next corollaries highlight what Theorem 4.31 says about such rings.

**Corollary 4.33** *A primitive ring $R$ has minimal left ideals if and only if $\overline{R}$ contains a nonzero finite rank operator.*

Theorem 4.31 together with Lemma 4.15 gives the following.

**Corollary 4.34** *A primitive ring $R$ having minimal left ideals contains a nonzero ideal (in fact, its socle), which is isomorphic to a dense ring of finite rank linear operators of a vector space over a division ring.*

The point here is, of course, that these operators are of finite rank. This brings them close to operators on finite dimensional spaces, and hence close to matrices.

A primitive ring having minimal left ideals thus contains a particularly nice ideal. Do we control the whole ring if we control one of its nonzero ideals? We certainly cannot claim this for general rings. Say, if a ring $R$ is equal to the direct product of

two unrelated rings $R_1$ and $R_2$, then the properties of the ideal $R_1 \times 0$ have nothing to do with the properties of the other part $0 \times R_2$. However, prime, and therefore primitive rings cannot be decomposed in such a way. Problems on prime rings can be often reduced to the same problems on any of their nonzero ideals.

Our final corollary is a special case of the Wedderburn-Artin theory (see Theorem 3.86).

**Corollary 4.35 (Wedderburn-Artin)** *A simple unital ring $R$ having minimal left ideals is isomorphic to $M_n(D)$ for some division ring $D$ and $n \in \mathbb{N}$.*

***Proof*** The simplicity of $R$ implies that $\operatorname{soc}(R) = R$. In particular, $1 \in \operatorname{soc}(R)$, and so $\overline{1}$ has finite rank by Theorem 4.31. Being the unity of a dense ring $\overline{R}$, $\overline{1}$ must be the identity map on $V$. This means that $V$ is finite dimensional over $\Delta$, and so $\overline{R} = \operatorname{End}_\Delta(V)$ by Corollary 4.14. Invoke Theorem 3.33 and the desired conclusion follows. □

Since left artinian rings have minimal left ideals (see Lemma 3.77), Corollary 4.35 implies that a simple left artinian (unital) ring is isomorphic to the matrix ring $M_n(D)$. This result may be viewed as the heart of the Wedderburn-Artin theory, and we have now presented its third proof. The first two proofs were given in Sect. 3.10 (see the final part of the proof of Theorem 3.83 and the last paragraph of that section).

Primitive rings having minimal left ideals form an important, easily approachable, and also somewhat exclusive subclass of primitive rings. Every such ring must in particular contain a nontrivial idempotent, unless it is a division ring. A primitive domain that is not a division ring therefore cannot have minimal left ideals. A concrete example is the Weyl algebra $\mathscr{A}_1$. Another important examples are free algebras, which will be introduced in Chap. 6.

## 4.5 Primitive Ideals

Our aim now is to briefly examine the following notion that links primitive rings with the central topic of the second part of this chapter, the Jacobson radical.

**Definition 4.36** An ideal $P$ of a ring $R$ is said to be a **primitive ideal** if $P$ is the annihilator of a simple $R$-module.

(A formally more appropriate term is a *left primitive ideal*—not because it was merely a left ideal, it certainly is two-sided, but because "module" means "left module" by our convention.)

Clearly, $R$ is a primitive ring if and only if 0 is a primitive ideal of $R$. More generally, the following holds.

**Lemma 4.37** *An ideal $P$ of a ring $R$ is a primitive ideal if and only if $R/P$ is a primitive ring.*

***Proof*** Let $P$ be a primitive ideal of $R$ and let $M$ be a simple $R$-module such that $P = \operatorname{ann}_R(M)$. The additive group of $M$ becomes a simple $R/P$-module by defining

$$(r + P)m = rm \quad \text{for all } r \in R, m \in M.$$

This operation is well-defined since $P \subseteq \operatorname{ann}_R(M)$. On the other hand, from $P \supseteq \operatorname{ann}_R(M)$ we infer that $M$ is a faithful $R/P$-module. The ring $R/P$ is therefore primitive.

The proof of the converse is analogous. Starting with a faithful simple $R/P$-module $N$, we note that by defining

$$rn = (r + P)n \quad \text{for all } r \in R, n \in N,$$

the additive group of $N$ becomes a simple $R$-module (the well-definedness is not an issue here). It is clear that $\operatorname{ann}_R(N) = P$, so $P$ is a primitive ideal. □

**Example 4.38** Let $R = R_1 \times \cdots \times R_n$, where each $R_i$ is a primitive ring. Then

$$P_i = R_1 \times \cdots \times R_{i-1} \times 0 \times R_{i+1} \times \cdots \times R_n \tag{4.6}$$

is a primitive ideal. Indeed, $R/P_i \cong R_i$ is a primitive ring. In the special case when each $R_i$ is a simple unital ring, Lemma 2.44 readily implies that the $P_i$'s are also the only primitive ideals of $R$. Moreover, in this case they are the maximal ideals.

Observe that this example in particular shows that in semisimple rings there is no difference between primitive ideals and maximal ideals. What is the connection between them in general? We first note that not every primitive ideal is maximal.

**Example 4.39** The zero ideal of a non-simple primitive ring is primitive, but not maximal. A concrete example of such a ring is $\operatorname{End}_\Delta(V)$ where $V$ is an infinite dimensional vector space over a division ring $\Delta$ (see Example 4.10).

If $R$ is a ring with zero multiplication, then its maximal ideals are not primitive. However, this is a rather special example. Maximal ideals usually are primitive. To show this, we need a little lemma.

**Lemma 4.40** *Let $W$ be an ideal of a ring $R$. If $R/W$ is a simple ring, then $W$ is a maximal ideal. Conversely, if $W$ is maximal and $R^2 \not\subseteq W$ (this is automatically fulfilled if $R$ is unital), then $R/W$ is a simple ring.*

***Proof*** Let $R/W$ be simple. If $J$ is an ideal of $R$ such that $W \subseteq J \subseteq R$, then $J/W$ is an ideal of $R/W$, so $J/W = 0$ or $J/W = R/W$. This readily implies $J = W$ or $J = R$, proving the maximality of $W$.

If $I$ is an ideal of $R/W$, then $Z = \{r \in R \mid r + W \in I\}$ is an ideal of $R$ which contains $W$. Assuming that $W$ is maximal it follows that $Z = W$ or $Z = R$, that is, $I = 0$ or $I = R/W$. Therefore, $R/W$ is a simple ring provided that $(R/W)^2 \neq 0$, i.e., $R^2 \not\subseteq W$. □

**Corollary 4.41** *Every maximal ideal of a unital ring $R$ is a primitive ideal.*

***Proof*** Since simple unital rings are primitive (Lemma 4.6), this follows from Lemmas 4.37 and 4.40. □

Maximal ideals of $\mathbb{Z}$ and $C[a, b]$, described in Example 3.46, are thus examples of primitive ideals. Actually, these are also the only examples, since, like in semisimple rings, the converse of Corollary 4.41 holds in commutative rings.

**Corollary 4.42** *Every primitive ideal of a commutative ring $R$ is a maximal ideal.*

***Proof*** If $P$ is a primitive ideal of $R$, then $R/P$ is a primitive commutative ring by Lemma 4.37, and hence a field by Lemma 4.7. Therefore, $P$ is maximal by Lemma 4.40. □

A *primitive ideal of an algebra $A$* is, of course, defined as the annihilator of a simple (algebra) $A$-module. As one would expect, all results from this section still hold if we replace "ring" by "algebra". Corollary 4.20 shows that finite dimensional primitive algebras are automatically simple, and so a similar argument as in the proof of Corollary 4.42 gives the following.

**Corollary 4.43** *Every primitive ideal of a finite dimensional algebra $A$ is a maximal ideal.*

Primitive ideals are also related to maximal left ideals.

**Lemma 4.44** *If $P$ is a primitive ideal of a ring $R$, then there exists a maximal left ideal $U$ of $R$ such that $P = \{x \in R \mid xR \subseteq U\}$. Conversely, if $U$ is a maximal left ideal of $R$ and $R^2 \not\subseteq U$ (this is automatically fulfilled if $R$ is unital), then $P = \{x \in R \mid xR \subseteq U\}$ is a primitive ideal of $R$.*

***Proof*** Let $M$ be a simple $R$-module such that $P = \operatorname{ann}_R(M)$. By Lemma 3.48, there exists a maximal left ideal $U$ of $R$ such that $M \cong R/U$. Therefore, $P = \operatorname{ann}_R(R/U) = \{x \in R \mid xR \subseteq U\}$.

Assume now that $U$ is a maximal left ideal of $R$ such that $R^2 \not\subseteq U$. Lemma 3.48 tells us that $R/U$ is a simple $R$-module. Consequently, $\operatorname{ann}_R(R/U) = \{x \in R \mid xR \subseteq U\} = P$ is a primitive ideal of $R$. □

**Corollary 4.45** *Every maximal left ideal $U$ of a unital ring $R$ contains a primitive ideal $P$.*

## 4.6 Introducing the Jacobson Radical

Recall that the radical of a finite dimensional algebra $A$ is defined as the (necessarily unique) maximal nilpotent ideal of $A$. A simple-minded attempt to use the same definition for general rings immediately fails, simply because rings may not have maximal nilpotent ideals (Example 2.12).

Anyway, generalizations should have some purpose. Our goal is to define the radical of a ring $R$ as an ideal consisting of all the most "undesired" elements in $R$, that is, the elements whose presence creates an obstacle for understanding the structure of $R$. Further, we wish the factor ring of $R$ modulo the radical to be "nice" in the sense that its radical is zero, so that we can say something about its structure.

As witnessed by Theorem 2.74, the radical of a finite dimensional algebra does satisfy these criteria: the corresponding factor algebra is the direct product of full matrix algebras over division algebras. Of course, we have to be realistic in our expectations when trying to extend this result to general rings. One cannot hope that the structure of "nice" rings is as simple as the structure of "nice" finite dimensional algebras.

There are different radicals defined for general rings, each of them having its own relevance. From the point of view of the structure theory, the following radical is arguably the most important.

**Definition 4.46** The **Jacobson radical of a ring** $R$, denoted $\mathrm{rad}(R)$, is the intersection of all the primitive ideals of $R$. If $R$ has no primitive ideals (i.e., $R$ has no simple modules), then we define $\mathrm{rad}(R) = R$.

In other words, $\mathrm{rad}(R)$ consists of all elements that annihilate every simple $R$-module. It is obviously a (two-sided) ideal of $R$. Since our modules are actually left modules, apparently a more appropriate name for $\mathrm{rad}(R)$ would be the *left Jacobson radical*. Fortunately, it turns out that the Jacobson radical is a left-right symmetric notion, so this terminological query is unnecessary.

The meaning of the Jacobson radical may not be evident from the definition. In what follows we will discover several properties and characterizations of $\mathrm{rad}(R)$, and thereby give a better insight into its nature. It truly is an amazing concept; we shall see that it touches several seemingly unrelated notions, from simple modules and primitive rings (and hence dense rings of linear operators) to maximal (two-sided and one-sided) ideals, nilpotency and invertibility.

Let us first clarify the relation to nilpotent ideals. Actually, it is not more demanding to consider more general nil ideals.

**Lemma 4.47** *Every nil left or right ideal of a ring $R$ is contained in $\mathrm{rad}(R)$.*

***Proof*** Let $I$ be a nil left (resp. right) ideal. If the lemma was not true, there would exist a simple $R$-module $M$, $u \in I$ and $m \in M$ such that $n = um \neq 0$. Since $Rn = M$ (Lemma 3.38), $xn = m$ for some $x \in R$. Hence, $v = xu \in I$ (resp. $v = ux \in I$) satisfies $vm = m$ (resp. $vn = n$), and hence $v^k m = m$ (resp. $v^k n = n$) for every $k \in \mathbb{N}$. But this is impossible since $v$ is nilpotent. □

Later we will see that even a domain may have a large Jacobson radical. Therefore, $\mathrm{rad}(R)$ is not always nil, let alone nilpotent.

It goes without saying that the **Jacobson radical of an algebra** $A$, $\mathrm{rad}(A)$, is defined as the intersection of all primitive (algebra) ideals of $A$. We may, on the other hand, treat an algebra merely as a ring. Fortunately, the Jacobson radical of the ring $A$ coincides with the Jacobson radical of the algebra $A$. We suppress the proof

of this fact (see, e.g., [5, p. 451]). We will not use it anyway. The only information about the Jacobson radical of an algebra that we need is that it contains all nilpotent ideals. This is clear from the proof of Lemma 4.47.

**Theorem 4.48** *If $A$ is a finite dimensional algebra, then $\mathrm{rad}(A)$ is the unique maximal nilpotent ideal of $A$.*

***Proof*** Let $N$ be the unique maximal nilpotent ideal of $A$ (its existence is explained in Sect. 2.1). We know that $N \subseteq \mathrm{rad}(A)$, so we only need to prove the converse inclusion.

Take $q \in \mathrm{rad}(A)$. We claim that $q + N \in \mathrm{rad}(A/N)$. If this was not true, there would be a simple $A/N$-module $M$ such that $(q + N)M \neq 0$. Note that the additive group of $M$ becomes a simple $A$-module by defining

$$xm = (x + N)m \quad \text{for all } x \in A, m \in M.$$

Since $q \in \mathrm{rad}(A)$, we have $qM = 0$. This contradicts $(q + N)M \neq 0$, and so our claim is proved.

Let us complete the proof by showing that $\mathrm{rad}(A/N) = 0$. By Theorem 2.74 we know that $A/N$ is isomorphic to a finite direct product of full matrix algebras over division algebras. We have thus arrived at the situation considered in Example 4.38. Defining primitive (in fact, maximal) ideals $P_i$ as in (4.6) we clearly have $\bigcap_i P_i = 0$. Consequently, $\mathrm{rad}(A/N) = 0$. □

The Jacobson radical of a finite dimensional algebra $A$ thus coincides with the radical of $A$ introduced in Sect. 2.1.

## 4.7 Quasi-Invertibility

The Jacobson radical is intimately connected with the concept of invertibility. This statement does not seem to make much sense for non-unital rings. However, we will deal with the following notions which make it possible to imitate invertibility.

**Definition 4.49** An element $y$ in a ring $R$ is said to be **left quasi-invertible** if there exists an $x \in R$ such that

$$x + y = xy. \tag{4.7}$$

We call $x$ a **left quasi-inverse** of $y$. Analogously, $x \in R$ is said to be **right quasi-invertible** if there exists a $y \in R$, called a **right quasi-inverse** of $x$, satisfying (4.7).

If $R$ is unital, then (4.7) can be written as

$$(1 - x)(1 - y) = 1.$$

Therefore,

$$x \text{ is a left quasi-inverse of } y \iff 1 - x \text{ is a left inverse of } 1 - y.$$

In particular,

$$y \text{ is left quasi-invertible} \iff 1 - y \text{ is left invertible.}$$

(Note that 1 is not left quasi-invertible, but 0 is.) A similar statement can be made for right quasi-invertibility.

There is another way of looking at these notions. Define $\circ : R \times R \to R$ by

$$x \circ y = x + y - xy.$$

This operation is associative. Indeed, one immediately checks that

$$(x \circ y) \circ z = x + y + z - xy - xz - yz + xyz = x \circ (y \circ z).$$

Clearly $x \circ 0 = 0 \circ x = x$. Accordingly, $R$ endowed with $\circ$ is a monoid with identity element 0. We can interpret (4.7) as that $x$ is a left inverse of $y$ with respect to $\circ$. Since the left and right inverses in a monoid coincide, $x \circ y = y \circ z = 0$ implies $x = z$. We can reword this as follows.

**Lemma 4.50** *If an element $y$ has a left quasi-inverse $x$ and a right quasi-inverse $z$, then $x = z$.*

**Definition 4.51** An element $y \in R$ is said to be **quasi-invertible** if there exists an $x \in R$ which is both a left and right quasi-inverse of $y$. In this case we call $x$ a **quasi-inverse** of $y$.

In other words, $x$ is a quasi-inverse of $y$ if it satisfies (4.7) and $xy = yx$. Actually, $x$ is *the* quasi-inverse of $y$; its uniqueness follows from Lemma 4.50. We also see that $y \in R$ is quasi-invertible if and only if it is both left and right quasi-invertible. If $R$ is unital, this is further equivalent to the invertibility of $1 - y$.

**Example 4.52** Every nilpotent element $y \in R$ is quasi-invertible. Indeed, its quasi-inverse is $-\sum_{n=1}^{\infty} y^n$ (the nilpotency of $y$ implies that this sum is actually finite). If $R$ is unital, we can rewrite this observation in the familiar form: $(1 - y)^{-1} = \sum_{n=0}^{\infty} y^n$.

**Definition 4.53** We say that $S \subseteq R$ is a **quasi-invertible set** if all elements in $S$ are quasi-invertible. Analogously we define a **left (and right) quasi-invertible set**.

**Lemma 4.54** *If a left ideal $L$ is left quasi-invertible, then $L$ is quasi-invertible.*

***Proof*** Take $\ell \in L$. By assumption, $\ell$ has a left quasi-inverse $k \in R$. From $k = k\ell - \ell$ we see that $k$ also lies in $L$. Thus, $k$ has a left quasi-inverse $m$. Since $\ell$ is a right quasi-inverse of $k$, Lemma 4.50 implies that $\ell = m$. This means that $k$ is the quasi-inverse of $\ell$. □

This lemma will eventually lead to the left-right symmetry of the Jacobson radical.

If $R$ is unital and $1 - a \in R$ is not left invertible, then $R(1 - a)$ is a proper left ideal of $R$ and is so contained in a maximal left ideal $U$ (Lemma 3.47), which gives rise to a simple $R$-module $R/U$ (Lemma 3.48). These observations can be extended to the non-unital case.

**Lemma 4.55** *Let $R$ be an arbitrary ring. If $a \in R$ is not left quasi-invertible, then there exists a maximal left ideal $U$ of $R$ such that $a \notin U$. Moreover, $R/U$ is a simple $R$-module and $a \notin \operatorname{ann}_R(R/U)$.*

***Proof*** The assumption that $a$ is not left quasi-invertible can be read as that $a$ does not lie in the left ideal $L = \{ra - r \mid r \in R\}$. We claim that $a$ also does not lie in any proper left ideal $L'$ containing $L$. Indeed, writing $r \in R$ as $r = -(ra - r) + ra$, we see that $R$ is the only left ideal of $R$ that contains $L \cup \{a\}$. This makes it possible for us to apply Zorn's lemma in a similar way as in the proof of Lemma 3.47. Let $\mathscr{S}$ be the set of all proper left ideals of $R$ that contain $L$. Partially order $\mathscr{S}$ by inclusion. If $\{K_i \mid i \in I\}$ is a chain in $\mathscr{S}$, then $K = \bigcup_{i \in I} K_i$ is its upper bound. Indeed, $K$ is a proper left ideal since $a \notin K$. Therefore, $\mathscr{S}$ has a maximal element $U$. Clearly, $U$ is a maximal left ideal and $a \notin U$. Furthermore, since $a^2 - a \in L \subseteq U$ it follows that $a^2 \notin U$. Lemma 3.48 now tells us that $R/U$ is a simple $R$-module. Finally, $a(a + U) \neq 0$ shows that $a \notin \operatorname{ann}_R(R/U)$. □

This lemma in particular shows that for every element that is not left quasi-invertible we can find a simple module which is not annihilated by this element. Elements from $\operatorname{rad}(R)$ are therefore necessarily left quasi-invertible, and hence quasi-invertible by Lemma 4.54. This observation can be sharpened as follows.

**Theorem 4.56** *Let $R$ be a ring and let $q \in R$. The following statements are equivalent:*

(i) *The ideal generated by $q$ is quasi-invertible.*
(ii) *The left ideal $Rq$ is quasi-invertible.*
(iii) $q \in \operatorname{rad}(R)$.

***Proof*** (i) $\Longrightarrow$ (ii). Trivial.

(ii) $\Longrightarrow$ (iii). Suppose $q \notin \operatorname{rad}(R)$. Then there exists a simple $R$-module $M$ such that $qm \neq 0$ for some $m \in M$. Accordingly, $Rqm = M$ by Lemma 3.38. Therefore, $\ell m = m$ for some $\ell \in Rq$. Since $\ell$ is quasi-invertible, there exists an $r \in R$ such that $r + \ell = r\ell$. Consequently, $m = \ell m = r\ell m - rm = 0$, which contradicts $qm \neq 0$.

(iii) $\Longrightarrow$ (i). As noticed before the statement of the theorem, Lemmas 4.54 and 4.55 show that $\operatorname{rad}(R)$ is a quasi-invertible ideal. Its subsets are therefore also quasi-invertible. □

By making obvious modifications in the proof, one shows that (i) is also equivalent to (ii') the *right* ideal $qR$ is quasi-invertible, and (iii') $q$ annihilates all simple right $R$-modules. We have thereby established that the *left* Jacobson

radical coincides with the *right* Jacobson radical (see the comment following Definition 4.46).

The following corollary is immediate.

**Corollary 4.57** *The Jacobson radical of a ring $R$ is a quasi-invertible ideal which contains every quasi-invertible one-sided ideal of $R$.*

Incidentally, Corollary 4.57 implies Lemma 4.47 since nilpotent elements are quasi-invertible (see Example 4.52).

**Corollary 4.58** *The Jacobson radical of a unital ring $R$ is equal to the intersection of all the maximal left ideals of $R$.*

***Proof*** Let $T$ denote the intersection of all the maximal left ideals of $R$. Clearly, $T$ is a left ideal. Lemma 4.55 along with Lemma 4.54 implies that $T$ is quasi-invertible. Hence $T \subseteq \text{rad}(R)$ by Corollary 4.57. The converse inclusion $\text{rad}(R) \subseteq T$ follows from Corollary 4.45. □

It is interesting in its own right that the intersection of all the maximal left ideals is a two-sided ideal. Of course we may replace "left" by "right" in the statement of Corollary 4.58.

For the reader's convenience, in the next corollary we gather together various characterizations of the Jacobson radical of a unital ring.

**Corollary 4.59** *Let $R$ be a nonzero unital ring. The following statements are equivalent for $q \in R$:*

(i) $q \in \text{rad}(R)$.
(ii) $qM = 0$ *for every simple $R$-module $M$.*
(iii) $q$ *lies in every ideal $P$ of $R$ such that $R/P$ is a primitive ring.*
(iv) $1 - xq$ *is invertible for all $x \in R$.*
(v) $1 - \sum_i x_i q y_i$ *is invertible for all $x_i, y_i \in R$.*
(vi) $q$ *lies in every maximal left ideal of $R$.*

One can add appropriate right versions of (ii)–(iv) and (vi).

## 4.8 Computing the Jacobson Radical

After discovering a variety of abstract descriptions of the Jacobson radical, it may still be unclear how to determine $\text{rad}(R)$ for a given ring $R$. There is no magical device for this, but we can gain intuition by examining some special situations.

Let us begin with two illustrative examples. Since we know that the Jacobson radical of a finite dimensional algebra is the maximal nilpotent ideal (Theorem 4.48), our first example is just a repetition of Example 2.14.

**Example 4.60** The Jacobson radical of $T_n(F)$, the algebra of all upper triangular $n \times n$ matrices over a field $F$, is the set of all strictly upper triangular matrices.

The ring in the next example has no nonzero nilpotent elements, in fact it is a domain. Yet its Jacobson radical is rather big.

**Example 4.61** Take a division ring $D$ and consider the ring $D[[\omega]]$ of formal power series. Let $T$ denote its ideal generated by $\omega$, i.e., $T$ is the set of all elements with zero constant term. Lemma 1.55 says that $f \in D[[\omega]]$ is not invertible if and only if $f \in T$. This readily implies that every proper one-sided ideal of $D[[\omega]]$ is contained in $T$. Therefore, $T$ is the unique maximal left ideal of $D[[\omega]]$. Corollary 4.58 shows that $\mathrm{rad}(D[[\omega]]) = T$.

The Jacobson radical is compatible with various ring constructions. Let us consider only the simplest construction, i.e., the direct product.

**Proposition 4.62** *Let $\{R_i \mid i \in I\}$ be a family of rings. Then*

$$\mathrm{rad}(\Pi_{i\in I} R_i) = \Pi_{i\in I}\mathrm{rad}(R_i).$$

***Proof*** Note that $(r_i)$ is quasi-invertible in $R = \Pi_{i\in I} R_i$ if and only if $r_i$ is quasi-invertible in $R_i$ for every $i$. Accordingly, $q = (q_i) \in R$ is such that $Rq$ is quasi-invertible in $R$ if and only $R_i q_i$ is quasi-invertible in $R_i$ for every $i \in I$. Thus, by Theorem 4.56, $q \in \mathrm{rad}(R)$ if and only if $q_i \in \mathrm{rad}(R_i)$ for every $i \in I$. □

Relating the Jacobson radical of a ring to the Jacobson radical of some of its subrings is another topic of interest. For ideals, the relation is simple and nice.

**Proposition 4.63** *If $I$ is an ideal of a ring $R$, then* $\mathrm{rad}(I) = I \cap \mathrm{rad}(R)$.

***Proof*** Let $y \in I \cap \mathrm{rad}(R)$. As an element of $\mathrm{rad}(R)$, $y$ is quasi-invertible in $R$. If $x$ is its quasi-inverse, then we see from $x = xy - y$ that $x$ also lies in $I$. This shows that $I \cap \mathrm{rad}(R)$ is a quasi-invertible ideal of the ring $I$. Therefore, $I \cap \mathrm{rad}(R) \subseteq \mathrm{rad}(I)$ by Corollary 4.57.

Proving the converse inclusion is a bit more tricky. Take $q \in \mathrm{rad}(I)$ and $r \in R$. By Theorem 4.56, it is enough to show that $u = rq$ is quasi-invertible in $R$. Since $u^2 = (rqr)q \in I \cdot \mathrm{rad}(I) \subseteq \mathrm{rad}(I)$, it follows that $u^2$ is quasi-invertible. We claim that this implies that $u$ is also quasi-invertible. In unital rings this follows easily from $1 - u^2 = (1-u)(1+u)$, but of course we do not wish to assume the existence of unity. At any rate, the proof is easy. A direct computation shows that if $v$ is the quasi-inverse of $u^2$, then $vu + v - u$ is a left quasi-inverse of $u$ and $uv + v - u$ is a right quasi-inverse of $u$. Lemma 4.50 thus shows that $u$ is quasi-invertible (and $uv = vu$). □

If the size of $\mathrm{rad}(R)$ measures the intricacy of the structure of $R$, the following definition considers the worse situation.

**Definition 4.64** A ring $R$ is said to be a **radical ring** if $\mathrm{rad}(R) = R$.

Note that a radical ring is necessarily non-unital.

**Example 4.65** Every nil ring is a radical ring by Lemma 4.47.

**Proposition 4.66** *The following statements are equivalent for a ring $R$:*

(i) *$R$ is a radical ring.*
(ii) *Every element in $R$ is quasi-invertible.*
(iii) *$R$ has no simple modules.*

***Proof*** From Corollary 4.57 we see that (i) and (ii) are equivalent. If $R$ is a radical ring, then $R$ cannot have simple modules since their annihilators are always proper subsets of $R$. Thus, (i) implies (iii), while the converse holds by Definition 4.46. □

**Proposition 4.67** *The Jacobson radical of an arbitrary ring $R$ is a radical ring, i.e.,* $\mathrm{rad}(\mathrm{rad}(R)) = \mathrm{rad}(R)$.

***Proof*** Take $\mathrm{rad}(R)$ for $I$ in Proposition 4.63. □

Examples 4.60 and 4.61 therefore also yield concrete examples of radical rings. The one from the first example, the ring of all strictly upper triangular matrices, is nilpotent.

## 4.9 Semiprimitive Rings

After considering the most unfavorable situation where $\mathrm{rad}(R) = R$, we now examine the opposite extreme.

**Definition 4.68** A ring $R$ is said to be a **semiprimitive ring** if $\mathrm{rad}(R) = 0$.

Another name for a semiprimitive ring is a **Jacobson semisimple ring**.

**Example 4.69** Every primitive ring is semiprimitive, since 0 is its primitive ideal. More generally, the direct product of primitive rings (in fact, even of semiprimitive rings) is semiprimitive by Proposition 4.62.

**Example 4.70** If $R$ is a nonzero commutative unital ring, then $\mathrm{rad}(R)$ is equal to the intersection of all the maximal ideals of $R$ by Corollary 4.58. Recall from Example 3.46 that $U$ is a maximal ideal of $R$ if and only if $R/U$ is a field. We thus have:

(a) Since the intersection of all the maximal ideals $p\mathbb{Z}$ of $\mathbb{Z}$ is 0 (here, $p$ is a prime number), $\mathbb{Z}$ is a semiprimitive ring.
(b) Similarly, since $\bigcap_{a \le c \le b} U_c = 0$, where $U_c = \{f \in C[a,b] \mid f(c) = 0\}$, $C[a,b]$ is a semiprimitive ring.

A semiprimitive ring $R$ cannot have nonzero nil ideals by Lemma 4.47. In particular, semiprimitive rings are thus semiprime. The converse is not true.

**Example 4.71** Let $D$ be a division ring. In Example 4.61 we saw that $D[[\omega]]$ has a big Jacobson radical. Thus, $D[[\omega]]$ is an example of a semiprime ring (in fact, a domain) that is not semiprimitive. The polynomial ring $D[\omega]$, however, is semiprimitive. Indeed, take $f \in \mathrm{rad}(D[\omega])$. Then $f$ is quasi-invertible by

Corollary 4.57. That is to say, $1 - f$ is invertible. Since polynomials of degree 0 are clearly the only invertible elements in $D[\omega]$, and since $f \in \mathrm{rad}(D[\omega])$ cannot be invertible, it follows that $f = 0$.

The reader might have noticed that we did not use the full force of the assumption that $D$ is a division ring. Instead of analyzing what else the above simple argument brings, we mention a much deeper result by S. A. Amitsur from 1956: Every ring $R$ has a nil ideal $N$ such that $\mathrm{rad}(R[\omega]) = N[\omega]$. Consequently, $R[\omega]$ is semiprimitive if $R$ has no nonzero nil ideal.

Amitsur's theorem is not the end of the story. A precise description of $N$ is still a mystery. It turns out that the validity of the statement "*If $R$ is nil, then $N = R$*" is equivalent to a positive answer to Köthe's problem.

It is not always easy to find out whether or not a ring is semiprimitive. Here is a famous old problem.

**Problem 4.72** Let $G$ be a group and let $F$ be a field with $\mathrm{char}(F) = 0$. Is the group algebra $F[G]$ semiprimitive?

The answer is positive if $G$ is finite. Indeed, this is just a rewording of Maschke's theorem since, as is evident from Theorem 4.48, a finite dimensional algebra is semiprime if and only if it is semiprimitive. Thus, we are asking whether Maschke's theorem can be generalized to infinite groups. It turns out that the problem can be reduced to the case where $F = \mathbb{Q}$.

Let us return to problems we can handle. At the beginning of Sect. 4.6, we said that a decent radical should have the property that the factor ring of $R$ modulo the radical has zero radical. We still owe the proof that the Jacobson radical does satisfy this property.

**Lemma 4.73** *The ring $R/\mathrm{rad}(R)$ is semiprimitive for every ring $R$.*

***Proof*** The case where $R$ is a radical ring is trivial. By Proposition 4.66, we may therefore assume that $R$ has a simple module $M$. The additive group of $M$ becomes a simple module over $R/\mathrm{rad}(R)$ by defining

$$(x + \mathrm{rad}(R))m = xm \quad \text{for all } x \in R, m \in M.$$

Checking this is straightforward, including the well-definedness which follows from $\mathrm{rad}(R)M = 0$. Now, if $q + \mathrm{rad}(R) \in \mathrm{rad}(R/\mathrm{rad}(R))$, then $qM = (q + \mathrm{rad}(R))M = 0$. Since $M$ is an arbitrary simple $R$-module, this gives $q \in \mathrm{rad}(R)$. □

Another debt to the reader is an explanation why semiprimitive rings are supposed to be "nice". Why would it be easier to deal with a semiprimitive ring than with any old ring? For primitive rings this is clear as we have the Jacobson density theorem. Direct products of primitive rings can therefore also be viewed as easily approachable. Unfortunately, not every semiprimitive ring can be represented as such a product (say, $\mathbb{Z}$ cannot). We are forced to deal with the following generalization of the direct product.

**Definition 4.74** Let $\{R_i \mid i \in I\}$ be a family of rings. A subring $R$ of the direct product $\Pi_{i\in I} R_i$ is called a **subdirect product** of the family $\{R_i \mid i \in I\}$ if for each $j \in I$, the map $\pi_j : R \to R_j$, $\pi_j((r_i)) = r_j$, is surjective.

If $I = \{1, 2\}$, this means that for every $r_1 \in R_1$ there exists an $r_2 \in R_2$ such that $(r_1, r_2) \in R$, and similarly, for every $r_2 \in R_2$ there exists an $r_1 \in R_1$ such that $(r_1, r_2) \in R$. For instance, every ring $R$ is isomorphic to the subdirect product of two copies of $R$ via $r \mapsto (r, r)$. Subdirect products are thus quite different from direct products.

**Lemma 4.75** *A nonzero ring $R$ is semiprimitive if and only if $R$ is isomorphic to a subdirect product of primitive rings.*

***Proof*** Let $R$ be semiprimitive. Denote the family of all the primitive ideals of $R$ by $\{P_i \mid i \in I\}$. Recall that the rings $R/P_i$ are primitive (Lemma 4.37). By assumption, $\bigcap_{i\in I} P_i = 0$. This shows that the homomorphism

$$R \to \Pi_{i\in I} R/P_i, \quad r \mapsto (r + P_i) \tag{4.8}$$

is injective. Its image is clearly a subdirect product of the family $\{R/P_i \mid i \in I\}$.

Conversely, take a family of primitive rings $\{R_i \mid i \in I\}$ and let $\varphi : R \to \Pi_{i\in I} R_i$ be an injective homomorphism such that $\pi_j(\varphi(R)) = R_j$ for every $j \in I$. Set

$$P_j = \ker \pi_j \varphi.$$

Since $R/P_j \cong R_j$ we see, again by Lemma 4.37, that $P_j$ is a primitive ideal of $R$. If $r \in \bigcap_{i\in I} P_i$, then $\varphi(r) = 0$ and hence $r = 0$. Thus, $\operatorname{rad}(R) \subseteq \bigcap_{i\in I} P_i = 0$. □

**Example 4.76** Let $R = A$ be a finite dimensional semiprimitive algebra. As mentioned above, $A$ is semiprime, and hence, by Wedderburn's theorem, isomorphic to a finite direct product of simple (= primitive in this context) algebras. Therefore, $A$ has only finitely many primitive ideals (see Example 4.38). Note that the homomorphism from (4.8) is surjective and hence an isomorphism.

**Example 4.77** From Example 4.70 and the proof of Lemma 4.75 we infer the following:

(a) $\mathbb{Z}$ is isomorphic to a subdirect product of the fields $\mathbb{Z}_p$, $p$ prime.
(b) $C[a, b]$ is isomorphic to a subdirect product of the fields $C[a, b]/U_c \cong \mathbb{R}$, $a \le c \le b$ (see Example 3.46).

Roughly speaking, Lemma 4.75 shows that a semiprimitive ring can be decomposed into primitive factors, however, in a fashion which is not always easily controllable. When making this decomposition we somehow lose track of the way the components are "glued together". The example with $C[a, b]$ is illustrative. The direct product of the fields $C[a, b]/U_c \cong \mathbb{R}$, $a \le c \le b$, can be identified with the ring of all functions from $[a, b]$ into $\mathbb{R}$. Thus, when applying the method of

Lemma 4.75 to $C[a, b]$ we lose the information that functions from our ring are continuous. The structure theory is powerful, but has limitations.

Our final result in this section summarizes some of the main features of the structure theory. It can be compared with Theorem 2.74, which gives an analogous, but of course much stronger, statement on finite dimensional algebras.

**Theorem 4.78** *If a ring $R$ is not a radical ring, then the ring $R/\mathrm{rad}(R)$ is isomorphic to a subdirect product of dense rings of linear operators of vector spaces over division rings.*

***Proof*** Combine Lemmas 4.73 and 4.75 with the Jacobson density theorem. □

## 4.10 Structure Theory in Action

The structure theory that we have developed is not only beautiful, it is truly applicable. The purpose of this final section is to indicate how the theory works in "real life".

Suppose we are given a problem on a ring $R$. The following procedure, indicated by Theorem 4.78, sometimes leads to its solution:

(a) By factoring out $\mathrm{rad}(R)$ one reduces the problem to the case where $R$ is semiprimitive.
(b) Using the representation of a semiprimitive ring $R$ as a subdirect product of primitive rings, one further reduces the problem to the case where $R$ is primitive.
    (In practice, one does not necessarily work with subdirect products, but uses the fact that the intersection of the primitive ideals of $R$ is 0. We have introduced subdirect products primarily to indicate the philosophical aspect of semiprimitivity.)
(c) A problem on a primitive ring $R$ can be usually solved by representing $R$ as a dense ring of linear operators of a vector space $V$ over a division ring $\Delta$.
(d) If $\dim_\Delta V = 1$, the density condition does not have the same meaning as otherwise. In this case $R$ is a division ring (isomorphic to $\Delta^{\mathrm{o}}$) and a different argument may be needed.

Needless to say, we cannot solve just any problem by following this procedure. There are, however, various important instances where at least some, if not all of the steps (a)–(d) can be realized. We will give an evidence of the feasibility of the procedure by presenting a proof of (yet another) theorem by Jacobson. This theorem was one of the first applications of the structure theory, and still is one of the most striking ones.

We first record two elementary lemmas. The reader is probably familiar with the first one. Anyway, we give the proof for the sake of completeness.

**Lemma 4.79** *If $K$ is a finite field with $\mathrm{char}(K) = p$, then there exists an $n \in \mathbb{N}$ such that $|K| = p^n$ and $k^{p^n} = k$ for every $k \in K$. Moreover, if $k_1, \ldots, k_{p^n-1}$ are*

*the nonzero elements in $K$, then*

$$\omega^{p^n} - \omega = (\omega - k_1)\dots(\omega - k_{p^n-1})\omega. \tag{4.9}$$

***Proof*** We may consider $K$ as a vector space over the prime subfield $F_0 \cong \mathbb{Z}_p$. If $K$ is $n$-dimensional over $F_0$, then $|K| = |F_0^n| = p^n$. The multiplicative group $K^* = K \setminus \{0\}$ thus has $p^n - 1$ elements, so $k^{p^n-1} = 1$ for every $k \in K^*$. Consequently, $k^{p^n} = k$ for every $k \in K$. That is to say, every element in $K$ is a root of the polynomial $\omega^{p^n} - \omega$. Since the degree of this polynomial is equal to $|K|$, (4.9) follows. □

The next lemma concerns powers of inner derivations in rings of prime characteristic. We denote the inner derivation $x \mapsto ax - xa$ by $\operatorname{ad} a$ (this notation is borrowed from the theory of Lie algebras). We first remark that in any ring $R$, regardless of its characteristic, for all $a, x \in R$ we have

$$(\operatorname{ad} a)^m(x) = \sum_{k=0}^{m} (-1)^k \binom{m}{k} a^{m-k} x a^k. \tag{4.10}$$

Using $\binom{m}{k-1} + \binom{m}{k} = \binom{m+1}{k}$, one easily proves (4.10) by induction on $m$.

**Lemma 4.80** *If $R$ is a ring with $\operatorname{char}(R) = p$, where $p$ is prime, then $(\operatorname{ad} a)^{p^n} = \operatorname{ad} a^{p^n}$ for all $a \in R$ and $n \in \mathbb{N}$.*

***Proof*** If $1 \le k \le p-1$, then $p$ does not divide $k!(p-k)!$, so it divides $\binom{p}{k} = \frac{p!}{k!(p-k)!}$. Therefore, (4.10) reduces to $(\operatorname{ad} a)^p = \operatorname{ad} a^p$ if $m = p$ (if $p = 2$, then this is true as $xa^2 = -xa^2$). This proves the claim of the lemma for $n = 1$. We may therefore assume the claim is true for $n - 1$. Consequently,

$$(\operatorname{ad} a)^{p^n} = \left((\operatorname{ad} a)^{p^{n-1}}\right)^p = \left(\operatorname{ad} a^{p^{n-1}}\right)^p = \operatorname{ad}\left(a^{p^{n-1}}\right)^p = \operatorname{ad} a^{p^n},$$

as desired. □

Before stating the Jacobson's theorem, a few words about the background. The reader is perhaps familiar with the notion of a *Boolean ring*. This is a ring in which every element is an idempotent (a simple concrete example is a direct product of copies of $\mathbb{Z}_2$). One of the first things one learns about such rings is that they are commutative. The proof is short: If $x$ and $y$ are elements of a Boolean ring $R$, then $x, y, x + y$ are idempotents, which readily implies $xy + yx = 0$; setting $y = x$ we obtain $\operatorname{char}(R) = 2$, so $xy + yx = 0$ can be written as $xy = yx$. This was extremely easy, but the fact itself may appear a bit surprising at first since idempotents in rings do not always commute. Now consider a more general condition where a ring $R$ satisfies $x^3 = x$ for every $x \in R$. Is $R$ commutative? It turns out that this is true indeed, but giving a proof may already be a small challenge for the reader. The case where $x^4 = x$ is yet more involved, but again the commutativity can be derived

by ad hoc arguments. If we wish to consider the general case where there exists an $n \geq 2$ such that

$$x^n = x \quad \text{for all } x \in R, \tag{4.11}$$

the need for a conceptual approach becomes apparent.

But why would anyone be interested in (4.11)? As noticed in Lemma 4.79, every finite field $F$ satisfies (4.11) with $n = |F|$. The very same proof shows that this holds for every finite division ring. Of course, such rings are automatically fields by Wedderburn's theorem (Theorem 1.54), but this is exactly the point we wish to make. We are wondering whether Wedderburn's theorem can be generalized to arbitrary rings via (4.11). And indeed it can. Even better, there is no need to fix $n$; it may depend on $x$.

**Theorem 4.81 (Jacobson)** *If a ring $R$ is such that for every $x \in R$ there exists an integer $n(x) \geq 2$ such that $x^{n(x)} = x$, then $R$ is commutative.*

***Proof*** We will follow the procedure outlined above.

(a) In the first step, we show that $\operatorname{rad}(R) = 0$ (so that there is no need to factor out $\operatorname{rad}(R)$). Choose $q \in \operatorname{rad}(R)$, and set $n = n(q)$. Then $q^{n-1}$ also lies in $\operatorname{rad}(R)$, so it is quasi-invertible (Corollary 4.57). Thus, there exists an $r \in R$ such that $q^{n-1} + r = rq^{n-1}$. Multiplying this identity from the right by $q$ and using $q^n = q$ we obtain $q = 0$.

(b) We now know that $R$ is semiprimitive. We may assume that $R \neq 0$, so that there is a family $\{P_i \mid i \in I\}$ of primitive ideals of $R$ such that $\bigcap_{i \in I} P_i = 0$. Therefore, $R$ is commutative if and only if each primitive ring $R/P_i$ is commutative. Since $R/P_i$ satisfies the same condition, $x^{n(x)} = x$ for every $x \in R/P_i$, we see that without loss of generality we may assume that $R$ itself is primitive.

(c) In view of the Jacobson density theorem, we may assume that $R$ is a dense ring of linear operators of a vector space $V$ over a division ring $\Delta$. Suppose there exist linearly independent vectors $u_1, u_2 \in V$. Choose $f \in R$ such that $f(u_1) = u_2$ and $f(u_2) = 0$. Then $f^n(u_1) = 0$ for every $n \geq 2$, while $f(u_1) \neq 0$. However, this contradicts our assumption that $f^{n(f)} = f$. The result is thus proved if $\dim_\Delta V \geq 2$.

It remains to treat the case where $\dim_\Delta V = 1$, i.e., the case where $R \cong \Delta^{\mathrm{o}}$ is a division ring. This is the difficult part of the proof.

(d) Let us write $D$ instead of $R$. We begin by pointing out two facts that follow immediately from the condition $x^{n(x)} = x$, $x \in D$. The first one is that every nonzero subring of $D$ is automatically a division ring (since the inverse of $x$ is a power of $x$), and the second one is that $D$ has finite characteristic $p$ (indeed, just take $2 = 1 + 1$ for $x$).

Suppose, on the contrary, that $D$ is not commutative. Take $a \in D \setminus Z(D)$. Since $D$ has finite characteristic and $a^{n(a)-1} = 1$, we see that the subring generated by $a$, let us call it $K$, is finite. As observed in the preceding paragraph,

$K$ is necessarily a field, so we can invoke Lemma 4.79. Thus, there exists an $n \in \mathbb{N}$ such that $|K| = p^n$ and, in particular, $a^{p^n} = a$. Consider $D$ as a vector space over $K$. The inner derivation $d = \operatorname{ad} a$ is clearly a $K$-linear map, so it can be considered as an element of the $K$-algebra $\operatorname{End}_K(D)$. We can therefore evaluate the polynomial from (4.9) at $d$. Since $d^{p^n} = d$ by Lemma 4.80, we obtain

$$(d - k_1) \dots (d - k_{p^n - 1})d = 0, \tag{4.12}$$

where the $k_i$'s are the nonzero elements of $K$. Note that, in this context, $k_i$ should be understood as the map $D \to D$, $x \mapsto k_i x$. Since $a \notin Z(D)$, $d$ is a nonzero map. From (4.12) it thus follows that at least one of the maps $d - k_i$ has a nontrivial kernel. This means that there is a $b \in D$ such that $ab - ba = d(b) = k_i b$ with $0 \neq k_i \in K$. Accordingly, $a$ and $b$ do not commute and $ba \in Kb$.

Let $D_0$ be the subring of $D$ generated by $a$ and $b$. As observed at the beginning, $D_0$ is in fact a division ring. Of course, it is noncommutative since $ab \neq ba$. From $ba \in Kb$ it clearly follows that $ba^r \subseteq Kb$ for every $r \in \mathbb{N}$. Hence $bK \subseteq Kb$, and consequently, $b^s K \subseteq Kb^s$ for every $s \in \mathbb{N}$. This shows that

$$K + Kb + \dots + Kb^{n(b)-2}$$

is a subring of $D$. Since it is a subset of $D_0$ and since it contains $a$ and $b$, it is actually equal to $D_0$. But then $D_0$ is a finite set. However, this contradicts Wedderburn's theorem on finite division rings (Theorem 1.54). □

The first three steps (a)–(c) were quite easy, but nevertheless fascinating. They illustrated how the structure theory can be used to reduce a problem on a general ring to a much simpler ring, in this case a division ring. After that we were left to ourselves, and so the last step (d) had to be more ingenious.

It should be mentioned that Jacobson's original proof was somewhat different. We have mainly followed the proof given by I. N. Herstein.

## Exercises

**4.1** Let $e$ be a nonzero idempotent in a ring $R$. Show that $eV$ is a simple $eRe$-module if $V$ is a simple $R$-module and $eV \neq 0$. Hence infer that $eRe$ is primitive if $R$ is primitive. Use this to show that the ring $M_n(\mathbb{Z})$ is prime but not primitive.

**4.2** Show that a ring $R$ is primitive if and only if $M_n(R)$ is primitive.

**4.3** Let $R$ be a primitive ring with a faithful simple $R$-module $V$. Show that $R$ is a division ring if and only if for all $r \in R$ and $v \in V$, $rv = 0$ implies $r = 0$ or $v = 0$.

**4.4** Let $R$ be a primitive ring. Suppose that $a, b \in R$ satisfy $(xa)^2 = (xb)^2$ for every $x \in R$. Show that either $a = b$ or $a = -b$.

**4.5** Extend Corollary 4.20 to left artinian rings. That is, show that the following statements are equivalent for a nonzero left artinian ring $R$:

(i) $R$ is primitive.
(ii) $R$ is prime.
(iii) $R$ is simple.
(iv) There exist an $n \in \mathbb{N}$ and a division ring $D$ such that $R \cong M_n(D)$.

**Remark** If $R$ is unital, then this follows from Theorem 3.86. However, we are not assuming that. A consequence of (iii) $\Longrightarrow$ (iv) thus is that a simple left artinian ring is automatically unital. This was mentioned, but not proved in the preceding chapter. Incidentally, Theorem 3.86 still needs the assumption that $R$ is unital since a simple ring with a minimal left ideal may not be left artinian, see Example 4.32 and Exercise 4.13.

**4.6** Give an example showing that the assumption that $F$ is algebraically closed in Burnside's theorem (Corollary 4.24) is necessary.

**4.7** Give an alternative proof of Lemma 1.42 by using the Artin-Whaples theorem (Corollary 4.25).

**4.8** Show that the Weyl algebra $\mathscr{A}_1$ is a dense algebra of linear operators of $F[\omega]$.

**Remark** Since $\mathscr{A}_1$ is primitive (Example 4.9), the Jacobson density theorem for algebras tells us that $\mathscr{A}_1$ is isomorphic to a dense algebra of linear operators of a vector space over some division algebra. What has to be shown is that $\mathscr{A}_1$ is a dense algebra of linear operators of the original space $F[\omega]$, i.e., the fulfillment of the conditions of Theorem 4.22 has to be verified.

**4.9** Let $R$ be a ring, $V$ be an $R$-module, and $\varphi$ be an automorphism of $R$. Note that the additive group of $V$ becomes an $R$-module if we define the module operation by $x \cdot v = \varphi(x)v$ for all $x \in R$, $v \in V$. If $V$ is faithful and simple, then so is the new module. Use this construction to show that the Weyl algebra $\mathscr{A}_1$ has nonisomorphic faithful simple modules.

**4.10** Let $R$ be a primitive ring having a minimal left ideal $L$. Show that every faithful simple $R$-module is isomorphic to $L$.

**Remark** This is a generalization of Lemma 3.42.

**4.11** Fields are the only commutative primitive rings (Lemma 4.7), and the center of a simple unital ring is necessarily a field (Lemma 1.28). However, the center of a unital primitive ring may not be a field. Find an example of such a ring.

*Hint:* If $R$ is a primitive ring having minimal left ideals, then every subring of $R$ that contains $\operatorname{soc}(R)$ is also primitive.

**4.12** Show that the sum of minimal left ideals of the upper triangular matrix ring $T_2(F)$ does not coincide with the sum of its minimal right ideals.

**4.13** Let $R$ be a primitive ring having minimal left ideals. Show that $\mathrm{soc}(R)$ is simple as a ring, and is contained in every nonzero ideal of $R$.

**4.14** Extend Theorem 4.48 to left artinian rings. That is, show that the Jacobson radical of a left artinian ring $R$ is nilpotent, and is hence the unique maximal nilpotent ideal of $R$.

*Hint:* First observe that there is an $n \in \mathbb{N}$ such that $\mathrm{rad}(R)^n = \mathrm{rad}(R)^{n+1} = \cdots$ Assuming that $\mathrm{rad}(R)^n \neq 0$, take a left ideal $L$ minimal with respect to the property that $\mathrm{rad}(R)^n L \neq 0$. If $\ell \in L$ is such that $\mathrm{rad}(R)^n \ell \neq 0$, then, by minimality, $\mathrm{rad}(R)^n \ell = L$, and hence $r\ell = \ell$ for some $r \in \mathrm{rad}(R)^n$, which leads to a contradiction.

**4.15** Let $R$ be a unital ring and let $M$ be a nonzero finitely generated unital $R$-module. Show that $\mathrm{rad}(R)M$ is a proper submodule of $M$.

**Remark** This result, known as **Nakayama's lemma**, is particularly well known in commutative algebra and algebraic geometry.

**4.16** A unital ring $R$ is said to be **local** if the set of all non-invertible elements in $R$ forms an ideal. Show that this ideal is exactly $\mathrm{rad}(R)$, and that $R$ is local if and only if $R/\mathrm{rad}(R)$ is a division ring. Verify that the following rings are local:

(a) The ring $D[[\omega]]$ where $D$ is a division ring (compare Example 4.61).
(b) The subring of $M_n(D)$, $D$ a division ring, consisting of upper triangular matrices all of whose diagonal entries are equal.
(c) $\mathbb{Z}_{p^n}$ where $p$ is prime and $n \in \mathbb{N}$.
(d) The unitization of a radical algebra.

**4.17** Show that the following statements are equivalent for an element $a$ of a unital ring $R$:

(i) $a$ is invertible.
(ii) $a + \mathrm{rad}(R)$ is invertible in $R/\mathrm{rad}(R)$.
(iii) $a + P$ is invertible in $R/P$ for every primitive ideal $P$ of $R$.

*Hint:* If $a$ does not have a left inverse, then there exists a maximal left ideal containing $a$. If $a$ has a left inverse $b$, then $(1 - ab)a = 0$.

**4.18** Let $a$ be an element in a ring $R$. Show that if there exists an $n \in \mathbb{N}$ such that $a^n$ is quasi-invertible, then so is $a$.

*Hint:* After examining the (intuitively more clear) case where $R$ is unital, one can easily guess how to handle the general case.

**4.19** Let $a, b$ be elements in a ring $R$. Show that $ab$ is quasi-invertible if and only if $ba$ is quasi-invertible.

*Hint:* Again it may be easier to first handle the case where $R$ is unital. Our statement then reads as that $1 - ab$ is invertible if and only if $1 - ba$ is invertible. This can be proved by finding a formula connecting $(1 - ab)^{-1}$ and $(1 - ba)^{-1}$. An interesting way to derive it is by "cheating" and using the familiar formula

$(1-q)^{-1} = \sum_{n=0}^{\infty} q^n$, even though the (infinite) geometric series makes no sense in general rings.

**4.20** Let $a, b$ be elements in a ring $R$. Show that $aRb$ is a quasi-invertible set if and only if $ba \in \mathrm{rad}(R)$.

**4.21** Let $a$ be a nonzero element in a ring $R$ such that $a^n = a$ for some $n \geq 2$. Show that $a \notin \mathrm{rad}(R)$. (In particular, $\mathrm{rad}(R)$ does not contain nonzero idempotents.)

**4.22** Show that for any ring $R$,

$$\mathrm{rad}\left(\begin{bmatrix} R & R \\ 0 & R \end{bmatrix}\right) = \begin{bmatrix} \mathrm{rad}(R) & R \\ 0 & \mathrm{rad}(R) \end{bmatrix}.$$

**Remark** This can be extended to upper triangular matrices over $R$ of any size.

**4.23** Show that for any ring $R$, $\mathrm{rad}(M_n(R)) = M_n(\mathrm{rad}(R))$.

*Hint:* If $M$ is a simple $R$-module, then $M^n$, the direct sum of $n$ copies of $M$, becomes a simple $M_n(R)$-module under the standard matrix operations (write the elements in $M^n$ as columns). Use this to show that $\mathrm{rad}(M_n(R)) \subseteq M_n(\mathrm{rad}(R))$. To prove the converse inclusion, show that $M_n(\mathrm{rad}(R))E_{ii}$ is a left quasi-regular left ideal of $M_n(R)$.

**4.24** Show that a nonzero ring $R$ is semiprimitive if and only if $R$ has a faithful semisimple module.

**4.25** Let $R$ be a semiprimitive ring. True or False:

(a) If for every primitive ideal $P$ of $R$ the ring $R/P$ has no nonzero nilpotent elements, then $R$ has no nonzero nilpotent elements.
(b) If for every primitive ideal $P$ of $R$ the ring $R/P$ has no nontrivial idempotents, then $R$ has no nontrivial idempotents.

**4.26** Let $R$ be a semiprimitive ring. Show that the following statements are equivalent for $a \in R$:

(i) For every $x \in R$ there exists an $m(x) \in \mathbb{N}$ such that $[a, x]^{m(x)} = 0$.
(ii) For every $x \in R$ there exists an $n(x) \in \mathbb{N}$ such that $(ax)^{n(x)} = (xa)^{n(x)}$.
(iii) $a \in Z(R)$.

**4.27** Show that an algebraic division algebra over a finite field is commutative.

*Hint:* One way of proving this result, which is also due to Jacobson, is to use Jacobson's Theorem 4.81, together with standard facts about fields and their extensions.

**4.28** Show that a ring $R$ is commutative if every commutator in $R$ is an idempotent.

**Remark** This is a very special case of Herstein's generalization of Theorem 4.81 which says that a ring $R$ is commutative if and only if for every pair $x, y \in R$ there exists an integer $n(x, y) \geq 2$ such that $[x, y]^{n(x,y)} = [x, y]$.

# Chapter 5
# Tensor Products in Noncommutative Algebra




Sometimes we find some mathematical concept difficult to comprehend at the beginning. Later, when we get used to it, it appears so natural to us that we do not understand anymore what appeared so mysterious about it before. Perhaps the tensor product is a concept with which many have such an experience.

Our intention is to give a gentle introduction to tensor products. After a preliminary consideration of tensor products of vector spaces, we will proceed with tensor products of algebras. This will yield a deeper insight into central simple algebras. In particular, we will prove general versions of Wedderburn's principal theorem and the Skolem-Noether theorem, whose special cases were derived by elementary means in early chapters. Thereby, we will illustrate the usefulness and strength of the approach based on tensor products.

## 5.1 The Concept of Tensor Product

Before starting with a rigorous treatment, we make some informal remarks regarding the definition of the tensor product.

Take two vector spaces $U$ and $V$ over a field $F$. Temporarily and only for simplicity, assume that they are finite dimensional, $\dim_F U = m$ and $\dim_F V = n$. The tensor product of $U$ and $V$ is then a vector space of dimension $mn$. One is usually interested in algebraic objects only up to isomorphism, and so might consider this description of the tensor product satisfactory. However, it does not really give a clue about the idea upon which this concept is based. The tensor product serves as a device transforming bilinear maps into linear ones. This ambiguous description in some sense better reflects the essence of this concept.

Let us indicate more clearly in what way substituting linear for bilinear maps is connected with an $mn$-dimensional space. Pick a basis $\{e_1, \ldots, e_m\}$ of $U$ and a basis $\{f_1, \ldots, f_n\}$ of $V$. Note that a *bilinear* map $\beta$ from $U \times V$ to another vector space

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$W$ is uniquely determined by $mn$ values $\beta(e_i, f_j)$, $1 \le i \le m$, $1 \le j \le n$. Now, take any vector space of dimension $mn$, choose its basis, and define a *linear* map $\overline{\beta}$ from this space to $W$ so that the elements from the chosen basis are mapped into the elements $\beta(e_i, f_j)$. It is clear that $\overline{\beta}$, together with the chosen basis, contains all information about $\beta$. Thus, instead of considering a bilinear map on $U \times V$, we can consider a linear map defined on a space of dimension $\dim_F U \cdot \dim_F V$.

We now proceed to the formal definition. Its main advantage comparing to the above construction is that it is independent of basis.

First, a few comments to avoid a possible misapprehension.

Given any set $X$, we can construct a vector space over $F$ having $X$ as a basis. As already explained at the beginning of Sect. 2.11, its elements are formal linear combinations of elements from $X$, i.e., they can be written as $\sum_{x \in X} \lambda_x x$ where all but finitely many $\lambda_x \in F$ are zero. Thus, for every element $w$ there exist a finite subset $\{x_1, \dots, x_n\}$ of $X$ and unique $\lambda_1, \dots, \lambda_n \in F$ such that $w = \lambda_1 x_1 + \cdots + \lambda_n x_n$. The addition and scalar multiplication are defined in the obvious way.

We will deal with the situation where $X$ is the Cartesian product $U \times V$ of spaces $U$ and $V$. Let us emphasize that here we consider $U \times V$ only as a set, not as a vector space. Therefore, for example, $(0, 0)$ is just an element from the standard basis of the space with basis $U \times V$, it is not its zero element. Further, $(u + u', v)$ is an element from this basis and is therefore different from $(u, v) + (u', v)$, which is the sum of two basis elements. Similarly, $\lambda(u, v)$ coincides with $(\lambda u, v)$ only when $\lambda = 1$. The space with basis $U \times V$ is thus huge, incomparably larger than $U$ and $V$.

Let us now state the definition. It concerns arbitrary vector spaces, not necessarily finite dimensional ones.

**Definition 5.1** Let $U$ and $V$ be vector spaces over $F$. Denote by $\mathscr{Y}$ the vector space with basis $U \times V$. Let $\mathscr{N}$ be the subspace of $\mathscr{Y}$ generated by all elements of the form

$$(\lambda u + \lambda' u', v) - \lambda(u, v) - \lambda'(u', v),$$

$$(u, \lambda v + \lambda' v') - \lambda(u, v) - \lambda'(u, v'),$$

$\lambda, \lambda' \in F, u, u' \in U, v, v' \in V$. The **tensor product** of $U$ and $V$ is the factor vector space $\mathscr{Y}/\mathscr{N}$. It is denoted by $U \otimes V$ (or by $U \otimes_F V$, when we wish to emphasize that we consider spaces over $F$).

This definition is of little practical use. The characteristic property of $U \otimes V$, described in the following theorem, is the one that matters.

**Theorem 5.2** *Let $U$ and $V$ be vector spaces over $F$. Then there exists a bilinear map $U \times V \to U \otimes V$, $(u, v) \mapsto u \otimes v$, such that:*

(a) *Every element in $U \otimes V$ is a sum of elements of the form $u \otimes v$, $u \in U$, $v \in V$.*

(b) *Given a bilinear map $\beta : U \times V \to W$, where $W$ is a vector space over $F$, there exists a unique linear map $\overline{\beta} : U \otimes V \to W$ such that $\overline{\beta}(u \otimes v) = \beta(u, v)$ for all $u \in U$, $v \in V$.*

*Moreover, properties (a) and (b) characterize $U \otimes V$ up to isomorphism.*

***Proof*** We define $u \otimes v$ as the coset $(u, v) + \mathscr{N}$. Considering the generators of $\mathscr{N}$, we see that the map $(u, v) \mapsto u \otimes v$ is bilinear. By definition, every element in $U \otimes V$ can be written as a linear combination of elements of the form $u \otimes v$. Since $\lambda(u \otimes v) = (\lambda u) \otimes v$, (a) follows.

Let $\beta : U \times V \to W$ be a bilinear map. Since $U \times V$ is a basis of $\mathscr{Y}$, there exists a linear map $B : \mathscr{Y} \to W$ such that

$$B\big((u, v)\big) = \beta(u, v)$$

for all $u \in U$, $v \in V$ (see Remark 3.31). The bilinearity of $\beta$ implies that $\mathscr{N} \subseteq \ker B$. Accordingly, we can define a linear map $\overline{\beta}$ on $\mathscr{Y}/\mathscr{N} = U \otimes V$ by

$$\overline{\beta}(x + \mathscr{N}) = B(x).$$

In particular, $\overline{\beta}(u \otimes v) = \beta(u, v)$. From (a) we see that $\overline{\beta}$ is the unique linear map from $U \otimes V$ to $W$ having this property.

To prove the last statement, assume that $T$ is another space for which there exists a bilinear map $U \times V \to T$, $(u, v) \mapsto u \odot v$, with properties (a) and (b). Since $(u, v) \mapsto u \otimes v$ is a bilinear map, there exists a linear map $\varphi : T \to U \otimes V$ satisfying $\varphi(u \odot v) = u \otimes v$. Analogously, there exists a linear map $\psi : U \otimes V \to T$ satisfying $\psi(u \otimes v) = u \odot v$. Consequently, $\varphi$ and $\psi$ are inverses of each other, and hence $T$ and $U \otimes V$ are isomorphic vector spaces. □

Bilinear maps appear so often in mathematics that we cannot avoid them. However, they may be clumsy to deal with. Theorem 5.2 shows that we can "glue" spaces $U$ and $V$ into the space $U \otimes V$ so that every bilinear map defined on $U \times V$ can be credibly represented by a linear map defined on $U \otimes V$. Linear maps are not only simpler than bilinear maps, they are the "proper" maps (i.e., the homomorphisms) in the context of vector spaces.

**Example 5.3** Let $A$ be an $F$-algebra. The multiplication on $A$ is by definition a bilinear map $(x, y) \mapsto xy$ from $A \times A$ to $A$. Sometimes it is more appropriate to consider multiplication as a linear map from $A \otimes A$ into $A$, determined by $x \otimes y \mapsto xy$.

Let us mention that one can also define tensor products of modules. For modules over commutative rings this can be done in essentially the same way as for vector spaces, the differences are merely terminological. More substantial changes are needed if rings are noncommutative.

## 5.2 Basic Properties of Tensor Products

The results of this and of the next couple of sections should convince the reader that the concept of the tensor product is natural. We will see that it is compatible with fundamental linear algebra concepts.

Let us first summarize the findings of the preceding section.

The tensor product of vector spaces $U$ and $V$ is a vector space $U \otimes V$ consisting of elements that can be written as

$$u_1 \otimes v_1 + u_2 \otimes v_2 + \cdots + u_n \otimes v_n, \quad u_i \in U, v_i \in V.$$

Such an expression is far from unique. After all, we have to take into account that the map $(u, v) \mapsto u \otimes v$ is bilinear, meaning that

$$(\lambda u + \lambda' u') \otimes v = \lambda(u \otimes v) + \lambda'(u' \otimes v),$$

$$u \otimes (\lambda v + \lambda' v') = \lambda(u \otimes v) + \lambda'(u \otimes v')$$

for all $\lambda, \lambda' \in F$, $u, u' \in U$, $v, v' \in V$. These formulas readily imply

$$u \otimes 0 = 0, \quad 0 \otimes v = 0$$

for all $u \in U$, $v \in V$. Last but not least, the tensor product has property (b) from Theorem 5.2, which is illustrated by the commutative diagram in Fig. 5.1. We call it the **universal property** of the tensor product. Note that $\otimes$ (i.e., the canonical map $(u, v) \mapsto u \otimes v$) and $\beta$ are bilinear maps, while $\overline{\beta}$ is linear.

Elements of the form $u \otimes v$ are called **simple tensors**. One should keep in mind that these are only generators of the space $U \otimes V$, not its typical elements. This warning may seem redundant at this point, but actually its disregard is a common source of errors. Standard tensor product considerations deal only with simple tensors, which may be misleading. Say, Theorem 5.2 describes $\overline{\beta}$ by $\overline{\beta}(u \otimes v) = \beta(u, v)$; this is all right since $\overline{\beta}$ is linear and so its action on $U \otimes V$ is uniquely determined by this rule. There are, however, other instances where information about simple tensors is not sufficient.

Let us examine further properties of tensor products. First, a word about notation: Throughout this and the next section, $U, U', V, V', W$ denote arbitrary vector spaces over the field $F$. The next lemma introduces the **tensor product of linear maps**.

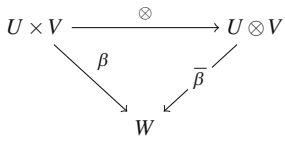



**Fig. 5.1** Universal property of tensor product

**Lemma 5.4** *If $\varphi : U \to U'$ and $\psi : V \to V'$ are linear maps, then there exists a unique linear map $U \otimes V \to U' \otimes V'$, which we denote by $\varphi \otimes \psi$, such that*

$$(\varphi \otimes \psi)(u \otimes v) = \varphi(u) \otimes \psi(v)$$

*for all $u \in U$, $v \in V$.*

***Proof*** The linearity of $\varphi$ and $\psi$, together with the bilinearity of $\otimes$, clearly imply that the map $(u, v) \mapsto \varphi(u) \otimes \psi(v)$ is bilinear. Theorem 5.2 thus shows that the linear map sending $u \otimes v$ into $\varphi(u) \otimes \psi(v)$ exists and is unique. □

The lemma essentially states that the map given by

$$\sum_i u_i \otimes v_i \mapsto \sum_i \varphi(u_i) \otimes \psi(v_i)$$

is well-defined. The given proof is quite illustrative. In order to establish something about tensor products one usually begins by introducing a bilinear map, and then, by the universal property, transforms it into a linear one. As a rule this quickly yields conclusions which may not be so easily obtained by other means.

It is clear that for linear maps with appropriate domains and codomains, we have

$$(\varphi \otimes \psi)(\varphi' \otimes \psi') = \varphi\varphi' \otimes \psi\psi'.$$

Consequently, if $\varphi$ and $\psi$ are isomorphisms, then so is $\varphi \otimes \psi$, and

$$(\varphi \otimes \psi)^{-1} = \varphi^{-1} \otimes \psi^{-1}.$$

This in particular shows the following:

$$U \cong U' \text{ and } V \cong V' \Longrightarrow U \otimes V \cong U' \otimes V'.$$

Let us also mention the equations

$$(\lambda\varphi + \lambda'\varphi') \otimes \psi = \lambda(\varphi \otimes \psi) + \lambda'(\varphi' \otimes \psi),$$

$$\varphi \otimes (\lambda\psi + \lambda'\psi') = \lambda(\varphi \otimes \psi) + \lambda'(\varphi \otimes \psi'),$$

which follow immediately from the definition.

In the next lemmas we prove that the tensor product is commutative, associative, and distributive. One should not understand this literally; such statements can only be true up to isomorphism. However, such abuse of terminology is quite standard.

**Lemma 5.5** $U \otimes V \cong V \otimes U$.

***Proof*** The map $U \times V \to V \otimes U$, $(u, v) \mapsto v \otimes u$, is bilinear, hence there exists a linear map $U \otimes V \to V \otimes U$, $u \otimes v \mapsto v \otimes u$. Similarly there is a linear map

$V \otimes U \to U \otimes V$, $v \otimes u \mapsto u \otimes v$. Obviously, these two maps are inverses of each other. □

**Lemma 5.6** $(U \otimes V) \otimes W \cong U \otimes (V \otimes W)$.

***Proof*** The bulk of the proof is to show that the rule $(u \otimes v) \otimes w \mapsto u \otimes (v \otimes w)$ determines a well-defined map from $(U \otimes V) \otimes W$ to $U \otimes (V \otimes W)$. Since the elements $(u \otimes v) \otimes w$ generate the vector space $(U \otimes V) \otimes W$ (why?), the symmetrically defined map $u \otimes (v \otimes w) \mapsto (u \otimes v) \otimes w$ is obviously its inverse.

Take $w \in W$. The map $U \times V \to U \otimes (V \otimes W)$, $(u, v) \mapsto u \otimes (v \otimes w)$, is bilinear, and hence $u \otimes v \mapsto u \otimes (v \otimes w)$ determines a well-defined map from $U \otimes V$ to $U \otimes (V \otimes W)$. This is true for every $w \in W$. Accordingly,

$$\Big(\sum_i u_i \otimes v_i, w\Big) \mapsto \sum_i u_i \otimes (v_i \otimes w)$$

is a well-defined bilinear map from $(U \otimes V) \times W$ to $U \otimes (V \otimes W)$. Therefore, the map determined by $(u \otimes v) \otimes w \mapsto u \otimes (v \otimes w)$ is indeed well-defined. □

One usually identifies $(U \otimes V) \otimes W$ and $U \otimes (V \otimes W)$, and denotes each of these two spaces simply by

$$U \otimes V \otimes W.$$

Alternatively one can introduce $U \otimes V \otimes W$ via trilinear maps; the resulting space is the same up to isomorphism. Of course, we also define the tensor product of more than three spaces. The $n$th **tensor power** of $U$ is the tensor product of $U$ with itself $n$ times. It is denoted by $U^{\otimes n}$.

**Lemma 5.7** $(U \oplus V) \otimes W \cong (U \otimes W) \oplus (V \otimes W)$.

***Proof*** Let $\pi_U : U \oplus V \to U$ and $\iota_U : U \to U \oplus V$ be given by $\pi_U(u, v) = u$ in $\iota_U(u) = (u, 0)$. Similarly we define $\pi_V : U \oplus V \to V$ and $\iota_V : V \to U \oplus V$. Now, define linear maps

$$\varphi : (U \oplus V) \otimes W \to (U \otimes W) \oplus (V \otimes W),$$
$$\psi : (U \otimes W) \oplus (V \otimes W) \to (U \oplus V) \otimes W$$

by

$$\varphi(x) = \big((\pi_U \otimes \mathrm{id}_W)(x), (\pi_V \otimes \mathrm{id}_W)(x)\big),$$

$$\psi(x_1, x_2) = (\iota_U \otimes \mathrm{id}_W)(x_1) + (\iota_V \otimes \mathrm{id}_W)(x_2).$$

We will show that $\varphi$ are $\psi$ are inverses of each other, and therefore isomorphisms.

The definitions tell us that

$$\varphi\big((u, v) \otimes w\big) = (u \otimes w, v \otimes w),$$
$$\psi(u \otimes w, v \otimes w') = (u, 0) \otimes w + (0, v) \otimes w'.$$

Consequently,

$$(\psi\varphi)\big((u, v) \otimes w\big) = (u, v) \otimes w,$$
$$(\varphi\psi)(u \otimes w, v \otimes w') = (u \otimes w, v \otimes w').$$

This shows that $\psi\varphi = \mathrm{id}_{(U\oplus V)\otimes W}$ and $\varphi\psi = \mathrm{id}_{(U\otimes W)\oplus(V\otimes W)}$. □

**Lemma 5.8** *$U \otimes F \cong U$ and $F \otimes V \cong V$.*

***Proof*** By bilinearity of $\otimes$, every element in $U \otimes F$ can be written as $u \otimes 1$. Therefore, $u \otimes 1 \mapsto u$ is an isomorphism from $U \otimes F$ onto $U$. Similarly, $1 \otimes v \mapsto v$ is an isomorphism from $F \otimes V$ onto $V$. □

## 5.3 Linear (In)dependence in Tensor Products

An element in $U \otimes V$ can be written as a sum of simple tensors in many different ways. Fortunately, this non-uniqueness has limitations.

**Lemma 5.9** *Let $e_1, \ldots, e_n \in U$ be linearly independent. If $v_1, \ldots, v_n \in V$ are such that*

$$e_1 \otimes v_1 + \cdots + e_n \otimes v_n = 0,$$

*then each $v_i = 0$.*

***Proof*** Let $f_i : U \to F$ be a linear functional such that $f_i(e_j) = \delta_{ij}$ for $j = 1, \ldots, n$ (see Theorem 3.30 and Remark 3.31). The map $(u, v) \mapsto f_i(u)v$ is bilinear, so there exists a linear map $\alpha_i : U \otimes V \to V$ satisfying $\alpha_i(u \otimes v) = f_i(u)v$. Consequently, $v_i = \sum_{j=1}^{n} \alpha_i(e_j \otimes v_j) = \alpha_i\Big(\sum_{j=1}^{n} e_j \otimes v_j\Big) = 0$. □

All results in this section will follow from this simple lemma. Let us first point out that in particular it shows that for all $u \in U$ and $v \in V$,

$$u \otimes v = 0 \Longrightarrow u = 0 \text{ or } v = 0.$$

Incidentally, the tensor product of modules over rings does not always enjoy this property.

We continue with a useful technical improvement of Lemma 5.9.

**Lemma 5.10** *Let $e_1, \ldots, e_n \in U$ be linearly independent. If $v_1, \ldots, v_n \in V$ are such that*

$$e_1 \otimes v_1 + \cdots + e_n \otimes v_n = w_1 \otimes z_1 + \cdots + w_m \otimes z_m$$

*for some $w_1 \ldots, w_m \in U$ and $z_1, \ldots, z_m \in V$, then each $v_i$ is a linear combination of $z_1, \ldots, z_m$.*

***Proof*** The set $\{e_1, \ldots, e_n\}$ can be extended to a basis of the linear span of the set $\{e_1, \ldots, e_n, w_1, \ldots, w_m\}$. Denote the additional vectors (if there are any) in this basis by $e_{n+1}, \ldots, e_p$. Write each $w_j$ as a linear combination of basis vectors. Then, using the bilinearity of $\otimes$, we see that the given equation can be rewritten as

$$e_1 \otimes v_1' + \cdots + e_n \otimes v_n' + e_{n+1} \otimes z_{n+1}' + \cdots + e_p \otimes z_p' = 0,$$

where each $v_i'$ is the sum of $v_i$ and a linear combination of the $z_j$'s. Lemma 5.9 tells us that $v_i' = 0$. □

Let us also record the obvious analogues of the above lemmas.

**Lemma 5.11** *Let $f_1, \ldots, f_n \in V$ be linearly independent. If $u_1, \ldots, u_n \in U$ are such that*

$$u_1 \otimes f_1 + \cdots + u_n \otimes f_n = 0,$$

*then each $u_i = 0$.*

**Lemma 5.12** *Let $f_1, \ldots, f_n \in V$ be linearly independent. If $u_1, \ldots, u_n \in U$ are such that*

$$u_1 \otimes f_1 + \cdots + u_n \otimes f_n = w_1 \otimes z_1 + \cdots + w_m \otimes z_m$$

*for some $w_1, \ldots, w_m \in U$ and $z_1 \ldots, z_m \in V$, then each $u_i$ is a linear combination of $w_1, \ldots, w_m$.*

We can now show that bases of $U$ and $V$ give rise to bases of $U \otimes V$.

**Theorem 5.13** *If $\{e_i \mid i \in I\}$ is a basis of $U$ and $\{f_j \mid j \in J\}$ is a basis of $V$, then $\{e_i \otimes f_j \mid i \in I, j \in J\}$ is a basis of $U \otimes V$.*

***Proof*** Writing $u \in U$ as a linear combination of the $e_i$'s and $v \in V$ as a linear combination of the $f_j$'s, it follows, by the bilinearity of $\otimes$, that $u \otimes v$ is a linear combination of the $e_i \otimes f_j$'s. Therefore, the set $\{e_i \otimes f_j \mid i \in I, j \in J\}$ generates $U \otimes V$.

It remains to show that this set is linearly independent. Take (finitely many) $\lambda_{ij} \in F$ that satisfy

$$\sum_{i,j} \lambda_{ij}(e_i \otimes f_j) = 0.$$

We can write this as

$$\sum_i e_i \otimes \Big( \sum_j \lambda_{ij} f_j \Big) = 0,$$

so Lemma 5.9 shows that $\sum_j \lambda_{ij} f_j = 0$. Since the $f_j$'s are linearly independent, $\lambda_{ij} = 0$ for all $i$ and $j$. □

The next two corollaries are immediate.

**Corollary 5.14** *If $\{e_i \mid i \in I\}$ is a basis of $U$, then every element in $U \otimes V$ can be written uniquely in the form $\sum_{i \in I} e_i \otimes v_i$.*

If $I$ is an infinite set, this should of course be understood as that all but finitely many $v_i$'s are zero.

**Corollary 5.15** *If $\{f_j \mid j \in J\}$ is a basis of $V$, then every element in $U \otimes V$ can be written uniquely in the form $\sum_{j \in J} u_j \otimes f_j$.*

**Remark 5.16** Another useful observation is that every nonzero element $x \in U \otimes V$ can be written as $x = \sum_{i=1}^{d} u_i \otimes v_i$ where $u_1 \ldots, u_d$ are linearly independent vectors in $U$ and $v_1 \ldots, v_d$ are linearly independent vectors in $V$. The proof is easy: just require the minimality of $d$ with respect to all possible expressions of $x$, and the linear independence of the $u_i$'s and of the $v_i$'s clearly follows.

Let now $I$ and $J$ be finite sets. Theorem 5.13 then yields the formula announced at the beginning of the chapter.

**Corollary 5.17** *If $U$ and $V$ are finite dimensional vector spaces over $F$, then*

$$\dim_F U \otimes V = \dim_F U \cdot \dim_F V.$$

Observe the analogy with the formula on direct sums:

$$\dim_F U \oplus V = \dim_F U + \dim_F V.$$

**Remark 5.18** Corollary 5.17 resembles another formula from basic linear algebra: The dimension of $\mathrm{Hom}_F(U, V)$, the vector space of all linear maps from $U$ to $V$, is equal to $\dim_F U \cdot \dim_F V$. The spaces $\mathrm{Hom}_F(U, V)$ and $U \otimes V$ are therefore isomorphic. What we wish to point out is a canonical isomorphism—yet not between exactly these two spaces.

Let $U^*$ be the dual space of $U$ (i.e., the space of all linear functionals on $U$). Instead of $U \otimes V$, we consider $V \otimes U^*$, which is a space of the same dimension. Given $v \in V$ and $f \in U^*$, define $T_{v,f} \in \mathrm{Hom}_F(U, V)$ by $T_{v,f}(u) = f(u)v$. Note that $T_{v,f}$ has rank one provided that $v \neq 0$ and $f \neq 0$, and conversely, every linear map $U \to V$ of rank one is of the form $T_{v,f}$ for some $0 \neq v \in V$ and $0 \neq f \in U^*$. The map $(v, f) \mapsto T_{v,f}$ is bilinear, hence there exists a linear map

$$\varphi : V \otimes U^* \to \mathrm{Hom}_F(U, V), \quad \varphi(v \otimes f) = T_{v,f}.$$

As every nonzero map in $\mathrm{Hom}_F(U, V)$ is a sum of maps of rank one, $\varphi$ is surjective. Consequently, $\varphi$ is an isomorphism since it is linear and the spaces $V \otimes U^*$ and $\mathrm{Hom}_F(U, V)$ have the same dimension.

One often identifies $v \otimes f$ with $T_{v,f}$, and thus considers simple tensors $v \otimes f$ as rank one maps given by

$$(v \otimes f)(u) = f(u)v.$$

This is commonly used regardless of whether $U$ and $V$ are finite dimensional or not.

## 5.4 Tensor Product of Algebras

After a linear algebra inset, we are now ready to return to themes that are in the spirit of this book. Let $A$ and $B$ be algebras over a field $F$. The vector space $A \otimes B$ can be turned into an algebra by defining multiplication in a simple, natural way.

**Lemma 5.19** *If $A$ and $B$ are $F$-algebras, then $A \otimes B$ is an $F$-algebra relative to the multiplication determined by*

$$(x \otimes y)(z \otimes w) = xz \otimes yw$$

*for all $x, z \in A$, $y, w \in B$.*

***Proof*** Take $z \in A$ and $w \in B$. Let $R_z$ and $R_w$ denote the corresponding right multiplication maps. By Lemma 5.4, we have the map $R_z \otimes R_w \in \mathrm{End}_F(A \otimes B)$ that sends $x \otimes y$ into $xz \otimes yw$. Since the map

$$A \times B \to \mathrm{End}_F(A \otimes B), \ \ (z, w) \mapsto R_z \otimes R_w,$$

is bilinear, there exists a linear map

$$\varphi : A \otimes B \to \mathrm{End}_F(A \otimes B), \ \ \varphi(z \otimes w) = R_z \otimes R_w.$$

We now define the product of $r, s \in A \otimes B$ as follows:

$$rs = \varphi(s)(r).$$

This multiplication is obviously bilinear, and for all $x, z \in A$, $y, w \in B$ we have

$$(x \otimes y)(z \otimes w) = \varphi(z \otimes w)(x \otimes y) = (R_z \otimes R_w)(x \otimes y) = xz \otimes yw.$$

Finally, from the associativity in $A$ and $B$ we immediately infer that

$$\Big((x \otimes y)(z \otimes w)\Big)(u \otimes v) = (x \otimes y)\Big((z \otimes w)(u \otimes v)\Big).$$

Since every element in $A \otimes B$ is a sum of simple tensors, this implies that our multiplication is associative. □

Lemma 5.19 essentially states that

$$\Big(\sum_i x_i \otimes y_i\Big)\Big(\sum_j z_j \otimes w_j\Big) = \sum_{i,j} x_i z_j \otimes y_i w_j$$

is a well-defined operation on $A \otimes B$. From now on we will always consider $A \otimes B$ as an algebra in which multiplication is defined by the above equation. We call $A \otimes B$ the **tensor product of algebras** $A$ and $B$.

If the vector spaces from Lemmas 5.5–5.8 are algebras, then the vector space isomorphisms from the proofs are multiplicative, and therefore algebra isomorphisms. Thus, the following hold for all $F$-algebras $A$, $B$, and $C$:

- $A \otimes B \cong B \otimes A$,
- $(A \otimes B) \otimes C \cong A \otimes (B \otimes C)$,
- $(A \times B) \otimes C \cong (A \otimes C) \times (B \otimes C)$,
- $A \otimes F \cong A$ and $F \otimes B \cong B$.

Note also that $A \cong A'$ and $B \cong B'$ implies $A \otimes B \cong A' \otimes B'$.

Let $S$ be a subalgebra of $A$ and $T$ be a subalgebra of $B$. Regarding $S$ and $T$ as algebras, we can form their tensor product $S \otimes T$, which is clearly canonically isomorphic to the subalgebra of $A \otimes B$ consisting of all elements of the form $\sum_i s_i \otimes t_i$, $s_i \in S$, $t_i \in T$. In this sense, we can consider $S \otimes T$ as a subalgebra of $A \otimes B$.

We continue with illuminating examples.

**Example 5.20** Let $G$ and $H$ be groups. What is the tensor product of group algebras $F[G]$ and $F[H]$? From Theorem 5.13 we know that the set $\{g \otimes h \mid g \in G, h \in H\}$ is a basis of $F[G] \otimes F[H]$. By definition, $G \times H$ is a basis of the group algebra $F[G \times H]$. Therefore, there exists a bijective linear map $F[G] \otimes F[H] \to F[G \times H]$ determined by $g \otimes h \mapsto (g, h)$. This map is clearly also multiplicative, so we have

$$F[G] \otimes F[H] \cong F[G \times H].$$

**Example 5.21** Let $A$ be an arbitrary $F$-algebra. We claim that

$$A \otimes F[\omega] \cong A[\omega].$$

Indeed, using Corollary 5.15 we see that every element in $A \otimes F[\omega]$ can be *uniquely* written as $\sum_i a_i \otimes \omega^i$. It is now straightforward to check that the map

$$A \otimes F[\omega] \to A[\omega], \quad \sum_i a_i \otimes \omega^i \mapsto \sum_i a_i \omega^i,$$

is an algebra isomorphism.

In particular,

$$F[\omega_1] \otimes F[\omega_2] \cong (F[\omega_1])[\omega_2] = F[\omega_1, \omega_2].$$

Note that, via this isomorphism, the element $f \otimes g \in F[\omega_1] \otimes F[\omega_2]$ corresponds to the polynomial $f(\omega_1)g(\omega_2) \in F[\omega_1, \omega_2]$. It is in fact common not to distinguish between the two, and thus consider $f \otimes g$ as the polynomial

$$(f \otimes g)(\omega_1, \omega_2) = f(\omega_1)g(\omega_2).$$

See also Exercise 5.6, which gives further insight.

**Example 5.22** For every $F$-algebra $A$ and every $n \geq 1$, we have

$$M_n(F) \otimes A \cong M_n(A). \tag{5.1}$$

The proof is similar to that from the preceding example. After noticing that every element in $M_n(F) \otimes A$ can be uniquely written as the sum of simple tensors $E_{ij} \otimes a_{ij}$, where $E_{ij}$ are the standard matrix units, it readily follows that

$$M_n(F) \otimes A \to M_n(A), \quad \sum_{i=1}^{n}\sum_{j=1}^{n} E_{ij} \otimes a_{ij} \mapsto \sum_{i=1}^{n}\sum_{j=1}^{n} a_{ij}E_{ij},$$

is an algebra isomorphism. This map should be considered canonical. One often identifies $(\lambda_{ij}) \otimes a \in M_n(F) \otimes A$ with the matrix in $M_n(A)$ whose $(i, j)$ entry is $\lambda_{ij}a$.

As an important special case of (5.1), we have

$$M_n(F) \otimes M_m(F) \cong M_{nm}(F). \tag{5.2}$$

More precisely, (5.2) follows from (5.1) along with $M_n(M_m(F)) \cong M_{nm}(F)$, which is essentially just a rewording of the well-known fact that matrices can be added and multiplied blockwise. Therefore, we omit a detailed proof.

Alternatively, we can state (5.2) as

$$\mathrm{End}_F(U) \otimes \mathrm{End}_F(V) \cong \mathrm{End}_F(U \otimes V),$$

where $U$ and $V$ are finite dimensional vector spaces over $F$.

**Example 5.23** Let $V$ be a vector space with a countably infinite basis $\{e_1, e_2, \dots\}$. Set $A = \mathrm{End}_F(V)$ and define $f_{ij} \in A$, $i, j = 1, 2$, according to

$$f_{11}(e_{2k-1}) = e_{2k-1}, \qquad f_{11}(e_{2k}) = 0,$$
$$f_{12}(e_{2k-1}) = 0, \qquad f_{12}(e_{2k}) = e_{2k-1},$$

$$f_{21}(e_{2k-1}) = e_{2k}, \qquad f_{21}(e_{2k}) = 0,$$
$$f_{22}(e_{2k-1}) = 0, \qquad f_{22}(e_{2k}) = e_{2k}$$

for all $k \in \mathbb{N}$. One can check that $\{f_{11}, f_{12}, f_{21}, f_{22}\}$ forms a set of $2 \times 2$ matrix units of $A$. Therefore, $A \cong M_2(f_{11}Af_{11})$ by Lemma 2.52 (and Remark 2.53). It is an easy exercise to show that $f_{11}Af_{11} \cong \mathrm{End}_F(V_1)$ where $V_1$ is the subspace of $V$ generated by all $e_{2k-1}$, $k \in \mathbb{N}$. As the spaces $V_1$ and $V$ are isomorphic, we actually have $\mathrm{End}_F(V_1) \cong A$, and hence $A \cong M_2(A)$. It is not difficult to generalize this to

$$A \cong M_n(A) \quad \text{for every } n \in \mathbb{N}.$$

We can now better understand and appreciate the uniqueness in Wedderburn's theorem (Theorem 3.73). Further, now we see that the tensor product of algebras does not obey the "cancellation law", i.e.,

$$S \otimes A \cong T \otimes A \not\Longrightarrow S \cong T.$$

Indeed, by (5.1) we have $F \otimes A \cong M_n(F) \otimes A$ for every $n \in \mathbb{N}$.

**Example 5.24** In Sect. 3.11, we introduced Weyl algebras $\mathscr{A}_n$, $n = 1, 2, \ldots$ We claim that for every $n \geq 2$ and $1 \leq k < n$,

$$\mathscr{A}_k \otimes \mathscr{A}_{n-k} \cong \mathscr{A}_n.$$

The proof is essentially easy but notationally heavy, so we will restrict ourselves to showing that $\mathscr{A}_1 \otimes \mathscr{A}_1 \cong \mathscr{A}_2$ and also omit some routine verifications.

Observe first that $\mathscr{A}_1$ is isomorphic to the subalgebra $\mathscr{B}_1$ of $\mathscr{A}_2$ generated by $D_1$ and $L_1$. The natural isomorphism $\varphi_1 : \mathscr{A}_1 \to \mathscr{B}_1$ sends $D$ to $D_1$ and $L$ to $L_1$. Similarly, $\mathscr{A}_1$ is isomorphic to the subalgebra $\mathscr{B}_2$ of $\mathscr{A}_2$ generated by $D_2$ and $L_2$ via the isomorphism $\varphi_2 : \mathscr{A}_1 \to \mathscr{B}_2$ that sends $D$ to $D_2$ and $L$ to $L_2$. The map $(x, y) \mapsto \varphi_1(x)\varphi_2(y)$ is bilinear, so there exists a linear map $\varphi : \mathscr{A}_1 \otimes \mathscr{A}_1 \to \mathscr{A}_2$ satisfying

$$\varphi(x \otimes y) = \varphi_1(x)\varphi_2(y).$$

As each of $D_1, L_1$ commutes with each of $D_2, L_2$, the elements from $\mathscr{B}_1$ commute with the elements from $\mathscr{B}_2$. This implies that $\varphi$ is multiplicative, and thus an algebra homomorphism. Recall from the proof of Theorem 3.89 that $\{L^m D^n \mid m, n \geq 0\}$ is a basis of $\mathscr{A}_1$. Similarly we see that $\{L_1^m D_1^n L_2^p D_2^q \mid m, n, p, q \geq 0\}$ is a basis of $\mathscr{A}_2$. Since

$$\varphi(L^m D^n \otimes L^p D^q) = L_1^m D_1^n L_2^p D_2^q$$

and $\{L^m D^n \otimes L^p D^q \mid m, n, p, q \geq 0\}$ is a basis of $\mathscr{A}_1 \otimes \mathscr{A}_1$ by Theorem 5.13, $\varphi$ is bijective.

## 5.5 Multiplication Algebra and Tensor Products

Let $A$ and $B$ be unital algebras. Then $1 \otimes 1$ is clearly the unity of $A \otimes B$. One usually identifies $A$ with $A \otimes 1 = \{a \otimes 1 \mid a \in A\}$ $(= A \otimes F)$ and $B$ with $1 \otimes B = \{1 \otimes b \mid b \in B\}$, and in this way considers $A$ and $B$ as subalgebras of $A \otimes B$. Obviously, the elements from $A$ commute with the elements from $B$, and the algebra $A \otimes B$ is generated by $A \cup B$. If we add the statement of Lemma 5.9 to these properties, we obtain a characterization of the tensor product of unital algebras.

**Lemma 5.25** *Suppose a unital algebra $\mathscr{T}$ has subalgebras $A$ and $B$ that contain the unity of $\mathscr{T}$ and satisfy:*

(a) $ab = ba$ *for all* $a \in A$, $b \in B$.
(b) $\mathscr{T}$ *is generated by* $A \cup B$.
(c) *For any linearly independent* $a_1, \ldots, a_n \in A$ *and any* $b_1, \ldots, b_n \in B$, $a_1 b_1 + \cdots + a_n b_n = 0$ *implies* $b_1 = \cdots = b_n = 0$.

*Then* $A \otimes B \cong \mathscr{T}$ *under the map* $a \otimes b \mapsto ab$.

***Proof*** The standard universal property argument shows that there exists a linear map $\varphi : A \otimes B \to \mathscr{T}$ such that $\varphi(a \otimes b) = ab$ for all $a \in A$, $b \in B$. According to (a), (b), and the assumption that $A$ and $B$ contain the unity of $\mathscr{T}$, every element in $\mathscr{T}$ is of the form $a_1 b_1 + \cdots + a_n b_n$ with $a_i \in A$ and $b_i \in B$. Therefore, $\varphi$ is surjective. Since every $x \in A \otimes B$ can be written as $x = a_1 \otimes b_1 + \cdots + a_n \otimes b_n$ with $a_1, \ldots, a_n$ linearly independent (see Remark 5.16), we see from (c) that $\varphi(x) = 0$ implies $x = 0$. Therefore, $\varphi$ is injective. Finally, using (a) we obtain

$$\varphi\big((a \otimes b)(a' \otimes b')\big) = \varphi(aa' \otimes bb') = aa'bb' = aba'b' = \varphi(a \otimes b)\varphi(a' \otimes b'),$$

which shows that $\varphi$ is an algebra isomorphism. □

**Remark 5.26** It is instructive to compare (a)–(c) with the analogous properties of the direct product. Suppose an algebra $\mathscr{S}$ has subalgebras $A$ and $B$ such that:

(a') $ab = ba = 0$ for all $a \in A$, $b \in B$.
(b') $\mathscr{S}$ is the sum of $A$ and $B$.
(c') If $a \in A$ and $b \in B$ are such that $a + b = 0$, then $a = b = 0$.

Then $A \times B \cong \mathscr{S}$ under the map $(a, b) \mapsto a + b$.

Recall that $M(A)$ stands for the multiplication algebra of the algebra $A$, and $A^{\mathrm{o}}$ stands for the opposite algebra of $A$. From the next proof it will be evident that the situation considered in Lemma 5.25 was indirectly encountered in Sect. 1.5 when studying $M(A)$.

**Theorem 5.27** *If $A$ is a central simple algebra, then $A \otimes A^{\mathrm{o}} \cong M(A)$ under the map $a \otimes b \mapsto L_a R_b$.*

***Proof*** The sets $\mathscr{L} = \{L_a \mid a \in A\}$ and $\mathscr{R} = \{R_b \mid b \in A\}$ of all left and right multiplication maps are subalgebras of $M(A)$. By definition, $M(A)$ is generated by $\mathscr{L} \cup \mathscr{R}$. Obviously, elements from $\mathscr{L}$ commute with elements from $\mathscr{R}$. It is clear that $a_1, \ldots, a_n \in A$ are linearly independent if and only if $L_{a_1}, \ldots, L_{a_n} \in \mathscr{L}$ are linearly independent. In this case, $L_{a_1}R_{b_1} + \cdots + L_{a_n}R_{b_n} = 0$ implies $b_i = 0$ (and hence $R_{b_i} = 0$) by Lemma 1.42. Lemma 5.25 therefore tells us that

$$\mathscr{L} \otimes \mathscr{R} \cong M(A) \text{ under the map } L_a \otimes R_b \mapsto L_a R_b.$$

We can polish this a little bit. Note that $A \cong \mathscr{L}$ under $a \mapsto L_a$. Since $R_{bb'} = R_{b'}R_b$, there is no reason why $A$ and $\mathscr{R}$ should be isomorphic. However, $A^{\mathrm{o}}$ and $\mathscr{R}$ are; $b \mapsto R_b$ is, of course, an isomorphism. Therefore,

$$A \otimes A^{\mathrm{o}} \cong \mathscr{L} \otimes \mathscr{R} \text{ under the map } a \otimes b \mapsto L_a \otimes R_b.$$

Composing the isomorphisms $a \otimes b \mapsto L_a \otimes R_b$ and $L_a \otimes R_b \mapsto L_a R_b$, we arrive at the desired conclusion. □

Lemma 1.42 can now be sharpened as follows.

**Corollary 5.28** *If $A$ is a central simple algebra and $a_i, b_i \in A$, then $\sum_{i=1}^n L_{a_i}R_{b_i} = 0$ if and only if $\sum_{i=1}^n a_i \otimes b_i = 0$.*

**Corollary 5.29** *If $A$ is a finite dimensional central simple algebra, then $A \otimes A^{\mathrm{o}} \cong M_n(F)$, where $n = [A : F]$.*

***Proof*** Apply Theorem 1.43, which states that $M(A) = \mathrm{End}_F(A)$, together with the standard fact $\mathrm{End}_F(A) \cong M_n(F)$. □

**Example 5.30** The conjugation $h \mapsto \overline{h}$ is an antiautomorphism of $\mathbb{H}$. Hence, $\mathbb{H}^{\mathrm{o}} \cong \mathbb{H}$ under the map $h \mapsto \overline{h}$, and therefore $\mathbb{H} \otimes_{\mathbb{R}} \mathbb{H}^{\mathrm{o}} \cong \mathbb{H} \otimes_{\mathbb{R}} \mathbb{H}$. As $\mathbb{H} \otimes_{\mathbb{R}} \mathbb{H}^{\mathrm{o}} \cong M_4(\mathbb{R})$ by Corollary 5.29, we have

$$\mathbb{H} \otimes_{\mathbb{R}} \mathbb{H} \cong M_4(\mathbb{R}).$$

## 5.6 Centralizers in Tensor Products

The center $Z(A)$ of an algebra $A$ is the set of elements in $A$ that commute with *all* elements in $A$. We now extend this concept by considering sets of elements in $A$ that commute with *some* elements in $A$.

**Definition 5.31** Let $S$ be a subset of an algebra $A$. The **centralizer of $S$ in $A$** is the set

$$C_A(S) = \{a \in A \mid as = sa \quad \text{for every } s \in S\}.$$

In the same way one defines the centralizer of a subset of a ring, but we shall deal only with centralizers in algebras. Let us also mention that we encountered centralizers of singletons $C_A(a)$ $(= C_A(\{a\})$ in the proof of Wedderburn's theorem on finite division rings.

The centralizer of any set $S$ is a subalgebra of $A$ which contains $Z(A)$. On the other hand, the centralizer of $S$ coincides with the centralizer of the subalgebra generated by $S$ and $Z(A)$. Therefore, one usually considers only centralizers of subalgebras containing the center.

Roughly speaking, a large subalgebra has a small centralizer, and a small subalgebra has a large centralizer. In more precise terms, the following holds:

$$S_1 \subseteq S_2 \Longrightarrow C_A(S_1) \supseteq C_A(S_2),$$

$$C_A(A) = Z(A) \quad \text{and} \quad C_A(Z(A)) = A.$$

Condition (a) in Lemma 5.25 can be read as $B \subseteq C_{\mathscr{J}}(A)$ (or $A \subseteq C_{\mathscr{J}}(B)$). This suggests that the concept of the centralizer plays a role in tensor products of algebras.

**Proposition 5.32** *Let $A$ and $B$ be unital algebras. If $S$ is a subalgebra of $A$ that contains the unity of $A$, and $T$ is a subalgebra of $B$ that contains the unity of $B$, then*

$$C_{A\otimes B}(S \otimes T) = C_A(S) \otimes C_B(T).$$

***Proof*** It is enough to prove $C_{A\otimes B}(S \otimes T) \subseteq C_A(S) \otimes C_B(T)$, since the converse inclusion is trivial. Take a nonzero $c \in C_{A\otimes B}(S \otimes T)$. By Remark 5.16, we can write $c = \sum_{i=1}^{d} y_i \otimes z_i$ where $y_1, \ldots, y_d$ are linearly independent elements in $A$ and $z_1, \ldots, z_d$ are linearly independent elements in $B$. Observe that $c(s \otimes 1) = (s \otimes 1)c$, $s \in S$, gives

$$\sum_{i=1}^{d} (y_i s - s y_i) \otimes z_i = 0.$$

Lemma 5.11 implies that $y_i s - s y_i = 0$ for every $i$ and every $s \in S$. Each $y_i$ therefore lies in $C_A(S)$. Analogously, by considering $c(1 \otimes t) = (1 \otimes t)c$, $t \in T$, we infer that each $z_i$ lies in $C_B(T)$. Thus, $c \in C_A(S) \otimes C_B(T)$. □

**Corollary 5.33** *If $A$ and $B$ are unital algebras, then $Z(A \otimes B) = Z(A) \otimes Z(B)$.*

***Proof*** Take $S = A$ and $T = B$. □

In the statement of the next corollary we identify $A$ with $A \otimes 1$ and $B$ with $1 \otimes B$.

**Corollary 5.34** *If $A$ and $B$ are central algebras, then $A \otimes B$ is also a central algebra, $C_{A\otimes B}(A) = B$ and $C_{A\otimes B}(B) = A$.*

***Proof*** The first assertion follows from Corollary 5.33, the second from Proposition 5.32 by taking $S = A$ and $T = F$ ($= F \cdot 1$), and the third from Proposition 5.32 by taking $S = F$ and $T = B$. □

**Corollary 5.35** *Let $A$ be a unital algebra and let $S$ be a subalgebra that contains the unity of $A$. If $B$ is a central algebra, then $C_{A\otimes B}(S \otimes B) \cong C_A(S)$. In particular, $C_{M_n(A)}(M_n(S)) \cong C_A(S)$ for every $n \in \mathbb{N}$.*

***Proof*** Proposition 5.32 shows that $C_{A\otimes B}(S \otimes B) = C_A(S) \otimes F \cong C_A(S)$. The last assertion follows by taking $M_n(F)$ for $B$ (see Example 5.22). □

If $S$ is a subalgebra of $A$, we can consider $A$ as a (left) $S$-module. Therefore, we can speak about $\mathrm{End}_S(A)$.

**Corollary 5.36** *Let $A$ be a finite dimensional central simple algebra. If $S$ is a subalgebra of $A$ that contains the unity of $A$, then $C_A(S) \otimes A^{\mathrm{o}} \cong \mathrm{End}_S(A)$ via the isomorphism $a \otimes b \mapsto L_a R_b$.*

***Proof*** Theorems 1.43 and 5.27 tell us that $a \otimes b \mapsto L_a R_b$ is an isomorphism from $A \otimes A^{\mathrm{o}}$ onto $\mathrm{End}_F(A)$. Accordingly, the centralizer of $S \otimes F$ in $A \otimes A^{\mathrm{o}}$ is isomorphic to the centralizer of $\{L_s \mid s \in S\}$ in $\mathrm{End}_F(A)$. The former equals $C_A(S) \otimes A^{\mathrm{o}}$ by Proposition 5.32, while the latter is nothing but $\mathrm{End}_S(A)$. □

## 5.7 Scalar Extension (The "Right" Approach)

We are now in a position to place the notion of the scalar extension, discussed in Sect. 2.9, on a solid foundation. Recall the basic idea behind the definition: Given an $F$-algebra $A$ and a field extension $K$ of $F$, we wish to construct a $K$-algebra $A_K$ that has essentially the same multiplication as $A$. We saw that by choosing and fixing a basis of $A$ this can be achieved in a straightforward manner. We will now show that by making use of the tensor product, this construction can be accomplished independently of basis.

A field extension $K$ of $F$ can be considered as an $F$-algebra. Therefore, we can form the $F$-algebra $K \otimes_F A$. The point is to convert it into a $K$-algebra.

**Lemma 5.37** *Let $A$ be an $F$-algebra and let $K$ be a field extension of $F$. Then $K \otimes_F A$ is a $K$-algebra relative to the scalar multiplication given by*

$$\lambda(\mu \otimes x) = (\lambda\mu) \otimes x, \quad \lambda, \mu \in K, \ x \in A.$$

***Proof*** We have to prove that this scalar multiplication is well-defined. This is easy. Take $\lambda \in K$. The left multiplication map $L_\lambda : K \to K$ is $F$-linear, and hence so is $L_\lambda \otimes \mathrm{id}_A : K \otimes_F A \to K \otimes_F A$. Our definition of the scalar multiplication can be written as

$$\lambda r = (L_\lambda \otimes \mathrm{id}_A)(r)$$

for $\lambda \in K$ and $r \in K \otimes_F A$. This shows that it is well-defined, and also that $\lambda(r+s) = \lambda r + \lambda s$ holds for $\lambda \in K$ and $r, s \in K \otimes_F A$. It is straightforward to check that we also have

$$(\lambda + \mu)r = \lambda r + \mu r, \quad (\lambda\mu)r = \lambda(\mu r), \quad \lambda(rs) = (\lambda r)s = r(\lambda s)$$

for $\lambda, \mu \in K, r, s \in K \otimes_F A$. □

**Definition 5.38** Let $A$ be an $F$-algebra and let $K$ be a field extension of $F$. The $K$-algebra $K \otimes_F A$ (defined in Lemma 5.37) is called the **scalar extension** of $A$ to $K$. It will be denoted by $A_K$.

The following assertions are immediate:

(a) $1 \otimes A$ is an $F$-subalgebra of $A_K$ isomorphic to $A$.
(b) $A_K$ is linearly spanned by $1 \otimes A$.
(c) If $\{e_i \mid i \in I\}$ is a basis of $A$, then $\{1 \otimes e_i \mid i \in I\}$ is a basis of $A_K$ (see Corollary 5.15).

Assume now that $A$ is finite dimensional. Then we see from (c) that

$$[A_K : K] = [A : F].$$

If $\{e_1, \ldots, e_d\}$ is a basis of $A$ with multiplication table

$$e_i e_j = \sum_{k=1}^{d} \alpha_{ijk} e_k, \quad \alpha_{ijk} \in F,$$

then the elements $\hat{e}_i = 1 \otimes e_i$ form a basis of $A_K$ with the same multiplication table:

$$\hat{e}_i \hat{e}_j = \sum_{k=1}^{d} \alpha_{ijk} \hat{e}_k.$$

Neglecting the insignificant notational differences, we thus see that Definition 5.38 coincides with the definition given in Sect. 2.9. Being independent of basis, the new definition is not only more clear and elegant, but also more suitable for practical use.

We already saw in Theorem 2.86 that scalar extensions can be used to transform finite dimensional central simple algebras into matrix algebras over fields. Let us consider this phenomenon from the conceptual standpoint.

**Definition 5.39** Let $A$ be a finite dimensional central simple $F$-algebra. A field extension $K$ of $F$ is said to be a **splitting field** for $A$ if $A_K \cong M_n(K)$ for some $n \in \mathbb{N}$.

We can now reword Theorem 2.86 as follows.

**Theorem 5.40** *The algebraic closure $\overline{F}$ of $F$ is a splitting field for every finite dimensional central simple $F$-algebra.*

For division algebras we have another, more intrinsic class of splitting fields.

**Theorem 5.41** *Every maximal subfield $K$ of a finite dimensional central division algebra $D$ is a splitting field for $D$.*

***Proof*** If $a \in C_D(K)$, then $a$ and $K$ generate a field and so $a \in K$ since $K$ is maximal. That is, $C_D(K) = K$. Similarly, $C_{D^o}(K) = K$. Taking $D^o$ for $A$ and $K$ for $S$ in Corollary 5.36, we thus obtain that $K \otimes_F D$ and $\mathrm{End}_K(D^o)$ are isomorphic as $F$-algebras via the isomorphism $k \otimes d \mapsto L_k R_d$. However, this map is $K$-linear, so they are actually isomorphic as $K$-algebras. Since $D^o$ is finite dimensional over $K$, we have $\mathrm{End}_K(D^o) \cong M_n(K)$. □

**Example 5.42** A standard example of a maximal subfield of $\mathbb{H}$ is $\mathbb{R} \oplus \mathbb{R}i$; however, in view of Theorem 5.40, this is hardly a new example of a splitting field for $\mathbb{H}$. If we take, for instance, the *rational quaternions* $\mathbb{Q} \oplus \mathbb{Q}i \oplus \mathbb{Q}j \oplus \mathbb{Q}k$, then Theorems 5.40 and 5.41 do yield different splitting fields.

## 5.8 Tensor Products of Simple Algebras and the Brauer Group

In light of Example 5.22, we can state the basic Wedderburn's structure theorem (Theorem 2.73) as follows: A finite dimensional algebra is simple if and only if it is isomorphic to the tensor product of a matrix algebra $M_n(F)$ and a division algebra $D$. This shows that, unlike the direct product, the more sophisticated tensor product may give rise to simple algebras. The next theorem examines when this can occur.

**Theorem 5.43** *Let $A$ be an arbitrary unital algebra and let $B$ be a central simple algebra. Then every ideal of $A \otimes B$ is of the form $I \otimes B$, where $I$ is an ideal of $A$. Accordingly, $A \otimes B$ is simple if and only if $A$ is simple.*

***Proof*** Let $U$ be an ideal of $A \otimes B$. It is immediate that

$$I = \{a \in A \mid a \otimes 1 \in U\}$$

is an ideal of $A$ and $I \otimes B \subseteq U$. To prove the converse inclusion, take $u \in U$. By Corollary 5.15, we can write $u = \sum_{j=1}^{n} a_j \otimes f_j$ for some $a_1, \ldots, a_n \in A$ and linearly independent $f_1, \ldots, f_n \in B$. Corollary 4.25 shows that there exist $b_i, c_i \in B$ such that $\varphi = \sum_{i=1}^{r} L_{b_i} R_{c_i}$ satisfies $\varphi(f_1) = 1$ and $\varphi(f_j) = 0$, $j \geq 2$. Hence,

$$U \ni \sum_{i=1}^{r} (1 \otimes b_i) u (1 \otimes c_i) = \sum_{i=1}^{r} \sum_{j=1}^{n} a_j \otimes b_i f_j c_i = \sum_{j=1}^{n} a_j \otimes \varphi(f_j) = a_1 \otimes 1.$$

Thus, $a_1 \in I$. Similarly we see that each $a_i \in I$, and therefore $u \in I \otimes B$. This proves that $U = I \otimes B$.

If $A$ is simple, then $I = 0$ or $I = A$ and hence $U = 0$ or $U = A \otimes B$, proving that $A \otimes B$ is simple. Conversely, assume that $A \otimes B$ is simple, and choose a nonzero ideal $I$ of $A$. Then $I \otimes B$ is a nonzero ideal of $A \otimes B$, so $I \otimes B = A \otimes B$. Given $a \in A$, we therefore have $a \otimes 1 = \sum_i a_i \otimes b_i$ for some $a_i \in I$ and $b_i \in B$. From Lemma 5.12 we see that $a$ is a linear combination of the $a_i$'s, so $a \in I$. Thus $I = A$, proving that $A$ is simple. □

**Corollary 5.44** *Let $A$ be a unital algebra. Then every ideal of $M_n(A)$ is of the form $M_n(I)$, where $I$ is an ideal of $A$. Accordingly, $M_n(A)$ is simple if and only if $A$ is simple.*

***Proof*** Take $M_n(F)$ for $B$ in Theorem 5.43. □

Corollary 5.44 can also be proved in a straightforward manner, avoiding tensor products.

**Corollary 5.45** *The tensor product of two central simple algebras is a central simple algebra.*

***Proof*** Combine Theorem 5.43 and Corollary 5.34. □

Let us look at tensor products of real division algebras (of nontrivial dimension).

**Example 5.46** We claim that

- $\mathbb{C} \otimes_{\mathbb{R}} \mathbb{C} \cong \mathbb{C} \times \mathbb{C}$,
- $\mathbb{C} \otimes_{\mathbb{R}} \mathbb{H} \cong M_2(\mathbb{C})$,
- $\mathbb{H} \otimes_{\mathbb{R}} \mathbb{H} \cong M_4(\mathbb{R})$.

The last statement was noticed in Example 5.30. We know already from Example 2.84 that the scalar extension of $\mathbb{H}$ to $\mathbb{C}$ is $M_2(\mathbb{C})$. Therefore, $\mathbb{C} \otimes_{\mathbb{R}} \mathbb{H}$ and $M_2(\mathbb{C})$ are isomorphic even as $\mathbb{C}$-algebras (and hence also as $\mathbb{R}$-algebras). We may also regard $\mathbb{C} \otimes_{\mathbb{R}} \mathbb{C}$ as a $\mathbb{C}$-algebra (in view of Lemma 5.37). As such it is isomorphic to $\mathbb{C} \times \mathbb{C}$ since they are both unital 2-dimensional $\mathbb{C}$-algebras containing an element which is not a scalar multiple of unity and whose square is $-1$ (i.e., $1 \otimes i$ and $(i, -i)$, respectively). Therefore, $\mathbb{C} \otimes_{\mathbb{R}} \mathbb{C}$ and $\mathbb{C} \times \mathbb{C}$ are isomorphic also as $\mathbb{R}$-algebras.

Let us summarize our findings. The tensor product of two finite dimensional simple algebras is not always simple. If, however, one of them is central, then it is. If both are central, then so is their tensor product. All this is nicely illustrated in Example 5.46.

From $\mathbb{H} \otimes_{\mathbb{R}} \mathbb{H} \cong M_4(\mathbb{R})$ we also see that the tensor product of two finite dimensional central division algebras is not necessarily a division algebra. However, it always is a central simple algebra, and such an algebra conceals within itself a central division algebra. That is, it is isomorphic, for some $n \in \mathbb{N}$, to $M_n(D)$, where $D$ is a central division algebra (Corollary 2.77). Moreover, $D$ is essentially unique (Theorem 3.73). Thus, ignoring the matrix part of the resulting central simple algebra, we can say that the tensor product of two finite dimensional central division

algebras gives rise to an algebra from the same class. Say, the product of $\mathbb{H}$ with itself brings out $\mathbb{R}$. (Maybe not a very exciting result, but what to expect in view of Frobenius' theorem?)

Let us make this precise. Fix a field $F$. Let $A$ and $A'$ be finite dimensional central simple $F$-algebras. Then, as we know, there exist $n, n' \in \mathbb{N}$ and central division algebras $D$, $D'$ such that $A \cong M_n(D)$ and $A' \cong M_{n'}(D')$. We shall say that $A$ and $A'$ are **equivalent**, and write $A \sim A'$, if $D \cong D'$. This definition is unambiguous since $D$ and $D'$ are uniquely determined up to isomorphism. Note that $\sim$ is indeed an equivalence relation in the class of all finite dimensional central simple $F$-algebras.

We denote the equivalence class of $A$ by $[A]$. If $D$ is a finite dimensional central division algebra, then $[D] = [M_2(D)] = [M_3(D)] = \dots$ As indicated above, among all representatives of this class we are interested only in $D$. The problem is that $D$ may be hidden, so we are forced to treat all representatives equally.

The next proposition speaks about the set of equivalence classes of finite dimensional central simple $F$-algebras. The fact that this is indeed a set requires a proof, but we omit it.

**Proposition 5.47** *The set of equivalence classes of finite dimensional central simple $F$-algebras, endowed with multiplication*

$$[A] \cdot [B] = [A \otimes B],$$

*is an abelian group.*

***Proof*** We first have to show that this multiplication is well-defined. That is, we have to prove that for all finite dimensional central simple algebras $A$, $A'$, $B$, $B'$,

$$A \sim A' \text{ and } B \sim B' \Longrightarrow A \otimes B \sim A' \otimes B'.$$

By definition, $A \sim A'$ means $A \cong M_n(D)$ and $A' \cong M_{n'}(D')$ with $D \cong D'$. But then we actually have $A' \cong M_{n'}(D)$. Similarly, $B \sim B'$ implies that $B \cong M_m(C)$ and $B' \cong M_{m'}(C)$ for some $m, m' \in \mathbb{N}$ and a central division algebra $C$. Since $D \otimes C$ is again a finite dimensional central simple algebra by Corollary 5.45, $D \otimes C \cong M_k(E)$ with $k \in \mathbb{N}$ and $E$ a central division algebra. Now, using the associativity and commutativity of tensor products, together with basic facts about matrix algebras represented as tensor products (explained in Example 5.22), we obtain

$$\begin{aligned} A \otimes B &\cong \big(M_n(F) \otimes D\big) \otimes \big(M_m(F) \otimes C\big) \\ &\cong \big(M_n(F) \otimes M_m(F)\big) \otimes \big(D \otimes C\big) \\ &\cong M_{nm}(F) \otimes \big(M_k(F) \otimes E\big) \\ &\cong M_{nmk}(F) \otimes E \\ &\cong M_{nmk}(E). \end{aligned}$$

Similarly we see that $A' \otimes B' \cong M_{n'm'k}(E)$. Accordingly, $A \otimes B \sim A' \otimes B'$.

The multiplication is associative and commutative since the tensor product is associative and commutative. Obviously, $[F]$ is the identity element. Finally, from $A \otimes A^{\mathrm{o}} \cong M_n(F)$ (Corollary 5.29) we infer that $[A]^{-1} = [A^{\mathrm{o}}]$. □

**Definition 5.48** The group from Proposition 5.47 is called the **Brauer group** of the field $F$. We denote it by $\mathbf{Br}(F)$.

The basic idea behind this concept is to use tensor products to classify finite dimensional central division algebras over a given field. The name was given after R. Brauer who made fundamental contributions to this area of research.

Computing $\mathbf{Br}(F)$ for a given field $F$ is not an easy task. At least we know the following:

- If $F$ is algebraically closed, then $\mathbf{Br}(F) = \{1\}$.
- If $F$ is finite, then $\mathbf{Br}(F) = \{1\}$.
- $\mathbf{Br}(\mathbb{R}) \cong \mathbb{Z}_2$.

These are restatements of Proposition 1.8, Theorem 1.54, and Corollary 1.27.

Brauer groups turn out to be important in different mathematical areas. Nevertheless, we will not discuss them in greater detail here, and only refer to [3] and [11] for possible further reading. One might protest that we have just warmed up, and besides the definition and elegant interpretations of some fundamental results from Chap. 1 we told nothing. However, we hope the reader will agree that the definition of the Brauer group is captivating in its own right. This short discussion was meant only as an invitation to a field of study.

## 5.9 Separable Algebras

Let $A$ be a finite dimensional algebra. Recall that the (Jacobson) radical of $A$ is the unique maximal nilpotent ideal (see Sects. 2.1 and 4.6). Like in Chap. 2, we denote it by $N$ (rather than $\mathrm{rad}(A)$). Recall also that the following conditions are equivalent if $A \neq 0$: (i) $A$ is semisimple, (ii) $A$ is semiprime, (iii) $N = 0$, and (iv) $A \cong A_1 \times \cdots \times A_r$ where $A_i$ are simple algebras isomorphic to matrix algebras over division algebras (see Theorems 2.71 and 3.71). If $A$ is commutative, then $A_i$ are fields, and we can add another equivalent condition that $A$ is reduced.

In Example 5.46 we showed that $\mathbb{C} \otimes_{\mathbb{R}} \mathbb{C} \cong \mathbb{C} \times \mathbb{C}$. The tensor product of fields thus is not necessarily a field. It is a bit more surprising that it is not necessarily semisimple. This follows from the next lemma. Its proof uses several standard results from commutative algebra surveyed in Prerequisites, as well as basic facts on scalar extensions from Sect. 5.7.

**Lemma 5.49** *A finite field extension $E$ of a field $F$ is separable if and only if $E_K = K \otimes_F E$ is a semisimple algebra for every field extension $K$ of $F$.*

***Proof*** In the first step of the proof, we show that if an element $a$ from some field extension of $F$ is algebraic over $F$ and $p(\omega) \in F[\omega]$ is the minimal polynomial of $a$ over $F$, then for every field extension $K$ of $F$ we have

$$F(a)_K = K \otimes_F F(a) \cong K[\omega]/(p(\omega)), \tag{5.3}$$

where $(p(\omega))$ is the principal ideal of $K[\omega]$ generated by $p(\omega)$. To this end, recall that $\{a^i \mid i = 0, 1, \ldots, m-1\}$, where $m$ is the degree of $p(\omega)$, is a basis of $F(a)$ over $F$, and hence $\{1 \otimes a^i \mid i = 0, 1, \ldots, m-1\}$ is a basis of $F(a)_K$ over $K$. This implies that $f(\omega) \mapsto f(1 \otimes a)$ is a surjective homomorphism from $K[\omega]$ onto $F(a)_K$. A standard application of the division algorithm shows that its kernel is $(p(\omega))$. Hence, (5.3) follows.

Assume now that $E$ is separable. By the primitive element theorem, there exists an $a \in E$ such that $E = F(a)$. Take a field extension $K$ of $F$. Viewing the minimal polynomial $p(\omega)$ of $a$ over $F$ as a polynomial in $K[X]$, we have $p(\omega) = p_1(\omega) \cdots p_n(\omega)$ where $p_i(\omega)$ are irreducible monic polynomials in $K[\omega]$. Note that $p_i(\omega)$ are distinct since $p(\omega)$ is separable. Therefore, they are relatively prime and hence

$$(p_i(\omega)) + (p_j(\omega)) = K[\omega]$$

whenever $i \neq j$. Observe that $f(\omega) \in K[\omega]$ is divisible by every $p_i(\omega)$ if and only if $f(\omega)$ is divisible by $p(\omega)$. Thus,

$$(p_1(\omega)) \cap \cdots \cap (p_n(\omega)) = (p(\omega)).$$

We may now use the Chinese reminder theorem to conclude that

$$K[\omega]/(p(\omega)) \cong K[\omega]/(p_1(\omega)) \times \cdots \times K[\omega]/(p_n(\omega)).$$

Each $K[\omega]/(p_i(\omega))$ is a field since $p_i(\omega)$ is irreducible over $K$. As $E = F(a)$, it follows from (5.3) that $E_K$ is semisimple.

To prove the converse, assume that $E$ is not separable. Then $E$ contains an element $a$ whose minimal polynomial $p(\omega)$ over F is not separable. Letting $K$ to be the splitting field for $p(\omega)$ over $F$, we can thus write $p(\omega) = g(\omega)^2 h(\omega)$ for some $g(\omega), h(\omega) \in K[\omega]$ with $g(\omega)$ nonconstant. Hence, $g(\omega)h(\omega) + (p(\omega))$ is a nonzero element of $K[\omega]/(p(\omega))$ having square zero. In light of (5.3), this means that $F(a)_K$ contains nonzero nilpotent elements. But then so does $E_K$, simply because $F(a) \subseteq E$ and hence $F(a)_K \subseteq E_K$. This proves that $E_K$ is not semisimple. □

**Definition 5.50** A finite dimensional algebra $A$ over a field $F$ is called **separable** if $A_K = K \otimes_F A$ is a semisimple algebra for every field extension $K$ of $F$.

Lemma 5.49 states that a finite field extension $E$ of $F$ is a separable algebra over $F$ if and only if $E$ is a separable field extension of $F$. This in particular shows

that semisimple algebras are not always separable. On the other, taking $F$ for $K$ in the definition we see that every separable algebra $A$ is semisimple, and hence isomorphic to the direct product $A_1 \times \cdots \times A_r$ where $A_i$ are simple algebras. Recall from Lemma 1.28 that their centers $Z(A_i)$ are fields. The next theorem shows that the separability of $A$ is equivalent to the separability of each $Z(A_i)$.

**Theorem 5.51** *A finite dimensional algebra $A$ over a field $F$ is separable if and only if $A \cong A_1 \times \cdots \times A_r$ where $A_i$ are simple algebras whose centers $Z(A_i)$ are separable field extensions of $F$.*

***Proof*** Assume that $A$ is separable. Let $A_i$ be simple algebras such that $A \cong A_1 \times \cdots \times A_r$. Take a field extension $K$ of $F$. The tensor product of algebras is distributive, so we have

$$A_K = K \otimes_F A \cong K \otimes_F A_1 \times \cdots \times K \otimes_F A_r. \tag{5.4}$$

Since $A_K$ is semisimple (equivalently, semiprime), so is each $K \otimes_F A_i$. Its center $Z(K \otimes_F A_i)$ is therefore also semisimple (equivalently, reduced). Now, Corollary 5.33 shows that

$$Z(K \otimes_F A_i) = K \otimes_F Z(A_i),$$

so Lemma 5.49 tells us that $Z(A_i)$ is a separable field extension of $F$.

Conversely, assume that $A \cong A_1 \times \cdots \times A_r$ with $A_i$ simple and $Z(A_i)$ a separable field extension of $F$. We want to show that $A_K$ is semisimple for every field extension $K$ of $F$. Since (5.4) holds, we may assume without loss of generality that $A$ is simple. Write $Z$ for $Z(A)$. We can consider $K \otimes_F Z$ and $A_K = K \otimes_F A$ as $Z$-algebras (via the scalar multiplications $z(k \otimes z') = k \otimes zz'$ and $z(k \otimes a) = k \otimes za$). Observe that the algebra $(K \otimes_F Z) \otimes_Z A$ is isomorphic to the algebra $A_K$ via $(k \otimes z) \otimes a \mapsto k \otimes za$. Let us show that $(K \otimes_F Z) \otimes_Z A$ is semisimple. We start by applying Lemma 5.49 to conclude that $K \otimes_F Z$ is isomorphic to $K_1 \times \cdots \times K_s$ where $K_i$ are fields. Again using that the tensor product is distributive, it follows that

$$(K \otimes_F Z) \otimes_Z A \cong (K_1 \otimes_Z A) \times \cdots \times (K_s \otimes_Z A).$$

Since $A$ is a central simple $Z$-algebra, Theorem 5.43 shows that each $K_i \otimes_Z A$ is a simple algebra. Consequently, $(K \otimes_F Z) \otimes_Z A$ is semisimple. □

If the field $F$ is perfect, $Z(A_i)$ are automatically separable field extensions of $F$. We thus have the following corollary.

**Corollary 5.52** *A finite dimensional semisimple algebra over a perfect field is separable.*

## 5.10 Wedderburn's Principal Theorem

Up to this point, this chapter was mostly devoted to building the tensor product machinery. In the remaining sections, we will prove three important, classical theorems whose statements do not involve tensor products, but proofs do.

In Chap. 2, we proved Wedderburn's principal theorem for algebras over algebraically closed fields (Corollary 2.76). The purpose of this section is to establish the general version of this theorem.

First, we need two easy lemmas. Before stating the first one, recall that if $N$ is the radical of a finite dimensional algebra $A$, then the algebra $A/N$ is semisimple (see Lemma 2.28).

**Lemma 5.53** *Let $A$ be a finite dimensional algebra with radical $N$. If $I$ is an ideal of $A$ such that the factor algebra $A/I$ is semisimple, then $I \supseteq N$. Moreover, if $I$ is nilpotent, then $I = N$.*

***Proof*** Observe that $(N + I)/I$ is a nilpotent ideal of $A/I$. As $A/I$ has no nonzero nilpotent ideals, $N + I = I$ and thus $I \supseteq N$. If $I$ is nilpotent, then $I = N$ since $N$ is a maximal nilpotent ideal. □

The second lemma presents the conclusion we wish to prove in three different, but equivalent ways.

**Lemma 5.54** *Let $A$ be a finite dimensional algebra with radical $N$ and let $\pi : A \to A/N$ be the canonical homomorphism. The following conditions are equivalent:*

(i) *$A$ has a subalgebra $A_0$ isomorphic to $A/N$.*
(ii) *$A$ has a subalgebra $A_0$ such that $A$ is the vector space direct sum of $A_0$ and $N$.*
(iii) *There exists an embedding $\varphi : A/N \to A$ such that $\pi\varphi = \mathrm{id}_{A/N}$.*

***Proof*** (i) $\Longrightarrow$ (ii). Since $A_0 \cong A/N$ is semisimple and $A_0 \cap N$ is a nilpotent ideal of $A_0$, $A_0 \cap N = 0$. As the dimension of $A$ is equal to the sum of the dimensions of $A/N \cong A_0$ and $N$, we have $A = A_0 \oplus N$.

(ii) $\Longrightarrow$ (iii). From $A = A_0 \oplus N$ it follows that $\pi|_{A_0}$, the restriction of $\pi$ to $A_0$, is an isomorphism from $A_0$ onto $A/N$. Its inverse $\varphi : A/N \to A_0$ thus satisfies $\pi\varphi = \mathrm{id}_{A/N}$.

(iii) $\Longrightarrow$ (i). Take $A_0 = \varphi(A/N)$. □

**Example 5.55** Let $A = T_n(F)$ be the algebra of all upper triangular matrices over $F$. We know from Example 2.29 that the radical $N$ of $A$ is the set of all strictly upper triangular matrices, and that $A/N \cong F^n$, the direct product of $n$ copies of $F$. Denoting the subalgebra of all diagonal matrices in $A$ by $A_0$, we clearly have $A_0 \cong A/N$, $A = A_0 \oplus N$, and $\varphi : A/N \to A$ defined by $\varphi(a + N) = d$, where $d$ is the diagonal part of the matrix $a$, is an embedding satisfying $\pi\varphi = \mathrm{id}_{A/N}$.

In our statement of Wedderburn's principal theorem, we will combine conditions (i) and (ii). Note that by $A = A_0 \oplus N$ we mean that $A$ is the vector space direct sum of $A_0$ and $N$.

**Theorem 5.56 (Wedderburn's Principal Theorem)** *Let $A$ be a finite dimensional algebra with radical $N$. If the algebra $A/N$ is separable, then $A$ contains a subalgebra $A_0$ such that $A_0 \cong A/N$ and $A = A_0 \oplus N$.*

***Proof*** We first consider the case where $N^2 = 0$. In the final part of the proof, we will show that the general case follows quickly from this one.

Pick a basis $\{e_1, \ldots, e_r\}$ of $N$ and extend it to a basis $\{e_1, \ldots, e_r, e_{r+1}, \ldots, e_d\}$ of $A$. Let $\alpha_{ijk}$ be elements in the base field $F$ such that $e_i e_j = \sum_{k=1}^{d} \alpha_{ijk} e_k$. Write $\overline{e}_i = e_i + N$ and observe that $\{\overline{e}_{r+1}, \ldots, \overline{e}_d\}$ is basis of $A/N$ and that $\overline{e}_i \overline{e}_j = \sum_{k=r+1}^{d} \alpha_{ijk} \overline{e}_k$.

Our goal is to show that $A$ satisfies the equivalent conditions of Lemma 5.54. Let us reformulate condition (iii) in terms of $\alpha_{ijk}$. Note first that a linear map $\varphi : A/N \to A$ satisfies $\pi\varphi = \mathrm{id}_{A/N}$ if and only if $\varphi(\overline{e_i}) - e_i \in N$, $i = r+1, \ldots, d$, meaning that

$$\varphi(\overline{e}_i) = e_i + \sum_{l=1}^{r} \xi_{il} e_l$$

for some $\xi_{il} \in F$. Further, $\varphi$ is multiplicative, and hence an algebra embedding, if and only if $\varphi(\overline{e}_i \overline{e}_j) = \varphi(\overline{e}_i)\varphi(\overline{e}_j)$, $i, j = r+1, \ldots, d$. Since

$$\varphi(\overline{e}_i \overline{e}_j) = \sum_{k=r+1}^{d} \alpha_{ijk} \varphi(\overline{e}_k) = \sum_{k=r+1}^{d} \alpha_{ijk} e_k + \sum_{k=r+1}^{d} \sum_{l=1}^{r} \alpha_{ijk} \xi_{kl} e_l$$

and, using $NA$, $AN \subseteq N$ and $N^2 = 0$,

$$\begin{aligned}
\varphi(\overline{e}_i)\varphi(\overline{e}_j) &= (e_i + \sum_{k=1}^{r} \xi_{ik} e_k)(e_j + \sum_{k=1}^{r} \xi_{jk} e_k) \\
&= \sum_{l=1}^{d} \alpha_{ijl} e_l + \sum_{k=1}^{r} \sum_{l=1}^{r} \xi_{ik} \alpha_{kjl} e_l + \sum_{k=1}^{r} \sum_{l=1}^{r} \xi_{jk} \alpha_{ikl} e_l \\
&= \sum_{l=1}^{r} \left( \alpha_{ijl} + \sum_{k=1}^{r} (\alpha_{kjl} \xi_{ik} + \alpha_{ikl} \xi_{jk}) \right) e_l + \sum_{k=r+1}^{d} \alpha_{ijk} e_k,
\end{aligned}$$

we see by comparing the coefficients of both expressions that the multiplicativity of $\varphi$ is equivalent to

$$\sum_{k=r+1}^{d} \alpha_{ijk}\xi_{kl} - \sum_{k=1}^{r}(\alpha_{kjl}\xi_{ik} + \alpha_{ikl}\xi_{jk}) = \alpha_{ijl}, \tag{5.5}$$

$i, j = r+1, \ldots, d$, $l = 1, \ldots, r$. We have thus shown that condition (iii) of Lemma 5.54 is equivalent to the existence of elements $\xi_{il} \in F$ satisfying (5.5). In this way, we transformed our problem into the problem of the existence of a solution of a certain system of linear equations.

Let $K$ be a field extension of $F$. Since $A/N$ is separable by assumption, the scalar extension algebra $(A/N)_K$ is semisimple. Note that $(A/N)_K$ is canonically isomorphic to $A_K/N_K$. Indeed, $\mathrm{id}_K \otimes \pi$ is a surjective $K$-algebra homomorphism from $A_K = K \otimes_F A$ onto $(A/N)_K = K \otimes_F (A/N)$ with kernel $N_K = K \otimes_F N$. Therefore, $A_K/N_K$ is semisimple. Since $N_K^2 = 0$, Lemma 5.53 shows that $N_K$ is the radical of $A_K$.

The elements $\hat{e}_i = 1 \otimes e_i$, $i = 1, \ldots, d$, form a basis of $A_K$ and satisfy $\hat{e}_i\hat{e}_j = \sum_{k=1}^{d} \alpha_{ijk}\hat{e}_k$ (see Sect. 5.7). Moreover, $\hat{e}_1, \ldots, \hat{e}_r$ form a basis of $N_K$, which was just shown to be the radical of $A_K$. The first part of the proof that led to (5.5) therefore also applies to $A_K$; accordingly, $A_K$ satisfies condition (iii) of Lemma 5.54 if and only if there exist elements $\Xi_{il} \in K$ satisfying

$$\sum_{k=r+1}^{d} \alpha_{ijk}\Xi_{kl} - \sum_{k=1}^{r}(\alpha_{kjl}\Xi_{ik} + \alpha_{ikl}\Xi_{jk}) = \alpha_{ijl}. \tag{5.6}$$

So far, $K$ was any field extension of $F$. If we choose $K$ to be algebraically closed (say, the algebraic closure of $F$), then Corollary 2.76 tells us that $A_K$ satisfies condition (i), and hence all conditions of Lemma 5.54. Therefore, (5.6) holds for some $\Xi_{il} \in K$.

Our goal, however, is to establish (5.5) with $\xi_{il} \in F$. This follows easily from (5.6). Indeed, by Theorem 3.30 we can choose a basis of $K$ over $F$ that contains 1. Representing each $\Xi_{il}$ as a linear combination of the basis elements and writing $\xi_{il} \in F$ for the coefficient at 1, it follows, by the linear independence of the basis elements, that (5.5) holds. This completes the proof for the case where $N^2 = 0$.

For the general case, we proceed by induction on $r = [N : F]$. If $r = 0$, then $N = 0$ and there is nothing to prove. Let $r > 0$. We may assume that $N^2 \neq 0$. By the isomorphism theorem (mentioned in Prerequisites), $(A/N^2)/(N/N^2) \cong A/N$. Lemma 5.53 therefore tells us $N/N^2$ is the radical of $A/N^2$. Since $[N/N^2 : F] < r$, the induction hypothesis implies that $A/N^2$ has a subalgebra $\tilde{A}_0$ isomorphic to $(A/N^2)/(N/N^2)$, and hence also isomorphic to $A/N$. Clearly, $A' = \{x \in A \mid x + N^2 \in \tilde{A}_0\}$ is a subalgebra of $A$ containing $N^2$ and satisfying $A'/N^2 = \tilde{A}_0$. As $\tilde{A}_0$ is semisimple, $N^2$ is the radical of $A'$ by Lemma 5.53. Observe that $N^2 \subsetneq N$, and hence $[N^2 : F] < r$ since $N$ is nilpotent. By the induction hypothesis, $A'$ has a subalgebra $A_0$ such that $A_0 \cong A'/N^2 = \tilde{A}_0 \cong A/N$. Thus, $A_0$ satisfies condition (i) of Lemma 5.54. □

**Remark 5.57** The part of the proof in which (5.5) was derived from (5.6) actually shows that if a system of linear equations with coefficients in the field $F$ has a solution in some field extension $K$, then it has a solution in $F$. This can also be proved by standard linear algebra arguments. However, it was convenient and instructive to apply Theorem 3.30.

What is the point of Theorem 5.56? We know that the algebra $A/N$ is semisimple. If, moreover, it is separable, then Theorem 5.56 says that $A/N \cong A_0$ can be viewed inside $A$, which makes calculations in $A$ very transparent. Indeed, writing elements in $A = A_0 \oplus N$ as $(a_0, n)$ with $a_0 \in A_0$ and $n \in N$, we have

$$\begin{aligned}(a_0, n) + (b_0, r) &= (a_0 + b_0, n + r),\\ (a_0, n)(b_0, r) &= (a_0 b_0, a_0 r + n b_0 + n r)\end{aligned} \tag{5.7}$$

(look again at Example 5.55 for a concrete illustration). We also know that $A_0$ is isomorphic to the direct product of matrix algebras over division algebras and that $N$ is nilpotent. With all this information, we can say that we have a considerable understanding and control over any finite dimensional algebra $A$ such that $A/N$ is separable. How restrictive is this condition? Corollary 5.52 shows that it is automatically fulfilled for algebras over perfect ideals. Thus, the following holds.

**Corollary 5.58** *Let $A$ be a finite dimensional algebra over a perfect field $F$, and let $N$ be the radical of $A$. Then $A$ contains a subalgebra $A_0$ such that $A_0 \cong A/N$ and $A = A_0 \oplus N$.*

Recall that perfect fields include fields of characteristic 0, algebraically closed fields, and finite fields. It is therefore fair to say that Wedderburn's principal theorem holds for most algebras of interest.

We show at the end that the assumption that $F$ is perfect cannot be removed.

**Example 5.59** A standard (and the simplest) example of an imperfect field is $F = \mathbb{Z}_2(\alpha)$, the field of rational functions over $\mathbb{Z}_2$ in the indeterminate $\alpha$. Let $I$ be the ideal of the polynomial algebra $F[\omega]$ generated by $(\omega^2 + \alpha)^2 = \omega^4 + \alpha^2$. Note that $A = F[\omega]/I$ is a 4-dimensional commutative $F$-algebra with basis $\{1, a, a^2, a^3\}$ where $a = \omega + I$. The elements in $N = \text{span}\{a^2 + \alpha, (a^2 + \alpha)a\}$ have square zero. Using that $\alpha$ is not a square of an element in $F$, one easily shows that the elements outside $N$ are not nilpotent. Therefore, $N$ is the radical of $A$. The element $b = a + N \in A/N$ satisfies $b^2 = \alpha$. On the other hand, $A$ has no element whose square is $\alpha$. Consequently, $A/N$ does not embed into $A$.

## 5.11 The Skolem-Noether Theorem

In Sect. 1.6, we proved that all automorphisms of finite dimensional central simple algebras are inner. As we mentioned then, this is a special case of the Skolem-Noether theorem. We are now in a position to establish the general version.

**Theorem 5.60 (Skolem-Noether Theorem)** *Let $A$ be a finite dimensional central simple algebra. If $S$ is a simple subalgebra of $A$ that contains the unity $1$ of $A$, then every homomorphism from $S$ to $A$ that maps $1$ into $1$ can be extended to an inner automorphism of $A$.*

***Proof*** We start with the case where $A = \mathrm{End}_F(V)$ with $V$ a finite dimensional vector space over the field $F$. Consider $V$ as a (unital) $A$-module in the standard way. Of course, $V$ is also a unital $S$-module. Observe that another way of making $V$ a unital $S$-module is by defining $s \cdot v = \varphi(s)v$. To avoid confusion, we denote $V$, when viewed as an $S$-module with respect to this operation, by $V_\varphi$. Now, since $V$ and $V_\varphi$ are of the same dimension over $F$ (as vector spaces over $F$ they are simply identical), Corollary 3.74 implies that they are isomorphic as $S$-modules. Let $a : V \to V_\varphi$ be an isomorphism. Then

$$a(sv) = s \cdot a(v) = \varphi(s)a(v) \tag{5.8}$$

for all $s \in S$ and $v \in V$. Also, since $V$ and $V_\varphi$ are identical as vector spaces, $a$ is an $F$-linear operator from $V$ to $V$. That is, $a \in A$, and (5.8) can be read as $as = \varphi(s)a$ for all $s \in S$. Since $a$ is bijective, it is invertible in $A$. Thus, $\varphi(s) = asa^{-1}$ for all $s \in S$. Clearly, $\varphi$ can be extended to the inner automorphism $x \mapsto axa^{-1}$ of $A$.

We proceed to the general case. The idea of the proof is to replace $A$ by $A \otimes A^{\mathrm{o}}$, which is isomorphic to the matrix algebra $M_n(F)$ (Corollary 5.29) and can therefore be identified with $\mathrm{End}_F(V)$ where $V$ is an $n$-dimensional vector space over $F$. Its subalgebra $S \otimes A^{\mathrm{o}}$ is simple (Theorem 5.43) and $\varphi \otimes \mathrm{id}_{A^{\mathrm{o}}}$ is a homomorphism from $S \otimes A^{\mathrm{o}}$ to $A \otimes A^{\mathrm{o}}$ that preserves the unity. By what was proved above, there exists an invertible $\mathbf{a} \in A \otimes A^{\mathrm{o}}$ such that

$$(\varphi \otimes \mathrm{id}_{A^{\mathrm{o}}})(s \otimes y) = \mathbf{a}(s \otimes y)\mathbf{a}^{-1} \tag{5.9}$$

for all $s \in S$ and $y \in A^{\mathrm{o}}$. Taking $s = 1$ it follows that $(1 \otimes y)\mathbf{a} = \mathbf{a}(1 \otimes y)$. That is, $\mathbf{a}$ lies in the centralizer of $1 \otimes A^{\mathrm{o}}$ in $A \otimes A^{\mathrm{o}}$. Proposition 5.32 therefore shows that $\mathbf{a} = a \otimes 1$ for some $a \in A$. Since $\mathbf{a}$ is invertible in $A \otimes A^{\mathrm{o}}$, $a$ is invertible in $A$. Taking $y = 1$ in (5.9), we obtain $\varphi(s) = asa^{-1}$ for all $s \in S$. □

The Skolem-Noether theorem turns out to be extremely useful. A small indication of its applicability is an alternative proof of Wedderburn's theorem on finite division rings, sketched in Exercise 5.22.

## 5.12 The Double Centralizer Theorem

Let $S$ be a subalgebra of an algebra $A$. The centralizer of its centralizer, $C_A(C_A(S))$, is called the **double centralizer** of $S$. It is obviously a subalgebra of $A$ that contains $S$. Quite possibly $S$ is its proper subset. Say, if $S$ does not contain $Z(A)$, then this is certainly the case. Another example: if $A = M_n(F)$ and $S = T_n(F)$, $n \geq 2$, then $C_A(S) = F = Z(A)$, and hence $C_A(C_A(S)) = A \supsetneq S$. On the other hand, if

$S = A$ or $S = Z(A)$, then $C_A(C_A(S)) = S$. A more illuminating situation where the equality holds is presented in the following example.

**Example 5.61** Considering central algebras $S$ and $T$ as subalgebras of $A = S \otimes T$, we have $C_A(S) = T$ and $C_A(T) = S$ by Corollary 5.34, and hence $C_A(C_A(S)) = S$ and $C_A(C_A(T)) = T$. If $S$ and $T$ are finite dimensional, we also have $[A : F] = [S : F] \cdot [C_A(S) : F]$ and similarly, $[A : F] = [T : F] \cdot [C_A(T) : F]$.

This example illustrates the next theorem in which, however, we do not assume that $S$ is central.

**Theorem 5.62 (Double Centralizer Theorem)** *Let $A$ be a finite dimensional central simple $F$-algebra. If $S$ is a simple subalgebra of $A$ that contains the unity of $A$, then:*

(a) *$C_A(S)$ is a simple algebra.*
(b) *$[A : F] = [S : F] \cdot [C_A(S) : F]$.*
(c) *$C_A(C_A(S)) = S$.*

***Proof*** We first note that (c) follows (a) and (b). Indeed, knowing that $C_A(S)$ is simple, we can replace $S$ by $C_A(S)$ in (b) to obtain

$$[A : F] = [C_A(S) : F] \cdot [C_A(C_A(S)) : F].$$

Comparing this equation with (b), we arrive at $[C_A(C_A(S)) : F] = [S : F]$. Since $S \subseteq C_A(C_A(S))$, (c) follows.

We divide the proof of (a) and (b) into two steps. In the first step we consider the case where $S$ is a division algebra, and in the second step we consider the general case.

**(I)** Assume that $S = D$ is a division algebra. Corollary 5.36 shows that $C_A(D) \otimes A^o \cong \mathrm{End}_D(A)$ and Theorem 3.33 shows that $\mathrm{End}_D(A) \cong M_s(D^o)$, where $s = \dim_D A$. Therefore, $\mathrm{End}_D(A)$ is a simple algebra. Using the easier "only if" part of Theorem 5.43, it follows that $C_A(D)$ is simple.

It remains to prove (b). On the one hand, we have

$$[\mathrm{End}_D(A) : F] = s^2[D : F],$$

and on the other hand,

$$[\mathrm{End}_D(A) : F] = [C_A(D) : F] \cdot [A : F].$$

Next, we claim that

$$s = \frac{[A : F]}{[D : F]}.$$

If $D$ was a field, this would follow from Lemma 1.14. Fortunately, the proof remains unchanged if $D$ is noncommutative. Combining the above three equations, we obtain (b) (for $S = D$).

**(II)** Assume now that $S$ is an arbitrary simple subalgebra of $A$. As we know from Wedderburn's theorem (Theorem 2.73), there exist an $r \in \mathbb{N}$ and a division algebra $D$ such that $S \cong M_r(D)$. This implies that $S$ contains a set of $r \times r$ matrix units $e_{ij}$. As $A$ has the same unity as $S$, these are also the matrix units of $A$. Lemma 2.52 therefore shows that $A \cong M_r(B)$, where $B = e_{11}Ae_{11} \supseteq e_{11}Se_{11} = D$. Note that $B$ is simple (Corollary 5.44) and central (Lemma 1.23). Since, by Corollary 5.35,

$$C_A(S) \cong C_B(D),$$

applying **(I)** (with $B$ playing the role of $A$) we obtain that $C_B(D)$ is a simple algebra and $[B : F] = [D : F] \cdot [C_B(D) : F]$. Thus, $C_A(S)$ is simple and

$$[A : F] = r^2[B : F] = r^2[D : F] \cdot [C_B(D) : F].$$

Since

$$r^2[D : F] = [S : F] \quad \text{and} \quad [C_B(D) : F] = [C_A(S) : F],$$

this proves (b). □

Adding the assumption that $S$ is central, we arrive at the situation considered in Example 5.61.

**Corollary 5.63** *Let $A$ be a finite dimensional central simple $F$-algebra. If $S$ is a central simple subalgebra of $A$ that contains the unity of $A$, then $A \cong S \otimes C_A(S)$.*

***Proof*** Consider the linear map $\varphi : S \otimes C_A(S) \to A$, $\varphi(s \otimes t) = st$. Since the elements from $S$ commute with the elements from $C_A(S)$, $\varphi$ is multiplicative. The algebra $S \otimes C_A(S)$ is simple (Theorem 5.43), so $\ker \varphi = 0$. The equation $[A : F] = [S : F] \cdot [C_A(S) : F]$ shows that $S \otimes C_A(S)$ and $A$ have the same dimension. Hence, $\varphi$ is an isomorphism. □

Corollary 5.63 deals with the case where the center of $S$ is minimal. Let us take a look at the opposite extreme. Assume that $S = K$ is a maximal subfield of a finite dimensional central division algebra $D$. As observed in the proof of Theorem 5.41, $C_D(K) = K$. Corollary 5.63 does not hold for $S = K$; after all, $K \otimes K$ is a commutative algebra. The only nontrivial information that we extract from the double centralizer theorem is $[D : F] = [K : F]^2$, which is nothing but Theorem 1.51.

## Exercises

**5.1** Let $U$ and $V$ be vector spaces of dimension at least 2. Show that not every element in $U \otimes V$ is a simple tensor.

**5.2** Let $U$ be a real vector space. Show that $\sum_{i=1}^{n} u_i \otimes u_i = 0$ in $U \otimes U$ implies that each $u_i = 0$.

**5.3** Let $U$ be a vector space over a field $F$ with $\mathrm{char}(F) \neq 2$. Denote by $S$ the subspace of $U \otimes U$ generated by all elements of the form $u \otimes u$, $u \in U$, and by $T$ the subspace generated by all elements of the form $u \otimes v - v \otimes u$, $u, v \in U$. Show that $U \otimes U = S \oplus T$. What is the dimension of $S$ and $T$ if $\dim_F U = n < \infty$?

*Hint:* Make use of the map $u \otimes v \mapsto v \otimes u$.

**5.4** Let $U, U', V, V'$ be vector spaces and let $\varphi : U \to U'$ and $\psi : V \to V'$ be linear maps. Show that $\ker(\varphi \otimes \psi) = \ker \varphi \otimes V + U \otimes \ker \psi$.

**5.5** Use the tensor product to construct a 9-dimensional algebra whose radical is 5-dimensional.

**5.6** For a field $F$ and a nonempty set $X$, the set $F^X$ of all functions from $X$ to $F$ is an $F$-algebra under the standard operations of functions. Show that the algebra $F^X \otimes F^Y$ is isomorphic to the subalgebra of $F^{X \times Y}$ consisting of sums of the functions of the form $(x, y) \mapsto f(x)g(y)$ where $f \in F^X$ and $g \in F^Y$.

**Remark** The function $(x, y) \mapsto f(x)g(y)$ is commonly denoted by $f \otimes g$. This exercise thus justifies this notation (see also Example 5.21).

**5.7** Let $S = (s_{ij}) \in M_m(F)$ and $T = (t_{ij}) \in M_n(F)$. The $mn \times mn$ matrix

$$S \otimes T = \begin{bmatrix} s_{11}T & s_{12}T & \dots & s_{1m}T \\ s_{21}T & s_{22}T & \dots & s_{2m}T \\ \vdots & \vdots & \ddots & \vdots \\ s_{m1}T & s_{m2}T & \dots & s_{mm}T \end{bmatrix} = \begin{bmatrix} s_{11}t_{11} & s_{11}t_{12} & \dots & s_{1m}t_{1n} \\ s_{11}t_{21} & s_{11}t_{22} & \dots & s_{1m}t_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ s_{m1}t_{n1} & s_{m1}t_{n2} & \dots & s_{mm}t_{nn} \end{bmatrix}$$

is called the **Kronecker** or **tensor product** of matrices $S$ and $T$. This definition is clearly in accordance with the convention from Example 5.22. Note that it is also compatible with the tensor product of linear maps in the following sense: If $S$ is the matrix corresponding to a linear map $f$ with respect to the basis $\{e_1, \dots, e_m\}$, and $T$ is the matrix corresponding to a linear map $g$ with respect to the basis $\{f_1, \dots, f_n\}$, then $S \otimes T$ is the matrix corresponding to $f \otimes g$ with respect to the basis $\{e_1 \otimes f_1, \dots, e_1 \otimes f_n, e_2 \otimes f_1, \dots, e_m \otimes f_n\}$. Show that $\mathrm{tr}(S \otimes T) = \mathrm{tr}(S)\mathrm{tr}(T)$ and $\det(S \otimes T) = \det(S)^n \det(T)^m$.

*Hint:* $S \otimes T = (S \otimes I_n)(I_m \otimes T)$ where $I_k$ stands for the $k \times k$ identity matrix.

**5.8** Show that nonzero algebras $A$ and $B$ must be unital if $A \otimes B$ is unital.

**5.9** Show that $p \in \mathbb{N}$ is prime if and only if there do not exist algebras $A \neq F$ and $B \neq F$ such that $A \otimes B \cong M_p(F)$.

**5.10** Let $A, A', B, B'$ be finite dimensional simple algebras. Suppose there exist $m, n \in \mathbb{N}$ such that $M_m(A) \cong M_n(B)$ and $M_n(A') \cong M_m(B')$. Show that $A \otimes A' \cong B \otimes B'$.

**5.11** Let $A$ be a central simple algebra over a field $F$ with $\mathrm{char}(F) \neq 2, 3$. Show that $a \in A$ satisfies $[a, [a, [a, x]]] = 0$ for all $x \in A$ if and only if there exists a $\lambda \in F$ such that $(a - \lambda)^2 = 0$. Analyze, in a similar fashion, the condition $[a, [a, [a, x]]] = [a, x]$.

*Hint:* Use Theorem 5.27.

**5.12** Let $A$ be a 4-dimensional central simple $F$-algebra. Show that $C_A(a) = F + Fa$ for every $a \in A \setminus F$.

*Hint:* Apply scalar extension to reduce the problem to the case where $A = M_2(F)$ and $F$ is algebraically closed (so that the Jordan normal form of $a$ is applicable).

**5.13** Let $A$ be a finite dimensional central simple algebra. Show that the linear span of all commutators in $A$ has codimension 1 in $A$.

**5.14** Let $A$ be an $\mathbb{R}$-algebra. Show that the set $A \times A$ equipped with the operations

$$\begin{aligned}(x, y) + (z, w) &= (x + z, y + w),\\ (\alpha + i\beta)(x, y) &= (\alpha x - \beta y, \beta x + \alpha y),\\ (x, y)(z, w) &= (xz - yw, xw + yz),\end{aligned}$$

where $x, y, z, w \in A$ and $\alpha, \beta \in \mathbb{R}$, is a $\mathbb{C}$-algebra isomorphic to $A_{\mathbb{C}}$.

**Remark** This construction provides a shortcut to the notion of the scalar extension for mathematicians working in areas in which $\mathbb{R}$ and $\mathbb{C}$ are the only fields that matter.

**5.15** Let $D_1$ and $D_2$ be finite dimensional central division $F$-algebras. As we know, $D_1 \otimes D_2$ is isomorphic to $M_m(D)$ for some $m \in \mathbb{N}$ and a central division algebra $D$. Examine $D_i^{\mathrm{o}} \otimes M_m(D)$ to show that $m$ divides $[D_i : F]$, $i = 1, 2$. Hence conclude that $D_1 \otimes D_2$ is a division algebra in case $[D_1 : F]$ and $[D_2 : F]$ are relatively prime.

**Remark** The latter result is usually used in the proof of the following important result: If $D$ is a finite dimensional central division $F$-algebra and $[D : F] = p_1^{k_1} \dots p_r^{k_r}$ where $p_i$ are different primes, then there exist central division algebras $D_1, \dots, D_r$ such that $D \cong D_1 \otimes \dots \otimes D_r$ and $[D_i : F] = p_i^{k_i}$.

**5.16** Let $A$ and $A'$ be finite dimensional central simple $F$-algebras. Show that:

(a) $A \sim A'$ if and only if there exist $m, m' \in \mathbb{N}$ such that $M_m(A) \cong M_{m'}(A')$.
(b) $A \cong A'$ if and only if $A \sim A'$ and $[A : F] = [A' : F]$.

**5.17** Let $A$ and $B$ be simple unital $F$-algebras. Show that the algebra $A \otimes B$ is semiprime if (and only if) its center is semiprime (or, equivalently, reduced).

*Hint:* Take a nilpotent ideal $I$ of $A \otimes B$, and consider an element $\sum_{i=1}^{n} a_i \otimes b_i \neq 0$ in I such that $n$ is minimal.

**5.18** Let $\mathrm{char}(F) = p > 0$. Suppose a unital $F$-algebra $A$ contains an element $a$ such that $a \notin F$ and $a^p \in F$. Show that $A \otimes A$ then contains a nonzero nilpotent

element. Use this to show that if $F = \mathbb{Z}_p(\alpha)$ and $A$ is the splitting field of $\omega^p - \alpha$ over $F$, then $A \otimes A$ is not semisimple (cf. Lemma 5.49).

**5.19** Show that a finite dimensional algebra $A$ over a field $F$ is separable if and only if the algebra $A \otimes A^{\mathrm{o}}$ is semisimple.

**5.20** Determine $A_0$ and $N$ from Wedderburn's principal theorem and write (5.7) for the case where:

(a) $A$ is the algebra of all matrices of the form $\left[\begin{smallmatrix} \alpha & \beta \\ 0 & \alpha \end{smallmatrix}\right]$ where $\alpha, \beta \in \mathbb{C}$.
(b) $A$ is the algebra of all matrices of the form $\left[\begin{smallmatrix} \alpha & \beta \\ 0 & 0 \end{smallmatrix}\right]$ where $\alpha, \beta \in \mathbb{C}$.
(c) $A$ is the unitization of the algebra from (b) (to what algebra is $A$ isomorphic?).

**5.21** Let $A$ be a finite dimensional real unital algebra. Show that if every ideal of $A$ has even dimension, then $A$ contains a subfield isomorphic to $\mathbb{C}$. Observe that together with the result of Exercise 1.4, this implies that the following conditions are equivalent for $A$: (i) Every left ideal of $A$ has even dimension, (ii) Every ideal of $A$ has even dimension, (iii) $A$ contains a subfield isomorphic to $\mathbb{C}$.

*Hint:* Observe that if $A = A_0 \oplus N$ is the decomposition from Wedderburn's principal theorem and $I$ is an ideal of $A_0$, then $I \oplus N$ is an ideal of $A$. Apply also Corollary 2.80 (a).

**5.22** Wedderburn's theorem stating that a finite division ring $D$ is commutative was proved in Chap. 1 (Theorem 1.54). We now sketch an alternative proof and ask the reader to fill in the details. Write $F$ for the center of $D$ and assume that $F \neq D$. Fix a maximal subfield $K$ of $D$. Using Theorem 1.51, we see that any other maximal subfield $K'$ has the same cardinality. Since finite fields of the same cardinality are isomorphic, the Skolem-Noether theorem implies that $K' = aKa^{-1}$ for some $a \in D$. This gives $D^* = \cup_{a \in D^*} aK^*a^{-1}$ where $D^*$ (resp. $K^*$) is the group of invertible elements in $D$ (resp. $K$). However, this is impossible, since if $H$ is a proper subgroup of a finite group $G$, then $\cup_{a \in G} aHa^{-1} \subsetneq G$. To prove this, recall from group theory that the **normalizer** of $H$ in $G$ is the subgroup $N_G(H) = \{g \in G \mid gHg^{-1} = H\}$. Note that the number of distinct conjugate subgroups $aHa^{-1}$ is equal to $[G : N_G(H)]$, and hence conclude that the number of elements different from 1 in $\cup_{a \in G} aHa^{-1}$ does not exceed $(|H| - 1)[G : N_G(H)]$. Finally, observe that this latter number is smaller than $|G| - 1$.

**5.23** Use the Skolem-Noether theorem to show that besides 0, $\mathbb{R}$, and $\mathbb{H}$, the only subalgebras of $\mathbb{H}$ are those of the form $h(\mathbb{R} \oplus \mathbb{R}i)h^{-1}$, $h \in \mathbb{H} \setminus \{0\}$.

**5.24** Let $D$ be a finite dimensional central division algebra over $F$. Suppose that $u, v \in D$ have the same minimal polynomial over $F$. Use the Skolem-Noether theorem to show that there exists a nonzero $a \in D$ such that $v = aua^{-1}$.

**Remark** This result is attributed to L. E. Dickson.

**5.25** Let $A$ be a finite dimensional central simple algebra over a field $F$ with the property that $\lambda^2 \neq -1$ for every $\lambda \in F$. Let $u \in A$ be such that $u^2 = -1$. Use the Skolem-Noether theorem to show that:

(a) There exists an invertible element $a \in A$ that anticommutes with $u$.
(b) For every $v \in A$ satisfying $v^2 = -1$ there exists an invertible element $b \in A$ such that $v = bub^{-1}$.

**5.26** A linear map $\delta$ from an algebra $A$ to itself is called a **generalized derivation** if $\delta(xyz) = \delta(xy)z - x\delta(y)z + x\delta(yz)$ for all $x, y, z \in A$. Show that every generalized derivation of a finite dimensional central simple algebra $A$ is the sum of a left multiplication map and a right multiplication map.

*Hint:* Observe that

$$\begin{bmatrix} x & 0 \\ 0 & x \end{bmatrix} \mapsto \begin{bmatrix} x & \delta(x) - \delta(1)x \\ 0 & x \end{bmatrix}$$

defines an algebra homomorphism from an isomorphic copy of $A$ in $M_2(A)$ to $M_2(A)$.

**5.27** Let $S$ be a subalgebra of an algebra $A$. Show that:

(a) If $S$ is commutative, then so is $C_A(C_A(S))$, and $C_A(C_A(S)) \subseteq C_A(S)$.
(b) If $S$ is the centralizer of some subset of $A$, then $S = C_A(C_A(S))$.

**5.28** Let $A$ be a finite dimensional central simple $\mathbb{R}$-algebra. Show that $A$ contains a subfield $K$ such that $K = C_A(K)$ if and only if $A$ is isomorphic to $\mathbb{R}$, $M_2(\mathbb{R})$, or $\mathbb{H}$.

**5.29** Let $A$ and $S$ be as in the double centralizer theorem. Note that $S$, $C_A(S)$, and $C_A(Z)$ are algebras over the center $Z$ of $S$. Show that $S \otimes_Z C_A(S) \cong C_A(Z)$.

# Chapter 6
# Noncommutative Polynomials




Polynomials in several variables are a fundamental topic of interest in commutative algebra. Their counterparts in noncommutative algebra are the noncommutative polynomials, i.e., polynomials in noncommuting indeterminates. Thus, if $\xi, \eta$ are such indeterminates, then $\xi\eta$ and $\eta\xi$ are different noncommutative polynomials. Operations on these polynomials are defined in the obvious way. For example, if $f = \xi\eta + \eta$ and $g = \eta\xi - \eta$, then

$$f + g = \xi\eta + \eta\xi, \quad fg = \xi\eta^2\xi + \eta^2\xi - \xi\eta^2 - \eta^2, \quad gf = \eta\xi^2\eta - \eta^2.$$

The purpose of this chapter is to give an introduction to a couple of topics related to noncommutative polynomials. There will be plenty of examples and simple observations, more than theorems and proofs. The emphasis will be on polynomial identities. These are noncommutative polynomials which vanish when replacing indeterminates with arbitrary elements from a given algebra. However, most of the deeper results on polynomial identities will be obtained in the next chapter, when having more tools at our disposal. The main goal of this chapter is to introduce an appropriate framework.

## 6.1 Free Algebras

After the above informal introduction, let us define noncommutative polynomials precisely.

Take a nonempty set $X$. It is convenient to consider it as an indexed set, so let us write $X = \{\xi_i \mid i \in I\}$. A finite sequence of elements from $X$ will be denoted by $\xi_{i_1}\xi_{i_2}\ldots\xi_{i_m}$ and called a **word**. The empty sequence is not excluded; we denote it by 1 and call it the **empty word**. Defining multiplication by juxtaposition, i.e.,

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M. Brešar, *Introduction to Noncommutative Algebra*, Universitext,
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$$(\xi_{i_1}\xi_{i_2}\dots\xi_{i_m})\cdot(\xi_{j_1}\xi_{j_2}\dots\xi_{j_n}) = \xi_{i_1}\xi_{i_2}\dots\xi_{i_m}\xi_{j_1}\xi_{j_2}\dots\xi_{j_n},$$

the set of all words becomes a monoid (with identity element 1), which we denote by $X^*$. It is called the **free monoid** on $X$. Viewing $\xi_i \in X$ as an element of $X^*$, we can write $\xi_i^2$ for $\xi_i\xi_i$, $\xi_i^3$ for $\xi_i\xi_i\xi_i$, etc. Accordingly, every $w \neq 1$ in $X^*$ can be written as $w = \xi_{i_1}^{k_1}\xi_{i_2}^{k_2}\dots\xi_{i_r}^{k_r}$, where $i_j \in I$ and $k_j \in \mathbb{N}$. Of course, $i_j$ can be equal to $i_k$ if $j \neq k$, but we can achieve that $i_j \neq i_{j+1}$.

Recall that, given a monoid and a field, one can form a monoid algebra over this field.

**Definition 6.1** Let $F$ be a field and $X$ a set. The **free algebra** on $X$ over $F$ is the monoid algebra of $X^*$ over $F$.

In short, this is the algebra whose basis is the set of all words formed from $X$ (including 1) and whose multiplication is determined by juxtaposition of words.

The free algebra on $X$ over $F$ will be denoted by $F\langle X\rangle$, or, if $X$ is a finite (resp. countably infinite) set, also by $F\langle\xi_1,\dots,\xi_n\rangle$ (resp. $F\langle\xi_1,\xi_2,\dots\rangle$). The elements in $X$ are called **indeterminates**. If $X$ has just two elements, then we will avoid indices and write $F\langle\xi,\eta\rangle$.

We call elements in free algebras **noncommutative polynomials**. However, we will usually omit "noncommutative" and simply call them **polynomials**. The classical polynomials in commuting indeterminates will also occasionally appear in our discussion, but there is little chance for confusion. In particular, the notation will be different. We will use $\omega$ for denoting commuting indeterminates, and $\xi$ or $\eta$ for noncommuting indeterminates.

The notions such as the **coefficient**, **constant term**, and **constant polynomial** are defined in the same way as for commutative polynomials.

We can replace the role of the field $F$ in the definition of $F\langle X\rangle$ with an arbitrary ring $R$ and form the ring $R\langle X\rangle$ of (noncommutative) polynomials with coefficients in $R$. The operations in $R\langle X\rangle$ are defined just as in $F\langle X\rangle$. In particular, the elements from $R$ commute with the indeterminates from $X$, but not necessarily between themselves. Studying $R\langle X\rangle$ at a basic level is not much harder than studying $F\langle X\rangle$ with $F$ a field. Nevertheless, we restrict ourselves to the latter, at least in this chapter. In the next one, we will actually encounter the case where $R = \mathbb{Z}$. One can also describe $\mathbb{Z}\langle X\rangle$ as the subring of $\mathbb{Q}\langle X\rangle$ consisting of polynomials with integer coefficients.

Let $A$ be another unital $F$-algebra. A *unital* algebra homomorphism (i.e., an algebra homomorphism that sends 1 to 1) from $F\langle X\rangle$ to $A$ is uniquely determined by the action on $X$. This is clear since $F\langle X\rangle$ is generated by the set $X \cup \{1\}$. What we wish to point out is a kind of converse: An arbitrary function $f$ from $X$ to $A$ can be extended to a unique unital algebra homomorphism $\overline{f}$ from $F\langle X\rangle$ into $A$. Indeed, there is only one way to define this extension, and it is well-defined and is a homomorphism. The diagram in Fig. 6.1 is thus commutative. This may be just a simple observation, but is of fundamental importance. We will refer to it as to the **universal property** of $F\langle X\rangle$. (Readers familiar with the categorical language will

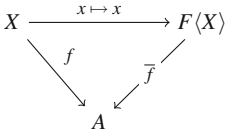



**Fig. 6.1** Universal property of $F\langle X\rangle$

recognize that this property shows that $F\langle X\rangle$ is a free object in the category of unital $F$-algebras.)

Every polynomial $f \in F\langle X\rangle$ involves only a finite number of indeterminates. If these are $\xi_{i_1}, \ldots, \xi_{i_n}$, then we write $f = f(\xi_{i_1}, \ldots, \xi_{i_n})$. Usually we will simply write $f = f(\xi_1, \ldots, \xi_n)$. Given such a polynomial and elements $x_1, \ldots, x_n \in A$, we define the **evaluation** $f(x_1, \ldots, x_n)$ in the obvious way. Say, if $f = \xi_1^2\xi_2\xi_1 - \xi_2\xi_3 + 1$, then $f(x_1, x_2, x_3) = x_1^2x_2x_1 - x_2x_3 + 1 \in A$, i.e., one just substitutes $x_i$ for $\xi_i$. In more formal terms, the universal property shows that the function $\xi_i \mapsto x_i$ gives rise to the unital homomorphism from $F\langle\xi_1, \ldots, \xi_n\rangle$ to $A$ which maps $f$ to $f(x_1, \ldots, x_n)$. If $A$ is not unital, then $f(x_1, \ldots, x_n)$ still makes sense provided that $f$ has zero constant term.

A polynomial is said to be a **monomial** if it is a nonzero scalar multiple of a word. Every nonzero polynomial $f \in F\langle X\rangle$ is a sum of monomials. More precisely, $f$ can be uniquely written as $f = \lambda_1 w_1 + \cdots + \lambda_m w_m$ where $w_1, \ldots, w_m$ are different words and the coefficients $\lambda_1, \ldots, \lambda_m$ are nonzero scalars in $F$. In this case, $f$ is the sum of the monomials $\lambda_i w_i$.

We define the **length** of a nonempty word $w = \xi_{i_1}\xi_{i_2}\ldots\xi_{i_m}$ as $\ell(w) = m$, and the length of the empty word as $\ell(1) = 0$. The **degree** of a nonzero polynomial $f = \lambda_1 w_1 + \cdots + \lambda_m w_m$ (where $\lambda_i \neq 0$) is defined as

$$\deg(f) = \max\{\ell(w_1), \ldots, \ell(w_m)\}.$$

If $\ell(w_1) = \cdots = \ell(w_m)$, i.e., if all monomials of $f$ have the same degree, then $f$ is said to be a **homogeneous polynomial**. For example, $\xi^2\eta - \xi\eta\xi + \eta^3$ is homogeneous of degree 3. We will be often concerned with homogeneous polynomials of the following type: $f = f(\xi_1, \ldots, \xi_n)$ is called a **multilinear polynomial** if each indeterminate $\xi_i$ appears in every monomial of $f$ exactly once. That is, $f$ is of the form

$$f = \sum_{\sigma \in S_n} \lambda_\sigma \xi_{\sigma(1)} \ldots \xi_{\sigma(n)},$$

where $\lambda_\sigma \in F$ and $S_n$ is the symmetric group on $\{1, \ldots, n\}$. Observe that for every $F$-algebra $A$, the map $(x_1, \ldots, x_n) \mapsto f(x_1, \ldots, x_n)$, $x_i \in A$, is multilinear.

Take nonzero polynomials $f = \lambda_1 w_1 + \cdots + \lambda_m w_m$ and $g = \mu_1 z_1 + \cdots + \mu_n z_n$, and assume, for notational simplicity, that $\deg(f) = \ell(w_1)$ and $\deg(g) = \ell(z_1)$. Note that $w_1 z_1 = w_k z_l$ occurs only when $k = l = 1$. Hence, $\deg(fg) = \ell(w_1 z_1) = \ell(w_1) + \ell(z_1)$. Thus,

$$\deg(fg) = \deg(f) + \deg(g)$$

holds for all nonzero $f, g \in F\langle X\rangle$ ("nonzero" can be omitted if one additionally defines $\deg(0) = -\infty$). An immediate consequence is that $F\langle X\rangle$ *is a domain*.

**Remark 6.2** A much deeper result is that $F\langle X\rangle$ *is a primitive algebra* unless $X$ is a singleton (note that $F\langle\xi\rangle$ is just the usual polynomial algebra $F[\xi]$, which is not primitive by Lemma 4.7). We shall not need this result, so we omit the proof. Let us just give its rough outline for the simplest case where $X = \{\xi, \eta\}$. Take a vector space $V$ over $F$ with basis $\{e_n \mid n \in \mathbb{N}\}$. Define $g, h \in \mathrm{End}_F(V)$ by $g(e_1) = 0$ and $g(e_n) = e_{n-1}$ for $n \neq 1$, and $h(e_n) = e_{n^2+1}$ for every $n$. Let $A$ be the subalgebra of $\mathrm{End}_F(V)$ generated by $g$ and $h$. It is not hard to prove that $A$ is a primitive algebra, in fact a dense algebra of linear operators of $V$. A slightly harder, but still elementary exercise is to prove that $F\langle\xi, \eta\rangle$ is isomorphic to $A$. The point is to show that the homomorphism $f(\xi, \eta) \mapsto f(g, h)$, which exists by the universal property, is injective. Full details can be found in [6, pp. 184–185].

## 6.2 Algebras Defined by Generators and Relations

Let $A$ be a unital algebra over a field $F$. Take a subset $X$ of $A$ that generates $A$. In principle, $X$ can be any set of generators, but in practice we are more interested in smaller sets. Forget for a moment that $X$ is a subset of an algebra and consider it merely as a set. Then we can form the free algebra $F\langle X\rangle$. The function $\xi \mapsto \xi$ from the set $X$ to the algebra $A$ can be, as pointed out in the preceding section, uniquely extended to a unital algebra homomorphism $\varphi : F\langle X\rangle \to A$. As its image contains $X$, $\varphi$ is surjective. We have thereby shown that *every unital algebra is a homomorphic image of a free algebra.* Setting $I = \ker\varphi$, we can rephrase this as follows: For every unital $F$-algebra $A$ there exists a free algebra $F\langle X\rangle$ and its ideal $I$ such that

$$A \cong F\langle X\rangle/I.$$

These simple observations indicate that free algebras play a special role in noncommutative algebra. They are hidden behind every algebra. When searching for counterexamples, one often starts by examining free algebras.

We now change the point of view and begin by taking a nonempty set $X$ and a subset $R$ of $F\langle X\rangle$. Now form the factor algebra $F\langle X\rangle/(R)$, where $(R)$ is the ideal of $F\langle X\rangle$ generated by $R$. As noticed above, every unital algebra is isomorphic to an algebra of this kind. This construction thus gives us endless opportunities to create interesting examples. Also, it makes it possible for us to get a clearer picture of some important algebras that we often come across.

Before proceeding to examples, we introduce some notation and terminology. Write $X = \{\xi_i \mid i \in I\}$ and $R = \{f_j = f_j(\xi_{i_1}, \ldots, \xi_{i_{n(j)}}) \mid j \in J\}$. Denote the coset

$\xi_i + (R)$ by $x_i$. Note that $f_j(x_{i_1}, \ldots, x_{i_{n(j)}}) = 0$ for every $j \in J$. Let us switch to a more suggestive notation and write $F\langle X\rangle/(R)$ as

$$F\langle x_i,\ i \in I \mid f_j(x_{i_1}, \ldots, x_{i_{n(j)}}) = 0,\ j \in J\rangle.$$

We say that this algebra is defined by the **generators** $x_i$ and **relations** $f_j$. Indeed the elements $x_i, i \in I$, together with 1 generate the algebra, and, unlike the generators $\xi_i$ of the free algebra $F\langle X\rangle$, they are not independent but related via the polynomials $f_j$. These may not be the only polynomials that relate the $x_i$'s. However, if $f = f(\xi_{i_1}, \ldots, \xi_{i_n}) \in F\langle X\rangle$ is such that $f(x_{i_1}, \ldots, x_{i_n}) = 0$, i.e.,

$$f(\xi_{i_1}, \ldots, \xi_{i_n}) + (R) = f(\xi_{i_1} + (R), \ldots, \xi_{i_n} + (R)) = 0,$$

then $f$ lies in the ideal of $F\langle X\rangle$ generated by the $f_j$'s.

**Example 6.3** Let

$$A = F\langle x, y \mid x^2 = 0,\ y^2 = 0,\ xy + yx = 1\rangle.$$

A concrete example of a pair of elements satisfying these relations are the matrix units $E_{12}$ and $E_{21}$ in $M_2(F)$. Let us show that $A$ is actually isomorphic to $M_2(F)$. We begin by noticing that $xyx = x$ and $yxy = y$. Hence it follows that every element in $A$ is a linear combination of $x, y, xy$, and $yx$. Suppose $\lambda_1 x+\lambda_2 y+\lambda_3 xy+\lambda_4 yx = 0$ for some $\lambda_i \in F$. Then the polynomial $f(\xi, \eta) = \lambda_1\xi+\lambda_2\eta+\lambda_3\xi\eta+\lambda_4\eta\xi$ lies in the ideal of $F\langle\xi, \eta\rangle$ generated by $R = \{\xi^2, \eta^2, \xi\eta+\eta\xi-1\}$. Accordingly, for every $F$-algebra $B$ and every pair of elements $a, b \in B$, the evaluation $f(a, b)$ lies in the ideal of $B$ generated by the set $\{a^2, b^2, ab+ba-1\}$. Choosing $B = M_2(F)$ and $a = E_{12}, b = E_{21}$, we see that each $\lambda_i = 0$. Consequently, $\{x, y, xy, yx\}$ is a basis of $A$. Note that its multiplication table is the same as that of $\{E_{12}, E_{21}, E_{11}, E_{22}\}$ in $M_2(F)$. Therefore, $A \cong M_2(F)$.

**Example 6.4** Assume that $\mathrm{char}(F) = 0$. Let us show that the first Weyl algebra $\mathcal{A}_1$ over $F$, introduced in Sect. 3.11, is isomorphic to

$$A = F\langle x, y \mid [x, y] = 1\rangle.$$

By the universal property of $F\langle\xi, \eta\rangle$, we can define the unital homomorphism $f(\xi, \eta) \mapsto f(D, L)$ from $F\langle\xi, \eta\rangle$ onto $\mathcal{A}_1$. Since $R = \{[\xi, \eta]-1\}$ is contained in its kernel, this homomorphism gives rise to the homomorphism $f(x, y) \mapsto f(D, L)$ from $A = F\langle\xi, \eta\rangle/(R)$ onto $\mathcal{A}_1$. We have to show that it is an isomorphism. To this end, we refer to the proof of the assertion (a) from the proof of Theorem 3.89. It is evident that it depends only on the formula $[D, L] = I$. Therefore, the same conclusion holds for every unital algebra generated by a pair of elements whose commutator equals 1. Accordingly, every element in $A$ can be written as a linear combination of the elements $y^m x^n, m, n \geq 0$. Using (b) from the proof of Theorem 3.89, we now see that our homomorphism is injective, and hence an isomorphism.

Similarly one shows that the $n$th Weyl algebra $\mathcal{A}_n$ is isomorphic to

$$F\langle x_i,\, y_i,\, i=1,\dots,n \,|\, [x_i, y_j]=\delta_{ij},\, [x_i, x_j]=0,\, [y_i, y_j]=0,\, i,j=1,\dots,n\rangle.$$

**Example 6.5** The algebra

$$A = F\langle x_i,\, i\in I \,|\, [x_i, x_j]=0,\, i,j\in I\rangle$$

is generated by the elements $x_i$ that commute with each other, and these commuting relations are the only defining relations. It should come as no surprise that $A$ is isomorphic to $F[\omega_i,\, i\in I]$, the algebra of polynomials in commuting indeterminates $\omega_i$. To give a formal proof, we use the universal property to define the unital homomorphism from $F\langle X\rangle$, $X=\{\xi_i \,|\, i\in I\}$, onto $F[\omega_i,\, i\in I]$, determined by $\xi_i\mapsto\omega_i$. Its kernel contains the set $R=\{\xi_i\xi_j-\xi_j\xi_i \,|\, i,j\in I\}$, and every element in $A=F\langle X\rangle/(R)$ can be written as $f+(R)$ where $f$ is such that in each of its monomials the indeterminates $\xi_i$ appear in the same fixed order. This readily implies that $A\cong F[\omega_i,\, i\in I]$.

**Example 6.6** Let $F$ be a field with $\mathrm{char}(F)\neq 2$. We call

$$G = F\langle x_i,\, i\in\mathbb{N} \,|\, x_ix_j+x_jx_i=0,\, i,j\in\mathbb{N}\rangle$$

the **Grassmann algebra** (or the **exterior algebra**) over $F$. Note that for $i=j$ the relation reads as $x_i^2=0$, $i\in\mathbb{N}$.

Observe that, for example, $x_1x_2x_1=-x_1^2x_2=0$; more generally, $x_iGx_i=0$. Note also that every product of different $x_i$'s can be written as plus or minus

$$x_{i_1}x_{i_2}\dots x_{i_n},\ i_1<i_2<\dots<i_n. \tag{6.1}$$

Accordingly, these elements, together with 1, linearly span $G$. Let us first show that each of them is nonzero.

It suffices to show that $x_1x_2\dots x_n\neq 0$. Suppose this not true. Then $\xi_1\xi_2\dots\xi_n$ lies in the ideal of $F\langle\xi_1,\xi_2,\dots\rangle$ generated by $R=\{\xi_i^2,\ \xi_i\xi_j+\xi_j\xi_i \,|\, i,j\in\mathbb{N}\}$. Thus, we can write $\xi_1\xi_2\dots\xi_n$ as a sum of terms of the form $m\xi_i^2m'$ and $p(\xi_i\xi_j+\xi_j\xi_i)p'$, where $m, m', p, p'$ are monomials. The linear span of all multilinear monomials, i.e., the space of all multilinear polynomials, has trivial intersection with the linear span of all monomials that are not multilinear. Hence, it follows that $\xi_1\xi_2\dots\xi_n$ is a linear combination of the polynomials

$$\xi_{\sigma(1)}\dots\xi_{\sigma(j-1)}\Big(\xi_{\sigma(j)}\xi_{\sigma(j+1)}+\xi_{\sigma(j+1)}\xi_{\sigma(j)}\Big)\xi_{\sigma(j+2)}\dots\xi_{\sigma(n)},$$

where $\sigma\in S_n$. Such a linear combination is a multilinear polynomial with the property that the sum of its coefficients corresponding to even permutations coincides with the sum of its coefficients corresponding to odd permutations. Since

$\xi_1\xi_2\ldots\xi_n$ does not share this property, we have arrived at a contradiction. Thus, the elements from (6.1) are indeed nonzero.

Suppose $\sum_{i=1}^{n}\lambda_i b_i = 0$ where $\lambda_i \in F\setminus\{0\}$ and each $b_i$ is either 1 or is as in (6.1). We may assume that there exists a $k\in\mathbb{N}$ such that $x_k$ appears in $b_n$, but not in $b_1$. Hence $\sum_{i=1}^{n-1}\lambda_i b_i x_k = 0$ and $b_1 x_k \neq 0$. Continuing this procedure we arrive at a contradiction that one of the $\lambda_i$'s is zero. Thus, the elements of the form (6.1), together with 1, form a basis of $G$.

It is obvious that $x_1x_2$ commutes with each $x_i$, and hence it lies in $Z(G)$, the center of $G$. The same reasoning shows that every element from (6.1) with $n$ even lies in $Z(G)$. Moreover, using the assumption that $\mathrm{char}(F)\neq 2$, it is easy to see that $Z(G)$ is actually the linear span of all such elements and 1. Write $G_0$ for $Z(G)$, and similarly, write $G_1$ for the linear span of all elements of the form (6.1) with $n$ odd. Note that $G = G_0\oplus G_1$, the vector space direct sum.

One often considers the Grassmann algebra in $n$ generators. We can describe it as the subalgebra of $G$ generated by $1, x_1, \ldots, x_n$. By counting the basis elements we see that its dimension is $2^n$.

The Grassmann algebra has a big center and plenty of nilpotent ideals, in contrast to algebras that we prefer in structure theory. However, $G$ turns out to be an extremely useful tool in various mathematical areas. A small evidence of its applicability will be given in the proof of the Amitsur-Levitzki theorem in Sect. 6.9.

## 6.3 Alternating Polynomials

In this section, we consider special polynomials that play an important role in the theory of polynomial identities, which is the theme of the subsequent sections. For simplicity, we will confine ourselves to multilinear polynomials.

We begin with an example. Consider the polynomial

$$h(\xi_1,\xi_2,\xi_3,\eta) = \xi_1\eta\xi_2\xi_3 - \xi_1\eta\xi_3\xi_2 + \xi_2\eta\xi_3\xi_1 - \xi_2\eta\xi_1\xi_3 + \xi_3\eta\xi_1\xi_2 - \xi_3\eta\xi_2\xi_1.$$

Note that by substituting $\xi_1$ for $\xi_2$ we get 0, i.e., $h(\xi_1,\xi_1,\xi_3,\eta) = 0$. Similarly, $h(\xi_1,\xi_2,\xi_1,\eta) = 0$ and $h(\xi_1,\xi_2,\xi_2,\eta) = 0$.

**Definition 6.7** A multilinear polynomial $f = f(\xi_1,\ldots,\xi_n,\eta_1,\ldots,\eta_r)\in F\langle X\rangle$ is said to be **alternating** in $\xi_1,\ldots,\xi_n$ if $f$ becomes zero whenever one substitutes $\xi_j$ for $\xi_i$, $1\le i<j\le n$.

Replacing $\xi_1$ by $\xi_1+\xi_2$ in $f(\xi_1,\xi_1,\xi_3,\ldots,\xi_n,\eta_1,\ldots,\eta_r) = 0$ gives

$$f(\xi_1,\xi_2,\xi_3,\ldots,\xi_n,\eta_1,\ldots,\eta_r) = -f(\xi_2,\xi_1,\xi_3,\ldots,\xi_n,\eta_1,\ldots,\eta_r).$$

Similarly, $f$ changes sign if we switch any pair of indeterminates $\xi_i$ and $\xi_j$. The name "alternating" comes from this condition, which is obviously equivalent to the definition provided that $\mathrm{char}(F)\neq 2$.

Our main reason for considering alternating polynomials is their connection to the notion of linear dependence.

**Lemma 6.8** *Let $A$ be an $F$-algebra and let $a_1, \ldots, a_n \in A$ be linearly dependent elements. If a multilinear polynomial $f = f(\xi_1, \ldots, \xi_n, \eta_1, \ldots, \eta_r) \in F\langle X\rangle$ is alternating in $\xi_1, \ldots, \xi_n$, then $f(a_1, \ldots, a_n, x_1, \ldots, x_r) = 0$ for all $x_1, \ldots, x_r \in A$.*

***Proof*** We may assume that $a_n = \sum_{i=1}^{n-1} \lambda_i a_i$ for some $\lambda_i \in F$. Consequently,

$$f(a_1, \ldots, a_n, x_1, \ldots, x_r) = \sum_{i=1}^{n-1} \lambda_i f(a_1, \ldots, a_{n-1}, a_i, x_1, \ldots, x_r) = 0$$

since $f(\xi_1, \ldots, \xi_i, \ldots, \xi_{n-1}, \xi_i, \eta_1, \ldots, \eta_r) = 0$ for $i = 1, \ldots, n-1$. □

Two families of alternating polynomials are of special importance. The first one consists of polynomials that are alternating in all indeterminates (i.e., there are no $\eta_j$'s).

**Definition 6.9** Let $n \geq 2$. The polynomial

$$s_n = s_n(\xi_1, \ldots, \xi_n) = \sum_{\sigma \in S_n} \operatorname{sgn}(\sigma) \xi_{\sigma(1)} \ldots \xi_{\sigma(n)}$$

is called the **standard polynomial** of degree $n$.

The reader's first thought may have been that this is similar to the definition of the determinant of a matrix. There are indeed several analogies.

Choose indices $i, j$ and a permutation $\sigma$. Write $\xi_{\sigma(1)} \ldots \xi_{\sigma(n)}$ as $m\xi_i m' \xi_j m''$ where $m, m', m''$ are monomials in the remaining indeterminates. Then the term $m\xi_j m' \xi_i m''$ corresponds to a permutation whose sign is opposite to that of $\sigma$. This readily implies that $s_n$ is alternating.

Note that $s_2 = \xi_1\xi_2 - \xi_2\xi_1$ and $s_3 = h(\xi_1, \xi_2, \xi_3, 1)$ where $h$ is the polynomial from the beginning of the section (substituting 1 for $\eta$ simply means "erasing" $\eta$ in the formula defining $h$).

The second family is similar, but involves the $\eta_j$'s.

**Definition 6.10** Let $n \geq 2$. The polynomial

$$c_n = c_n(\xi_1, \ldots, \xi_n, \eta_1, \ldots, \eta_{n-1}) = \sum_{\sigma \in S_n} \operatorname{sgn}(\sigma) \xi_{\sigma(1)} \eta_1 \xi_{\sigma(2)} \eta_2 \ldots \eta_{n-1} \xi_{\sigma(n)}$$

is called the $n$**th Capelli polynomial**.
(Warning: some authors use slightly different definitions.)

Thus, $c_2 = \xi_1\eta_1\xi_2 - \xi_2\eta_1\xi_1$, and one obtains $c_3$ by writing $\eta_1$ for $\eta$ and inserting $\eta_2$ between the last two factors of each term of $h$.

Observe that

$$s_n(\xi_1, \ldots, \xi_n) = c_n(\xi_1, \ldots, \xi_n, 1, \ldots, 1).$$

It is easy to see that $c_n$ is alternating in $\xi_1, \ldots, \xi_n$. Let us also mention the easily derived formula

$$c_n = \sum_{i=1}^{n} (-1)^{i-1} \xi_i \eta_1 c_{n-1}(\xi_1, \ldots, \xi_{i-1}, \xi_{i+1}, \ldots, \xi_n, \eta_2, \ldots, \eta_{n-1}), \tag{6.2}$$

which holds for every $n \geq 3$. Similarly, for every $n \geq 3$ we have

$$s_n = \sum_{i=1}^{n} (-1)^{i-1} \xi_i s_{n-1}(\xi_1, \ldots, \xi_{i-1}, \xi_{i+1}, \ldots, \xi_n). \tag{6.3}$$

## 6.4 Polynomial Identities: Definition and Examples

Much of the rest of the book is devoted to the following notion.

**Definition 6.11** A polynomial $f = f(\xi_1, \ldots, \xi_n) \in F\langle \xi_1, \xi_2, \ldots \rangle$ is said to be a **polynomial identity** (or simply an **identity**) of an $F$-algebra $A$ if $f(x_1, \ldots, x_n) = 0$ for all $x_1, \ldots, x_n \in A$. In this case, we also say that $A$ **satisfies** $f$.

The reader should not confuse this with algebras defined by generators and relations, where polynomials vanish only on some of the generators. The condition that $f$ is an identity of $A$ is quite restrictive, provided, of course, that $f$ is a nonzero polynomial.

**Definition 6.12** If a nonzero polynomial is a polynomial identity of $A$, then $A$ is called a PI-**algebra**.

One can define polynomial identities for rings, not only for algebras. We will in fact deal with PI-rings in the next chapter. In this one we consider only algebras in order to make the exposition simpler.

There are two basic examples of PI-algebras.

**Example 6.13** Saying that $A$ is commutative is the same as saying that $A$ satisfies the identity $[\xi_1, \xi_2]$ $(= \xi_1\xi_2 - \xi_2\xi_1)$. Every commutative algebra is thus a PI-algebra.

One way of looking at the notion of a polynomial identity is thus that it is a generalization of commutativity. The second basic example is of different nature.

**Example 6.14** Every finite dimensional algebra is a PI-algebra. If $[A : F] = d$, then every multilinear polynomial that is alternating in $d + 1$ indeterminates is an identity of $A$ by Lemma 6.8. In particular, $A$ satisfies $s_{d+1}$.

Thus, $M_n(F)$ is a PI-algebra. In fact, a considerable part of the theory of polynomial identities is centered around matrices. Roughly speaking, the theory has two branches: *combinatorial* and *structural*. The goal of the first one is the understanding of the set of all identities of a given algebra $A$, with special emphasis on the case where $A = M_n(F)$. Concerning the second one, let us only mention at this point that a prime PI-algebra can be embedded, in a particularly nice way, into a matrix algebra over a field extension of the base field. This indicative result is a byproduct of the structure theory, which will be exposed in the next chapter (see Corollary 7.59 and Remark 7.60). The combinatorial aspect will be briefly touched upon in the rest of this chapter.

There are other examples of PI-algebras, quite different from commutative and finite dimensional ones.

**Example 6.15** The commutator $[x, y]$ of any two elements $x, y$ in the Grassmann algebra $G$ lies in $Z(G) = G_0$. It is enough to show this for basis elements $x = x_{i_1} \dots x_{i_n}$ and $y = x_{j_1} \dots x_{j_m}$. Note that $[x, y]$ can be nonzero only if both $n$ and $m$ are odd, and in this case it indeed lies in $G_0$. Thus, $G$ satisfies the identity $[[\xi_1, \xi_2], \xi_3]$.

On the other hand, if $\mathrm{char}(F) = 0$, then none of the standard polynomials is an identity of $G$. To show this, we first observe that

$$x_{\sigma(1)} \dots x_{\sigma(r)} = \mathrm{sgn}(\sigma) x_1 \dots x_r \tag{6.4}$$

for every permutation $\sigma$ on $\{1, \dots, r\}$. Indeed, writing $\sigma$ as a product of transpositions, we see that it is enough to prove (6.4) for the case where $\sigma$ itself is a transposition, and in this case the formula is pretty obvious. Consequently, $s_r(x_1, \dots, x_r) = r! x_1 \dots x_r \neq 0$ for every $r \geq 2$. From Example 6.14 we therefore infer that $G$ cannot be embedded into any algebra that is finite dimensional over some field extension of $F$; in fact, from Theorem 6.44 below we see that it cannot even be embedded into $M_n(C)$ for any commutative algebra $C$.

**Example 6.16** If $A$ is a nilpotent algebra, then $A$ satisfies the identity $\xi_1 \dots \xi_n$ for some $n \in \mathbb{N}$. A concrete example is the algebra of all *strictly* upper triangular matrices over a commutative algebra $C$. Now consider $T_n(C)$, the algebra of all upper triangular $n \times n$ matrices over $C$. Note that the commutator of any two matrices from $T_n(C)$ is a strictly upper triangular matrix. Therefore, $T_n(C)$ satisfies $[\xi_1, \eta_1] \dots [\xi_n, \eta_n]$.

**Example 6.17** Suppose $\xi^2$ is an identity of $A$. This does not necessarily mean that $A^2 = 0$ (for example, consider the 3-dimensional subalgebra $Fx_1 + Fx_2 + Fx_1x_2$ of the Grassmann algebra $G$). However, $A^3 = 0$, provided that $\mathrm{char}(F) \neq 2$. The proof is easy. We have $x^2 = 0$ for every $x \in A$. Replacing $x$ by $x + y$, and using $x^2 = y^2 = 0$, we get $xy + yx = 0$ for all $x, y \in A$ (i.e., $\xi\eta + \eta\xi$ is an identity of $A$). Consequently, for all $x, y, z \in A$ we have

$$(xy)z = -z(xy) = -(zx)y = y(zx) = (yz)x = -x(yz),$$

yielding $xyz = 0$.

Much more can be said. If $\mathrm{char}(F) = 0$ and $A$ satisfies the identity $\xi^n$, then $A$ is a nilpotent algebra, i.e., $A$ satisfies $\xi_1\xi_2 \dots \xi_d$ for some $d$, depending on $n$. This result is called the **Nagata-Higman theorem**. The proof uses elementary tools, but we omit it (it can be found, for example, in [14]). Finding the exact lower bound for $d$ is an open problem.

**Example 6.18** Let $n \geq 2$. Theorem 4.81 implies that an algebra satisfying the identity $\xi^n - \xi$ is commutative. Concrete examples are Boolean rings and finite fields. Of course, Theorem 4.81 says more than that. However, the condition $x^{n(x)} = x$ cannot be interpreted in terms of polynomial identities.

How to obtain new PI-algebras from old? In the next two examples, we give some answers to this question. The first example, as trivial as it is, is of fundamental importance.

**Example 6.19** If $f$ is an identity of the algebra $A$, then $f$ is also an identity of every subalgebra of $A$ as well as of every homomorphic image of $A$. In particular, a subalgebra or a homomorphic image of a PI-algebra is again a PI-algebra.

**Example 6.20** Let $\{A_i \mid i \in I\}$ be a family of algebras. If each $A_i$ satisfies the same identity $f$, then so does the direct product $\Pi_{i\in I} A_i$. On the other hand, if each $A_i$ is a PI-algebra, then there is no reason to believe that $\Pi_{i\in I} A_i$ is a PI-algebra too. If, however, $I = \{1, \dots, n\}$ is a finite set, then this is true. Namely, if $f_i$ is an identity of $A_i$, then $f_1 \cdots f_n$ is an identity of $A_1 \times \cdots \times A_n$. Not only the direct product, even the tensor product of two PI-algebras is again a PI-algebra. However, this is anything but obvious. It was proved by A. Regev in the early 1970's.

All these examples may give the wrong impression that most algebras are PI-algebras. Infinite dimensional algebras with small centers, e.g., infinite dimensional central algebras, are actually only rarely PI-algebras. This will become clear as the theory unfolds. Let us mention, at this point, just an obvious example of an algebra without nonzero identities.

**Example 6.21** The free algebra $F\langle \xi_1, \xi_2, \dots \rangle$ is not a PI-algebra.

## 6.5 Linearization

The purpose of this section is to get familiar with the **process of linearization**. The concept of this process is applicable to various situations throughout mathematics. Our goal, however, is to show that it can be used to recast an arbitrary polynomial identity into a multilinear one.

We have already met a simple example of linearization in Example 6.17: If $A$ satisfies $\xi^2$, then $A$ also satisfies the multilinear identity $\xi\eta + \eta\xi$. The proof is easy, one just replaces an arbitrary element by the sum of two arbitrary elements. This simple idea is the essence of "linearization" and can be pushed further.

**Example 6.22** Consider a slightly more complicated situation where $A$ satisfies $f(\xi) = \xi^3$. All relevant calculations can be made in a free algebra, so we can ignore $A$ for a while. Let us define two new polynomials $g = g(\xi_1, \xi_2)$ and $h = h(\xi_1, \xi_2, \xi_3)$ as follows:

$$g = f(\xi_1 + \xi_2) - f(\xi_1) - f(\xi_2),$$

$$h = g(\xi_1, \xi_2 + \xi_3) - g(\xi_1, \xi_2) - g(\xi_1, \xi_3).$$

Of course, $h$ can be explicitly expressed by $f$:

$$h = f(\xi_1+\xi_2+\xi_3) - f(\xi_1+\xi_2) - f(\xi_1+\xi_3) - f(\xi_2+\xi_3) + f(\xi_1) + f(\xi_2) + f(\xi_3).$$

However, to compute $h$, it is easier to first compute $g$,

$$g = \xi_1\xi_2^2 + \xi_2\xi_1\xi_2 + \xi_2^2\xi_1 + \xi_1^2\xi_2 + \xi_1\xi_2\xi_1 + \xi_2\xi_1^2,$$

and hence derive

$$h = \sum_{\sigma \in S_3} \xi_{\sigma(1)}\xi_{\sigma(2)}\xi_{\sigma(3)}.$$

Since $A$ satisfies $f$, $A$ also satisfies $h$. We have thus found a multilinear polynomial identity of $A$, but at the cost of increasing the number of indeterminates.

To consider arbitrary polynomials, we first introduce an auxiliary notion. The **degree of the polynomial** $f$ *in* $\xi_i$ is the maximal number of occurrences of $\xi_i$ in the monomials of $f$. For example, $f = \xi_1^2\xi_2\xi_3\xi_1 + \xi_2\xi_3\xi_2$ has degree 3 in $\xi_1$, degree 2 in $\xi_2$, and degree 1 in $\xi_3$.

**Theorem 6.23** *If an algebra $A$ satisfies a nonzero polynomial identity, then $A$ also satisfies a nonzero multilinear polynomial identity of the same or lower degree.*

***Proof*** Let $f = f(\xi_1, \ldots, \xi_n)$ be a nonzero identity of $A$. Denote by $d_i$ the degree of $f$ in $\xi_i$. The proof is by induction on $d = \max\{d_1, \ldots, d_n\}$.

If $d = 1$, then each $\xi_i$ appears in every monomial of $f$ at most once—"at most" and not "exactly", so $f$ is not necessarily multilinear. But this can be easily remedied. If, say, $\lambda\xi_1 \ldots \xi_m$, where $\lambda$ is a nonzero scalar, is a monomial of $f$ of minimal degree, then we just substitute 0 for the remaining indeterminates, and hence obtain a nonzero multilinear identity $f(\xi_1, \ldots, \xi_m, 0, \ldots, 0)$ of degree less than or equal to $\deg(f)$.

Let $d > 1$. Without loss of generality, we may assume that there exists a $k \leq n$ such that $d_k = \cdots = d_n = d$ and $d_i < d$ for $i < k$. Define a new polynomial which involves an additional indeterminate, $g = g(\xi_1, \ldots, \xi_n, \xi_{n+1})$, by

$$g = f(\xi_1, \ldots, \xi_{n-1}, \xi_n + \xi_{n+1}) - f(\xi_1, \ldots, \xi_{n-1}, \xi_n) - f(\xi_1, \ldots, \xi_{n-1}, \xi_{n+1}).$$

Note that $g$ is also an identity of $A$. Let us write $f = \sum_i \lambda_i w_i$, where the $w_i$'s are pairwise different words and the $\lambda_i$'s are nonzero scalars. Then $g$ can be written as $g = \sum_i \lambda_i g_i$, where $g_i$ is obtained from $w_i$ in the same way as $g$ is obtained from $f$. If $\xi_n$ does not occur in $w_i$, then $g_i = -w_i$. If $\xi_n$ occurs only once in $w_i$, then $g_i = 0$. If $\xi_n$ occurs at least twice in $w_i$, then $g_i$ is the sum of all possible words obtained by replacing at least one, but not all, of the $\xi_n$'s in $w_i$ by $\xi_{n+1}$. Thus, if in any of these words one substitutes $\xi_n$ for $\xi_{n+1}$, then one gets back the word $w_i$. The words appearing in $g_i$ are therefore different from the words appearing in $g_{i'}$ whenever $i' \neq i$. As $d > 1$, the indices $i$ such that $\xi_n$ occurs at least twice in $w_i$ do exist. This shows that $g \neq 0$.

The following conclusions can be drawn from the preceding paragraph:

- $g$ is a nonzero identity of $A$.
- $\deg(g) \leq \deg(f)$.
- For $j = 1, \ldots, n-1$, the degree of $g$ in $\xi_j$ is less than or equal to $d_j$.
- The degree of $g$ in $\xi_n$ and $\xi_{n+1}$ is $d - 1$.

Now repeat this procedure, first with $g$ in place of $f$ and $\xi_{n-1}$ in place of $\xi_n$, and then with other indeterminates down to $\xi_k$. At the end, we arrive at a nonzero identity having degree at most $d - 1$ in every indeterminate. □

This theorem is of crucial importance if one is interested in the structure of an algebra satisfying some nonzero polynomial identity. With regard to particular identities, it should be remarked that in the finite characteristic case some information can be lost when performing the linearization process. This is evident from the following simple example.

**Example 6.24** Return to our first example where an $F$-algebra $A$ satisfies $\xi^2$. Linearizing, we obtain that it also satisfies $\xi\eta + \eta\xi$. If $\mathrm{char}(F) \neq 2$, then the two identities are equivalent, they imply each other. If $\mathrm{char}(F) = 2$, then the multilinear one only tells us that the algebra is commutative, while the original one shows much more, every element has square zero.

## 6.6 Stable Identities

Let $f \in F\langle \xi_1, \xi_2, \ldots \rangle$ be an identity of an $F$-algebra $A$. Take an extension field $K$ of $F$. Is $f$ an identity of $A_K = K \otimes A$, the scalar extension of $A$ to $K$? Since $A_K$ has the same multiplication table as $A$, an affirmative answer seems plausible. However, the next simple example shows that this may not be true even for $A = F$.

**Example 6.25** Let $F$ be a finite field with $n$ elements. Then $f = \xi^n - \xi$ is an identity of $F$ (see Lemma 4.79). Since $f$ cannot have more than $n$ zeros in any field, $f$ is not an identity of any extension field $K$ of $F$ that properly contains $F$.

The main goal of this section is to show that the above question does have an affirmative answer if $F$ is infinite. Moreover, we may replace the role of the field $K$ by an arbitrary commutative algebra $C$.

**Definition 6.26** An identity $f$ of an $F$-algebra $A$ is said to be **stable** (for $A$) if $f$ is an identity of $C \otimes A$ for every commutative $F$-algebra $C$.

Example 6.25 thus shows that $\xi^n - \xi$ is not a stable identity for $A = F$ if $|F| = n$. Let us turn to positive results.

**Lemma 6.27** *Every multilinear identity is stable.*

***Proof*** Let $f = f(\xi_1, \ldots, \xi_n)$ be a multilinear polynomial, and consider, for arbitrary $A$ and $C$, $f(\overline{x}_1, \ldots, \overline{x}_n)$ with $\overline{x}_i \in C \otimes A$. Each $\overline{x}_i$ is a sum of simple tensors, and so, by multilinearity of $f$, $f(\overline{x}_1, \ldots, \overline{x}_n)$ can be written as a sum of elements of the form $f(c_1 \otimes x_1, \ldots, c_n \otimes x_n)$. Since $C$ is commutative, we have

$$f(c_1 \otimes x_1, \ldots, c_n \otimes x_n) = c_1 \cdots c_n \otimes f(x_1, \ldots, x_n),$$

which is 0 if $f$ is an identity of $A$. □

The classical polynomials in commuting indeterminates play a similar role in commutative algebra as the noncommutative polynomials do in noncommutative algebra. The algebra of commutative polynomials $F[\Omega]$ satisfies the *universal property* in the context of commutative unital algebras, the *evaluation* of a commutative polynomial in elements from a commutative algebra can be defined in the obvious way, and every commutative unital algebra is a homomorphic image of $F[\Omega]$ for some set $\Omega$. A detailed explanation of all this is essentially the same as for $F\langle X\rangle$ in Sects. 6.1 and 6.2. At any rate, these are just elementary observations—but very useful ones, as we shall see in the rest of the chapter.

**Lemma 6.28** *Let $f = f(\xi_1, \ldots, \xi_n)$ be an identity of an $F$-algebra $A$. If $f$ is an identity of $F[\omega_1, \ldots, \omega_s] \otimes A$ for every $s \in \mathbb{N}$, then $f$ is stable for $A$.*

***Proof*** Let $C$ be a commutative unital $F$-algebra. Since every commutative algebra is a subalgebra of a commutative unital algebra (say, of its unitization), it suffices to show that $f$ is an identity of $C \otimes A$. Take arbitrary elements $\overline{x}_i = \sum_{j=1}^{m_i} c_{ij} \otimes x_{ij}$, $i = 1, \ldots, n$, in $C \otimes A$. Let us set

$$\tilde{x}_i = \sum_{j=1}^{m_i} \omega_{ij} \otimes x_{ij} \in F[\Omega] \otimes A,$$

where $\Omega = \{\omega_{ij} \mid i = 1, \ldots, n,\ j = 1, \ldots, m_i\}$. By assumption, $f(\tilde{x}_1, \ldots, \tilde{x}_n) = 0$. Therefore, if $\varphi$ is the (unital) homomorphism from $F[\Omega]$ to $C$ sending $\omega_{ij}$ to $c_{ij}$, then $f(\overline{x}_1, \ldots, \overline{x}_n) = (\varphi \otimes \mathrm{id}_A)\big(f(\tilde{x}_1, \ldots, \tilde{x}_n)\big) = 0$. □

We warn the reader that henceforth our argumentation will be somewhat less formal than in the proof of Lemma 6.28. We will avoid speaking about the universal

property and homomorphisms from $F[\Omega]$, but simply substitute elements from commutative algebras for indeterminates in equations involving polynomials.

**Theorem 6.29** *Let $F$ be an infinite field. Then every polynomial identity of an arbitrary $F$-algebra $A$ is stable.*

***Proof*** Let $f = f(\xi_1, \ldots, \xi_n)$ be an identity of $A$. Take an $s \in \mathbb{N}$ and arbitrary $\hat{x}_i = \sum_j p_{ij} \otimes x_{ij} \in F[\omega_1, \ldots, \omega_s] \otimes A$, $i = 1, \ldots, n$. By Lemma 6.28, it is enough to show that $f(\hat{x}_1, \ldots, \hat{x}_n) = 0$. Suppose this is not true. Then we can write

$$f(\hat{x}_1, \ldots, \hat{x}_n) = \sum_k q_k \otimes a_k, \tag{6.5}$$

where $q_1 \neq 0$ and the $a_k$'s are linearly independent (see Remark 5.16). Since $F$ is infinite, there exist $\lambda_1, \ldots, \lambda_s \in F$ such that $q_1(\lambda_1, \ldots, \lambda_s) \neq 0$ (the proof is a straightforward induction on $s$; for $s = 1$ just use the fact that the number of zeros of a polynomial cannot exceed its degree). Now set $\alpha_{ij} = p_{ij}(\lambda_1, \ldots, \lambda_s)$ and $\beta_k = q_k(\lambda_1, \ldots, \lambda_s)$. Thus, $\beta_1 \neq 0$. Substituting $\lambda_i$ for $\omega_i$ in (6.5), we obtain

$$f\Big(\sum_j \alpha_{1j} \otimes x_{1j}, \ldots, \sum_j \alpha_{nj} \otimes x_{nj}\Big) = \sum_k \beta_k \otimes a_k. \tag{6.6}$$

Since $\alpha_{ij}, \beta_k \in F$, we have

$$\sum_j \alpha_{ij} \otimes x_{ij} = 1 \otimes \Big(\sum_j \alpha_{ij} x_{ij}\Big) \quad \text{and} \quad \sum_k \beta_k \otimes a_k = 1 \otimes \Big(\sum_k \beta_k a_k\Big).$$

We can therefore rewrite (6.6) as

$$1 \otimes f\Big(\sum_j \alpha_{1j} x_{1j}, \ldots, \sum_j \alpha_{nj} x_{nj}\Big) = 1 \otimes \Big(\sum_k \beta_k a_k\Big).$$

However, this is a contradiction since $f$ is an identity of $A$ and so the left-hand side is 0, while the right-hand side is not 0 because $\beta_1 \neq 0$ and the $a_k$'s are linearly independent. □

Theorem 6.29 can be proved in a more direct manner, avoiding the polynomial algebra. However, we wanted to point out the general principle. If one wishes to prove a result for general commutative algebras, it is sometimes enough to prove it for polynomial algebras. This approach will be used a few more times in the rest of the chapter.

By $\mathrm{Id}(A)$ we denote the set of all polynomial identities of the $F$-algebra $A$.

**Corollary 6.30** *If $A$ is a central simple $F$-algebra of dimension $n^2$, then* $\mathrm{Id}(A) = \mathrm{Id}(M_n(F))$.

***Proof*** If $F$ is finite, then $A$, being finite dimensional, is finite too. Since $A$ is simple, Corollary 2.81 tells us that $A \cong M_{n'}(F')$ for some $n' \in \mathbb{N}$ and a finite field $F'$. However, $A$ is central, so $F' = F$ and consequently $n' = n$. Thus, there is nothing to prove in this case.

Let $F$ be infinite. Take a splitting field $K$ for $A$. If $f$ is an identity of $A$, then, by Theorem 6.29, $f$ is also an identity of $A_K = K \otimes A \cong M_n(K)$, and hence also of $M_n(F)$ (simply because $F \subseteq K$). Conversely, if $f$ is an identity of $M_n(F)$, then, again by Theorem 6.29, $f$ is also an identity of $K \otimes M_n(F) \cong M_n(K)$, and hence, referring to $M_n(K) \cong K \otimes A$, also of $1 \otimes A \cong A$. □

**Example 6.31** The real algebras $\mathbb{H}$ and $M_2(\mathbb{R})$ satisfy the same identities.

## 6.7 T-ideals

Throughout this section, $X$ denotes the countably infinite set $\{\xi_1, \xi_2, \dots\}$ of noncommuting indeterminates. For any $F$-algebra $A$, $\mathrm{Id}(A)$ is clearly an ideal of the free algebra $F\langle X\rangle$. One can say more about it, as we shall soon see.

We restrict our discussion to nonzero unital algebras, so that evaluations make sense for polynomials with nonzero constant term. This will simplify the exposition.

**Definition 6.32** An ideal $I$ of $F\langle X\rangle$ is called a **T-ideal** if $\varphi(I) \subseteq I$ for every algebra endomorphism $\varphi$ of $F\langle X\rangle$.

**Lemma 6.33** *The following statements are equivalent for an ideal $I$ of $F\langle X\rangle$:*

(i) *$I$ is a T-ideal.*
(ii) *If $f = f(\xi_1, \dots, \xi_n) \in I$, then $f(g_1, \dots, g_n) \in I$ for all $g_1, \dots, g_n \in F\langle X\rangle$.*

***Proof*** If $\varphi$ is an endomorphism of $F\langle X\rangle$ and $g_i = \varphi(\xi_i)$, then

$$\varphi\big(f(\xi_1, \dots, \xi_n)\big) = f(g_1, \dots, g_n)$$

for every $f \in F\langle X\rangle$. On the other hand, for any $g_i \in F\langle X\rangle$, there exists an endomorphism $\varphi$ of $F\langle X\rangle$ satisfying $\varphi(\xi_i) = g_i$. The lemma readily follows from these two observations. □

If $A$ is a nonzero unital algebra, then $\mathrm{Id}(A)$ is obviously a proper T-ideal of $F\langle X\rangle$. The next lemma in particular shows that the converse is also true. That is, for every proper $T$-ideal $I$ of $F\langle X\rangle$, there exists a nonzero unital $F$-algebra $A$ such that $I = \mathrm{Id}(A)$.

**Lemma 6.34** *If $I$ is a proper T-ideal of $F\langle X\rangle$, then $I = \mathrm{Id}(F\langle X\rangle/I)$.*

***Proof*** Take an arbitrary polynomial $f = f(\xi_1, \dots, \xi_n) \in F\langle X\rangle$, and consider its evaluations on $F\langle X\rangle/I$. Observe that

$$f(g_1 + I, \dots, g_n + I) = f(g_1, \dots, g_n) + I$$

for all $g_1, \dots, g_n \in F\langle X\rangle$. From Lemma 6.33 we thus infer that $f \in I$ if and only if $f$ is an identity of $F\langle X\rangle/I$. □

Assume temporarily that $I$ is just a nonempty subset of $F\langle X\rangle$. The **variety** determined by $I$, denoted by $\mathcal{V}(I)$, is the class of all unital algebras $A$ such that $I \subseteq \mathrm{Id}(A)$; that is, every polynomial from $I$ is an identity of $A$.

**Example 6.35** The variety $\mathcal{V}\big(\{\xi_1\xi_2 - \xi_2\xi_1\}\big)$ is the class of all commutative unital algebras.

Note that the variety determined by the set $I$ coincides with the variety determined by the T-ideal generated by $I$. Therefore, from now on we assume again that $I$ itself is a (proper) T-ideal.

Lemma 6.34 shows that the algebra $F\langle X\rangle/I$ belongs to the variety $\mathcal{V}(I)$. The next lemma further shows that this algebra is a distinguished representative of this variety.

**Lemma 6.36** *Let $I$ be a proper T-ideal of $F\langle X\rangle$. If $A \in \mathcal{V}(I)$ then for any $a_i \in A$, $i \in \mathbb{N}$, there exists a unique unital homomorphism $\varphi : F\langle X\rangle/I \to A$ such that $\varphi(\xi_i + I) = a_i$, $i \in \mathbb{N}$.*

***Proof*** Take the unital homomorphism from $F\langle X\rangle$ to $A$ given by $\xi_i \mapsto a_i$, observe that $I$ lies in its kernel since $A \in \mathcal{V}(I)$, and the existence of $\varphi$ follows. The uniqueness is obvious. □

Because of this property, we call $F\langle X\rangle/I$ the **relatively free algebra** (with respect to $I$).

To each unital algebra $A$, we can assign the T-ideal $\mathrm{Id}(A)$ and the corresponding relatively free algebra $F\langle X\rangle/\mathrm{Id}(A)$.

**Example 6.37** If $A$ is not a PI-algebra, then $\mathrm{Id}(A) = \{0\}$. The corresponding relatively free algebra is thus the free algebra $F\langle X\rangle$.

Our next goal is to describe the relatively free algebra corresponding to a finite dimensional central simple algebra over an infinite field $F$. To this end, we first introduce a seemingly unrelated notion.

Let $\Omega = \{\omega_{jk}^{(i)} \mid j,k = 1,\dots,n,\ i = 1,2,\dots\}$ be a set of commuting indeterminates. For each $i$, we set

$$g^{(i)} = \begin{bmatrix} \omega_{11}^{(i)} & \omega_{12}^{(i)} & \dots & \omega_{1n}^{(i)} \\ \omega_{21}^{(i)} & \omega_{22}^{(i)} & \dots & \omega_{2n}^{(i)} \\ \vdots & \vdots & \ddots & \vdots \\ \omega_{n1}^{(i)} & \omega_{n2}^{(i)} & \dots & \omega_{nn}^{(i)} \end{bmatrix} \in M_n\big(F[\Omega]\big).$$

We call $g^{(i)}$ an $n \times n$ **generic matrix**. The entries of a generic matrix are thus distinct (independent) commuting indeterminates. The subalgebra of $M_n(F[\Omega])$ generated by all $n \times n$ generic matrices $g^{(i)}$, $i \in \mathbb{N}$, and the identity matrix is called the **algebra of $n \times n$ generic matrices**. We denote it by $\mathrm{GM}_n(F)$.

**Proposition 6.38** *Let $F$ be an infinite field and let $A$ be a central simple $F$-algebra of dimension $n^2$. Then*

$$F\langle X\rangle/\mathrm{Id}(A) \cong \mathrm{GM}_n(F),$$

*i.e., the relatively free algebra corresponding to $A$ is isomorphic to the algebra $\mathrm{GM}_n(F)$ of $n \times n$ generic matrices.*

***Proof*** Recall from Corollary 6.30 that $\mathrm{Id}(A) = \mathrm{Id}(M_n(F))$. We may therefore assume that $A = M_n(F)$.

Take an identity $f$ of $M_n(F)$. Since $F$ is infinite, Theorem 6.29 tells us that $f$ is also an identity of $F[\Omega] \otimes M_n(F) \cong M_n\big(F[\Omega]\big)$ (see Example 5.22). But then $f$ is also an identity of the subalgebra $\mathrm{GM}_n(F)$ of $M_n\big(F[\Omega]\big)$. Accordingly, $\mathrm{GM}_n(F) \in \mathcal{V}(I)$ where $I = \mathrm{Id}(M_n(F))$.

We may now use Lemma 6.36 to obtain a unital homomorphism $\varphi : F\langle X\rangle/I \to \mathrm{GM}_n(F)$ satisfying

$$\varphi(\xi_i + I) = g^{(i)}, \quad i = 1, 2, \ldots$$

It is clear that $\varphi$ is surjective. Let us prove that it is also injective. Take $f = f(\xi_1, \ldots, \xi_n)$ such that $\varphi(f + I) = 0$, i.e., $f(g^{(1)}, \ldots, g^{(n)}) = 0$. Since the entries of the matrices $g^{(i)}$ are independent commuting indeterminates, we can replace them by arbitrary elements in $F$ (see the preceding section). This shows that $f \in I$. Therefore, $f$ is injective and hence an isomorphism. □

**Remark 6.39** From the proof it is evident that $\mathrm{Id}(A) = \mathrm{Id}(M_n(F)) = \mathrm{Id}(\mathrm{GM}_n(F))$.

In 1987, A. R. Kemer proved that every $T$-ideal of $F\langle X\rangle$ is finitely generated (as a T-ideal) if $\mathrm{char}(F) = 0$. This is one of the deepest results in the theory of polynomial identities. It is known as the solution of **Specht's problem**.

## 6.8 The Characteristic Polynomial

This section may appear as an intruder in the chapter. The main reason for its inclusion is that it is needed for the proof of the Amitsur-Levitzki theorem in the next section. On the other hand, although the results of the section do not directly involve polynomial identities, they consider some other kind of identities that have many interactions with them. At any rate, the topics of this section are of general interest and importance.

Some of the readers may have already met what we are about to say in a linear algebra course, yet probably at a lower level of generality. A typical course considers only matrices with entries from a field. Many concepts, however, still make sense if a field is replaced by a more general ring. Say, the determinant of a matrix $A \in$

$M_n(C)$, which we denote by $\det(A)$, can be defined in the usual way as long as $C$ is a commutative ring. Accordingly, we can define the **characteristic polynomial** of $A$ as

$$p_A(\omega) = \det(A - \omega I) \in C[\omega].$$

For any polynomial $q \in C[\omega]$, we define the evaluation $q(A)$ in the same way as when $C$ is a field. The **Cayley-Hamilton theorem** states that

$$p_A(A) = 0.$$

Quite possibly the reader is familiar with this theorem only in the case where $C$ is a field. However, from many of its proofs it is evident that $C$ can be any commutative ring. On the other hand, the Cayley-Hamilton theorem for fields easily implies the Cayley-Hamilton theorem for commutative rings. Indeed, assuming the truth of the theorem for fields, it follows that it is also true for commutative domains since they can be embedded into their fields of quotients. In particular, the theorem holds for the ring $\mathbb{Z}[\omega_{11}, \omega_{12}, \ldots, \omega_{nn}]$. Thus, if $g = (\omega_{ij})$ is a generic matrix, then $p_g(g) = 0$. Now take any commutative ring $C$, choose $c_{ij} \in C$, $1 \leq i, j \leq n$, substitute $c_{ij}$ for $\omega_{ij}$ in $p_g(g) = 0$, and the validity of the Cayley-Hamilton theorem for $C$ follows.

For any matrix $A \in M_n(C)$, we define $\operatorname{tr}(A)$, the trace of $A$, in the usual way as the sum of the diagonal entries. Our goal is to express the coefficients of $p_A$ through the traces of the powers of $A$. To this end, we need formulas on symmetric polynomials in commuting indeterminates with integer coefficients, which were discovered already by Isaac Newton.

Recall that a polynomial is said to be **symmetric** if it remains unchanged after permuting the indeterminates. Important examples are **elementary symmetric polynomials**, defined by $e_0 = 1$ and

$$e_k = e_k(\omega_1, \ldots, \omega_n) = \sum_{1 \leq i_1 < i_2 < \cdots < i_k \leq n} \omega_{i_1} \omega_{i_2} \ldots \omega_{i_k},$$

$k = 1, \ldots, n$, i.e., $e_k$ is the sum of all distinct products of $k$ distinct indeterminates. In particular, $e_1$ is the sum of all $\omega_i$'s, and $e_n$ is their product.

Let $\omega$ be another indeterminate. Note that

$$(-1)^n(\omega - \omega_1)(\omega - \omega_2)\ldots(\omega - \omega_n) = \sum_{j=0}^{n} (-1)^j e_{n-j} \omega^j. \tag{6.7}$$

Elementary symmetric polynomials are thus intimately connected with the roots of polynomials. Another reason for their importance is that they generate the algebra of all symmetric polynomials, but we will not need this fact. Our goal is to connect the elementary symmetric polynomials with the symmetric polynomials

$$p_j = p_j(\omega_1, \ldots, \omega_n) = \sum_{i=1}^{n} \omega_i^j, \quad j \geq 1.$$

One can verify that

$$e_1 = p_1, \;\; 2e_2 = e_1 p_1 - p_2, \;\; 3e_3 = e_2 p_1 - e_1 p_2 + p_3, \tag{6.8}$$

which indicates the general rule.

**Lemma 6.40 (Newton's Formulas)** *For $k = 1, \ldots, n$, we have*

$$ke_k = \sum_{j=1}^{k} (-1)^{j-1} e_{k-j} p_j. \tag{6.9}$$

***Proof*** From (6.7) it follows that

$$\sum_{j=0}^{n} (-1)^j e_{n-j} \omega_i^j = 0, \quad i = 1, \ldots, n.$$

Summing up over all $i$ we obtain (6.9) for $k = n$.

The case where $k < n$ follows easily from this one, we are only facing a problem in notation. The one that we are using suggests that $n$ is fixed and $k$ varies. While this setting is natural in view of applications, in the proof that follows it is convenient to regard $n$ as varying. Let us therefore write, just for the purpose of this proof, $e_{k,n}$ for $e_k$ and $p_{j,n}$ for $p_j$.

Our goal is to show that the polynomial

$$P = ke_{k,n} - \sum_{j=1}^{k} (-1)^{j-1} e_{k-j,n} p_{j,n}$$

is 0. Observe that for any $i$, $j \leq k$,

$$e_{i,n}(\omega_1, \ldots, \omega_k, 0, \ldots, 0) = e_{i,k}(\omega_1, \ldots, \omega_k),$$

$$p_{j,n}(\omega_1, \ldots, \omega_k, 0, \ldots, 0) = p_{j,k}(\omega_1, \ldots, \omega_k).$$

Since (6.9) holds for $n = k$, it follows that $P(\omega_1, \ldots, \omega_k, 0, \ldots, 0) = 0$. This means that $P$ does not contain nonzero monomials in $\omega_1, \ldots, \omega_k$. Similarly, by setting zeros at other places, we see that $P$ does not contain nonzero monomials in any set of $k$ indeterminates. However, from the definition of $P$ it is evident that its nonzero monomials could only be of degree $k$, and therefore cannot involve more than $k$ indeterminates. Hence, $P = 0$. □

In the characteristic zero case we can therefore express the $e_k$'s through the $p_j$'s. Say, from (6.8) we obtain

$$e_1 = p_1, \quad e_2 = \frac{1}{2}\big(p_1^2 - p_2\big), \quad e_3 = \frac{1}{6}\big(p_1^3 - 3p_1p_2 + 2p_3\big).$$

In general, for each $k = 1, \ldots, n$ there exists a polynomial $q_k = q_k(\omega_1, \ldots, \omega_k) \in \mathbb{Q}[\omega_1, \omega_2, \ldots]$ such that

$$e_k = q_k(p_1, \ldots, p_k). \tag{6.10}$$

For our goals it is important to note that $q_k$ has zero constant term. We also remark that if $\lambda\omega_1^{m_1} \ldots \omega_k^{m_k}$, $\lambda \in \mathbb{Q}$, $m_j \geq 0$, is a monomial of $q_k$, then $m_1 + 2m_2 + \cdots + km_k = k$. For convenience, we also set $q_0 = 1$, so that (6.10) holds for $k = 0, 1, \ldots, n$.

**Theorem 6.41** *If $C$ is a commutative $\mathbb{Q}$-algebra, then for every $A \in M_n(C)$ we have*

$$p_A(\omega) = \sum_{j=0}^{n} (-1)^j q_{n-j}\Big(\mathrm{tr}(A), \mathrm{tr}\big(A^2\big), \ldots, \mathrm{tr}\big(A^{n-j}\big)\Big)\omega^j.$$

***Proof*** A standard reasoning shows that it suffices to consider the case where $A$ is a generic matrix. We may therefore assume that $C = \mathbb{Q}[\omega_{11}, \omega_{12} \ldots, \omega_{nn}]$. In particular, $C$ is a commutative domain and as such it can be embedded into a field (e.g., its field of quotients), which can be further embedded into an algebraically closed field (e.g., its algebraic closure). But then there is no loss of generality in assuming that $C$ itself is an algebraically closed field. Hence,

$$p_A(\omega) = (-1)^n(\omega - \lambda_1)\ldots(\omega - \lambda_n),$$

where the $\lambda_i$'s are the eigenvalues of $A$. From the Jordan normal form of $A$ it readily follows that $\mathrm{tr}(A^i) = p_i(\lambda_1, \ldots, \lambda_n)$, $i = 1, \ldots, n$. Using (6.7), we now obtain

$$\begin{aligned} p_A(\omega) &= \sum_{j=0}^{n} (-1)^j e_{n-j}(\lambda_1, \ldots, \lambda_n)\omega^j \\ &= \sum_{j=0}^{n} (-1)^j q_{n-j}\big(p_1(\lambda_1, \ldots, \lambda_n), \ldots, p_{n-j}(\lambda_1, \ldots, \lambda_n)\big)\omega^j \\ &= \sum_{j=0}^{n} (-1)^j q_{n-j}\Big(\mathrm{tr}(A), \ldots, \mathrm{tr}\big(A^{n-j}\big)\Big)\omega^j, \end{aligned}$$

as desired. □

For example, for $n = 2$ we have

$$p_A(\omega) = \omega^2 - \mathrm{tr}(A)\omega + \frac{1}{2}\Big(\mathrm{tr}(A)^2 - \mathrm{tr}\big(A^2\big)\Big) \tag{6.11}$$

(and hence $\det(A) = \frac{1}{2}\Big(\mathrm{tr}(A)^2 - \mathrm{tr}\big(A^2\big)\Big)$ in this case). In general, the coefficient at $\omega^j$ is a $\mathbb{Q}$-linear combination of expressions of the form

$$\mathrm{tr}(A)^{m_1}\mathrm{tr}\big(A^2\big)^{m_2}\ldots\mathrm{tr}\big(A^{n-j}\big)^{m_{n-j}}$$

where $m_1 + 2m_2 + \cdots + (n-j)m_{n-j} = n - j$.

**Corollary 6.42** *Let $C$ be a commutative $\mathbb{Q}$-algebra. If $A \in M_n(C)$ is such that $\mathrm{tr}(A) = \mathrm{tr}(A^2) = \cdots = \mathrm{tr}(A^n) = 0$, then $A^n = 0$.*

***Proof*** Since the polynomials $q_k$, $k = 1, \ldots, n$, have zero constant terms, it follows that $p_A(\omega) = (-1)^n\omega^n$. □

## 6.9 The Amitsur-Levitzki Theorem

The matrix algebra $M_n(F)$ has dimension $n^2$ and hence satisfies $s_{n^2+1}$ (see Example 6.14). From (6.3) we therefore see that it also satisfies $s_{n^2+2}$, $s_{n^2+3}$, etc. But does it satisfy $s_k$ with $k < n^2 + 1$? Moreover, what is the minimal degree of a nonzero identity of $M_n(F)$? We first approach these questions from the opposite direction.

**Lemma 6.43** *A nonzero polynomial of degree less than $2n$ is not an identity of $M_n(F)$.*

***Proof*** By Theorem 6.23, it suffices to consider multilinear polynomials. Take a nonzero multilinear polynomial $f$ of degree $2n - 1$. Thus,

$$f = \sum_{\sigma\in S_{2n-1}} \lambda_\sigma \xi_{\sigma(1)}\ldots\xi_{\sigma(2n-1)}$$

and, without loss of generality, $\lambda_1 \neq 0$. The sequence of standard matrix units

$$E_{11}, E_{12}, E_{22}, \ldots, E_{n-1,n}, E_{nn}$$

has the property that their product in the given order is $E_{1n}$, while the product in any other order is zero. Therefore, $f(E_{11}, E_{12}, \ldots, E_{nn}) = \lambda_1 E_{1n} \neq 0$. Similarly, by taking an appropriate subsequence, we see that a nonzero multilinear polynomial of degree less than $2n - 1$ is not an identity of $M_n(F)$. □

Thus, the best we can hope for is that $M_n(F)$ satisfies $s_{2n}$—and this is indeed the case! This was proved by S. A. Amitsur and J. Levitzki in 1950. The charming proof that we are about to give is due to S. Rosset.

**Theorem 6.44 (Amitsur-Levitzki)** *The standard polynomial $s_{2n}$ is an identity of $M_n(C)$ for every commutative algebra $C$.*

***Proof*** By assumption, $C$ is an algebra over some field $F$. Let $F_0$ be the prime subfield of $F$ and consider $C$ as an algebra over $F_0$. Since $M_n(C) \cong C \otimes_{F_0} M_n(F_0)$ (Example 5.22) and since multilinear identities are stable (Lemma 6.27), it is enough to prove that $s_{2n}$ is an identity of $M_n(F_0)$. Suppose this is true for $F_0 = \mathbb{Q}$. Then $s_{2n}$ is also an identity of $M_n(\mathbb{Z})$. However, $M_n(\mathbb{Z}_p)$ is a homomorphic image of $M_n(\mathbb{Z})$. Indeed, if $\varphi$ is the canonical homomorphism from $\mathbb{Z}$ onto $\mathbb{Z}_p$, then $(m_{ij}) \mapsto \big(\varphi(m_{ij})\big)$ is a homomorphism from $M_n(\mathbb{Z})$ onto $M_n(\mathbb{Z}_p)$. Consequently, $f$ is also an identity of $M_n(\mathbb{Z}_p)$. We have thus reduced our problem to proving that $s_{2n}$ is an identity of $M_n(\mathbb{Q})$.

Take $A_1, \ldots, A_{2n} \in M_n(\mathbb{Q})$. Our goal is to show that $s_{2n}(A_1, A_2, \ldots, A_{2n}) = 0$. We will accomplish this with the help of the Grassmann algebra $G = G_0 \oplus G_1$ over $\mathbb{Q}$ with generators $x_1, x_2, \ldots$ (see Example 6.6). For $B = (b_{ij}) \in M_n(\mathbb{Q})$ and $x \in G$, we define $Bx$ to be the matrix $(b_{ij}x) \in M_n(G)$. Observe that the product $(Bx)(B'x')$ of two such matrices is equal to $(BB')(xx')$.

We set

$$A = A_1x_1 + A_2x_2 + \cdots + A_{2n}x_{2n} \in M_n(G).$$

From $x_iGx_i = 0$ we easily infer that

$$A^{2n} = \sum_{\sigma \in S_{2n}} A_{\sigma(1)}A_{\sigma(2)} \ldots A_{\sigma(2n)}x_{\sigma(1)}x_{\sigma(2)} \ldots x_{\sigma(2n)}.$$

Recall that $x_{\sigma(1)} \ldots x_{\sigma(r)} = \operatorname{sgn}(\sigma)x_1 \ldots x_r$ (see (6.4)). Consequently,

$$A^{2n} = s_{2n}(A_1, A_2, \ldots, A_{2n})x_1x_2 \ldots x_{2n}.$$

Hence we are reduced to proving that $A^{2n} = 0$, i.e., $\big(A^2\big)^n = 0$.

Note that $A^2$ lies in $M_n(G_0)$. As $G_0$ is a commutative algebra, Corollary 6.42 tells us that it suffices to show that $\big(A^2\big)^k = A^{2k}$ has zero trace for $k = 1, \ldots, n$. To this end, notice that $A^{2k-1} = \sum_j B_j y_j$ for some $B_j \in M_n(\mathbb{Q})$ and $y_j \in G_1$. Consequently,

$$\operatorname{tr}\big(A^{2k}\big) = \operatorname{tr}\big(AA^{2k-1}\big) = \operatorname{tr}\Big(\sum_{i,j} A_iB_jx_iy_j\Big) = \sum_{i,j} \operatorname{tr}\big(A_iB_j\big)x_iy_j.$$

On the other hand,

$$\operatorname{tr}\big(A^{2k}\big) = \operatorname{tr}\big(A^{2k-1}A\big) = \sum_{i,j} \operatorname{tr}\big(B_j A_i\big) y_j x_i .$$

Comparing both expressions and using $\operatorname{tr}\big(A_i B_j\big) = \operatorname{tr}\big(B_j A_i\big)$ (a general property of the trace) and $x_i y_j = -y_j x_i$ (a consequence of $y_j \in G_1$), we obtain $\operatorname{tr}\big(A^{2k}\big) = 0$. □

## Exercises

**6.1** Suppose that $f \in F\langle \xi, \eta \rangle$ commutes with $\xi\eta$. Show that there exists a $p \in F[\omega]$ such that $f = p(\xi\eta)$.

**Remark** This is a simple special case of a theorem by G. Bergman: If $f, g \in F\langle X \rangle$ commute, then there exist $h \in F\langle X \rangle$ and $p, q \in F[\omega]$ such that $f = p(h)$ and $g = q(h)$.

**6.2** Find an automorphism of $F\langle \xi, \eta \rangle$ different from the identity.

**6.3** Show that $F\langle \xi_1, \xi_2, \dots \rangle$ can be embedded into $F\langle \xi, \eta \rangle$ via $\xi_i \mapsto \xi^i \eta$.

**6.4** A linear subspace $J$ of an algebra $A$ is called a **Jordan subalgebra** of $A$ if $u \circ v = uv + vu \in J$ for all $u, v \in J$. For example, the set of symmetric matrices in $M_n(F)$, $n \geq 2$, is a Jordan subalgebra, but not a subalgebra of $M_n(F)$. Elements of the Jordan subalgebra of $F\langle X \rangle$ generated by $X \cup \{1\}$ are called **Jordan polynomials**. For example, if $\operatorname{char}(F) \neq 2$ then $\xi_1^2 = \frac{1}{2}\xi_1 \circ \xi_1$ and $\xi_1\xi_2\xi_1 = \frac{1}{2}\xi_1 \circ (\xi_1 \circ \xi_2) - \frac{1}{4}(\xi_1 \circ \xi_1) \circ \xi_2$ are Jordan polynomials. Show that $\xi_1^n$, $\xi_1\xi_2\xi_3 + \xi_3\xi_2\xi_1$, $(\xi_1\xi_2 - \xi_2\xi_1)^2$ are also Jordan polynomials. Find a Jordan subalgebra $J$ of the Grassmann algebra $G$ such that $u_1u_2u_3u_4 + u_4u_3u_2u_1 \notin J$ for some $u_1, u_2, u_3, u_4 \in J$. Hence, conclude that $\xi_1\xi_2\xi_3\xi_4 + \xi_4\xi_3\xi_2\xi_1$ is not a Jordan polynomial, and neither is $\xi_1\xi_2 \dots \xi_n + \xi_n\xi_{n-1} \dots \xi_1$ for any $n \geq 4$.

**Remark** A nonassociative (= not necessarily associative) algebra $J$ is called a **Jordan algebra** if $xy = yx$ and $x^2(yx) = (x^2y)x$ for all $x, y \in J$. Every Jordan subalgebra of an associative algebra is a Jordan algebra under the product $\circ$. Such Jordan algebras are called **special**. Jordan algebras that are not special are called **exceptional**. They exist, but are rare.

**6.5** Analogously, a **Lie subalgebra** of an algebra $A$ is defined as a linear subspace $L$ of $A$ such that $[u, v] \in L$ for all $u, v \in L$. For example, the set of all trace zero matrices in $M_n(F)$, $n \geq 2$, is a Lie subalgebra, but not a subalgebra of $M_n(F)$. A **Lie polynomial** is an element of the Lie subalgebra of $F\langle X \rangle$ generated by $X$. Find a multilinear Lie polynomial of degree 3 which is simultaneously a Jordan polynomial. Show that $\xi_1\xi_2 \dots \xi_n - \xi_n\xi_{n-1} \dots \xi_1$ is not a Lie polynomial for $n \geq 3$.

**Remark** A nonassociative algebra $L$ is called a **Lie algebra** if $x^2 = 0$ and $(xy)z + (zx)y + (yz)x = 0$ for all $x, y, z \in L$. A Lie subalgebra of an

associative algebra is a Lie algebra under the product $[\,.\,,\,.\,]$. Conversely, the **Poincaré-Birkhoff-Witt theorem** shows that every Lie algebra is isomorphic to a Lie subalgebra of an associative algebra. (The product in Lie algebras is actually usually denoted by $[\,.\,,\,.\,]$, so the defining identities are written as $[x, x] = 0$ and $[[x, y], z] + [[z, x], y] + [[y, z], x] = 0$; the latter is called the **Jacobi identity**.)

**6.6** Describe the $\mathbb{R}$-algebras $\mathbb{C}$ and $\mathbb{R} \times \mathbb{R}$ in terms of generators and relations.

**6.7** To which well-known algebra is $F\langle x, y \mid x^2 = y^2 = -1,\ xy + yx = 0\rangle$ isomorphic if (a) $F = \mathbb{R}$, (b) $F = \mathbb{C}$?

**6.8** The algebra $F\langle x, x' \mid xx' = x'x = 1\rangle$, more commonly denoted by $F[x, x^{-1}]$, is called the algebra of **Laurent polynomials** (a glance at Sect. 1.9 reveals the reason for this name). Find its invertible elements and show that it is isomorphic to the group algebra $F[\mathbb{Z}]$.

**6.9** Let $q \in F^*$, not a root of unity. Show that the algebra $F\langle x, x', y, y' \mid xx' = x'x = 1,\ yy' = y'y = 1,\ yx = qxy\rangle$ (more commonly denoted by $F_q[x^{\pm 1}, y^{\pm 1}]$) is a simple noetherian domain.

**6.10** Show that $F\langle x, y \mid xy = 1\rangle$ is a primitive algebra with nonzero socle and has a left zero-divisor that is not a right zero-divisor.

*Hint:* Consider $e = 1 - yx$.

**6.11** Prove that $F\langle \xi_1, \xi_2\rangle \otimes F\langle \eta_1, \eta_2\rangle \cong F\langle x_1, x_2, y_1, y_2 \mid [x_i, y_j] = 0,\ i, j = 1, 2\rangle$.

**6.12** Find the Jacobson radical of the Grassmann algebra $G$.

**6.13** Show that a finite dimensional algebra over a finite field satisfies a nonzero identity in one indeterminate.

**Remark** A solution is indirectly given in Example 7.62 below.

**6.14** Let $f_i(\xi_1, \ldots, \xi_n) \in F\langle X\rangle$, $i = 1, \ldots, m$, be linearly independent polynomials. Show that an $F$-algebra $A$ must be a PI-algebra if for any $x_1, \ldots, x_n \in A$, the evaluations $f_i(x_1, \ldots, x_n)$, $i = 1, \ldots, m$, are linearly dependent in $A$.

**6.15** Prove the Nagata-Higman theorem for semiprime algebras. That is, prove that $\xi^n$ is not an identity of a nonzero semiprime algebra $A$ over a field $F$ with $\mathrm{char}(F) = 0$.

*Hint:* Use the Vandermonde determinant argument on the equation $(x + \lambda y)^n = 0$, where $x, y \in A$ and $\lambda \in F$.

**6.16** We say that $h(\xi_1, \ldots, \xi_n) \in F\langle X\rangle$ is homogeneous in $\xi_i$ if each monomial of $h$ has the same degree in $\xi_i$. If $h$ is homogeneous in every indeterminate $\xi_i$, then we call it a **multihomogeneous polynomial**. Note that every polynomial $f \in F\langle X\rangle$ can be written as $f = \sum_i f_i$ with $f_i$ multihomogeneous. Show that if the field $F$ is infinite, then $f$ is an identity of the $F$-algebra $A$ if and only if each $f_i$ is an identity of $A$.

**6.17** An $F$-algebra $A$ is said to be an **algebraic algebra of bounded degree** if there exists an $n \in \mathbb{N}$ such that for every $x \in A$ there is an $f(\omega) \in F[\omega]$ of degree $n$ satisfying $f(x) = 0$. Show that the class of algebraic algebras of bounded degree properly contains the class of finite dimensional algebras, and is properly contained in the class of PI-algebras.

**6.18** Show that the algebra of all finite rank operators on an infinite dimensional vector space over $F$ is an algebraic algebra, but not a PI-algebra.

**6.19** Let $A$ be an algebra and let $M$ be an $A$-bimodule. Equip $A \times M$ with pointwise addition and scalar multiplication, and the product $(x, m)(y, n) = (xy, xn + my)$. Check that in this way $A \times M$ becomes an algebra; let us denote it by $A_M$. Show that $A$ is a PI-algebra if and only if $A_M$ is a PI-algebra.

**6.20** Find an identity of minimal degree of the subalgebra of $M_n(F)$ consisting of all matrices that have zeros in all columns except in the first one.

**6.21** Show that, in general, $\mathrm{Id}(A) \neq \mathrm{Id}(A[\omega])$, although $A$ and $A[\omega]$ satisfy the same multilinear identities.

**6.22** Let $f = f(\xi_1, \ldots, \xi_n) \in F\langle X\rangle$ be a multilinear polynomial and let $k \in \mathbb{N}$. Prove that $\mathrm{tr}\big(f(A_1, \ldots, A_n)\big) = 0$ for all $A_1, \ldots, A_n \in M_k(F)$ if and only if there exist an $f_0 \in \mathrm{Id}(M_k(F))$ and $g_i, h_i \in F\langle X\rangle$ such that $f = f_0 + \sum_i [g_i, h_i]$.
*Hint:* Use $m\xi_n m' = m'm\xi_n + [m\xi_n, m']$ for monomials $m, m'$.

**6.23** Show that every generic matrix in $M_n\big(F[\Omega]\big)$ is invertible in $M_n\big(F(\Omega)\big)$. Hence conclude that if $F$ is an infinite field, $m \in F\langle X\rangle$ is a nonzero monomial, and $f \in F\langle X\rangle$ is such that $mf \in \mathrm{Id}(M_n(F))$, then $f \in \mathrm{Id}(M_n(F))$.

**Remark** At the end of the book, we will show that much more is true; see Theorem 7.63.

**6.24** Let $A$ be a nonzero finite dimensional semiprime algebra. Show that there exists an $n \in \mathbb{N}$ such that $A$ satisfies $s_{2n}$, but does not satisfy a nonzero identity of degree less than $2n$.

**6.25** Given $A, B \in M_2(F)$, let

$$P_{A,B}(\xi, \eta) = \xi\eta + \eta\xi - \mathrm{tr}(B)\xi - \mathrm{tr}(A)\eta + \mathrm{tr}(A)\mathrm{tr}(B) - \mathrm{tr}(AB) \in F\langle \xi, \eta\rangle.$$

Recalling the form (6.11) of the characteristic polynomial, we see that by applying the linearization method to the Cayley-Hamilton theorem we obtain $P_{A,B}(A, B) = 0$. Show that the identity

$$\sum_{\sigma \in S_4} \mathrm{sgn}(\sigma) P_{A_{\sigma(1)}A_{\sigma(2)}, A_{\sigma(3)}A_{\sigma(4)}}(A_{\sigma(1)}A_{\sigma(2)}, A_{\sigma(3)}A_{\sigma(4)}) = 0$$

yields the Amitsur-Levitzki theorem for $n = 2$ (provided that $\mathrm{char}(F) \neq 2$, but from the first paragraph of our proof of the Amitsur-Levitzki theorem it is evident that this can be assumed without loss of generality).

**Remark** We have confined ourselves to $n = 2$ for clarity. The Amitsur-Levitzki theorem can be, in a similar fashion, derived from the Cayley-Hamilton theorem for any $n$. This was discovered by Yu. P. Razmyslov, and is actually one of the standard proofs of the Amitsur-Levitzki theorem. A deeper result, obtained independently by Razmyslov and C. Procesi, shows that in fact *every* polynomial identity of $M_n(F)$, $\mathrm{char}(F) = 0$, is a consequence of the Cayley-Hamilton identity $p_A(A) = 0$. Moreover, the same is true for the more general **trace identities**. We omit giving their definition, but just give a hint by saying that the characteristic polynomial $p_A$ gives rise to a trace identity that is not a polynomial identity.

# Chapter 7
# Rings of Quotients and Structure of PI-Rings




A commutative domain can be embedded into its field of quotients. One usually learns this simple and very useful fact in an introductory abstract algebra course. It is therefore natural to seek analogies in noncommutative rings.

The lack of commutativity causes considerable difficulties. Nevertheless, various generalizations of the construction of the field of quotients exist in noncommutative ring theory. We will discuss a few of them: rings of central quotients, classical rings of quotients, and Martindale rings of quotients.

Although the issue now is the noncommutativity, we cannot entirely escape from the commutative context. The centers of the constructed rings will be of crucial importance for us. The extended centroid of a prime ring, i.e., a field defined as the center of the Martindale ring of quotients, will enable us to extend a part of the theory of central simple algebras to general prime rings. We will also use it as a principal tool in the second part of the chapter, devoted to the structure theory of prime rings satisfying (generalized) polynomial identities. As a rough summary, it can be stated that prime PI-rings are close to finite dimensional simple algebras.

## 7.1 Rings of Central Quotients

A glance at the construction of the field of quotients of a commutative domain shows that it highly depends on commutativity. There is no obvious way to avoid it. However, in this first section we will take an easy way; we are going to use essentially the same construction for a noncommutative ring $R$, but allow only the elements from the center $Z = Z(R)$ as "denominators". More precisely, we will assume that the nonzero elements from $Z$ satisfy the (obviously necessary) condition from the next definition, and enlarge $R$ in such a way that these elements become invertible in the new, larger ring.

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**Definition 7.1** An element $a$ in a ring $R$ is said to be **regular** if $a \neq 0$ and $a$ is not a zero-divisor.

**Constructing the Ring of Central Quotients $Q_Z(R)$** We assume that $R$ is any ring such that its center $Z$ is nonzero and all elements in $Z \setminus \{0\}$ are regular; in particular, $Z$ is a commutative domain.

We claim that

$$(r, z) \sim (r', z') \iff rz' = r'z$$

defines an equivalence relation on the set $R \times \big(Z \setminus \{0\}\big)$. Only the transitivity is not entirely obvious. Thus, assume that $(r, z) \sim (r', z')$ and $(r', z') \sim (r'', z'')$, i.e., $rz' = r'z$ and $r'z'' = r''z'$. Then $(rz'' - r''z)z' = r'zz'' - r'z''z = 0$, and so $(r, z) \sim (r'', z'')$ follows from the regularity of $z'$.

Denote by $rz^{-1}$ the equivalence class of $(r, z)$. Thus, $rz^{-1} = r'z'^{-1}$ if and only if $rz' = r'z$; in particular, $rz(zw)^{-1} = rw^{-1}$. Let $Q_Z(R)$ denote the set of all equivalence classes, equipped with addition and multiplication defined by

$$rz^{-1} + sw^{-1} = (rw + sz)(zw)^{-1},$$
$$rz^{-1} \cdot sw^{-1} = rs(zw)^{-1}.$$

To prove that these operations are well-defined, we must show that $(r, z) \sim (r', z')$ and $(s, w) \sim (s', w')$ imply $(rw + sz, zw) \sim (r'w' + s'z', z'w')$ and $(rs, zw) \sim (r's', z'w')$. This is straightforward. It should come as no surprise that $Q_Z(R)$, equipped with these operations, is a ring. The verification requires a little patience, but no more than the verification that $\mathbb{Q}$ is a ring.

**Definition 7.2** The ring $Q_Z(R)$ is called the **ring of central quotients** of $R$.

No matter whether $R$ is unital or not, $Q_Z(R)$ is. Its unity is $z_0 z_0^{-1}$, where $z_0$ is an arbitrary element in $Z \setminus \{0\}$. The center of $Q_Z(R)$, which we denote by $\widehat{Z}$, is easily seen to consist of elements of the form $zw^{-1}$, $z, w \in Z$, $w \neq 0$. If $z \neq 0$, then $wz^{-1}$ is the inverse of $zw^{-1}$. Therefore, $\widehat{Z}$ is a field; of course, $\widehat{Z}$ is nothing but the field of quotients of $Z$. Every ring whose center is a field can be considered as a central algebra over its center. Thus, $Q_Z(R)$ is a central algebra over $\widehat{Z}$.

One immediately checks that

$$\iota : R \to Q_Z(R), \quad \iota(r) = (rz_0)z_0^{-1},$$

is an embedding of $R$ into $Q_Z(R)$ (note that the definition of $\iota$ is independent of the choice of $z_0 \in Z \setminus \{0\}$). It is convenient to identify $R$ with its isomorphic copy $\iota(R)$ inside $Q_Z(R)$, and thus consider $R$ as a subring of $Q_Z(R)$. The expression $rz^{-1}$, so far used just as the notation for the equivalence class, can then be interpreted as the product of $r \in R$ and the inverse of $z \in Z \setminus \{0\}$.

Intuitively, $Q_Z(R)$ is the smallest ring containing $R$ and having the property that every element in $Z \setminus \{0\}$ is invertible in $Q_Z(R)$. The next proposition states this in more formal terms.

**Proposition 7.3** *If $T$ is a unital ring and $\varphi : R \to T$ is a homomorphism such that $\varphi(z)$ is invertible in $T$ for every $z \in Z \setminus \{0\}$, then $\varphi$ can be uniquely extended to a homomorphism $\overline{\varphi} : Q_Z(R) \to T$.*

***Proof*** It is straightforward to check that $\overline{\varphi}(rz^{-1}) = \varphi(r)\varphi(z)^{-1}$ is a well-defined homomorphism which extends $\varphi$. The uniqueness is clear. □

Let us add that if $\varphi$ is injective and every element in $T$ can be written as $\varphi(r)\varphi(z)^{-1}$, then $\overline{\varphi}$ is an isomorphism.

The next theorem summarizes our discussion up to this point.

**Theorem 7.4** *Let $R$ be a ring such that its center $Z$ is nonzero and all elements in $Z \setminus \{0\}$ are regular. The ring $Q_Z(R)$ has the following properties:*

(a) *$Q_Z(R)$ is a unital ring containing $R$ as a subring.*
(b) *Every element in $Z \setminus \{0\}$ is invertible in $Q_Z(R)$.*
(c) *Every element in $Q_Z(R)$ is of the form $rz^{-1}$, where $r \in R$ and $z \in Z \setminus \{0\}$.*

*Moreover, these properties characterize $Q_Z(R)$ up to isomorphism.*

It is easy to see that if $R$ is a prime ring with nonzero center $Z$, then $Q_Z(R)$ exists (i.e., nonzero elements in $Z$ are regular) and is a prime ring.

We continue with examples. The first two are obvious, but worth mentioning.

**Example 7.5** If the center of a unital ring $R$ is a field, then $Q_Z(R) = R$.

**Example 7.6** If $R$ is a commutative domain, then $Q_Z(R)$ is the field of quotients of $R$.

**Example 7.7** We claim that $Q_Z(M_n(\mathbb{Z})) \cong M_n(\mathbb{Q})$. Obviously, $M_n(\mathbb{Q})$ has property (a). Commuting with matrix units we see that the center of $M_n(\mathbb{Z})$ consists of matrices $nI$ with $n \in \mathbb{Z}$ (see the proof of Lemma 1.23). Therefore, $M_n(\mathbb{Q})$ has property (b). Finally, by making use of a common denominator of the matrix entries, we see that it also has property (c).

**Example 7.8** A small variation of the preceding example: If $R = \left[\begin{smallmatrix} \mathbb{Z} & 2\mathbb{Z} \\ \mathbb{Z} & \mathbb{Z} \end{smallmatrix}\right]$, i.e., the subring of $M_2(\mathbb{Z})$ consisting of matrices whose $(1, 2)$ entry is an even integer, then $Q_Z(R) = M_2(\mathbb{Q})$. The point we wish to make here is that even if a ring is a bit "odd", its ring of central quotients may be "spotless".

**Example 7.9** If $R$ is the ring of quaternions with integer coefficients, $R = \mathbb{Z} \oplus \mathbb{Z}i \oplus \mathbb{Z}j \oplus \mathbb{Z}k$, then $Q_Z(R)$ is the ring of quaternions with rational coefficients, $Q_Z(R) = \mathbb{Q} \oplus \mathbb{Q}i \oplus \mathbb{Q}j \oplus \mathbb{Q}k$. Indeed, $Z(R) = \mathbb{Z}$ and checking (a)–(c) is immediate.

Certain modifications of our construction of $Q_Z(R)$ are also important and can be found in standard textbooks, usually under the name "(central) localization". We will not need them, but let us just give a clue what changes can be made. The role of

$Z \setminus \{0\}$ can be replaced by any subset $S$ of $Z$ that is closed under multiplication, and instead of requiring that the elements from $S$ are regular, one defines the equivalence relation $\sim$ on $R \times S$ as follows: $(r, z) \sim (r', z') \iff (rz' - r'z)z'' = 0$ for some $z'' \in S$.

## 7.2 Classical Rings of Quotients

Given a ring $R$, is it possible to construct a larger ring $Q$ such that all regular elements in $R$, not only those from the center, are invertible in $Q$? More specifically, can we construct $Q$ in such a way that a version of Theorem 7.4, in which the set of all regular elements plays the role of $Z \setminus \{0\}$, holds for $Q$? The answer is not straightforward. It certainly is positive for some rings (like commutative domains, for example). But it can also be negative, as we will see.

Let us make precise what ring $Q$ we wish to have. To avoid trivialities, we tacitly assume that our rings contain at least one regular element.

**Definition 7.10** Let $R$ be a ring and let $S$ be the set of all regular elements in $R$. A ring $Q$ is called a **right classical ring of quotients** of $R$ if it has the following properties:

(a) $Q$ is a unital ring containing $R$ as a subring.
(b) Every element in $S$ is invertible in $Q$.
(c) Every element in $Q$ is of the form $rs^{-1}$, where $r \in R$ and $s \in S$.

It is clear that elements of $R$ that are not regular cannot be invertible in any larger ring. The existence of a right classical ring of quotients of $R$ is thus the best we can hope for if we wish to make as many elements in $R$ invertible as possible.

A **left classical ring of quotients** of $R$ is defined analogously, one just substitutes $s^{-1}r$ for $rs^{-1}$ in (c). Up until the final remark of the section, however, we will consider only right classical rings of quotients.

**Example 7.11** Let $R$ be any of the rings from Examples 7.6–7.9, i.e., a commutative domain, $M_n(\mathbb{Z})$, $\left[\begin{smallmatrix} \mathbb{Z} & 2\mathbb{Z} \\ \mathbb{Z} & \mathbb{Z} \end{smallmatrix}\right]$, or $\mathbb{Z} \oplus \mathbb{Z}i \oplus \mathbb{Z}j \oplus \mathbb{Z}k$. It is easy to verify that the ring of central quotients $Q_Z(R)$ is also a right classical ring of quotients of $R$.

**Example 7.12** If every regular element in $R$ is invertible, then $R$ is its own right classical ring of quotients. Recall from Proposition 1.10 (a) that every finite dimensional unital algebra has this property.

Suppose $R$ has a right classical ring of quotients $Q$. Given $r \in R$ and $s \in S$, we have $s^{-1}r \in Q$ by (a) and (b), and so (c) implies that there exist an $r_1 \in R$ and an $s_1 \in S$ such that $s^{-1}r = r_1 s_1^{-1}$. Multiplying by $s$ from the left and by $s_1$ from the right, we obtain $rs_1 = sr_1$. Thus, the following holds:

$$rS \cap sR \neq \emptyset \quad \text{for all } r \in R \text{ and } s \in S. \tag{7.1}$$

It is easy to find rings in which (7.1) is not fulfilled, and hence rings not having right classical ring of quotients (see Example 7.18 below). The main goal of this section is to prove that (7.1) is also sufficient, not only necessary, condition for a ring $R$ to have a right classical ring of quotients. This result is basically due to O. Ore; more precisely, he proved it, in 1931, for the special case where $R$ is a domain. The condition (7.1) is called the **right Ore condition**.

Unlike in the preceding section, the noncommutativity is now a serious obstacle. There are two standard ways to overcome it. The first one, used already by Ore, is based on a careful modification of the construction from the preceding section. While this approach is intuitively clear, working out the details is extremely tedious. We will therefore choose the other way. To gain intuition, we first sketch it for the seminal case where $R = \mathbb{Z}$.

Via the regular representation, we may identify $\mathbb{Z}$ with the maps $x \mapsto nx$, $n \in \mathbb{Z}$, from $\mathbb{Z}$ to itself. Our goal now is to describe $\mathbb{Q}$ without explicit use of fractions, i.e., by staying inside the framework of the ring $\mathbb{Z}$. Consider, for example, the map $f : 2\mathbb{Z} \to \mathbb{Z}$, $f(2x) = x$. Note that $f$ halves each number from its domain, so it carries information about the rational number $\frac{1}{2}$. Similarly, $g : m\mathbb{Z} \to \mathbb{Z}$, $g(mx) = nx$, where $m \in \mathbb{N}$ and $n \in \mathbb{Z}$, corresponds to $\frac{n}{m} \in \mathbb{Q}$ (in the sense that $g(u) = \frac{n}{m}u$ for all $u \in m\mathbb{Z}$). Given another map of this kind, $g' : m'\mathbb{Z} \to \mathbb{Z}$, $g'(m'x) = n'x$, we can consider $g + g'$ and $gg'$ as maps defined on $mm'\mathbb{Z}$ (possibly one can choose larger domains, but this is irrelevant for us). Note that $g+g'$ corresponds to $\frac{n}{m} + \frac{n'}{m'}$, and $gg'$ to $\frac{n}{m} \cdot \frac{n'}{m'}$. By identifying maps that correspond to the same rational number, we obtain in this way an alternative definition of the field $\mathbb{Q}$. Perhaps an overcomplicated one, but having an advantage of being suitable for generalizations to noncommutative rings. The crucial point is that the above maps are $\mathbb{Z}$-module homomorphisms from ideals of $\mathbb{Z}$ to $\mathbb{Z}$.

We proceed to the general case.

**Constructing the Right Classical Ring of Quotients $Q_{rc}(R)$** Let $R$ be a ring satisfying the right Ore condition (7.1). Denote by $\mathscr{I}$ the set of all right ideals of $R$ that contain at least one regular element. A right ideal can be considered as a right $R$-module. Consider the set of all pairs $(f, I)$ where $I \in \mathscr{I}$ and $f : I \to R$ is a right $R$-module homomorphism. Define an equivalence relation on this set as follows: $(f, I) \sim (g, J)$ if $f$ and $g$ coincide on some $K \in \mathscr{I}$ such that $K \subseteq I \cap J$. In order to verify the transitivity, it is enough to notice that $I_1 \cap I_2 \in \mathscr{I}$ if both $I_1, I_2 \in \mathscr{I}$. Let us therefore prove this. Take $s_i \in I_i \cap S$, $i = 1, 2$, and use (7.1) to find an $s \in S$ and an $r \in R$ such that $s_1 s = s_2 r$. Hence $I_1 \cap I_2$ contains a regular element, namely $s_1 s$, so it belongs to $\mathscr{I}$.

Write $[f, I]$ for the equivalence class of $(f, I)$. Let $Q_{rc}(R)$ denote the set of all equivalence classes, endowed with addition and multiplication

$$[f_1, I_1] + [f_2, I_2] = [f_1 + f_2, I_1 \cap I_2],$$

$$[f_1, I_1][f_2, I_2] = [f_1 f_2, f_2^{-1}(I_1)].$$

We must prove that these operations are well-defined. First of all, $f_2^{-1}(I_1) = \{x \in I_2 \mid f_2(x) \in I_1\}$ indeed lies in $\mathscr{I}$. Indeed, from (7.1) we infer that given $s_i \in I_i \cap S$, $i = 1, 2$, there exist an $s \in S$ and an $r \in R$ such that $f_2(s_2)s = s_1 r$, showing that the right ideal $f_2^{-1}(I_1)$ contains the regular element $s_2 s$. Assume now that $(f_1, I_1) \sim (g_1, J_1)$ and $(f_2, I_2) \sim (g_2, J_2)$. Thus, there exist $K_i \in \mathscr{I}$ such that $K_i \subseteq I_i \cap J_i$ and $f_i = g_i$ on $K_i, i = 1, 2$. Consequently, $f_1 + f_2 = g_1 + g_2$ on $K_1 \cap K_2 \in \mathscr{I}$ and hence $(f_1 + f_2, I_1 \cap I_2) \sim (g_1 + g_2, J_1 \cap J_2)$, which proves that addition is well-defined. Concerning multiplication, note that $f_1 f_2 = g_1 g_2$ on $f_2^{-1}(K_1) \cap K_2 \in \mathscr{I}$ and hence $(f_1 f_2, f_2^{-1}(I_1)) \sim (g_1 g_2, g_2^{-1}(J_1))$. Thus, multiplication is well-defined too. Verifying that $Q_{rc}(R)$ is a ring with zero element $[0, R]$ and unity $[\mathrm{id}_R, R]$ is straightforward.

This construction is the basic example of what is called a **noncommutative localization**. Its general version considers an arbitrary set closed under multiplication instead of the set of all regular elements. This does not make the construction substantially more complicated, but we will not go into this matter.

We are now in a position to prove the result announced above.

**Theorem 7.13 (Ore)** *A ring $R$ has a right classical ring of quotients if and only if $R$ satisfies the right Ore condition.*

***Proof*** We only have to prove the "if" part. Assume that $R$ satisfies the right Ore condition and let us show that $Q_{rc}(R)$ *is a right classical ring of quotients of* $R$.

We embed $R$ into $Q_{rc}(R)$ via

$$\iota : R \to Q_{rc}(R), \quad \iota(r) = [L_r, R];$$

here, as always, $L_r$ stands for a left multiplication map. Let us only check the injectivity. If $[L_r, R] = 0$, then $L_r = 0$ on some $I \in \mathscr{I}$. Since $I$ contains a regular element, it follows that $r = 0$.

Let $s \in S$. Observe that the map $f_s : sR \to R$, $f_s(sx) = x$, is well-defined. Moreover, $f_s$ is a right $R$-module homomorphism. Note that $sR \in \mathscr{I}$ since $s^2 \in sR \cap S$. The reader can readily verify that $[f_s, sR]$ is the inverse of $\iota(s)$. Thus, $\iota(s)$ is invertible.

Now take an arbitrary $[f, I] \in Q_{rc}(R)$. Pick an $s \in I \cap S$ and set $r = f(s) \in R$. We have $f(sx) = rx = L_r f_s(sx)$ for all $x \in R$. Hence,

$$[f, I] = [f, sR] = [L_r f_s, sR] = [L_r, R][f_s, sR] = \iota(r)\iota(s)^{-1}.$$

Identifying $R$ with $\iota(R)$, we thus see that $Q_{rc}(R)$ has all the desired properties. □

The ring $Q_{rc}(R)$ has property analogous to that satisfied by $Q_Z(R)$ (see Proposition 7.3).

**Proposition 7.14** *Let $R$ be a ring satisfying the right Ore condition. If $T$ is a unital ring and $\varphi : R \to T$ is a homomorphism such that $\varphi(s)$ is invertible in $T$ for every regular element $s \in R$, then $\varphi$ can be uniquely extended to a homomorphism $\overline{\varphi} : Q_{rc}(R) \to T$.*

***Proof*** Take an $[f, I] \in Q_{rc}(R)$. Pick an $s \in I \cap S$ and define $\overline{\varphi} : Q_{rc}(R) \to T$ by

$$\overline{\varphi}([f, I]) = \varphi(f(s))\varphi(s)^{-1}.$$

One can immediately check that $\overline{\varphi}(\iota(r)) = \varphi(r)$, so $\overline{\varphi}$ extends $\varphi$. However, we must first prove that $\overline{\varphi}$ is well-defined. Assume, therefore, that $(f, I) \sim (g, J)$. Our goal is to show that $\varphi(f(s))\varphi(s)^{-1} = \varphi(g(t))\varphi(t)^{-1}$ for every $t \in J \cap S$. Let $K \in \mathscr{I}$ be such that $K \subseteq I \cap J$ and $f = g$ on $K$. It suffices to show that $\varphi(f(s))\varphi(s)^{-1} = \varphi(f(u))\varphi(u)^{-1}$ holds for $u \in K \cap S$. Indeed, by analogy we then also have $\varphi(g(t))\varphi(t)^{-1} = \varphi(g(u))\varphi(u)^{-1}$, and so the desired conclusion follows from $f(u) = g(u)$. By regularity of $s$ and $u$ we have $sR, uR \in \mathscr{I}$ and hence $sR \cap uR \in \mathscr{I}$. Therefore, there exist $x, y \in R$ such that $v = sx = uy \in S$. Since $\varphi(s)$ and $\varphi(v)$ are invertible in $T$ by assumption, $v = sx$ gives $\varphi(s)^{-1} = \varphi(x)\varphi(v)^{-1}$. Consequently,

$$\varphi(f(s))\varphi(s)^{-1} = \varphi(f(s))\varphi(x)\varphi(v)^{-1} = \varphi(f(v))\varphi(v)^{-1}.$$

Similarly, from $v = uy$ we infer $\varphi(f(u))\varphi(u)^{-1} = \varphi(f(v))\varphi(v)^{-1}$. By comparing, we obtain $\varphi(f(s))\varphi(s)^{-1} = \varphi(f(u))\varphi(u)^{-1}$.

The additivity of $\overline{\varphi}$ is clear. To prove the multiplicativity, take $[f_i, I_i] \in Q_{rc}(R)$, $i = 1, 2$. Pick an $s_1 \in I_1 \cap S$. By definition,

$$\overline{\varphi}([f_1, I_1][f_2, I_2]) = \varphi(f_1(f_2(s_2)))\varphi(s_2)^{-1},$$

where $s_2$ is an arbitrary element in $f_2^{-1}(I_1) \cap S$; since $[f_1, I_1] = [f_1, s_1 R]$, we may choose an $s_2 \in f_2^{-1}(s_1 R)$. Thus, $f_2(s_2) = s_1 x$ and hence $\varphi(x) = \varphi(s_1)^{-1}\varphi(f_2(s_2))$, yielding

$$\varphi(f_1(f_2(s_2))) = \varphi(f_1(s_1))\varphi(x) = \varphi(f_1(s_1))\varphi(s_1)^{-1}\varphi(f_2(s_2)).$$

This readily implies that $\overline{\varphi}([f_1, I_1][f_2, I_2]) = \overline{\varphi}([f_1, I_1])\overline{\varphi}([f_2, I_2])$. Thus, $\overline{\varphi}$ is a homomorphism extending $\varphi$. The uniqueness is obvious because of condition (c) from Definition 7.10. □

**Corollary 7.15** *If a ring satisfies the right Ore condition, then its right classical ring of quotients is unique up to isomorphism.*

**Remark 7.16** Similar results of course hold for the *left* classical ring of quotients. One defines the **left Ore condition** as $Sr \cap Rs \neq \emptyset$ for all $r \in R$ and $s \in S$, and constructs the left classical ring of quotients $Q_{lc}(R)$ of $R$ by using left ideals and right multiplication maps for defining the canonical embedding. Proposition 7.14 together with its left analogue easily implies that if $R$ has both a left and right classical ring of quotients, then they are isomorphic. In other words, if both the left and right Ore condition are fulfilled, then $Q_{rc}(R) \cong Q_{lc}(R)$.

## 7.3 Ore Domains

In this section, we take a closer look at the classical situation where $R$ is a domain. The right Ore condition (7.1) then gets a simpler form:

$$rR \cap sR \neq 0 \quad \text{for all } r, s \in R \setminus \{0\}. \tag{7.2}$$

**Definition 7.17** A nonzero domain satisfying the right Ore condition (7.2) is called a **right Ore domain**.

Theorem 7.13 implies that a ring $R$ is a right Ore domain if and only if $R$ is a subring of a division ring $D$ ($=Q_{rc}(R)$) such that every element in $D$ is of the form $rs^{-1}$ with $r, s \in R$, $s \neq 0$.

Not every domain is right Ore.

**Example 7.18** Let $R = F\langle \xi, \eta \rangle$. Since we obviously have $\xi R \cap \eta R = 0$, $R$ is not a right Ore domain (and neither a left Ore domain, i.e., the condition $Rr \cap Rs \neq 0$ for all $r, s \in R \setminus \{0\}$ is not fulfilled).

A domain that is neither right nor left Ore may still be embedded into a division ring, but then condition (c) of Definition 7.10 is not fulfilled. The free algebra $F\langle \xi, \eta \rangle$ turns out to be an example of such a domain. On the other hand, there do exist domains that cannot be embedded into division rings. For details, we refer the reader to [8, Chap. 9].

The connection between $F\langle \xi, \eta \rangle$ and the nonfulfillment of the right Ore condition is even tighter.

**Proposition 7.19** *Let $A$ be a nonzero unital algebra over a field $F$. If $A$ is a domain that is not right Ore, then $A$ contains a subalgebra isomorphic to $F\langle \xi, \eta \rangle$.*

***Proof*** By assumption, there exist $r, s \in A \setminus \{0\}$ such that $rA \cap sA = 0$. Let $A_0$ be the unital subalgebra of $A$ generated by $r$ and $s$. We claim that the unital homomorphism $f(\xi, \eta) \mapsto f(r, s)$ from $F\langle \xi, \eta \rangle$ onto $A_0$ is injective, and hence an isomorphism. Suppose this is not true. Then there exists a nonzero polynomial $f(\xi, \eta)$ such that $f(r, s) = 0$. We may assume that $f$ is of minimal degree. Let us write $f = \lambda + \xi g + \eta h$, where $\lambda$ is the constant term of $f$ and $g, h \in F\langle \xi, \eta \rangle$. There is no loss of generality in assuming that $h \neq 0$. Setting $u = g(r, s)$ and $v = h(r, s)$, we thus have $\lambda + ru + sv = 0$. Multiplying on the right by $r$ gives $r(\lambda + ur) + svr = 0$, and hence $r(\lambda + ur) = -svr \in rA \cap sA = 0$. Since $\deg(h) < \deg(f)$, $v \neq 0$. As $s$ and $r$ are also nonzero and $A$ is a domain, we have arrived at a contradiction. □

Trivial examples of right Ore domains are division rings and commutative domains. As another simple example, we mention the ring of quaternions with integer coefficients (see Examples 7.9 and 7.11). The next proposition gives rise to less obvious examples.

**Proposition 7.20 (Goldie)** *A right noetherian domain $R$ is a right Ore domain.*

***Proof*** Take $r, s \in R \setminus \{0\}$, and set $J_n = rsR + r^2sR + \cdots + r^nsR$. Since the $J_n$'s obviously form an ascending chain of right ideals, there exists an $n \in \mathbb{N}$ such that $J_n = J_{n+1} = \ldots$ Pick any $x_0 \in R \setminus \{0\}$. Then $r^{n+1}sx_0$ is a nonzero element from $J_{n+1}$, and hence also from $J_n$. Thus there exists a $k \leq n$ such that

$$r^k sx_k + r^{k+1}sx_{k+1} + \cdots + r^n sx_n = r^{n+1}sx_0$$

for some $x_k, \ldots, x_n \in R$ with $x_k \neq 0$. The cancellation property of $R$ yields

$$sx_k + rsx_{k+1} + \cdots + r^{n-k}sx_n = r^{n+1-k}sx_0.$$

Therefore, $sx_k \neq 0$ lies in $rR$. This shows that (7.2) holds. □

**Example 7.21** The Weyl algebra $\mathscr{A}_n$ is a noetherian domain (see Sect. 3.11). Therefore, $Q_{rc}(\mathscr{A}_n)$ exists and is a division ring. Moreover, $Q_{rc}(\mathscr{A}_n) \cong Q_{lc}(\mathscr{A}_n)$ by Remark 7.16 and the left version of Proposition 7.20.

Proposition 7.20 is a small extract from the theory developed by A. W. Goldie in the late 1950s. His main result characterizes rings whose right classical ring of quotients is isomorphic to $M_{n_1}(D_1) \times \cdots \times M_{n_r}(D_r)$ where the $D_i$'s are division rings. A corollary of special importance is that right noetherian semiprime rings do have this property. Goldie's theory is treated in many books, including [4], [8], [9], and [12].

## 7.4 Martindale Rings of Quotients

In this section, we take another step forward in the study of rings of quotients—or perhaps a step backwards, as the next construction is similar to but slightly simpler than the construction of $Q_{rc}(R)$. Roughly speaking, we will just replace the role of the right ideals containing regular elements by the (easier to handle) nonzero two-sided ideals. In spite of the technical similarities, the philosophy behind these two constructions is different. The goal of the one we are about to introduce is not making regular elements invertible, but merely providing a larger ring that can be used as a tool for solving problems concerning the original ring.

The field of quotients of a commutative domain will again serve as a prototype. We will assume that our rings are prime. This is a natural setting; a commutative ring is a domain if and only if it is prime, and the role of prime rings among general rings is parallel to the role of domains among commutative rings. With a little extra effort, we could handle more general semiprime rings; however, we will stick with the classical prime ring situation, as considered in the seminal work of W. S. Martindale 3rd from 1969.

**Constructing the Right Martindale Ring of Quotients $Q_r(R)$** Let $R \neq 0$ be a prime ring. The set $\mathscr{J}$ of all nonzero ideals of $R$ is closed under products, and hence

also under finite intersections. We endow the set of all pairs $(f, I)$, where $I \in \mathscr{J}$ and $f : I \to R$ is a right $R$-module homomorphism, by the following relation, which is readily seen to be equivalence: $(f, I) \sim (g, J)$ if $f$ and $g$ coincide on some $K \in \mathscr{J}$ such that $K \subseteq I \cap J$.

Write $[f, I]$ for the equivalence class of $(f, I)$, and denote by $Q_r(R)$ the set of all equivalence classes, endowed with addition and multiplication

$$[f_1, I_1] + [f_2, I_2] = [f_1 + f_2, I_1 \cap I_2],$$
$$[f_1, I_1][f_2, I_2] = [f_1 f_2, I_2 I_1]$$

(note that $f_1 f_2$ is indeed defined on $I_2 I_1$ since $f_2(I_2 I_1) = f_2(I_2) I_1 \subseteq I_1$). To show that these operations are well-defined, assume that $(f_1, I_1) \sim (g_1, J_1)$ and $(f_2, I_2) \sim (g_2, J_2)$. This means that there exist $K_i \in \mathscr{J}$ such that $K_i \subseteq I_i \cap J_i$ and $f_i = g_i$ on $K_i$, $i = 1, 2$. Then $f_1 + f_2 = g_1 + g_2$ on $K_1 \cap K_2 \in \mathscr{J}$ and $f_1 f_2 = g_1 g_2$ on $K_2 K_1 \in \mathscr{J}$. Thus, $(f_1 + f_2, I_1 \cap I_2) \sim (g_1 + g_2, J_1 \cap J_2)$ and $(f_1 f_2, I_2 I_1) \sim (g_1 g_2, J_2 J_1)$. One immediately checks that $Q_r(R)$ is a ring with zero element $[0, R]$ and unity $[\mathrm{id}_R, R]$.

**Definition 7.22** The ring $Q_r(R)$ is called the **right Martindale ring of quotients** of $R$.

The **left Martindale ring of quotients** $Q_l(R)$ is constructed analogously through the left $R$-module homomorphisms. In general, $Q_r(R) \ncong Q_l(R)$.

**Theorem 7.23** *Let $R$ be a nonzero prime ring and let $\mathscr{J}$ be the set of all nonzero ideals of $R$. The ring $Q_r(R)$ has the following properties:*

(a) *$Q_r(R)$ is a unital ring containing $R$ as a subring.*
(b) *For every $q \in Q_r(R)$, there exists an $I \in \mathscr{J}$ such that $qI \subseteq R$.*
(c) *For every $q \in Q_r(R)$ and every $I \in \mathscr{J}$, $qI = 0$ implies $q = 0$.*
(d) *If $I \in \mathscr{J}$ and $f : I \to R$ is a right $R$-module homomorphism, then there exists a $q \in Q_r(R)$ such that $f(x) = qx$ for all $x \in I$.*

*Moreover, these properties characterize $Q_r(R)$ up to isomorphism.*

***Proof*** We already know that $Q_r(R)$ is a unital ring. Note that

$$\iota : R \to Q_r(R), \quad \iota(r) = [L_r, R],$$

where $L_r$ is a left multiplication map, is an embedding of $R$ into $Q_r(R)$ (the injectivity follows from the primeness of $R$). Identifying $R$ with its isomorphic copy $\iota(R)$, we may consider $R$ as a subring of $Q_r(R)$. Thus, (a) holds.

Take a $q = [f, I] \in Q_r(R)$. For every $x \in I$, we have

$$qx = [f, I][L_x, R] = [f L_x, RI] = [L_{f(x)}, R] = f(x) \in R.$$

This proves (b). Moreover, observe that $qI = 0$ implies $f(I) = 0$ and hence $q = [f, I] = 0$. To prove (c), however, we have to show that for every $I' \in \mathscr{J}$, $qI' = 0$ implies $q = 0$. To this end, note that $qI' = 0$ yields $(qI)I' = 0$ since $II' \subseteq I'$. However, $qI \subseteq R$ and so, as $R$ is prime, $qI = 0$, which, as observed, implies the desired conclusion that $q = 0$. Verifying (d) is also easy: If $I \in \mathscr{J}$ and $f : I \to R$ is a right $R$-module homomorphism, then, as noticed at the beginning of this paragraph, $q = [f, I] \in Q_r(R)$ satisfies $f(x) = qx$ for all $x \in I$.

Let $Q$ be an arbitrary ring with properties (a)–(d). Take a $q \in Q$. By (b), there exists an $I \in \mathscr{J}$ such that $qI \subseteq R$. Therefore, we can define the right $R$-module homomorphism $f : I \to R$ by $f(x) = qx$. If $I' \in \mathscr{J}$ also satisfies $qI' \subseteq R$, then $f$ coincides with $f' : I' \to R$, $f(x') = qx'$, on $I \cap I' \in \mathscr{J}$. Hence we see that the map

$$\varphi : Q \to Q_r(R), \quad \varphi(q) = [f, I],$$

is well-defined. It is easy to check that $\varphi$ is a ring homomorphism. The injectivity of $\varphi$ follows from (c), and the surjectivity from (d). Therefore, $Q \cong Q_r(R)$. □

Further properties of $Q_r(R)$ can be extracted from the basic properties (a)–(d). For example, (c) can be extended as follows.

**Lemma 7.24** *For every $q_1, q_2 \in Q_r(R)$ and $I \in \mathscr{J}$, $q_1 I q_2 = 0$ implies $q_1 = 0$ or $q_2 = 0$. Accordingly, $Q_r(R)$ is a prime ring.*

***Proof*** By (b), we may choose $I_i \in \mathscr{J}$ such that $q_i I_i \subseteq R$, $i = 1, 2$. Therefore, $(q_1 I_1)I(q_2 I_2) \subseteq (q_1 I q_2)I_2 = 0$ implies that $q_1 I_1 = 0$ or $q_2 I_2 = 0$ (see Remark 2.25). From (c) we thus obtain $q_1 = 0$ or $q_2 = 0$. □

For further reference we record a straightforward extension of (b).

**Lemma 7.25** *Given $q_1, \ldots, q_n \in Q_r(R)$, there exists an $I \in \mathscr{J}$ such that $q_i I \subseteq R$, $i = 1 \ldots, n$.*

***Proof*** By (b), for each $i$ there exists an $I_i \in \mathscr{J}$ such that $q_i I_i \subseteq R$. Therefore, $I = I_1 \cap \cdots \cap I_n$ has the desired property. □

**Example 7.26** If $R$ is a commutative domain, then $Q_r(R)$ is isomorphic to the field of quotients $Q_Z(R)$ of $R$. To prove this, we must show that $Q_Z(R)$ has properties (a)–(d). Only (d) is not entirely obvious. Assume, therefore, that $f : I \to R$ is a right $R$-module homomorphism. Take a nonzero $a \in I$. For every $x \in I$, we have $f(x)a = f(xa) = f(ax) = f(a)x$ , and hence $f(x) = qx$ where $q = f(a)a^{-1} \in Q_Z(R)$.

For a commutative domain $R$, we thus have $Q_Z(R) \cong Q_{cr}(R) \cong Q_r(R)$. The same is true if $R$ is any of the rings from Examples 7.7–7.9, i.e., $M_n(\mathbb{Z})$, $\left[\begin{smallmatrix} \mathbb{Z} & 2\mathbb{Z} \\ \mathbb{Z} & \mathbb{Z} \end{smallmatrix}\right]$, or $\mathbb{Z} \oplus \mathbb{Z}i \oplus \mathbb{Z}j \oplus \mathbb{Z}k$. This follows from Corollary 7.61 below, but can also be proved from scratch.

**Example 7.27** If $R$ is a simple unital ring, then $Q_r(R) = R$. Indeed, $R$ obviously has properties (a)–(c), and checking (d) is also immediate: If $f : R \to R$ is a right $R$-module homomorphism, then $f(x) = f(1x) = f(1)x$ for all $x \in R$.

**Example 7.28** Let $V$ be an infinite dimensional vector space over a field $F$. Denote by $K$ the subring of $\mathrm{End}_F(V)$ consisting of all finite rank operators. Clearly, every operator in $K$ is a sum of rank one operators $v \otimes h$, defined by $(v \otimes h)(u) = h(u)v$; here, $v \in V$ and $h \in V^*$, the dual of $V$ (see Remark 5.18). It is an easy exercise to show that $K$ is a simple ring without unity. Let $R$ be any ring such that $K \subseteq R \subseteq \mathrm{End}_F(V)$. Note that $R$ is prime and that $K$ is its **minimal ideal**, that is, a nonzero ideal that does not properly contain a nonzero ideal of $R$. Let us show that

$$Q_r(R) \cong \mathrm{End}_F(V).$$

We must verify that $\mathrm{End}_F(V)$ has the four properties (a)–(d). The first three are easy: (a) is trivial, (b) follows from the fact that $K$ is an ideal of $\mathrm{End}_F(V)$, and (c) from the fact that every nonzero ideal of $R$ contains $K$. To verify (d), take a nonzero ideal $I$ of $R$ and a right $R$-module homomorphism $f : I \to R$. Pick a $v_0 \in V$ and $h_0 \in V^*$ so that $h_0(v_0) = 1$. Define $q : V \to V$ by $q(v) = f(v \otimes h_0)(v_0)$. Obviously, $q(v + v') = q(v) + q(v')$. Since

$$f(\lambda k)k' = f(\lambda kk') = f(k(\lambda k')) = \lambda f(k)k'$$

for all $\lambda \in F$ and $k, k' \in K$, i.e., $\big(f(\lambda k) - \lambda f(k)\big)K = 0$, it follows that $f(\lambda k) = \lambda f(k)$, $k \in K$. This yields $q(\lambda v) = \lambda q(v)$, meaning that $q \in \mathrm{End}_F(V)$. Next, for all $v \in V$ and $h \in V^*$,

$$\begin{aligned} f(v \otimes h) &= f(v \otimes h_0 \cdot v_0 \otimes h) = f(v \otimes h_0) \cdot v_0 \otimes h \\ &= f(v \otimes h_0)(v_0) \otimes h = q(v) \otimes h \\ &= q \cdot v \otimes h. \end{aligned}$$

Thus, $f(k) = qk$ for every $k \in K$, and hence $f(x)k = f(xk) = qxk$ for all $x \in I$ and $k \in K$. That is, $(f(x) - qx)K = 0$, which yields $f(x) = qx$. This completes the proof.

## 7.5 The Extended Centroid

Throughout this section, $R$ stands for a nonzero prime ring.

**Definition 7.29** The center of $Q_r(R)$ is called the **extended centroid** of $R$.

The extended centroid of $R$ will be denoted by $C$. As usual, $Z$ will stand for the center of $R$ and, as in the preceding section, $\mathscr{J}$ for the set of all nonzero ideals of $R$. We will continuously refer to properties (a)–(d) of Theorem 7.23.

**Lemma 7.30** *If $q \in Q_r(R)$ is such that $qr = rq$ for all $r \in R$, then $q \in C$. Accordingly, $Z$ is a subring of $C$.*

***Proof*** Pick a $q' \in Q_r(R)$. We must show that $[q, q'] = 0$. By (b), there exists an $I \in \mathscr{J}$ such that $q'I \subseteq R$. For every $x \in I$, $q$ commutes with $x$ and $q'x$, so we have $qq'x = q'xq = q'qx$. Thus, $[q, q']I = 0$, and hence $[q, q'] = 0$ by (c). □

If the elements in $Q_r(R)$ correspond to right $R$-module homomorphisms, then the elements in $C$ correspond to $R$-bimodule homomorphisms—these are, of course, maps that are both left and right $R$-module homomorphisms.

**Lemma 7.31** *If $f : I \to R$, where $I \in \mathscr{J}$, is an $R$-bimodule homomorphism, then there exists a $\lambda \in C$ such that $f(x) = \lambda x$ for all $x \in I$.*

***Proof*** Since $f$ is, in particular, a right $R$-module homomorphism, by (d) there exists a $q \in Q_r(R)$ such that $f(x) = qx$, $x \in I$. On the other hand, since $f$ is also a left $R$-module homomorphism, it follows that $q(rx) = r(qx)$ for all $r \in R$ and $x \in I$. That is, $[q, r]I = 0$, and so $[q, r] = 0$ by (c). Therefore, $\lambda = q \in C$ by Lemma 7.30. □

Conversely, if $\lambda \in C$ and $I \in \mathscr{J}$ are such that $\lambda I \subseteq R$, then $x \mapsto \lambda x$ is an $R$-bimodule homomorphism from $I$ to $R$. The extended centroid, although defined as the center of the right Martindale ring of quotients, is thus a left-right symmetric notion.

**Remark 7.32** Suppose $R$ is an algebra over a field $F$. Every $\alpha \in F$ gives rise to the $R$-bimodule homomorphism $x \mapsto \alpha x$ from $R$ to $R$. Using Lemma 7.31, one therefore easily infers that $F$ embeds canonically in $C$. Accordingly, $F$ can be considered as a subfield of $C$.

**Theorem 7.33** *The extended centroid $C$ of a nonzero prime ring $R$ is a field.*

***Proof*** Take a $\lambda \neq 0$ in $C$ and choose $I \in \mathscr{J}$ satisfying (b). Clearly, $\lambda I \in \mathscr{J}$ by (c). We claim that $f : \lambda I \to R$, $f(\lambda x) = x$, is well-defined. Indeed, $\lambda x = 0$ obviously implies $\lambda I_x = 0$, where $I_x$ is the ideal of $R$ generated by $x$, and hence $x = 0$ by (c). Since $f$ is an $R$-bimodule homomorphism, Lemma 7.31 yields the existence of a $\mu \in C$ such that $f(y) = \mu y$, $y \in \lambda I$. Writing $y = \lambda x$, $x \in I$, we thus obtain $\mu\lambda x = x$. That is, $(\mu\lambda - 1)I = 0$, and so $\mu\lambda = 1$ by (c). Thus, $\lambda$ is invertible. □

We may therefore regard $Q_r(R)$ as a $C$-algebra.

**Definition 7.34** The $C$-subalgebra of $Q_r(R)$ generated by $R$ is called the **central closure** of $R$.

We denote the central closure of $R$ by $R_C$. Obviously, $R_C$ consists of elements of the form $\sum_i \lambda_i r_i$, where $\lambda_i \in C$, $r_i \in R$. Thus, $R_C$ is, like $C$, a left-right

symmetric object. The following symmetric version of Lemma 7.25 therefore holds for elements in $R_C$.

**Lemma 7.35** *Given $q_1, \ldots, q_n \in R_C$, there exists an $I \in \mathscr{J}$ such that $q_i I \subseteq R$ and $I q_i \subseteq R$, $i = 1 \ldots, n$.*

***Proof*** It is enough to prove the lemma for $n = 1$. Namely, as in the proof of Lemma 7.25, we can take the intersection of ideals corresponding to particular elements. Thus, pick a $q = \sum_i \lambda_i r_i \in R_C$. By Lemma 7.25, there exists an $I \in \mathscr{J}$ such that $\lambda_i I \subseteq R$ for each $i$. This clearly implies that $qI \subseteq R$ and, since $\lambda_i$ commutes with elements from $I$, also $Iq \subseteq R$. □

Assume that $R$ has a nonzero center $Z$. Then the field of quotients $\widehat{Z}$ of $Z$ may be viewed as a subfield of $C$. For clarity we record the following simple lemma.

**Lemma 7.36** *If $Z \neq 0$, then $Q_Z(R)$ can be embedded into $R_C$. Moreover, if $\widehat{Z} = C$, then $Q_Z(R) \cong R_C$.*

***Proof*** The first statement follows from Proposition 7.3, and the second one is obvious. □

Rings from Example 7.26 satisfy the condition $\widehat{Z} = C$. In general, $\widehat{Z}$ is not equal to $C$, not even when $Z$ is already a field and thus equal to $\widehat{Z}$.

**Example 7.37** Let $F_0$ be a subfield of $F$. Consider the special case of Example 7.28 where $R = K + F_0 \cdot 1$ (here, 1 is the identity operator). Note that $Z = F_0 \cdot 1$, and hence $\widehat{Z} = Z \cong F_0$. In Example 7.28, we have shown that $Q_r(R) \cong \mathrm{End}_F(V)$, which implies that $C \cong F$. Accordingly, $R_C = K + F \cdot 1$.

This example indicates the intrinsic nature of the notion of the extended centroid. Namely, in the framework of the $F$-linear operators it is more natural to deal with $F$ than with its subfields.

**Definition 7.38** If $R = R_C$, then $R$ is said to be **centrally closed**.

We remark that $Z \neq 0$ implies $C \subseteq R_C$. Indeed, $\lambda = (\lambda z^{-1})z \in R_C$ for every $\lambda \in C$, where $z \in Z \setminus \{0\}$. Therefore, in this case $R$ is centrally closed if and only if $Z = C$. However, $R$ may be centrally closed also when $Z = 0$.

**Example 7.39** Every simple ring $R$, regardless of whether its center is 0 or not, is centrally closed since $\lambda R \subseteq R$ for all $\lambda \in C$ by (b). A concrete example of a simple ring with zero center is the ring $K$ of finite rank operators from Example 7.28.

By a **centrally closed prime algebra** we will mean a centrally closed prime ring viewed as an algebra over its extended centroid. As pointed out above, the center of such an algebra $A$ is either 0 or coincides with the extended centroid. In the latter case, $A$ is a central algebra. The notion of a centrally closed prime algebra can be thus considered as a generalization of the notion of a central simple algebra. An example of a centrally closed prime algebra that is not simple is $\mathrm{End}_F(V)$ where $V$ is an infinite dimensional vector space over $F$ (see Example 7.28).

**Lemma 7.40** *The central closure $R_C$ of any prime ring $R$ is a centrally closed prime algebra over $C$.*

***Proof*** Since $R_C \supseteq R \in \mathscr{J}$, the primeness of $R_C$ follows from Lemma 7.24. Let us denote the extended centroid of $R_C$ by $E$. By Remark 7.32, we may consider $C$ as a subfield of $E$. The lemma will be established by showing that $E = C$. Take an $\epsilon \in E$. By (b), there exists a nonzero ideal $U$ of $R_C$ such that $\epsilon U \subseteq R_C$. Note that $I = \{x \in U \cap R \mid \epsilon x \in R\}$ is an ideal of $R$. We claim that $I \neq 0$. Indeed, choose $u \neq 0$ in $U$ and use Lemma 7.25 to obtain an $J \in \mathscr{J}$ such that $uJ \subseteq R$ and $(\epsilon u)J \subseteq R$. Then $uJ$ is contained in $I$ and is not 0 by (c). We can now define an $R$-bimodule homomorphism $f : I \to R$, $f(x) = \epsilon x$. By Lemma 7.31, there exists a $\lambda \in C$ such that $\epsilon x = \lambda x$, $x \in I$. This readily implies that $\epsilon y = \lambda y$ for every $y$ from the subspace $\widetilde{I}$ of $R_C$ generated by $I$. Observe that $\widetilde{I}$ is an ideal of $R_C$, so (c) yields $\epsilon = \lambda \in C$. □

## 7.6 Linear (In)dependence in Prime Rings

Numerous applications of the Martindale rings of quotients stem from the results we are about to establish. To get a feel for the topic, we first examine the condition studied in the next lemma in division rings.

Let $D$ be a division ring. Suppose $a, b \in D$ satisfy $axb = bxa$ for every $x \in D$. A natural possibility where this occurs is that $a$ and $b$ are linearly dependent over the center of $D$. Actually, this is the only possibility. Indeed, assuming that $a \neq 0$ and then multiplying $axb = bxa$ by $a^{-1}$ on both sides we obtain $xba^{-1} = a^{-1}bx$, which readily implies that $\lambda = ba^{-1} (= a^{-1}b)$ lies in the center of $D$ and satisfies $b = \lambda a$.

We thus have a simple connection between the multiplicative structure of $D$ ($axb = bxa$ for every $x$) and the linear structure of $D$ ($a$ and $b$ are linearly dependent). This little observation can be generalized in various directions.

We use the same notation as in the preceding sections. In particular, $R$ always denotes a nonzero prime ring and $\mathscr{J}$ the set of all nonzero ideals of $R$. By (b) and (c) we refer to conditions of Theorem 7.23.

**Lemma 7.41** *Let $a, b \in Q_r(R)$ and let $I \in \mathscr{J}$. If $axb = bxa$ for all $x \in I$, then $a$ and $b$ are linearly dependent over $C$.*

***Proof*** By Lemma 7.25, there exists a $J \in \mathscr{J}$ such that $aJ \subseteq R$ and $bJ \subseteq R$. By replacing the role of $I$ by $I \cap J$ we see that there is no loss of generality in assuming that $aI \subseteq R$ and $bI \subseteq R$.

We may assume that $a \neq 0$. Then $aI \neq 0$ by (c), and hence $K = IaI \in \mathscr{J}$. We claim that the map $f : K \to R$,

$$f\Big(\sum_i x_i a y_i\Big) = \sum_i x_i b y_i, \quad x_i, y_i \in I,$$

is well-defined. Assume that $\sum_i x_i a y_i = 0$. Multiplying on the right by $zb$, $z \in R$, and using $a(y_i z)b = b(y_i z)a$, it follows that $\left(\sum_i x_i b y_i\right) za = 0$. Hence $\sum_i x_i b y_i = 0$ since $R$ is prime. Note also that $f$ indeed maps to $R$ because $bI \subseteq R$. Obviously, $f$ is an $R$-bimodule homomorphism, so it follows from Lemma 7.31 that there exists a $\lambda \in C$ such that $f(w) = \lambda w$ for every $w \in K$. In particular, $xby = \lambda xay$ for all $x, y \in I$. That is, $I(b - \lambda a)I = 0$, and hence $b = \lambda a$ by Lemma 7.24. □

Even in the basic case where $a, b \in R$ and $I = R$, the involvement of the extended centroid $C$ in Lemma 7.41 is indispensable, i.e., it cannot be replaced by the center $Z$ even when $Z$ is a field. Indeed, a glance at Example 7.37 convinces us of that.

The next lemma treats a considerably more general condition. However, the proof can be easily reduced to the situation treated in Lemma 7.41, as we will see in a moment.

**Lemma 7.42** *Let $a_i, b_i \in Q_r(R)$ and $I \in \mathscr{J}$ be such that*

$$\sum_{i=1}^{n} a_i x b_i = 0 \quad \textit{for all } x \in I.$$

*If $a_1, \ldots, a_n$ are linearly independent over $C$, then each $b_i = 0$. Similarly, if $b_1, \ldots, b_n$ are linearly independent over $C$, then each $a_i = 0$.*

***Proof*** We only prove the first statement. The second statement can be handled similarly. Thus, assume that $a_1, \ldots, a_n$ are linearly independent. The $n = 1$ case follows from Lemma 7.24. We may therefore assume that the desired conclusion holds if the number of summands is less than $n$. Suppose that, say, $b_n \neq 0$. By (b), we can choose a $J \in \mathscr{J}$ such that $b_n J \subseteq R$. Then $xb_n y \in I$ whenever $x \in I$ and $y \in J$, so that $\sum_{i=1}^{n} a_i (xb_n y) b_i = 0$. Since the last term, $a_n x b_n y b_n$, is equal to $-\left(\sum_{i=1}^{n-1} a_i x b_i\right) y b_n$, we can rewrite this identity as

$$\sum_{i=1}^{n-1} a_i x (b_n y b_i - b_i y b_n) = 0 \quad \text{for all } x \in I,\ y \in J.$$

The induction hypothesis yields $b_n y b_i - b_i y b_n = 0$ for all $y \in J$ and $i = 1, \ldots, n-1$. As $b_n \neq 0$ by assumption, Lemma 7.41 tells us that there exist $\lambda_i \in C$ such that $b_i = \lambda_i b_n$, $i = 1, \ldots, n-1$. We can therefore rewrite $\sum_{i=1}^{n} a_i x b_i = 0$ as $\left(\sum_{i=1}^{n} \lambda_i a_i\right) x b_n = 0$, where $\lambda_n = 1$. Hence $\sum_{i=1}^{n} \lambda_i a_i = 0$ by Lemma 7.24, contradicting the linear independence of the $a_i$'s. □

Lemma 7.42 will play a crucial role in the rest of the chapter. Let us point out that this lemma generalizes Lemma 1.42, which was used as one of the main tools in the first chapters.

The following theorem is a refined version of Lemma 7.42.

**Theorem 7.43** *Let $a_i, b_i, c_j, d_j \in Q_r(R)$ and $I \in \mathscr{J}$ be such that*

$$\sum_{i=1}^{n} a_i x b_i = \sum_{j=1}^{m} c_j x d_j \quad \text{for all } x \in I.$$

*If $a_1, \ldots, a_n$ are linearly independent over $C$, then each $b_i$ is a $C$-linear combination of $d_1, \ldots, d_m$. Similarly, if $b_1, \ldots, b_n$ are linearly independent over $C$, then each $a_i$ is a $C$-linear combination of $c_1, \ldots, c_m$.*

***Proof*** We only prove the first statement. The proof is practically the same as that of Lemma 5.10, but we give it anyway. Extend $\{a_1, \ldots, a_n\}$ to a basis of the $C$-linear span of $\{a_1, \ldots, a_n, c_1, \ldots, c_m\}$. Denote the additional basis elements (if there are any) by $a_{n+1}, \ldots, a_p$, and write each $c_j$ as a $C$-linear combination of $a_1, \ldots, a_p$. Our identity can then be rewritten as $\sum_{i=1}^{p} a_i x b_i' = 0$, where $b_i'$, $i \leq n$, is the sum of $b_i$ and a linear combination of the $d_j$'s. Lemma 7.42 shows that each $b_i' = 0$. □

Mentioning a result on tensor products (Lemma 5.10) in the last proof was no coincidence. The above results can be interpreted in terms of tensor products and multiplication algebras. The next theorem is a generalization of Theorem 5.27 which considers central simple algebras.

**Theorem 7.44** *If $A$ is a centrally closed prime algebra, then $A \otimes A^{\mathrm{o}} \cong M(A)$ under the map $a \otimes b \mapsto L_a R_b$.*

***Proof*** The proof is literally the same as that of Theorem 5.27. The only difference is that one has to refer to Lemma 7.42 instead of to Lemma 1.42. □

The next theorem and its corollary make a passage to the topics of the next sections. The theorem shows that the linear dependence of elements in $Q_r(R)$ can be characterized through an identity involving the Capelli polynomials $c_m$ (introduced in Sect. 6.3).

**Theorem 7.45** *Elements $a_1, \ldots, a_m \in Q_r(R)$, $m \geq 2$, are linearly dependent over $C$ if and only if $c_m(a_1, \ldots, a_m, x_1, \ldots, x_{m-1}) = 0$ for all $x_1, \ldots, x_{m-1} \in R$.*

***Proof*** Since $c_m$ is alternating in the first $m$ indeterminates, the "only if" part follows from Lemma 6.8. The "if" part will be proved by induction on $m$. The $m = 2$ case is exactly the content of Lemma 7.41 (with $I = R$). We may thus assume that there exist $x_2, \ldots, x_{m-1} \in R$ such that $c_{m-1}(a_1, \ldots, a_{m-1}, x_2, \ldots, x_{m-1}) \neq 0$. Applying the formula (6.2) from Sect. 6.3 we see that $c_m(a_1, \ldots, a_m, x_1, \ldots, x_{m-1}) = 0$ can be written as

$$\sum_{i=1}^{m} (-1)^{i-1} a_i x_1 c_{m-1}(a_1, \ldots, a_{i-1}, a_{i+1}, \ldots, a_m, x_2, \ldots, x_{m-1}) = 0.$$

The linear dependence of $a_1, \ldots, a_m$ therefore follows from Lemma 7.42. □

Note that the theorem remains valid, with the same proof, if the $x_i$'s are taken from a nonzero ideal of $R$.

**Corollary 7.46** *For every $n \geq 2$, the Capelli polynomial $c_{n^2}$ is an identity of every proper subalgebra of $M_n(F)$, but not of $M_n(F)$ itself.*

***Proof*** A proper subalgebra $B$ of $M_n(F)$ has dimension less than $n^2$, so $c_{n^2}$ is an identity of $B$ by the "only if" part of Theorem 7.45 (the extended centroid of $M_n(F)$ is of course $F$, see Example 7.27). Conversely, choosing linearly independent $a_1, \ldots, a_{n^2} \in M_n(F)$, it follows by the "if" part that there exist $x_i \in M_n(F)$ such that $c_{n^2}(a_1, \ldots, a_{n^2}, x_1, \ldots, x_{n^2-1}) \neq 0$. □

## 7.7 Prime GPI-rings

We now change the subject of discussion. The theory developed in the previous sections will still be present, but just as a tool used in the proofs. The central topic from now on will be the structure of rings satisfying polynomial identities and their generalization called generalized polynomial identities. Although it may appear strange at first glance, we will first treat the latter and after that the former. This order is more suitable for our approach.

Informally, one can think of a **generalized polynomial identity (GPI)** as of a polynomial identity in which some of the indeterminates are replaced by fixed elements. Thus, there exist elements $a_{i_k}$ such that

$$\sum a_{i_0} x_{j_1} a_{i_1} \ldots a_{i_{n-1}} x_{j_n} a_{i_n} = 0 \tag{7.3}$$

holds for any elements $x_{j_l}$.

**Example 7.47** Let $R$ be the ring from Example 7.28, i.e., $R$ is any ring such that $K \subseteq R \subseteq \mathrm{End}_F(V)$ where $V$ is an infinite dimensional vector space over $F$ and $K$ is the subring of $\mathrm{End}_F(V)$ consisting of all finite rank operators. In fact, what we are about to say also makes sense if $V$ is finite dimensional (in this case we clearly have $R = \mathrm{End}_F(V) \cong M_n(F)$), we are now just more interested in rings that do not satisfy nontrivial polynomial identities. Take any rank one operator $a \in K$. One can readily check that for every $x \in R$ there exists a $\lambda_x \in F$ such that $axa = \lambda_x a$. Given $x, y \in R$, we thus have $(axa)ya = \lambda_x \lambda_y a = ay(axa)$. Thus,

$$axaya = ayaxa \quad \text{for all } x, y \in R. \tag{7.4}$$

This is a model of a generalized polynomial identity.

We will confine ourselves to prime rings. Otherwise $axb = 0$, $x \in R$, was a nontrivial generalized polynomial identity, which would make the subject rather muddled.

The standard definition of a prime GPI-ring requires the introduction of an appropriate generalization of a free algebra, which provides a suitable setting for "generalized polynomials". To make our exposition as simple as possible, we will avoid this and give a rather straightforward definition which involves only multilinear identities, i.e., such that each $x_j$ appears in each term of (7.3) exactly once (as for example in (7.4)). More precisely, by a **multilinear generalized polynomial identity** of a prime ring $R$ we mean an identity of the form

$$\sum_{\sigma\in S_n}\sum_{i=1}^{n_\sigma} a_{0i}^{\sigma}x_{\sigma(1)}a_{1i}^{\sigma}x_{\sigma(2)}a_{2i}^{\sigma}\dots a_{n-1,i}^{\sigma}x_{\sigma(n)}a_{ni}^{\sigma}=0 \tag{7.5}$$

for all $x_1,\dots,x_n \in R$, where $a_{ti}^{\sigma}$ are fixed elements in $Q_r(R)$. Assuming that the $a_{ti}^{\sigma}$'s belong to $R$ does not make the treatment much easier, and because of applications it is really useful to involve $Q_r(R)$.

Applying the linearization process, which works just as well as for ordinary polynomial identities, it is possible to show that restricting to multilinear identities does not cause loss of generality; our definition of a prime GPI-ring will in fact be equivalent to the standard one (as given in the treatise [2] of the theory of generalized identities).

Before giving this definition, let us ask ourselves what should be required in order to consider the identity (7.5) as nontrivial, in the sense that it can yield some information about the structure of $R$. Assuming that every $a_{ti}^{\sigma}\neq 0$ is certainly not enough. Say, $ax-xa=0$ only means that $a$ is in the center of $R$ and says nothing about $R$. More generally, from Lemma 7.42 (and Theorem 7.43) we know that the identities of the form $\sum_i a_i x b_i = 0$ are merely consequences of the linear dependence relations among the $a_i$'s and the $b_i$'s. A similar statement can be made for any multilinear generalized polynomial identity in which the $x_i$'s appear in the same order in each of the terms, i.e., $\sum_i a_{0i}x_1a_{1i}x_2a_{2i}\dots a_{n-1,i}x_na_{ni}=0$ (cf. Theorem 7.45 and its proof). Therefore, the least one has to require in order to regard (7.5) as nontrivial is that one of the summations

$$\sum_{i=1}^{n_\sigma} a_{0i}^{\sigma}x_{\sigma(1)}a_{1i}^{\sigma}x_{\sigma(2)}a_{2i}^{\sigma}\dots a_{n-1,i}^{\sigma}x_{\sigma(n)}a_{ni}^{\sigma} \tag{7.6}$$

is not always zero (i.e., is not an identity itself). As we shall see, this is already sufficient. Let us call (7.6) the $\sigma$-**term** of (7.5). Every multilinear generalized polynomial identity is thus the sum of its $\sigma$-terms.

**Definition 7.48** A nonzero prime ring $R$ is said to be a **GPI-ring** if there exists a multilinear generalized polynomial identity of $R$ such that one of its $\sigma$-terms is not an identity of $R$. For convenience, we also consider $R=0$ as a GPI-ring.

The ring $R$ from Example 7.47 is thus a GPI-ring since $axaya\neq 0$ for some $x,y\in R$. We remark that the center of $R$ may be 0 (say, if $R=K$), and that the extended centroid of $R$ may properly contain the field of quotients of the center of $R$

(Example 7.37). This shows that prime GPI-rings do not share all the nice properties of prime PI-rings that will be studied in the next sections; see Theorem 7.56 and Corollary 7.57.

The study of generalized polynomial identities was initiated by S. A. Amitsur. In the first paper on the subject, in 1965, he described primitive GPI-rings. Our goal in this section is to prove a more general result on prime GPI-rings, established by W. S. Martindale in 1969.

As always, $C$ stands for the extended centroid of a prime ring $R$, $R_C$ for the central closure of $R$, and $Q_r(R)$ for the right Martindale ring of quotients of $R$.

**Lemma 7.49** *If $R$ is a nonzero prime GPI-ring, then there exist $a, b \in R \setminus \{0\}$ such that $\dim_C aR_Cb < \infty$.*

***Proof*** Among multilinear generalized polynomial identities of $R$ whose $\sigma$-terms are not all identities, choose one with the minimal number of summands. There must be at least two permutations, $\sigma_1$ and $\sigma_2$, such that the corresponding $\sigma_1$-term and $\sigma_2$-term are not identities. As $\sigma_1^{-1} \neq \sigma_2^{-1}$, we may assume without loss of generality that $\sigma_1^{-1}(1) < \sigma_1^{-1}(2)$ and $\sigma_2^{-1}(2) < \sigma_2^{-1}(1)$. As usual, let $x_1, \ldots, x_n$ be the variables in our identity. Denote by $f_1(x_1, \ldots, x_n)$ the sum of all $\sigma$-terms such that $\sigma^{-1}(1) < \sigma^{-1}(2)$, and by $f_2(x_1, \ldots, x_n)$ the sum of all $\sigma$-terms such that $\sigma^{-1}(2) < \sigma^{-1}(1)$. Thus, $f_1(x_1, \ldots, x_n)$ consists of summands in which $x_1$ appears before $x_2$, and $f_2(x_1, \ldots, x_n)$ consists of summands in which $x_2$ appears before $x_1$. Of course, $f_1(x_1, \ldots, x_n) + f_2(x_1, \ldots, x_n) = 0$ for all $x_1, \ldots, x_n \in R$. In view of the minimality assumption, there exist $w_1, \ldots, w_n \in R$ such that $f_1(w_1, \ldots, w_n) \neq 0$.

Note that the identity $f_1(x, y, w_3, \ldots, w_n) = -f_2(x, y, w_3, \ldots, w_n)$, where $x, y$ are arbitrary and the $w_i$'s are fixed, can be written as

$$\sum_{i=1}^{l} p_i x q_i y r_i = \sum_{j=1}^{m} s_j y t_j x u_j \quad \text{for all } x, y \in R \tag{7.7}$$

and some $p_i, q_i, r_i, s_j, t_j, u_j \in Q_r(R)$. We may assume that $\{p_1, \ldots, p_k\}$ is a maximal linearly independent subset of $\{p_1, \ldots, p_l\}$. Expressing each $p_i$, $i > k$, as a linear combination of $p_1, \ldots, p_k$, we see that (7.7) gets the form

$$\sum_{i=1}^{k} p_i x h_i(y) = \sum_{j=1}^{m} s_j y t_j x u_j \quad \text{for all } x, y \in R, \tag{7.8}$$

where $h_i$ belong to $M(Q_r(R))$, the multiplication algebra of $Q_r(R)$. We are now in a position to apply Theorem 7.43. Accordingly, each $h_i(y)$ is a $C$-linear combination of $u_1, \ldots, u_m$. This holds for every $y \in R$, but since the elements in $A = R_C$ are $C$-linear combinations of the elements in $R$, it actually holds for every $y \in A$. As the summations in (7.8) are not 0 if we take $x = w_1$ and $y = w_2$, $h_i(y) \neq 0$ for some $i$ and $y$.

To summarize, the algebra $M(Q_r(R))$ contains nonzero elements that map $A$ into some finite dimensional spaces. Among them choose $h$ so that it can be written as $h = \sum_{i=1}^{d} L_{a_i} R_{b_i}$ with $d$ minimal. We have to prove that $d = 1$. Suppose this is not true, i.e., $d > 1$. We will derive a contradiction by a similar trick as in the proof of Lemma 7.42. First observe that the minimality of $d$ implies that the sets $\{a_1, \ldots, a_d\}$ and $\{b_1, \ldots, b_d\}$ are linearly independent. By Lemma 7.41, there exists an $r \in R$ such that $b_d r b_1 - b_1 r b_d \neq 0$. Note that $h' = h R_{b_d r} - R_{r b_d} h \in M(Q_r(R))$ can be written as $h' = \sum_{i=1}^{d-1} L_{a_i} R_{b_i'}$ where $b_i' = b_d r b_i - b_i r b_d$. However, this contradicts the assumption that $d$ is minimal. Indeed, $\dim_C h(A) < \infty$ implies $\dim_C h'(A) < \infty$, and the linear independence of $a_1, \ldots, a_{d-1}$ together with $b_1' \neq 0$ implies $h' \neq 0$ by Lemma 7.42.

Thus, there exist nonzero $a_1, b_1 \in Q_r(R)$ such that $\dim_C a_1 A b_1 < \infty$. This readily implies that $\dim_C a_1 r A b_1 s < \infty$ for any $r, s \in R$. By Theorem 7.23 (b)–(c), we may choose $r$ and $s$ so that $a = a_1 r$ and $b = b_1 s$ are nonzero elements in $R$. □

We are now just a step away from Martindale's theorem. However, let us pause for a moment and examine the special case where $R$ is a simple unital ring. This is needed for applications to PI-rings in the next section.

**Lemma 7.50** *A simple unital GPI-ring $R$ is a finite dimensional algebra over its center $Z$.*

***Proof*** Recall that $Q_r(R) = R$ and hence $C = Z$ (see Example 7.27). By Lemma 7.49, there exist $a, b \in R \setminus \{0\}$ such that $L_a R_b$ is a finite rank operator. Since $R$ is simple and unital, $\sum_j u_j a v_j = \sum_k w_k b z_k = 1$ for some $u_j, v_j, w_k, z_k \in R$. Consequently, $\sum_{j,k} L_{u_j} R_{z_k} (L_a R_b) L_{v_j} R_{w_k}$ is the identity operator and has finite rank. This means that $[R : Z] < \infty$. □

**Theorem 7.51 (Martindale)** *A nonzero prime ring $R$ is a GPI-ring if and only if $A = R_C$ is a primitive ring containing an idempotent $e$ such that $Ae$ is a minimal left ideal of $A$ and $eAe$ is a finite dimensional division algebra over $C$.*

***Proof*** We start with the easier "if" part. Let $d = [eAe : C]$. Then

$$s_{d+1}(ex_1e, \ldots, ex_{d+1}e) = 0$$

for all $x_1, \ldots, x_{d+1} \in R$ (see Example 6.14). This is a generalized polynomial identity whose 1-term $ex_1ex_2e \ldots ex_ne$ is not identically zero since $R$ is prime.

We proceed to the "only if" part. By Lemma 7.49, there exist $a, b \in R \setminus \{0\}$ such that $\dim_C aAb < \infty$. If $L \neq 0$ is a left ideal of $A$ such that $L \subseteq Ab$ and $M \neq 0$ is a right ideal of $A$ such that $M \subseteq aA$, then $ML \subseteq aAb$ and hence $\dim_C ML < \infty$; moreover, $ML \neq 0$ because $A$ is prime (Lemma 7.40). Choose $L$ and $M$ so that $ML$ has minimal dimension. We claim that $AML$ is a minimal left ideal of $A$. Let $L'$ be a left ideal such that $0 \neq L' \subseteq AML$. Then $L' \subseteq L$, hence $ML' \subseteq ML$, and so $ML' = ML$ by the dimension assumption. Consequently, $L' \supseteq AML' = AML$, proving that $AML$ is indeed minimal. We can now invoke Lemma 2.61 to obtain an

idempotent $e \in A$ such that $Ae = AML$ and $eAe$ is a division algebra. Moreover, using $e \in AML \subseteq AaAb$ it can be easily shown that $eAe$ is finite dimensional over $C$. Finally, Lemma 4.5 tells us that $A$ is a primitive ring. □

The results from Sect. 4.4 therefore hold for the central closure $R_C$ of a prime GPI-ring $R$. Moreover, the associated division ring of $R_C$ is a finite dimensional algebra in this context. A general prime GPI-ring is therefore quite close to the one considered in Example 7.47.

## 7.8 Primitive PI-rings

We proceed with polynomial identities. So far we have considered them only in algebras, not yet in rings. Thus, the first question that presents itself is how to define a polynomial identity of a ring.

An obvious choice is to give the same definition as for algebras, but requiring that the noncommutative polynomial in question has coefficients in $\mathbb{Z}$ (instead of in $F$). This is all right, but the problem, then, is identifying identities that should be considered nontrivial. For instance, if a ring $R$ has characteristic $p$, then the polynomial $p\xi$ vanishes on $R$. As there is no reason why an algebra of characteristic $p$ should be a PI-algebra, we cannot regard $p\xi$, in spite of being a nonzero polynomial in $\mathbb{Z}\langle\xi\rangle$, as a nontrivial identity of $R$.

Any definition of a nontrivial polynomial identity of a ring is therefore somewhat technical. One possibility is to admit those polynomials whose coefficient at one of their monomials of highest degree is 1. Assuming that such a polynomial is an identity and then employing the linearization process, one derives a multilinear identity in $\mathbb{Z}\langle\xi_1, \xi_2, \dots\rangle$ such that one of its coefficients is 1. This can be easily shown by inspecting the proof of Theorem 6.23, but there is no absolute need for us to bother with details. Let us simply restrict our attention to multilinear identities. Unlike in Chap. 6, we are now interested in the structure of rings satisfying some nontrivial polynomial identity, rather than in particular polynomial identities. We can therefore afford this restriction without loss of generality.

**Definition 7.52** A ring $R$ is said to be a **PI-ring** if there exists an

$$f = \sum_{\sigma \in S_n} k_\sigma \xi_{\sigma(1)} \dots \xi_{\sigma(n)} \in \mathbb{Z}\langle\xi_1, \xi_2, \dots\rangle$$

such that $k_1 = 1$ and $f$ is a **(polynomial) identity** of $R$, i.e., $f(x_1, \dots, x_n) = 0$ for all $x_1, \dots, x_n \in R$.

Of course, it is only important that one of the $k_\sigma$'s equals 1. It is just convenient to require that $k_1 = 1$. This does not harm the generality since we can always rename the indeterminates.

If $R$ is an algebra over a field, then $R$ is a PI-algebra in the sense of Definition 6.12 if and only if $R$ is a PI-ring in the sense of Definition 7.52. This is not obvious. However, we omit the proof.

We will consider only *prime PI-rings*. This will make it possible for us to develop a decent structure theory. Without any restriction on rings we would face examples such as nilpotent rings, Boolean rings, the Grassmann algebra, etc. That is too much to handle.

We will study general prime PI-rings in the next section, while this one is devoted to a smaller class of primitive PI-rings. What are examples of such rings? Recall that there are two basic examples of PI-algebras: commutative algebras and finite dimensional algebras. Since a commutative primitive ring is a field (Lemma 4.7) and hence a 1-dimensional algebra over itself, the only immediate examples of primitive PI-rings are finite dimensional primitive (and hence simple, see Corollary 4.20) algebras. We will show that these obvious examples are, in fact, the only examples.

For simple unital rings, which form a subclass of primitive rings (see Lemma 4.6), this follows immediately from Lemma 7.50. This is because a prime PI-ring $R$ is also a GPI-ring. Indeed, the identity $\sum_{\sigma\in S_n} k_\sigma x_{\sigma(1)}\dots x_{\sigma(n)} = 0$, $k_1 = 1$, can be regarded as a multilinear generalized polynomial identity (7.5) with $a_{ti}^{\sigma}$ being integer multiples of 1; if $R \neq 0$ then its 1-term $x_1 \dots x_n$ is not an identity since $R$, as a prime ring, is not nilpotent. Thus, the following holds.

**Lemma 7.53** *A simple unital PI-ring $R$ is a finite dimensional algebra over its center $Z$.*

The assumption that $R$ is unital is actually redundant. Theorem 7.56 below shows that a simple PI-ring has nonzero center, from which one easily deduces that it is automatically unital. Anyway, Lemma 7.53 in its present form is sufficient for our purposes. We will use it not only in the proof of the next theorem, but also in the next section. Since this lemma was indirectly derived from Lemma 7.49 which concerns prime GPI-rings, one might wonder whether the proof of the latter becomes simpler if we adapt it to the case where $R$ is a simple unital PI-ring. However, besides notational simplifications and the omission of the last paragraph, there are no essential differences.

The following theorem was proved in 1948 by I. Kaplansky in his pioneering paper on polynomial identities.

**Theorem 7.54 (Kaplansky)** *A primitive PI-ring $R$ is a finite dimensional simple algebra over its center $Z$.*

***Proof*** In light of the Jacobson density theorem (Theorem 4.17), we may assume that $R$ is a dense ring of linear operators of a vector space $V$ over a division ring $\Delta$. We will now mimic the proof of Lemma 6.43. Let $f$ be a polynomial as in Definition 7.52. Suppose there exist linearly independent vectors $u_0, u_1, \dots, u_n \in V$. By density, we can choose $h_1, \dots, h_n \in R$ so that $h_i(u_j) = \delta_{ij}u_{j-1}$, $1 \leq i, j \leq n$. Then $h_1 \dots h_n(u_n) = u_0$, while for any product of the $h_i$'s in a different order we have $h_{i_1}h_{i_2}\dots h_{i_n}(u_n) = 0$. Hence, $f(h_1, \dots, h_n)(u_n) = u_0 \neq 0$, a contradiction.

Consequently, $\dim_\Delta V \leq n$, and so $R = \mathrm{End}_\Delta(V)$ (Corollary 4.14). Thus, $R$ is a simple unital ring (see Theorem 3.33 and Lemma 1.18). Now apply Lemma 7.53. □

## 7.9 Prime PI-rings

The class of prime PI-rings is substantially larger than the class of primitive PI-rings. First of all, it contains all commutative domains. Rings from Examples 7.7–7.9, i.e., $M_n(\mathbb{Z})$, $\left[\begin{smallmatrix}\mathbb{Z} & 2\mathbb{Z}\\ \mathbb{Z} & \mathbb{Z}\end{smallmatrix}\right]$, and $\mathbb{Z} \oplus \mathbb{Z}i \oplus \mathbb{Z}j \oplus \mathbb{Z}k$, are all examples of prime PI-rings. Indeed, proving that they are prime is easy, and, as subrings of finite dimensional algebras ($M_n(\mathbb{Q})$, $M_2(\mathbb{Q})$, and $\mathbb{H}$, respectively), they satisfy standard polynomials and are hence PI-rings. A description of prime PI-rings is therefore necessarily somewhat entangled, especially when compared with that of primitive PI-rings. Examples just given indicate that rings of quotients naturally come into play in this context.

Our first lemma already says a lot, yet it does not give the full picture. We use the standard notation: $R$ is a nonzero prime ring with center $Z$, extended centroid $C$, and central closure $R_C$.

**Lemma 7.55** *If $R$ is a prime PI-ring, then $A = R_C$ is a finite dimensional central simple algebra.*

***Proof*** Let $U$ be a nonzero ideal of $A$. Every multilinear identity of $R$ is obviously also an identity of $A$, and hence also of $U$. Let $f = \sum_{\sigma \in S_n} k_\sigma \xi_{\sigma(1)} \dots \xi_{\sigma(n)} \in \mathbb{Z}\langle \xi_1, \xi_2, \dots \rangle$, $k_1 = 1$, be a multilinear identity of $U$ of minimal degree $n$. Write

$$f = g\xi_n + \sum_i g_i \xi_n m_i, \tag{7.9}$$

where $g, g_i$ are multilinear polynomials and each $m_i$ is a monomial with leading coefficient 1 and of degree at least 1. Since $g$ has degree $n-1$ and $\xi_1 \dots \xi_{n-1}$ is one of its monomials, $g$ is not an identity of $U$. Let $u_1, \dots, u_{n-1} \in U$ be such that $u = g(u_1, \dots, u_{n-1}) \neq 0$. According to (7.9), $f(u_1, \dots, u_{n-1}, x) = 0$ can be written as $ux = \sum v_i x w_i$ for all $x \in U$ and some $v_i \in A \cup \mathbb{Z}$, $w_i \in U$. This in particular holds for all $x$ in $R \cap U$, which is an ideal of $R$ different from 0 (the latter follows from Theorem 7.23). Writing $ux$ as $ux1$, we thus infer from Theorem 7.43 that 1 is a $C$-linear combination of the $w_i$'s. Therefore, $1 \in A$ and hence $C \subseteq A$, yielding $1 \in \sum_i Cw_i \subseteq U$. Thus $U = A$, showing that $A$ is a simple unital algebra. By Lemma 7.53, $A$ is finite dimensional over its center (which is equal to $C$ by Lemma 7.30). □

Given an arbitrary ring $R$ and its proper ideal $I$, it is quite likely that $I$ does not contain nonzero elements from the center $Z$ of $R$. For example, this is certainly the case if $Z$ is a field. On the other hand, take $M_n(\mathbb{Z})$, which is a standard example of a prime PI-ring. One can easily check that its nonzero ideals have the form $M_n(k\mathbb{Z})$ for

some $k \in \mathbb{N}$ (see Corollary 5.44). Each of them contains nonzero integer multiples of the identity matrix. Thus, nonzero ideals of $M_n(\mathbb{Z})$ intersect the center of $M_n(\mathbb{Z})$ nontrivially. We will now show that this is true for every prime PI-ring.

**Theorem 7.56** *If $I \neq 0$ is an ideal of a prime PI-ring $R$, then $I \cap Z \neq 0$.*

***Proof*** Take an arbitrary nonzero $C$-linear functional $\gamma$ on $A = R_C$. Since we know by Lemma 7.55 that $A$ is a finite dimensional central simple algebra, Theorem 1.43 shows that there exists a $g \in M(A)$ such that $g(x) = \gamma(x)1$, $x \in A$. Choose $a_i, b_i \in A$ so that $g = \sum_{i=1}^{n} L_{a_i} R_{b_i}$ and the sets $\{a_1, \ldots, a_n\}$ and $\{b_1, \ldots, b_n\}$ are linearly independent (see Remark 1.41). Lemma 7.35 implies that there exists a nonzero ideal $J$ of $R$ such that $a_i J \subseteq R$ and $J b_i \subseteq R$ for every $i$. Then $K = JIJ$ is also a nonzero ideal of $R$ and satisfies $g(K) \subseteq I \cap C$. Since $g(K) \neq 0$ by Lemma 7.42, it follows that $I \cap C \neq 0$. Of course, $I \cap C = I \cap Z$ since $I \subseteq R$. □

Theorem 7.56 actually holds even for semiprime rings. This was proved by L. H. Rowen in 1973 (for the proof, see for example [12, p. 419]).

**Corollary 7.57** *If $R$ is a prime PI-ring, then $Z \neq 0$ and $C = \widehat{Z}$, the field of quotients of $Z$. Moreover, if $Z$ is a field, then $R$ is a finite dimensional central simple algebra.*

***Proof*** Taking $R$ for $I$ in Theorem 7.56 we obtain $Z \neq 0$. Let $\lambda \in C$. By Theorem 7.23 (b), there exists a nonzero ideal $I$ of $R$ such that $\lambda I \subseteq R$. Using Theorem 7.56 again we obtain a nonzero $z \in I \cap Z$. Hence, $\lambda z \in R \cap C = Z$, and so $\lambda = z' z^{-1}$ with $z, z' \in Z$. This proves that $C = \widehat{Z}$. Finally, if $Z$ is a field, then $Z = \widehat{Z} = C$, and hence $R = R_C$ is a finite dimensional central simple algebra by Lemma 7.55. □

The next theorem sharpens Lemma 7.55. It is usually attributed to E. Posner who proved a slightly weaker version in 1960.

**Theorem 7.58 (Posner)** *If $R$ is a prime PI-ring, then $Q_Z(R)$ is a finite dimensional central simple algebra.*

***Proof*** Corollary 7.57, together with Lemma 7.36, shows that $Q_Z(R) \cong R_C$. Now use Lemma 7.55. □

By Wedderburn's theorem, $Q_Z(R)$ is isomorphic $M_n(D)$ for some finite dimensional central division algebra $D$ and $n \in \mathbb{N}$. The point of Theorem 7.58 is that every prime PI-ring is very close to such an algebra. Namely, every element in $Q_Z(R)$ can be written as $rz^{-1}$ where $r \in R$ and $z \in Z$. For example, every matrix in $M_n(\mathbb{Q})$ is of the form $\frac{1}{z}T$, where $z \in \mathbb{Z} \setminus \{0\}$ and $T \in M_n(\mathbb{Z})$.

The converse to Theorem 7.58 is trivially true: If $Q_Z(R)$ is finite dimensional, then it satisfies a standard identity and is hence a PI-ring, and so is its subring $R$. A somewhat rougher characterization of prime PI-rings reads as follows.

**Corollary 7.59** *A prime ring $R$ is a PI-ring if and only if $R$ can be embedded into $M_m(K)$ for some $m \in \mathbb{N}$ and some field $K$.*

***Proof*** The "if" part is clear. Let $R$ be a PI-ring. Then $Q_Z(R)$ is a finite dimensional simple algebra over its center $\widehat{Z}$. Let $K$ be a splitting field for $Q_Z(R)$ (for example, the algebraic closure of $\widehat{Z}$). The scalar extension of $Q_Z(R)$ to $K$ is then isomorphic to $M_m(K)$. □

**Remark 7.60** If $R$ is a prime PI-algebra over a field $F$, then $F$ is a subfield of $\widehat{Z}$ (see Remark 7.32), and so $K$ is an extension field of $F$.

Corollary 7.59 implies that every prime PI-ring $R$ satisfies a standard polynomial. Indeed, if $R$ embeds in $M_m(K)$, then, by the Amitsur-Levitzki theorem, $R$ satisfies $s_{2m}$.

As observed in the proof of Theorem 7.58, $Q_Z(R) \cong R_C$ holds for a prime PI-ring $R$. For clarification, we add the following corollary.

**Corollary 7.61** *If $R$ is a prime PI-ring, then $Q_Z(R) \cong Q_r(R) \cong Q_{cr}(R)$.*

***Proof*** It is clear that $Q_Z(R)$ satisfies conditions (a)–(c) of Theorem 7.23. It remains to prove (d). Take a nonzero ideal $I$ of $R$ and a right $R$-module homomorphism $f : I \to R$. By Theorem 7.56, there exists a nonzero $c \in I \cap Z$. We have $f(c)x = f(cx) = f(xc) = f(x)c$ for every $x \in I$, and so $f(x) = c^{-1}f(c)x$. Since $c^{-1}f(c) \in Q_Z(R)$, this proves that $Q_Z(R) \cong Q_r(R)$.

To prove that $Q_{cr}(R) \cong Q_Z(R)$, we have to verify that $Q_Z(R)$ satisfies conditions of Definition 7.10. Only the fulfillment of (b) is not entirely obvious. Take a regular element $s$ in $R$. Then $s$ is clearly also regular in $Q_Z(R)$. But then, since $Q_Z(R)$ is a finite dimensional unital algebra, $s$ is invertible in $Q_Z(R)$ (see Proposition 1.10 (a)). This proves (b). □

## 7.10 The Universal Division Algebra

In the remaining two sections, we will present some applications of the theory developed in this chapter.

We begin this section with a question: If polynomials $f, g \in F\langle X\rangle$, where $X = \{\xi_1, \xi_2, \dots\}$, are such that $fg$ is an identity of $M_n(F)$, is then one of $f$ and $g$ an identity of $M_n(F)$? Taking a moment to think about this question, the reader may find it intriguing.

In general, the answer is negative.

**Example 7.62** Suppose $F$ is a finite field. Then $M_n(F)$ is finite too. Write $M_n(F) = \{x_1, \dots, x_t\}$. Since the set of powers of each $x_i$ is finite, there exist natural numbers $r_i < s_i$ such that $x_i^{r_i} = x_i^{s_i}$. Therefore, $M_n(F)$ satisfies the identity

$$(\xi^{s_1} - \xi^{r_1}) \cdots (\xi^{s_t} - \xi^{r_t}) = (\xi^{s_1 - r_1} - 1) \cdots (\xi^{s_t - r_t} - 1)\xi^{r_1 + \cdots + r_t}.$$

However, neither $(\xi^{s_1 - r_1} - 1) \cdots (\xi^{s_t - r_t} - 1)$ nor $\xi^{r_1 + \cdots + r_t}$ is an identity of $M_n(F)$. (See also Exercise 7.21, which provides a more specific identity of $M_n(F)$ with $|F| = q$.)

We will show, however, that the answer is positive if the field $F$ is infinite. To this end, we first recast the question.

Recall from Proposition 6.38 that if $F$ is infinite, then the relatively free algebra $F\langle X\rangle/\mathrm{Id}(M_n(F))$ is isomorphic to the algebra of $n \times n$ generic matrices $\mathrm{GM}_n(F)$; this is the subalgebra of $M_n(F[\Omega])$, $\Omega = \{\omega_{jk}^{(i)} \mid j,k = 1,\ldots,n,\ i = 1,2,\ldots\}$, generated by all $n \times n$ generic matrices $g^{(i)} = (\omega_{jk}^{(i)})$, $i = 1,2,\ldots$, and the identity matrix. Our question can therefore be reformulated as whether $\mathrm{GM}_n(F)$ has no zero-divisors, i.e., whether it is a domain. In 1955, S. A. Amitsur gave an affirmative answer.

**Theorem 7.63 (Amitsur)** *If $F$ is an infinite field, then $\mathrm{GM}_n(F)$ is a domain.*

***Proof*** By Lemma 2.27, it suffices to show that $\mathrm{GM}_n(F)$ is prime and reduced.

We first show that $\mathrm{GM}_n(F)$ is prime. Let $F(\Omega)$ be the field of rational functions in $\Omega$ (i.e., $F(\Omega)$ is the field of quotients of $F[\Omega]$). Consider $\mathrm{GM}_n(F)$ as a subring of the $F(\Omega)$-algebra $M_n(F(\Omega))$. Denote by $\overline{\mathrm{GM}_n(F)}$ the $F(\Omega)$-subalgebra of $M_n(F(\Omega))$ generated by $\mathrm{GM}_n(F)$. Clearly, $\overline{\mathrm{GM}_n(F)}$ is just the linear span of $\mathrm{GM}_n(F)$. Corollary 7.46 tells us that the Capelli polynomial $c_{n^2}$ is not an identity of $M_n(F)$. Consequently, it is not an identity of $\mathrm{GM}_n(F)$, and hence neither of $\overline{\mathrm{GM}_n(F)}$. Using Corollary 7.46 again, it follows that $\overline{\mathrm{GM}_n(F)} = M_n(F(\Omega))$. Now, suppose that $a, b \in \mathrm{GM}_n(F)$ satisfy $axb = 0$ for every $x \in \mathrm{GM}_n(F)$. Then this also holds for every $x \in \overline{\mathrm{GM}_n(F)} = M_n(F(\Omega))$, and hence $a = 0$ or $b = 0$ since $M_n(F(\Omega))$ is prime. This proves that $\mathrm{GM}_n(F)$ is prime too.

It remains to prove that $\mathrm{GM}_n(F)$ is reduced. That is, we must show that if $f^2$ is an identity of $M_n(F)$, then so is $f$. We start the proof by invoking Corollary 1.62, which tells us that there exists a division algebra $D$ over $F$ such that the dimension of $D$ over its center $Z$ is $n^2$. Since identities of $M_n(F)$ are stable (Theorem 6.29), $f^2$ is also an identity of $M_n(Z) \cong Z \otimes M_n(F)$. Corollary 6.30 shows that $M_n(Z)$ and $D$ have the same identities. In particular, $f^2$ is an identity of $D$. However, since $D$ has no nonzero nilpotents, this readily implies that $f$ is an identity of $D$. Consequently, $f$ is also an identity of $M_n(Z)$, and hence also of $M_n(F)$. □

It is standard to transform problems on division algebras to problems on more familiar matrix algebras. What is particularly remarkable about the proof just given is that it shows that the other way around is also possible. Indeed, in its second part, a problem on matrix algebras has been transformed to the same problem on division algebras.

We can now apply Posner's theorem (Theorem 7.58) to $\mathrm{GM}_n(F)$ since it is a prime PI-ring (having, incidentally, the same identities as $M_n(F)$—see Remark 6.39)). Now, as $\mathrm{GM}_n(F)$ is a domain, so is $Q_Z(\mathrm{GM}_n(F))$. Being a finite dimensional central simple algebra, $Q_Z(\mathrm{GM}_n(F))$ is therefore a division algebra.

**Definition 7.64** The ring $Q_Z(\mathrm{GM}_n(F))$ is called the **universal division algebra of degree** $n$ and is denoted by $\mathrm{UD}_n(F)$.

As the name suggests, the dimension of $\mathrm{UD}_n(F)$ over its center $Z = Z(\mathrm{UD}_n(F))$ is $n^2$. To prove this, first note that since $\mathrm{GM}_n(F)$ satisfies the standard identity $s_{2n}$,

so does $\mathrm{UD}_n(F)$. Using Corollary 6.30 together with Lemma 6.43, we thus see that $[\mathrm{UD}_n(F) : Z] \leq n^2$. On the other hand, since $\mathrm{UD}_n(F)$ contains $\mathrm{GM}_n(F) \supseteq M_n(F)$, it does not satisfy identities of degree less than $2n$. This implies, in a similar way, that $[\mathrm{UD}_n(F) : Z] \geq n^2$. Consequently,

$$[\mathrm{UD}_n(F) : Z(\mathrm{UD}_n(F))] = n^2.$$

We have thus found new examples of central division algebras of dimension $n^2$ for any $n$. One should say, however, that $\mathrm{UD}_n(F)$ is a valuable theoretical tool and is therefore much more than just an example.

## 7.11 Central Polynomials

As usual, we denote the set $\{\xi_1, \xi_2, \dots\}$ by $X$.

**Definition 7.65** Let $A$ be an algebra over a field $F$. A polynomial $f(\xi_1, \dots, \xi_n) \in F\langle X\rangle$ with zero constant term is said to be a **central polynomial** for $A$ if $f(x_1, \dots, x_n)$ lies in the center of $A$ for all $x_1, \dots, x_n \in A$, but $f$ is not an identity of $A$.

In other words, $f(\xi_1, \dots, \xi_n)$ is a central polynomial if $f$ is not an identity, but $[f, \xi_{n+1}]$ is. This shows that central polynomials can exist only on PI-algebras. The condition that $f$ must have zero constant term excludes the trivial possibility where $f$ is a sum of a constant polynomial and an identity.

**Example 7.66** The polynomial $[\xi_1, \xi_2]$ is a central polynomial for the Grassmann algebra $G$; see Example 6.15.

**Example 7.67** The polynomial $[\xi_1, \xi_2]^2$ is a central polynomial for the algebra $M_2(F)$. One can check this by a short calculation. On the other hand, this follows from the Cayley-Hamilton theorem applied to the commutator of two $2 \times 2$ matrices. Indeed, the characteristic polynomial of $A \in M_2(F)$ is $p_A(\omega) = \omega^2 - \mathrm{tr}(A)\omega + \det(A)$, and $\mathrm{tr}(A) = 0$ if $A$ is a commutator. A similar, but multilinear polynomial $[\xi_1, \xi_2][\xi_3, \xi_4] + [\xi_3, \xi_4][\xi_1, \xi_2]$ is also central for $M_2(F)$.

There is no obvious way to generalize Example 7.67 to matrices of larger size. In fact, finding examples of central polynomials for $M_n(F)$ is not an easy task. However, they do exist. This was proved independently by E. Formanek and Yu. P. Razmyslov in the early 1970s.

We will give the proof of the Formanek-Razmyslov theorem only under the additional assumption that $F$ is infinite. In this way, we will be able to demonstrate the power of the abstract theory that we developed. Constructive proofs, such as given by Formanek and Razmyslov, work regardless of the cardinality of $F$. On the other hand, the case where $F$ is finite can be reduced to the case where $F$ is infinite. However, providing all the details would take too much space.

First, we reformulate the problem of the existence of central polynomials as follows.

**Lemma 7.68** *Central polynomials for an $F$-algebra $A$ exist if and only if the center of the relatively free algebra $F\langle X\rangle/\mathrm{Id}(A)$ has elements different from scalar multiples of unity.*

***Proof*** If $f$ is a central polynomial, then $f+\mathrm{Id}(A)$ lies in the center of $F\langle X\rangle/\mathrm{Id}(A)$ and is not a scalar multiple of 1. Conversely, if $f+\mathrm{Id}(A)$ is a central element and is not a scalar multiple of 1, then $f-\alpha_0$, where $\alpha_0$ is the constant term of $f$, is a central polynomial for $A$. □

**Theorem 7.69 (Formanek-Razmyslov)** *For every $n\in\mathbb{N}$, there exists a central polynomial for $M_n(F)$.*

***Proof*** As mentioned, we restrict ourselves to the case where $F$ is infinite. By Theorem 7.63, $\mathrm{GM}_n(F)$ is then a prime PI-ring (even a domain). Note that $\mathrm{GM}_n(F)$ is not a simple ring since the $F$-subalgebra generated by all generic matrices $g^{(i)}$ (but without the identity matrix) is its proper ideal. Theorem 7.56 therefore shows that the center of $\mathrm{GM}_n(F)$ contains elements different from scalar multiples of 1. Since $\mathrm{GM}_n(F)\cong F\langle X\rangle/\mathrm{Id}(M_n(F))$ by Proposition 6.38, the desired conclusion follows from Lemma 7.68. □

The existence of central polynomials and universal division algebras opens new horizons. Indeed, there is much more to say. However, every book must have an end.

## Exercises

**7.1** Let $R$ be a ring such that its center $Z$ is nonzero and all elements in $Z\setminus\{0\}$ are regular. Show that $Q_Z(S)\cong Q_Z(R)$ for every ring $S$ such that $R\subseteq S\subseteq Q_Z(R)$.

**7.2** Find an example of a ring $R$ such that $R\subsetneq Q_Z(R)\subsetneq Q_{rc}(R)$.

**7.3** Let $F\subseteq K$ be fields and let $T$ be a ring such that $F\subseteq T\subseteq K$ and $K$ is the field of quotients of $T$. Note that $R=\left[\begin{smallmatrix}F & T\\ 0 & T\end{smallmatrix}\right]$ is a ring under the standard matrix operations. Prove that $R$ satisfies the right Ore condition and describe its right classical ring of quotients. Show by an example that $R$ does not necessarily satisfy the left Ore condition.

**7.4** Show that every PI-algebra $A$ satisfies a nonzero identity in two indeterminates. Hence derive that if $A$ is a domain, then it is a right (and left) Ore domain.

**Remark** The latter can be also deduced from Proposition 7.19, as well as from the structure theory (see Corollary 7.61).

**7.5** Let $I$ be a nonzero ideal of a prime ring $R$. Show that $Q_r(I)\cong Q_r(R)$.

**7.6** Show that an automorphism of a nonzero prime ring $R$ can be uniquely extended to an automorphism of $Q_r(R)$.

**7.7** Let $\varphi$ be an automorphism of a prime ring $R$ and let $0 \neq q \in Q_r(R)$ be such that $q\varphi(x) = xq$ for all $x \in R$. Show that $q$ is invertible in $Q_r(R)$ (and so $\varphi(x) = q^{-1}xq$ for all $x \in R$).

**7.8** Let $R \neq 0$ be a prime ring. Denote by $Q_s(R)$ the set of all $q \in Q_r(R)$ for which there exists a nonzero ideal $J$ of $R$ such that $Jq \subseteq R$. Show that $Q_s(R)$ is a subring of $Q_r(R)$ which contains $R$ and whose center is the extended centroid $C$ (thus, $R \subseteq R_C \subseteq Q_s(R) \subseteq Q_r(R)$). We call $Q_s(R)$ the **symmetric Martindale ring of quotients** of $R$. Note that $q$ from the preceding exercise actually lies in $Q_s(R)$. Show that if $R$ is a domain, then so is $Q_s(R)$.

**7.9** Show that $\xi$ is a zero-divisor in $Q_r(F\langle\xi, \eta\rangle)$. Thus, $Q_r(F\langle\xi, \eta\rangle)$ is not a domain and it properly contains $F\langle\xi, \eta\rangle$. Show, however, that $F\langle\xi, \eta\rangle$ is centrally closed.

**7.10** Let $R$ be a ring. Note that the set of all $R$-bimodule homomorphisms from $R$ to $R$ forms a ring under the standard operations. We denote it by $W$ and call the **centroid** of $R$. Show that:

(a) If $R$ is unital, then $W$ is isomorphic to the center $Z$ of $R$.
(b) If $R^2 = R$, then $W$ is commutative.
(c) If $R$ simple, then $W$ is isomorphic to the extended centroid $C$ of $R$.
(d) If $R$ is prime, then $W$ can be embedded into $C$; show by an example that $W$ is not necessarily isomorphic to $C$.

**7.11** Let $d$ and $d'$ be derivations of a ring $R$. Check that $dd' - d'd$ is again a derivation, while there is no reason to believe that $dd'$ is a derivation. In fact, show that if $R$ is prime and $\mathrm{char}(R) \neq 2$, then $dd'$ is a derivation only when $d = 0$ or $d' = 0$. The assumption on $\mathrm{char}(R)$ is necessary; show that if $R$ is semiprime, then $d^2$ is a derivation if and only if $2d = 0$.

**Remark** The first of these two results was obtained by E. Posner in 1957, before the discovery of the extended centroid. However, the proof becomes shorter by making use of it.

**7.12** Let $R$ be a prime ring with extended centroid $C$. Suppose that a nonzero derivation $d$ of $R$ and $q \in Q_r(R) \setminus C$ satisfy $d(x)q = -qd(x)$ for all $x \in R$. Show that there exists a $\lambda \in C$ such that $d(x) = \lambda[q, x]$ for all $x \in R$, and that $q^2 \in C$.

**7.13** Let $R$ be a ring. A biadditive map $D : R \times R \to R$ is called a **biderivation** if the maps $x \mapsto D(x, y)$ and $x \mapsto D(y, x)$ are derivations for every $y \in R$. Suppose that $R$ is prime and noncommutative. Show that the extended centroid $C$ contains an element $\lambda$ such that $D(x, y) = \lambda[x, y]$ for all $x, y \in R$. Hence derive that every additive map $f : R \to R$ satisfying $f(x)x = xf(x)$ for every $x \in R$ is of the form $f(x) = \lambda x + \mu(x)$, where $\lambda \in C$ and $\mu : R \to C$.

*Hint:* Begin by computing $D(xy, zw)$ in two different ways.

**7.14** Let $R$ be a domain. Show that if $R$ is a GPI-ring, then it is a PI-ring.

**7.15** Let $R$ be a simple unital ring. Prove that $R$ is isomorphic to the ring of $n \times n$ matrices over a field if and only if there exists a nonzero $a \in R$ such that $axaya = ayaxa$ for all $x, y \in R$.

**7.16** Let $R$ be a prime ring and let $a_1, a_2, a_3 \in R$. Prove that $a_1 x a_2 x a_3 = 0$ for all $x \in R$ implies that at least one $a_i$ is 0. Show by an example that $a_1 x a_2 x a_3 x a_4 = 0$ for all $x \in R$ does not, in general, imply that one of the $a_i$'s is 0.

*Hint:* Theorem 7.51 suggests where to look for examples. However, they exist only under a strong restriction on $\mathrm{char}(R)$.

**7.17** True or False:

(a) Every nonzero ideal of a prime PI-ring contains a regular element.
(b) Every nonzero ideal of a prime GPI-ring contains a regular element.
(c) Every left zero-divisor in a prime PI-ring is also a right zero-divisor.
(d) Every left zero-divisor in a prime GPI-ring is also a right zero-divisor.

**7.18** True or False:

(a) If a nonzero ideal of a prime ring $R$ is a PI-ring, then $R$ is a PI-ring.
(b) If a nonzero left ideal of a prime ring $R$ is a PI-ring, then $R$ is a PI-ring.

**7.19** Show that a prime PI-ring with finite center is isomorphic to the ring of $n \times n$ matrices over a finite field.

**7.20** Let $R$ be prime PI-ring, let $\widehat{Z}$ be the field of quotients of the center $Z$ of $R$, and let $n \in \mathbb{N}$. Prove that the following statements are equivalent:

(i) $Q_Z(R)$ is a central simple algebra over $\widehat{Z}$ with $\dim_{\widehat{Z}} Q_Z(R) = n^2$.
(ii) $R$ satisfies $s_{2n}$, but does not satisfy nonzero identities of degree less than $2n$.
(iii) There exists a field $K$ such that $R$ can be embedded into $M_n(K)$, and $M_n(K)$ satisfies the same multilinear identities as $R$ (accordingly, $R$ cannot be embedded in $M_{n-1}(H)$ for any commutative ring $H$).
(iv) The degree of algebraicity over $\widehat{Z}$ of every element in $R$ is at most $n$, and there exist elements in $R$ whose degree of algebraicity is $n$.

**7.21** Let $n \in \mathbb{N}$ and let $F$ be a finite field with $q$ elements. Show that there exists an $r \in \mathbb{N}$ such that $(\xi^{q^r} - \xi)^n$ is an identity of $M_n(F)$.

**7.22** Let $F$ be an infinite field. Suppose $f(\xi_1, \ldots, \xi_m) \in F\langle \xi_1, \xi_2, \ldots \rangle$ is not an identity of $M_n(F)$. Show that there exist $a_i \in M_n(F)$ such that the matrix $f(a_1, \ldots, a_m)$ is invertible.

*Hint:* The universal division algebra.

**7.23** A polynomial $f(\xi_1, \ldots, \xi_m) \in F\langle \xi_1, \xi_2, \ldots \rangle$ is called a **weak identity** of $M_n(F)$ if $f(a_1, \ldots, a_m) = 0$ for all trace zero matrices $a_i \in M_n(F)$. Find a weak identity of $M_2(F)$ of degree 3.

*Hint:* Example 7.67.

**7.24** Show that $[\xi_1, \xi_2]$ cannot be a central polynomial for a prime algebra.

**Remark** This can be proved by applying the theory, but also directly, by elementary means.

**7.25** Show that a central polynomial for $M_n(F)$ is an identity of $M_{n-1}(F)$.

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